

SEMI E30-1103

GENERIC MODEL FOR COMMUNICATIONS AND CONTROL OF MANUFACTURING EQUIPMENT (GEM)

This standard was technically approved by the Global Information & Control Committee and is the direct responsibility of the Japanese Information & Control Committee. Current edition approved by the Japanese Regional Standards Committee on August 8, 2003. Initially available at www.semi.org October 2003; to be published November 2003. Originally published in 1992; previously published July 2003.

CONTENTS

1 Introduction

- 1.1 Revision History
- 1.2 Scope
- 1.3 Intent

Figure 1.1, GEM Scope

- 1.4 Overview

Figure 1.2, GEM Components

- 1.5 Applicable Documents

2 Definitions

3 State Models

- 3.1 State Model Methodology

- 3.2 Communications State Model

Figure 3.0, Example Equipment Component Overview

Figure 3.2.1, Communications State Diagram

Table 3.2, Communications State Transition Table

- 3.3 Control State Model

Figure 3.3, Control State Model

Table 3.3, CONTROL State Transition Table

- 3.4 Equipment Processing States

Figure 3.4, Processing State Diagram

Table 3.4, Processing State Transition Table

4 Equipment Capabilities and Scenarios

- 4.1 Establish Communications

- 4.2 Data Collection

Figure 4.2.1, Limit Combination Illustration: Control Application

Figure 4.2.2, Elements of One Limit

Figure 4.2.3, Limit State Model

Table 4.2, Limit State Transition Table

- 4.3 Alarm Management

Figure 4.3, State Diagram for Alarm ALIDn

Table 4.3.1, Alarm State Transition Table

Table 4.3.2

- 4.4 Remote Control

- 4.5 Equipment Constants

- 4.6 Process Program Management

- 4.7 Material Movement

- 4.8 Equipment Terminal Services

- 4.9 Error Messages

- 4.10 Clock

- 4.11 Spooling

Figure 4.11, Spooling State Diagram

Table 4.11, Spooling State Transition

- 4.12 Control

5 Data Items

- 5.1 Data Item Restrictions

- 5.2 Variable Item List

6 Collection Events

Table 6.1, GEM Defined Collection Events

7 SECS-II Message Subset

STREAM 1: Equipment Status

STREAM 2: Equipment Control and Diagnostics

STREAM 5: Exception (Alarm) Reporting

STREAM 6: Data Collection

STREAM 7: Process Program Load

STREAM 9: System Errors

STREAM 10: Terminal Services

STREAM 14: Object Services

STREAM 15: Recipe Management

8 GEM Compliance

8.1 Fundamental GEM Requirements

Figure 8.1, GEM Requirements and Capabilities

Table 8.1, Fundamental GEM Requirements

8.2 GEM Capabilities

Table 8.2, Section References for GEM Capabilities

8.3 Definition of GEM Compliance

8.4 Documentation

Figure 8.2, Host View of GEM

Table 8.3, GEM Compliance Statement

Table 8.4, SML Notation

A. Application Notes

A.1 Factory Operational Script

A.1.1 Anytime Capabilities

A.1.2 System Initialization and Synchronization

A.1.3 Production Set-Up

A.1.4 Processing

A.1.5 Post-Processing

A.2 Equipment Front Panel

A.2.1 Displays and Indicators

A.2.2 Switches/Buttons

A.3 Examples of Equipment Alarms

Table A.3, Alarm Examples Per Equipment Configuration

A.4 Trace Data Collection Example

A.5 Harel Notation

Figure A.5.1, Harel Statechart Symbols

Figure A.5.2, Example of OR Substates

Figure A.5.3, Example of AND Substates

A.5.1 State Definitions

A.5.2 Transition Table

Table A.5, Transition Table for Motor Example

A.6 Example Control Model Application

A.7 Examples of Limits Monitoring

A.7.1 Introduction

A.7.2 Examples

Figure A.7.1, Valve Monitoring Example

Figure A.7.2, Environment Monitoring Example

Figure A.7.3, Calibration Counter Example

A.8 Recipe Parameter Modification for Process and Equipment Control

A.8.1 Introduction

A.8.2 Equipment Constants

A.8.3 Example

Figure A.8.1, CMP Single Wafer “Polishing” System with Host Recipe Parameter Modification Capability

Index

SEMI E30-1103

GENERIC MODEL FOR COMMUNICATIONS AND CONTROL OF MANUFACTURING EQUIPMENT (GEM)

This standard was technically approved by the Global Information & Control Committee and is the direct responsibility of the Japanese Information & Control Committee. Current edition approved by the Japanese Regional Standards Committee on August 8, 2003. Initially available at www.semi.org October 2003; to be published November 2003. Originally published in 1992; previously published July 2003.

1 Introduction

1.1 Revision History — This is the first release of the GEM standard.

1.2 Scope — The scope of the GEM standard is limited to defining the behavior of semiconductor equipment as viewed through a communications link. The SEMI E5 (SECS-II) standard provides the definition of messages and related data items exchanged between host and equipment. The GEM standard defines which SECS-II messages should be used, in what situations, and what the resulting activity should be. Figure 1.1 illustrates the relationship of GEM, SECS-II and other communications alternatives.

The GEM standard does NOT attempt to define the behavior of the host computer in the communications link. The host computer may initiate any GEM message scenario at any time and the equipment shall respond as described in the GEM standard. When a GEM message scenario is initiated by either the host or equipment, the equipment shall behave in the manner described in the GEM standard when the host uses the appropriate GEM messages.

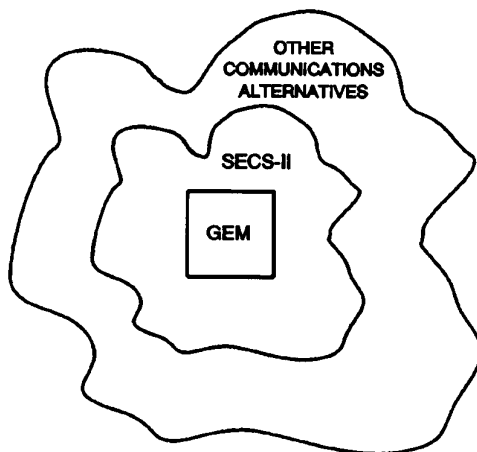


Figure 1.1
GEM Scope

The capabilities described in this standard are specifically designed to be independent of lower-level communications protocols and connection schemes

(e.g., SECS-I, SMS, point-to-point, connection-oriented or connectionless). Use of those types of standards is not required or precluded by this standard.

NOTICE: This standard does not purport to address safety issues, if any, associated with its use. It is the responsibility of the users of this standard to establish appropriate safety and health practices and determine the applicability of regulatory or other limitations prior to use.

1.3 Intent — GEM defines a standard implementation of SECS-II for all semiconductor manufacturing equipment. The GEM standard defines a common set of equipment behavior and communications capabilities that provide the functionality and flexibility to support the manufacturing automation programs of semiconductor device manufacturers. Equipment suppliers may provide additional SECS-II functionality not included in GEM as long as the additional functionality does not conflict with any of the behavior or capabilities defined in GEM. Such additions may include SECS-II messages, collection events, alarms, remote command codes, processing states, variable data items (data values, status values or equipment constants), or other functionality that is unique to a class (etchers, steppers, etc.) or specific instance of equipment.

GEM is intended to produce economic benefits for both device manufacturers and equipment suppliers. Equipment suppliers benefit from the ability to develop and market a single SECS-II interface that satisfies most customers. Device manufacturers benefit from the increased functionality and standardization of the SECS-II interface across all manufacturing equipment. This standardization reduces the cost of software development for both equipment suppliers and device manufacturers. By reducing costs and increasing functionality, device manufacturers can automate semiconductor factories more quickly and effectively. The flexibility provided by the GEM standard also enables device manufacturers to implement unique automation solutions within a common industry framework.

The GEM standard is intended to specify the following:

- A model of the behavior to be exhibited by semiconductor manufacturing equipment in a SECS-II communication environment,
- A description of information and control functions needed in a semiconductor manufacturing environment,
- A definition of the basic SECS-II communications capabilities of semiconductor manufacturing equipment,
- A single consistent means of accomplishing an action when SECS-II provides multiple possible methods, and
- Standard message dialogues necessary to achieve useful communications capabilities.

The GEM standard contains two types of requirements:

- fundamental GEM requirements and
- requirements of additional GEM capabilities.

The fundamental GEM requirements form the foundation of the GEM standard. The additional GEM capabilities provide functionality required for some types of factory automation or functionality applicable to specific types of equipment. A detailed list of the fundamental GEM requirements and additional GEM capabilities can be found in Chapter 8, GEM Compliance. Figure 1.2 illustrates the components of the GEM standard.

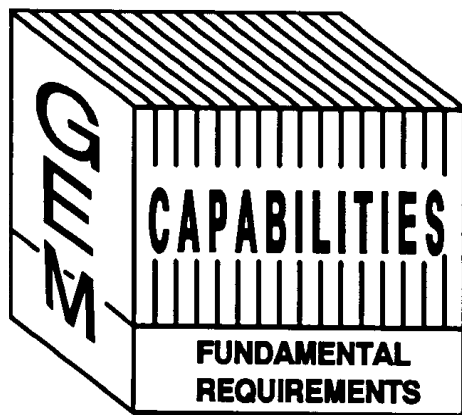


Figure 1.2
GEM Components

Equipment suppliers should work with their customers to determine which additional GEM capabilities should be implemented for a specific type of equipment. Because the capabilities defined in the GEM standard were specifically developed to meet the factory automation requirements of semiconductor manufacturers, it is anticipated that most device manufacturers will require most of the GEM capabilities that apply to a particular type of equipment. Some device manufacturers may not require all the GEM capabilities due to differences in their factory automation strategies.

1.4 Overview — The GEM standard is divided into sections as described below.

Section 1 — Introduction

This section provides the revision history, scope and intent of the GEM standard. It also provides an overview of the structure of the document and a list of related documents.

Section 2 — Definitions

This section provides definitions of terms used throughout the document.

Section 3 — State Models

This section describes the conventions used throughout this document to depict state models. It also describes the basic state models that apply to all semiconductor manufacturing equipment and that pertain to more than a single capability. State models describe the behavior of the equipment from a host perspective.

Section 4 — Capabilities and Scenarios

This section provides a detailed description of the communications capabilities defined for semiconductor manufacturing equipment. The description of each capability includes the purpose, definitions, requirements, and scenarios that shall be supported.

Section 5 — Data Definitions

This section provides a reference to the Data Item Dictionary and Variable Item Dictionary found in SEMI Standard E5. The first subsection shows those data items from SECS-II which have been restricted in their use (i.e., allowed formats). The second subsection lists variable data items that are available to the host for data collection and shows any restrictions on their SECS-II definitions.

Section 6 — Collection Events

This section provides a list of required collection events and their associated data.

Section 7 — SECS Message Subset

This section provides a composite list of the SECS-II messages required to implement all capabilities defined in the GEM standard.

Section 8 — GEM Compliance

This section describes the fundamental GEM requirements and additional GEM capabilities and provides references to other sections of the standard where detailed requirements are located. This section also defines standard terminology and documentation that can be used by equipment suppliers and device manufacturers to describe compliance with this standard.

Section A — Application Notes

These sections provide additional explanatory information and examples.

Section A.1 — Factory Operational Script

This section provides an overview of how the required SECS capabilities may be used in the context of a typical factory operation sequence. This section is organized according to the sequence in which actions are typically performed.

Section A.2 — Equipment Front Panel

This section provides guidance in implementing the required front panel buttons, indicators, and switches as defined in this document. A summary of the front panel requirements is provided.

Section A.3 — Examples of Equipment Alarms

This section provides examples of alarms related to various equipment configurations.

Section A.4 — Trace Data Collection Example

This section provides an example of trace initialization by the host and the periodic trace data messages that might be sent by the equipment.

Section A.5 — Harel Notation

This section explains David Harel's "Statechart" notation that is used throughout this document to depict state models.

Section A.6 — Example Control Model Application

This section provides one example of a host's interaction with an equipment's control model.

Section A.7 — Examples of Limits Monitoring

This section contains four limits monitoring examples to help clarify the use of limits and to illustrate typical applications.

1.5 Applicable Documents

1.5.1 SEMI Standards — The following SEMI standards are related to the GEM standard. The specific portions of these standards referenced by GEM constitute provisions of the GEM standard.

SEMI E4 — SEMI Equipment Communications Standard 1 — Message Transfer (SECS-I)

SEMI E5 — SEMI Equipment Communications Standard 2 — Message Content (SECS-II)

SEMI E13 — Standard for SEMI Equipment Communication Standard Message Service (SMS)

SEMI E23 — Specification for Cassette Transfer Parallel I/O Interface

1.5.2 Other References

Harel, D., "Statecharts: A Visual Formalism for Complex Systems," *Science of Computer Programming* 8 (1987) 231-274¹.

NOTICE: As listed or revised, all documents cited shall be the latest publications of adopted standards.

2 Definitions

2.1 alarm — An alarm is related to any abnormal situation on the equipment that may endanger people, equipment, or material being processed. Such abnormal situations are defined by the equipment manufacturer based on physical safety limitations. Equipment activities potentially impacted by the presence of an alarm shall be inhibited.

2.1.1 Note that exceeding control limits associated with process tolerance does not constitute an alarm nor do normal equipment events such as the start or completion of processing.

2.2 capabilities — Capabilities are operations performed by semiconductor manufacturing equipment. These operations are initiated through the communications interface using sequences of SECS-II messages (or scenarios). An example of a capability is the setting and clearing of alarms.

2.3 collection event — A collection event is an event (or grouping of related events) on the equipment that is considered to be significant to the host.

2.4 communication failure — A communication failure is said to occur when an established communications link is broken. Such failures are protocol specific. Refer to the appropriate protocol standard (e.g., SEMI E4 or

¹ Elsevier Science, P.O. Box 945, New York, NY 10159-0945, <http://www.elsevier.nl/homepage/browse.htm>

SEMI E37) for a protocol-specific definition of communication failure.

2.5 communication fault — A communication fault occurs when the equipment does not receive an expected message, or when either a transaction timer or a conversation timer expires.

2.6 control — To control is to exercise directing influence.

2.7 equipment model — An equipment model is a definition based on capabilities, scenarios, and SECS-II messages that manufacturing equipment should perform to support an automated manufacturing environment. (See also Generic Equipment Model.)

2.8 event — An event is a detectable occurrence significant to the equipment.

2.9 GEM compliance — The term “GEM Compliance” is defined with respect to individual GEM capabilities to indicate adherence to the GEM standard for a specific capability. Section 8 includes more detail on GEM Compliance.

2.10 Generic Equipment Model — The Generic Equipment Model is used as a reference model for any type of equipment. It contains functionality that can apply to most equipment, but does not address unique requirements of specific equipment.

2.11 host — The SEMI E4 and E5 standards define Host as “the intelligent system that communicates with the equipment.”

2.12 message fault — A message fault occurs when the equipment receives a message that it cannot process because of a defect in the message.

2.13 operational script — An operational script is a collection of scenarios arranged in a sequence typical of actual factory operations. Example sequences are system initialization powerup, machine setup, and processing.

2.14 operator — A human who operates the equipment to perform its intended function (e.g., processing). The operator typically interacts with the equipment via the equipment supplied operator console.

2.15 process unit — A process unit refers to the material that is typically processed as a unit via single run command, process program, etc. Common process units are wafers, cassettes, magazines, and boats.

2.16 processing cycle — A processing cycle is a sequence wherein all of the material contained in a typical process unit is processed. This is often used as a measure of action or time.

2.17 scenario — A scenario is a group of SECS-II messages arranged in a sequence to perform a capability. Other information may also be included in a scenario for clarity.

2.18 SECS-I — SEMI Equipment Communications Standard 1 (SEMI E4). This standard specifies a method for a message transfer protocol with electrical signal levels based upon EIA RS232-C.

2.19 SECS-II — SEMI Equipment Communications Standard 2 (SEMI E5). This standard specifies a group of messages and the respective syntax and semantics for those messages relating to semiconductor manufacturing equipment control.

2.20 SMS — SECS Message Service. An alternative to SECS-I to be used when sending SECS-II formatted messages over a network.

2.21 state model — A State Model is a collection of states and state transitions that combine to describe the behavior of a system. This model includes definition of the conditions that delineate a state, the actions/reactions possible within a state, the events that trigger transitions to other states, and the process of transitioning between states.

2.22 system default — Refers to state(s) in the equipment behavioral model that are expected to be active at the end of system initialization. It also refers to the value(s) that specified equipment variables are expected to contain at the end of system initialization.

2.23 system initialization — The process that an equipment performs at power-up, system activation, and/or system reset. This process is expected to prepare the equipment to operate properly and according to the equipment behavioral models.

2.24 user — A human or humans who represent the factory and enforce the factory operation model. A user is considered to be responsible for many setup and configuration activities that cause the equipment to best conform to factory operations practices.

3 State Models

The following sections contain state models for semiconductor manufacturing equipment. These state models describe the behavior of the equipment from a host perspective in a compact and easy to understand format. State models for different equipment will be identical in some areas (e.g., communications), but may vary in other areas (e.g., processing). It is desirable to divide the equipment into parallel components that can be modeled separately and then combined. An example of a component overview of an equipment is provided as Figure 3.0.

Equipment manufacturers must document the operational behavior of their equipment using state model methodology. State models are discussed in Sections 3.1 and A.5 and in a referenced article. Documentation of a state model shall include the following three elements:

- A *state diagram* showing the possible states of the system or components of a system and all of the possible transitions from one state to another. The states and transitions must each be labeled. Use of the Harel notation (see A.5) is recommended.
- A *transition table* listing each transition, the beginning and end states, what stimulus triggers the transition, and any actions taken as a result of the transition.
- A *definition of each state* specifying system behavior when that state is active.

Examples of the above elements are provided in Section A.5.

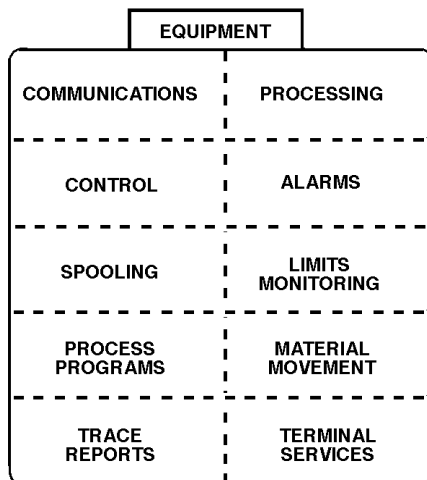


Figure 3.0
Example Equipment Component Overview

The benefits of providing state models are:

1. State machine models are a useful specification tool,
2. A host system can anticipate machine behavior based upon the state model,
3. End-users and equipment programmers have a common description of machine behavior from which to work,
4. “Legal” operations can be defined pertaining to any machine state,

5. External event notifications can be related to internal state transitions,
6. External commands can be related to state transitions,
7. State model components describing different aspects of machine control can be related to one another (example: processing state model with material transport state model; processing state model with internal machine safety systems).

3.1 State Model Methodology — To document the expected functionality of the various capabilities described in this document, the “Statechart” notation developed by David Harel has been adopted. An article by Harel is listed in Section 1.5 and should be considered “must” reading for a full understanding of the notation. The convention used in this and following sections is to describe the dynamic functionality of a capability with three items: a textual description of each state or substate defined, a table that describes the possible transitions from one state to another, and a graphical figure that uses the symbols defined by Harel to illustrate the relationships of the states and transitions. The combination of these items define the state model for a system or component. A summary of the Harel notation and a more detailed description of the text, table, and figure used to define behavior with this methodology is contained in the Application Note A.5.

The basic unit of a state model is the state. A state is a static set of conditions. If the conditions are met, the state is current. These conditions might involve sensor readings, switch positions, time of day, etc. Also part of a state definition is a description of reactions to specific stimuli (e.g., if message S_x, F_y is received, generate reply message $S_x, F_y + 1$). Stimuli may be quite varied but for semiconductor equipment would include received SECS messages, expired timers, operator input at an equipment terminal, and changes in sensor readings.

To help clarify the interpretation of this document and the state models described herein, it is useful to distinguish between a state and an event and the relationship of one to the other. An event is dynamic rather than static. It represents a change in conditions, or more specifically, the awareness of such a change. An event might involve a sensor reading exceeding a limit, a switch changing position, or a time limit exceeded.

A change to a new active state (state transition) must always be prompted by a change in conditions, and thus an event. In addition, a state transition may itself be termed an event. In fact, there are many events that may occur on an equipment, so it is important to classify

events based on whether they can be detected and whether they are of interest. In this document, the term event has been more narrowly defined as a detectable occurrence that is significant to the equipment.

A further narrowing of the definition of event is represented by the term “collection event,” which is an event (or group of related events) on the equipment that is considered significant to the host. It is these events that (if enabled) are reported to the host. By this definition, the list of collection events for an equipment would typically be only a subset of total events. The state models in this document are intended to be limited to the level of detail in which the host is interested. Thus, all state transitions defined in this standard, unless otherwise specified, shall correspond to collection events.

3.2 Communications State Model — The Communications State Model defines the behavior of the equipment in relation to the existence or absence of a communications link with the host. Section 4.1 expands on this section by defining the Establish Communications capability. This model pertains to a logical connection between equipment and host rather than a physical connection.

3.2.1 Terminology — The terms communication failure, connection transaction failure, and communication link are defined for use within this document only and should not be confused with the same or similar terms used elsewhere.

- See SEMI E4 (SECS-I) or SEMI E37 (HSMS) for a protocol specific definitions of communications failure.
- A connection transaction failure occurs when attempting to establish communications and is caused by
 - a communication failure,
 - the failure to receive an S1,F14 reply within a reply timeout limit, or
 - receipt of S1,F14 that has been improperly formatted or with COMMACK² not set to 0.
- A reply timeout period begins after the successful transmission of a complete primary message for which a reply is expected. (See SEMI E4 (SECS-I) or SEMI E37 (HSMS) for a protocol-specific definition of reply timeout.)

² Establish Communications Acknowledge Code, defined in Section 4.1. See the SEMI E5 Standard for further definition of this Data Item.

- A communication link is established following the first successful completion of any one S1,F13/F14 transaction with an acknowledgement of “accept”. The establishment of this link is logical rather than physical.
- Implementations may have mechanisms which allow outgoing messages to be stored temporarily prior to being sent. The noun queue is used to cover such stored messages. They are queued when placed within the queue and are dequeued by removing them from this storage.
- Send includes “queue to send” or “begin the process of attempting to send” a message. It does not imply the successful completion of sending a message.
- The host may attempt to establish communications with equipment at any time due to the initialization of the host or by independent detection of a communications failure by the host. Thus, the host may initiate an S1,F13/F14 transaction at any time.

3.2.2 CommDelay Timer — The CommDelay timer represents an internal timer used to measure the interval between attempts to send S1,F13. The length of this interval is equal to the value in the EstablishCommunicationsTimeout. The CommDelay timer is not directly visible to the host.

EstablishCommunicationsTimeout is the user-configurable equipment constant that defines the delay, in seconds, between attempts to send S1,F13. This value is used to initialize the CommDelay timer.

The CommDelay timer is initialized to begin timing. The CommDelay timer is initialized only when the state WAIT DELAY is entered.

The CommDelay timer is expired when it “times out,” and the time remaining in the interval between attempts to send is zero. When the timer expires during the state WAIT DELAY, it triggers a new attempt to send S1,F13 and the transition to the state WAIT CRA³.

3.2.3 Conventions

- The attempt to send S1,F13 is made only upon transit into the state WAIT CRA. The CommDelay Timer should be set to “expired” at this time.
- The CommDelay timer is initialized only upon transit into the state WAIT DELAY. A next

³ CRA is the mnemonic defined for Establish Communications Request Acknowledge (S1,F14).

attempt to send S1,F13 shall occur only upon a transit to the state WAIT CRA.

3.2.4 Communication States — There are two major states of SECS communication, DISABLED and ENABLED. The system default state must be user-configurable at the equipment (e.g., via a jumper setting or non-volatile memory variable).

Once system initialization has been achieved, the operator shall be able to change the communication state selection at any time via equipment terminal functions or momentary switch. A two-position type switch must not be used due to possible conflict with the system default.

The ENABLED state has two substates, NOT COMMUNICATING and COMMUNICATING, described below. The equipment must inform the operator of the current communication state via continuous display at the equipment, including the NOT COMMUNICATING and COMMUNICATING sub-states.

In the event of a connection transaction failure, a user-configurable equipment constant EstablishCommunicationsTimeout is used to establish the interval between attempts to send an S1,F13 (Establish Communications Request) while in the NOT COMMUNICATING sub-state.

Figure 3.2.1 shows the relationship between the superstates and substates of the Communications State Model. A description of the events triggering state transitions and the actions taken is given in Table 3.2.

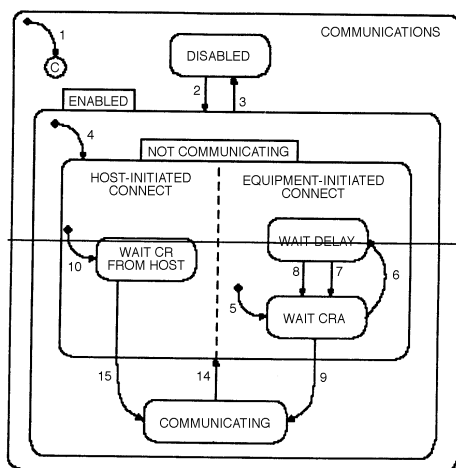


Figure 3.2.1
Communications State Diagram

The states of the Communications State Model are defined as follows:

DISABLED

In this state SECS-II communication with a host computer is non-existent. If the operator switches from ENABLED to DISABLED, all SECS-II communications must cease immediately. Any messages queued to send shall be discarded, and all further action on any open transactions and conversations shall be terminated.⁴ Handling of messages currently being transmitted is an issue for lower level message transfer protocols and is not addressed in this standard.

The DISABLED state is a possible system default.

ENABLED

ENABLED has two substates, COMMUNICATING and NOT COMMUNICATING. Whenever communications are enabled, either during system initialization or through operator selection, the substate of NOT COMMUNICATING is active until communications are formally established. Lower-level protocols (such as SECS-I) are assumed to be functioning normally in that they are capable of supporting the communication of SECS-II syntax.

The ENABLED state is a possible system default.

ENABLED/NOT COMMUNICATING

No messages other than S1,F13, S1,F14, and S9,Fx shall be sent while this substate is active. The equipment shall discard any messages received from the host other than S1,F13 or S1,F14 (Establish Communications Acknowledge). It shall also periodically attempt to establish communication with a host computer by issuing an S1,F13 until communications are successfully established. However, only one equipment-initiated S1,F13 transaction may be open at any time.

The NOT COMMUNICATING state has two AND substates, HOST-INITIATED CONNECT and EQUIPMENT-INITIATED CONNECT, both of which are active whenever the equipment is NOT COMMUNICATING. These two substates clarify the behavior of the equipment in the event that both the equipment and the host attempt to establish communications during the same period of time⁵.

⁴ Refer to SEMI E5, Section 5, for definitions of SECS-II transaction and conversation protocols.

⁵ Note that in the Harel notation, an exit from any AND substate is an exit from the parent state and thus from all other AND substates of that parent substate.

NOT COMMUNICATING/EQUIPMENT-INITIATED CONNECT

This state has two substates, WAIT CRA and WAIT DELAY. Upon any entry to the NOT COMMUNICATING state, whenever EQUIPMENT-INITIATED CONNECT first becomes active, a transition to WAIT CRA occurs, the CommDelay timer is set to “expired,” and an immediate attempt to send S1,F13 is made.

NOT COMMUNICATING/EQUIPMENT-INITIATED CONNECT/WAIT CRA

An Establish Communications Request has been sent. The equipment waits for the host to acknowledge the request.

NOT COMMUNICATING/EQUIPMENT-INITIATED CONNECT/WAIT DELAY

A connection transaction failure has occurred. The CommDelay timer has been initialized. The equipment waits for the timer to expire.

NOT COMMUNICATING/HOST-INITIATED CONNECT

This state describes the behavior of the equipment in response to a host-initiated S1,F13 while NOT COMMUNICATING is active.

NOT COMMUNICATING/HOST-INITIATED CONNECT/WAIT CR FROM HOST

The equipment waits for an S1,F13 from the host. If an S1,F13 is received, the equipment attempts to send an S1,F14 with COMMACK = 0.

ENABLED/COMMUNICATING

Communications have been established. The equipment may receive any message from the host, including S1,F13. When the equipment is COMMUNICATING, SECS communications with a host computer must be maintained. This state remains active until communications are disabled or a communication failure occurs. If the equipment receives S1,F13 from the host while in the COMMUNICATING substate, it should respond with S1,F14 with COMMACK set to zero. If the equipment receives S1,F14 from a previously sent S1,F13, no action is required.

In the event of communication failure, the equipment shall return to the NOT COMMUNICATING substate and attempt to re-establish communications with the host.

It is possible that the equipment may be waiting for an S1,F14 from the host in EQUIPMENT-

INITIATED CONNECT/WAIT CRA at the time an S1,F13 is received from the host in HOST-INITIATED CONNECT/WAIT CR FROM HOST. When this situation occurs, both equipment and host have an open S1,F13/S1,F14 transaction. Since communications are successfully established on the successful completion of any S1,F13/S1,F14 transaction, either of these two transactions may be the first to complete successfully and to cause the transition from NOT COMMUNICATING state to COMMUNICATING. In this event, the other transaction shall remain open regardless of the transition to COMMUNICATING until it is closed in a normal manner.

If the equipment has not yet sent⁶ an S1,F14 to a previously received S1,F13 at the time when COMMUNICATING becomes active, the S1,F14 response shall be sent in a normal manner. A failure to send the S1,F14 is then treated as any other communication failure.

If the equipment-initiated S1,F13/S1,F14 is still open when the transition to COMMUNICATING occurs, subsequent failure to receive a reply from the host is considered a communication fault by equipment. An S9,F9 should be sent when a transaction timer timeout occurs⁷. (See Section 4.9 for definitions of communication faults and message faults, as well as detail on Stream 9 Error Messages.)

3.2.5 State Transitions — Table 3.2 contains a full description of the state transitions depicted in Figure 3.2.1.

When the operator switches from the DISABLED state to the ENABLED state, no collection event shall occur, since no messages can be sent until communications have been established. The process of establishing communications serves to notify the host that communications are ENABLED. No other collection events are defined for the Communications State Model.

⁶ This includes transmissions that may have started but not yet successfully completed at the time that the transition to COMMUNICATING occurs.

⁷ The existence of a transaction timer is not a requirement in some protocols, such as SMS (SEMI E13).

Table 3.2 Communications State Transition Table

#	Current State	Trigger	New State	Action	Comment
1	(Entry to COMMUNICATIONS)	System initialization.	System Default	None.	The system default may be set to DISABLED or ENABLED.
2	DISABLED	Operator switches from DISABLED to ENABLED.	ENABLED	None.	SECS-II communications are enabled.
3	ENABLED	Operator switches from ENABLED to DISABLED.	DISABLED	None.	SECS-II communications are prohibited.
4	(Entry to ENABLED)	Any entry to ENABLED state.	NOT COMMUNICATING	None.	May enter from system initialization to ENABLED or through operator switch to ENABLED.
5	(Entry to EQUIPMENT-INITIATED CONNECT)	(Any entry to NOT COMMUNICATING)	WAIT CRA	Initialize communications. Set CommDelay timer "expired." Send S1,F13.	Begin the attempt to establish communications.
6	WAIT CRA	Connection transaction failure.	WAIT DELAY	Initialize CommDelay timer. Dequeue all messages queued to send.	If appropriate, dequeued messages shall be placed in spool buffer in the order generated. Wait for timer to expire.
7	WAIT DELAY	CommDelay timer expired.	WAIT CRA	Send S1,F13	Wait for S1,F14. May receive S1,F13 from Host.
8	WAIT DELAY	Received a message other than S1,F13.	WAIT CRA	Discard message. No reply. Set CommDelay timer "expired". Send S1,F13.	Indicates opportunity to establish communications.
9	WAIT CRA	Received expected S1,F14 with COMMACK = 0.	COMMUNICATING	None.	Communications are established.
10	(Entry to HOST-INITIATED CONNECT)	(Any entry to NOT COMMUNICATING)	WAIT CR FROM HOST	None.	Wait for S1,F13 from Host.
14	COMMUNICATING	Communication failure. (See SEMI E4 or SEMI E37 for a protocol-specific definition of communication failure.)	NOT COMMUNICATING	Dequeue all messages queued to send.	Dequeued messages may be placed in spool buffer as appropriate.
15	WAIT CR FROM HOST	Received S1,F13.	COMMUNICATING	Send S1,F14 with COMMACK = 0.	Communications are established.

3.3 Control State Model — The CONTROL state model defines the level of cooperation between the host and equipment. It also specifies how the operator may interact at the different levels of host control. While the COMMUNICATIONS state model addresses the ability for the host and equipment to exchange messages, the CONTROL model addresses the equipment's responsibility to act upon messages that it receives.

The CONTROL model provides the host with three basic levels of control. In the highest level (REMOTE), the host may control the equipment to the full extent possible. The middle level (LOCAL) allows the host full access to information, but places some limits on how the host can affect equipment operation. In the lowest level (OFF-LINE), the equipment allows no host control⁸ and only very limited information.⁹

The control model and communications model (when implemented) do not interact directly. That is, no action or state of one model directly causes a change in behavior of the other. It is true, however, that when the communication state is NOT COMMUNICATING then most message transaction are not functional. When messages cannot be transmitted, the control capabilities and all other GEM capabilities are affected.

Refer to Figure 3.3 as the CONTROL substates and state transitions are defined.

OFF-LINE

When the OFF-LINE state is active, operation of the equipment is performed by the operator at the operator console. While the equipment is OFF-LINE, message transfer is possible. However the use of messaging for any automation purpose is severely restricted. While the OFF-LINE state is active, the equipment will only respond to those messages used for the establishment of communications or a host request to activate the ON-LINE state.

While OFF-LINE, the equipment will respond with an Sx,F0 to any primary message from the host other than S1,F13 or S1,F17. It will process and respond to S1,F13 and S1,F17. S1,F17 is used by the host to request the equipment to transition to the ON-LINE state. The equipment will accept this request and send a positive response only when the HOST OFF-LINE state is active (see transition 11 definition below).

⁸ The host may establish communications. This does not affect equipment operation and for that reason is not termed a control operation.

⁹ The host may determine the equipment identification via the S1F13/F14 transaction.

While the OFF-LINE state is active, the equipment shall attempt to send no primary message other than S1,F13,¹⁰ S9,Fx,¹¹ and S1,F1 (see ATTEMPT ON-LINE substate). If the equipment receives a reply message from the host other than S1,F14 or S1,F2, this message is discarded.

No messages enter the spool when the system is OFF-LINE. Spooling may be active when the Communications State of NOT COMMUNICATING is active. This might occur during OFF-LINE, but since the equipment will not attempt to send messages except as mentioned in the previous paragraph¹², no messages will enter the spool.

OFF-LINE has three substates: EQUIPMENT OFF-LINE, ATTEMPT ON-LINE, and HOST OFF-LINE.

OFF-LINE/EQUIPMENT OFF-LINE

While this state is active, the system maintains the OFF-LINE state. It awaits operator instructions to attempt to go ON-LINE.

OFF-LINE/ATTEMPT ON-LINE

While the ATTEMPT ON-LINE state is active, the equipment has responded to an operator instruction to attempt to go to the ON-LINE state. Upon activating this state, the equipment attempts to send an S1,F1 to the host.

Note that when this state is active, the system does not respond to operator actuation of either the ON-LINE or the OFF-LINE switch.

¹⁰ Sending of S1,F13 is based upon the COMMUNICATIONS state model.

¹¹ S9,Fx messages may be issued only in response to the messages to which the equipment will normally respond while OFF-LINE (i.e. S1,F13 and S1,F17).

¹² The equipment may send S1,F1 or S1,F13, but since Stream 1 messages are not eligible for spooling, they will not enter the spool either.

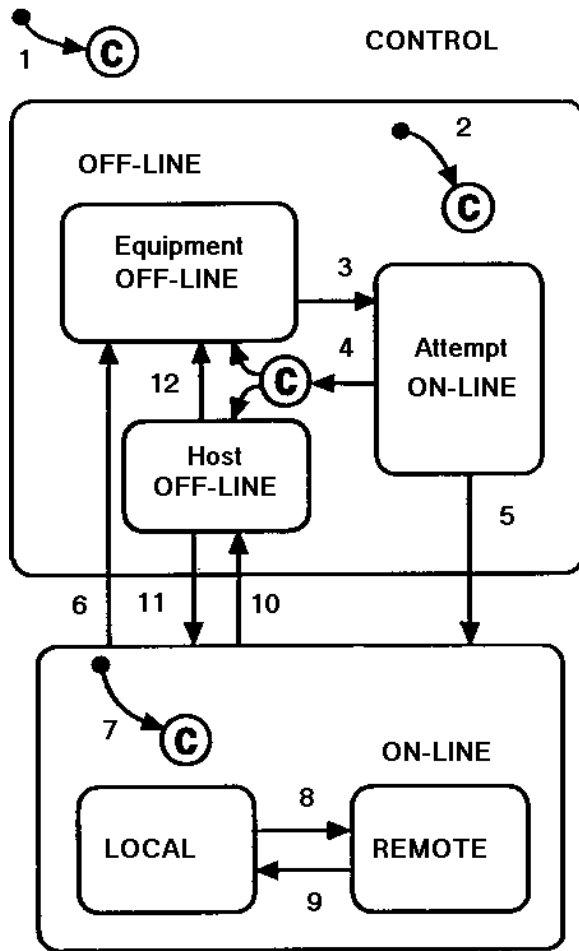


Figure 3.3
CONTROL State Model

OFF-LINE/HOST OFF-LINE

While the HOST OFF-LINE state is active, the operator's intent is that the equipment be ON-LINE. However, the host has not agreed. Entry to this state may be due to a failed attempt to go ON-LINE or to the host's request that the equipment go OFF-LINE from ON-LINE (see the transition table for more detail). While this state is active, the equipment shall positively respond to any host's request to go ON-LINE (S1,F17). Such a request shall be denied when the HOST OFF-LINE state is not active.

ON-LINE

While the ON-LINE state is active, SECS-II messages may be exchanged and acted upon. Capabilities that may be available to the host should be similar to those available from the operator console wherever practical.

The use of Sx,F0 messages is not required while the ON-LINE state is active. Their use is discouraged in this case. The only allowed use is to close open transactions in conjunction with message faults.

ON-LINE/LOCAL

Operation of the equipment is implemented by direct action of an operator. All operation commands shall be available for input at the local operator console of the equipment.

The host shall have the following capabilities and restrictions when the LOCAL state is active:

- The host shall be prohibited from the use of remote commands that cause physical movement or which initiate processing. During processing, the host shall be prohibited from the use of any remote command that affects that process.
- During processing, the host shall be prohibited from modifying any equipment constants that affect that process. Other equipment constants shall be changeable during processing. The host shall be able to modify all available equipment constants when no processing is in progress.
- The host shall be capable of initiating the upload and download of recipes to/from the recipe storage area on the equipment. The host shall be capable of selecting recipes for execution so long as this action does not affect any currently executing recipe.
- The host shall be able to configure automatic data reporting capabilities including alarms, event reporting, and trace data reporting. The host shall receive all such reports at the appropriate times.
- The host shall be able to inquire for data from the equipment, including status data, equipment constants, event reports, process program directories, and alarms.
- The equipment shall be able to perform Terminal Services as defined in GEM.

The host shall be allowed any other capabilities that were not specifically restricted in the above items as long as the LOCAL state is active.

NOTE 2: Capabilities mentioned above which are not implemented on a specific equipment may be ignored in this context.

ON-LINE/REMOTE

For equipment which supports the GEM capability of remote control (see Section 4.4), while the REMOTE state is active, the host shall have access, through the communications interface, to the necessary commands

to operate the equipment through the full process cycle in an automated manner. The equipment does not restrict any host capabilities when REMOTE is active. The degree of control executed by the host may vary from factory to factory. In some cases, the operator maybe required to interact during remotely controlled processes. This interaction may involve set-up operations, operator assist situations, and others. This state is intended to be flexible enough to accommodate these different situations.

To support the different factory automation policies and procedures, it shall be possible to configure the equipment to restrict the operator in specific non-emergency procedures. These restrictions shall be configurable so that the equipment may be set up to allow the operator to perform necessary functions without contention with the host. The categories for configuration shall include (but are not limited to):

- change equipment constants (process-related),
- change equipment constants (non-process-related),
- initiate process program download,
- select process program,
- start process program,
- pause/resume process program,
- operator assist,
- material movement to/from equipment,
- equipment-specific commands (on a command-by-command basis if needed).

NOTE 3: Capabilities mentioned above which are not implemented on a specific equipment may be ignored in this context.

No capabilities that are available to the operator when the LOCAL state is active should be unconditionally restricted when the REMOTE state is active. The supplier may provide for configurable restriction of operator capabilities not included in the list above if desired. No configurability is necessary for any operator functions not available to the host.

The control functions must be shared to some degree between the host and the local operator. At the very least, the operator must have the capability to change the CONTROL state, actuate an Emergency Stop, and interrupt processing (e.g., STOP, ABORT, or PAUSE).

All of these capabilities except Emergency Stop may be access-limited.¹³

The host software should be designed to be compatible with the capabilities allotted to the operator.

¹³ Definition of the method of limiting operator access (password, key, etc.) to a capability is not within the scope of this document.

Table 3.3 CONTROL State Transition Table

#	Current State	Trigger	New State	Action	Comments
1	(Undefined)	Entry into CONTROL state (system initialization).	CONTROL (Substate conditional on configuration).	None	Equipment may be configured to default to ON-LINE or OFF-LINE . (See NOTE 1.)
2	(Undefined)	Entry into OFF-LINE state.	OFF-LINE (Substate conditional on configuration.)	None	Equipment may be configured to default to any substate of OFF-LINE.
3	EQUIPMENT OFF-LINE	Operator actuates ON-LINE switch.	ATTEMPT ON-LINE	None	Note that an S1,F1 is sent whenever ATTEMPT ON-LINE is activated.
4	ATTEMPT ON-LINE	S1,F0.	New state conditional on configuration.	None	This may be due to a communication failure (See NOTE 2), reply timeout, or receipt of S1,F0. Configuration may be set to EQUIPMENT OFF-LINE or HOST OFF-LINE.
5	ATTEMPT ON-LINE	Equipment receives expected S1,F2 message from the host.	ON-LINE	None	Host is notified of transition to ON-LINE at transition 7.
6	ON-LINE	Operator actuates OFF-LINE switch.	EQUIPMENT OFF-LINE	None	“Equipment OFF-LINE” event occurs. (See NOTE 3.) Event reply will be discarded while OFF-LINE is active.
7	(Undefined)	Entry to ON-LINE state.	ON-LINE (Substate conditional on REMOTE/LOCAL switch setting.)	None	“Control State LOCAL” or “Control State REMOTE” event occurs. Event reported based on actual ON-LINE substate activated.
8	LOCAL	Operator sets front panel switch to REMOTE.	REMOTE	None	“Control State REMOTE” event occurs.
9	REMOTE	Operator sets front panel switch to LOCAL.	LOCAL	None	“Control State LOCAL” event occurs.
10	ON-LINE	Equipment accepts “Set OFF-LINE” message from host (S1,F15).	HOST OFF-LINE	None	“Equipment OFF-LINE” event occurs.
11	HOST OFF-LINE	Equipment accepts host request to go ON-LINE (S1,F17).	ON-LINE	None	Host is notified to transition to ON-LINE at transition 7.
12	HOST OFF-LINE	Operator actuates OFF-LINE switch.	EQUIPMENT OFF-LINE	None	“Equipment OFF-LINE” event occurs.

NOTE 1: The configuration mentioned for transitions 1 and 2 should be a single setting. This would provide the user with a choice of entering the EQUIPMENT OFF-LINE, ATTEMPT ON-LINE, HOST OFF-LINE, or ON-LINE states.

NOTE 2: Communication failures are protocol specific. Refer to the appropriate protocol standard (e.g., SEMI E4 or SEMI E37) for a protocol-specific definition of communication failure.

NOTE 3: Any host initiated transaction open at the equipment must be completed. This may happen either by sending the appropriate reply to the host prior to sending the event message or by sending an Sx,F0 message following the event message (i.e., after the transaction).

Table 3.4 Processing State Transition Table

#	Current State	Trigger	New State	Action	Comments
1	INIT	Equipment initialization complete.	IDLE	None	None
2	IDLE	Commit has been made to set up.	SETUP	None	None
3	SETUP	All setup activity has completed and the equipment is ready to receive a START command.	READY	This activity is equipment-specific.	None
4	READY	Equipment has received a START command from the host or operator console.	EXECUTING	This activity is equipment-specific.	None
5	EXECUTING	The processing task has been completed.	IDLE	None	None
6	PROCESSING ACTIVE	Equipment has received a STOP command from host or operator console.	IDLE	None	None
7	PROCESSING ACTIVE	Equipment has received an ABORT command from host or operator console.	IDLE	This activity is equipment-specific.	None
8	PROCESS	The equipment decides to PAUSE due to a condition such as alarm.	PAUSE	This activity is equipment-specific.	For this type of problem, an operator assist is usually required.
9	PROCESS	Equipment has received a PAUSE command from host or operator console.	PAUSE	This activity is equipment-specific.	None
10	PAUSE	Equipment has received a RESUME command from host or operator console.	Previous PROCESS substate	This activity is equipment-specific.	None

4 Equipment Capabilities and Scenarios

This section describes the details of the capabilities required by GEM and provides scenarios for their use. Capabilities are operations performed by semiconductor manufacturing equipment. These operations are initiated through the communications interface using SECS-II messages. A scenario is a group of SECS-II messages arranged in a sequence to perform a capability. Other information may be included with the scenario for clarity. For each capability, the reader is provided with a statement of purpose, pertinent definitions, a detailed description, requirements, and scenarios.

The following capabilities are discussed:

- Establish Communications
- Event Notification

Dynamic Event Report Configuration

Variable Data Collection

Trace Data Collection

Limits Monitoring

Status Data Collection

On-line Identification

Alarm Management

Remote Control

Equipment Constants

Process Program Management

Material Movement

Equipment Terminal Services

Error Messages

Clock

Spooling

Control

4.1 Establish Communications — The Establish Communications capability provides a means of formally establishing communications following system initialization or any loss of communications between communicating partners, and thus of notifying the communication partner that a period of non-communication has occurred.

4.1.1 Purpose — Communications between host and equipment are formally established through use of the Establish Communications Request/Establish Communications Acknowledge transaction.

The use of S1,F1/F2 for this purpose is ambiguous since the transaction can be used for other purposes and may occur at any time.

The S1,F13/F14 transaction, used in conjunction with the Communications State Model, provides a means for equipment to notify the host, or the host to notify the equipment, that there has been a period of inability to communicate. The successful completion of this transaction also signals a possible need for synchronization activities between host and equipment.

4.1.2 Definitions

COMMACK — Acknowledge code returned in the Establish Communications Acknowledge message. See the SEMI E5 Standard for a full definition of this data item.

EstablishCommunicationsTimeout — An equipment constant used to initialize the interval between attempts to re-send an Establish Communications Request. This value specifies the number of seconds for the interval. See the SEMI E5 Standard for a full definition of this variable data item.

4.1.3 Description — There are potential problems when one side of the communications link fails and the other side does not detect it. From the point of view of the host, a loss of communications has many possible causes. In some cases, host-controlled settings on the equipment may need to be reset. In other cases, the equipment may have continued an automatic processing sequence during the period of no communication and may have changed states. The definition of a formal protocol for establishing communications alerts the host to the need to synchronize itself with the equipment's current status.

Equipment shall consider communications as formally established whenever either of the following conditions have been satisfied:¹⁴

- Communications Request has been sent to the host and an Establish Communications Acknowledge has been received within the transaction timeout period and with an acknowledge code of Accept, or
- Communications Request has been received from the host, and an Establish Communications Acknowledge response has been successfully sent with an acknowledge code of Accept.

When the equipment sends an Establish Communications Request to the host, this notifies the host of the possible need to synchronize itself with the equipment.

When the equipment is attempting to establish communications, an Establish Communications Request shall be sent periodically until communications have been formally established as described above. The interval between attempts must be user-configurable and begins as soon as a connection transaction failure is detected (see Section 3.2).

Attempting to establish communications is not a low-level connectivity issue, but rather a logical application issue used by either party to notify its partner that the host may need to perform synchronization activities with the equipment.

4.1.4 Requirements

- Equipment must support the Communication State Model (see Section 3.2).
- Equipment must provide the EstablishCommunicationsTimeout equipment constant described above.

¹⁴ Satisfaction of either of these conditions will result in a transition to the COMMUNICATING substrate. See Section 3.2 for further detail.

4.1.5 Scenarios

4.1.5.1 Host Attempts to Establish Communications

COMMENT	HOST	EQUIPMENT	COMMENT
			Communications state is Enabled (any substate)
Establish Communications Request	S1,F13-->		
		<--S1,F14	Reply COMMACK = Accept and Communications state = COMMUNICATING

4.1.5.2 Equipment Attempts to Establish Communications and Host Acknowledges

COMMENT	HOST	EQUIPMENT	COMMENT
			Communications State = NOT COMMUNICATING
			[LOOP]
			[LOOP]--SEND
		<--S1,F13	Establish Communications Request
Establish Communications Acknowledge	S1,F14-->		
			[IF] S1,F14 rcvd w/o timeouts
			[THEN] exit_loop--SEND
			[ELSE] Delay for interval in Establish Communications- Timeout
			[ENDIF]
			[END_LOOP]--SEND
			[IF] COMMACK = Accept
			[THEN] Communications state= Communicating
			exit_loop--
			[ELSE] Reset timer for delay, and delay for interval specified in EstablishCommunications- Timeout
			[ENDIF]
			[END_LOOP]

4.1.5.3 Simultaneous Attempts to Establish Communications — For equipment that supports interleaving, it is possible that either the host or equipment could send an Establish Communications Request before receiving the request from its partner. As communications are established by the successful acceptance of any one Establish Communications Request, it is immaterial who sends the request first. The roles of host and equipment may be reversed.

Equipment Receives S1,F14 From Host Before Sending S1,F14:

COMMENT	HOST	EQUIPMENT	COMMENT
	Communications State = NOT COMMUNICATING		
		<--S1,F13	Establish Communications Request
Establish Communications Request	S1,F13-->		
Reply COMMACK = Accept	S1,F14-->		S1,F14 received from Host and Communications established ¹⁵ and Communications state = COMMUNICATING
		<--S1,F14	Reply COMMACK = Accept ¹⁶

Equipment Sends S1,F14 To Host Before Receiving S1,F14:

COMMENT	HOST	EQUIPMENT	COMMENT
	Communications State = NOT COMMUNICATING		
		<--S1,F13	Establish Communications Request
Establish Communications Request	S1,F13-->		
		<--S1,F14	Reply COMMACK = Accept ¹⁵ and Communications established ¹⁵ and Communications state = COMMUNICATING
Reply COMMACK = Accept	S1,F14-->		S1,F14 received from Host

4.2 Data Collection — Data collection allows the host to monitor equipment activity via event reporting, trace data reporting, limits monitoring, and query of selected status or other variable data.

4.2.1 Event Data Collection — Event data collection provides a dynamic and flexible method for the user to tailor the equipment to meet individual needs with respect to data representation and presentation to the host. The event-based approach to data collection provides automatic notification to the host of equipment activities and is useful in monitoring the equipment and in maintaining synchronization with the equipment.

Event data collection may be broken into two logical parts: host notification when an event occurs and dynamic configuration of the data attached to the event notification.

4.2.1.1 Event Notification — This section describes the method of notifying the host when equipment collection events occur.

4.2.1.1.1 Purpose — This capability provides data to the host at specified points in equipment operation. This asynchronous reporting eliminates the need for the host to poll the equipment for this information. Events on the equipment may trigger activity on the part of the host. Also, knowledge of the occurrence of events

¹⁵ Communications are established at the successful completion of the S1,F13/14 transaction where COMMACK is set to zero.

¹⁶ Communications are established on the successful transmission of S1,F14, even if there is an open S1,F13.

related to the equipment state models allows the host to track the equipment state. An equipment's behavior is related to its current state. Thus, this capability helps the host understand how an equipment will behave and how it will react to host behavior.

4.2.1.1.2 Definitions

Collection Event — An event (or grouping of related events) on the equipment that is considered significant to the host.

Collection Event ID (CEID) — A unique identifier of a collection event. See the SEMI E5 Standard for a full definition of this data item.

Event — A detectable occurrence significant to the equipment.

Report — A set of variables predefined by the equipment or defined by the host via S2,F33/F34.

4.2.1.1.3 Detailed Description — The equipment supplier must provide a set of predefined collection events. Specific collection events are required by individual capabilities and state models. Examples of collection events include:

- The completion of each action initiated by a host requested command,
- Selected processing and material handling activities,
- Operator action detected by the equipment,
- A state transition,
- The setting or clearing of an alarm condition on the equipment, and
- Exception conditions not considered alarms.

See Section 6 for a list of required collection events.

The reporting of a collection event may be disabled per event by the user to eliminate unwanted messages. An event report message shall be sent to the host upon the occurrence of a particular collection event if the

collection event (CEID) has been enabled. Attached to each event message is one or more event reports which contain variable data. Section 4.2.1.2 describes the capability which allows for the dynamic customization of event reports. The values of any data contained in an event report message must be current upon the occurrence of the event. This implies that event reports be built at the time of the event occurrence.

The equipment shall also provide the S6,F15/F16 transaction to allow the host to request the data from a specific event report.

4.2.1.1.4 Requirements

- The equipment supplier shall provide documentation of all collection events defined on the equipment and the conditions for each event to occur.
- The equipment supplier shall provide unique CEIDs for each of the various collection events that are available for reporting.
- The equipment supplier shall provide a method for enabling and disabling the reporting of each event. This method shall either be available via the host interface (see Section 4.2.1.2) or the equipment's operator console.
- For each event, the equipment supplier shall provide either
 - a default set of report(s) linked to the event which contain data pertinent to that event, or
 - the ability for the user to configure the data linked to that event via the equipment's operator console or host interface (see Section 4.2.1.2).

4.2.1.1.5 Scenarios

Collection Event Occurs on the Equipment:

COMMENT	HOST	EQUIPMENT	COMMENT
Multi-block grant		<--S6,F5 S6,F6-->	[IF] Event Report is Multi-block [THEN] send Multi-block inquire
Host acknowledges Event Report		<--S6,F11 S6,F12-->	[ENDIF] Equipment sends Event Report

Host Requests Event Report:

COMMENT	HOST	EQUIPMENT	COMMENTS
Host requests an event report S6,F15-->	<--S6,F16	Equipment sends event report.	
4.2.1.2 <i>Dynamic Event Report Configuration</i> — This section describes a capability which allows the host to dynamically modify the equipment event reporting setup. 4.2.1.2.1 <i>Purpose</i> — This capability is defined to provide the data reporting flexibility required in some manufacturing environments. It allows the host to increase or decrease the data flow according to need. For example, if the performance of an equipment degrades, the data flow from that equipment may be increased to help diagnose the problem. 4.2.1.2.2 <i>Definitions</i> <i>EventsEnabled</i> — A variable data item that consists of a list of currently enabled collection events (CEIDs). See SEMI E5 for a full definition of this variable data item. <i>Report ID (RPTID)</i> — A unique identifier of a specific report. See SEMI E5 for a full definition of this data item. <i>Variable Data (V)</i> — A data item containing status (SV), data (DVVAL), or constant (ECV) values. See SEMI E5 for a full definition of this data item. <i>Variable Data ID (VID)</i> — A unique identifier of a variable data item. The set of VID's is the union of all SVID's, ECID's, and ID's for DVVAL's (DVNAME's). See SEMI E5 for a full definition of this data item. 4.2.1.2.3 <i>Detailed Description</i> — The equipment shall support the following event report configuration functionality through the SECS-II interface: — Host definition/deletion of custom reports, — Host linking/unlinking of defined reports to specified collection events, and — Host enabling/disabling the reporting of specified collection events. NOTE 4: The equipment may also supply alternative means for defining reports and linking reports to events (e.g., via the operator console). Implementation of alternate means is not required.			The equipment can be instructed by the host to enable or disable reporting of collection events on an individual or collective basis. A status value (SV) shall be available that consists of a list of enabled collection events. (See Section 5.2, Variable Item List, EventsEnabled variable.) Reports may be attached to an event report message (S6,F11). These reports are specifically linked to the desired event and typically contain variable data relating to that event. The reports may be provided by the equipment supplier or created by the user. The user must be able to create reports and link them to events via the SECS-II interface. The data reported in the event report messages may consist of Status Values (SV's), Equipment Constant Values (ECV's), or Data Values (DVVAL's). Note that data values shall be valid and current on certain events and certain states and might not be current at other times. The implementor shall document when a data value will be current and available for reporting. 4.2.1.2.4 <i>Requirements</i> — The equipment manufacturer must provide documentation of all variable data available from the equipment. This is to include variable name, variable type or class (SV, ECV, DVVAL), units, format codes, possible range of values, and a description of the meaning and use of this variable. — The equipment manufacturer must provide unique VID's for the various variable data (V) available for data collection in the equipment. For example, this means that no SV shall have a VID which is the same as the VID of any ECV or DVVAL. — All variable data must be available for report definition and event data collection. See Section 5.2, Variable Item List, for a list of required variable data. — All report definitions, report-to-event links, and enable/disable status of event reports must be retained in non-volatile storage.

4.2.1.2.5 Scenario

Collection Event Reporting Set-up:

COMMENT	HOST	EQUIPMENT	COMMENT
[IF] Define Report is Multi-block [THEN] send Multi-block inquire S2,F39-->		<-- S2,F40	Multi-block grant.
[ENDIF] Send report definitions S2,F33-->			DATAIDs, RPTIDs, and VIDs received.
		<-- S2,F34	DRACK ¹⁷ = 0 the reports are OK
[IF] Link Event/Report is Multi-block [THEN] send Multi-block inquire S2,F39-->		<-- S2,F40	Multi-block grant.
[ENDIF] Link reports to events S2,F35-->			CEIDs and the corresponding RPTIDs are received.
		<-- S2,F36	LRACK = 0 the event linkages are acceptable.
Enable specific collection S2,F37--> events			Enable/ disable codes (CEEDs) and the respective event reporting CEIDs received.
		<-- S2,F38	ERACK = 0 OK, will generate the specified reports when the appropriate collection events happen

4.2.2 Variable Data Collection

4.2.2.1 *Purpose* — This capability allows the host to query for the equipment data variables and is useful during initialization and synchronization.

4.2.2.2 Definitions

Report ID (RPTID) — A unique identifier of a specific report. See SEMI E5 for a full definition of this data item.

Variable Data (V) — A variable data item containing status, discrete, or constant data. See SEMI E5 for a full definition of this data item.

4.2.2.3 *Detailed Description* — The host may request a report containing data variables from the equipment by specifying the RPTID. It is assumed that the report has been previously defined (e.g., using the Define Report S2,F33 transaction (see Section 4.2.1)). The values of any status variables (SV's) and equipment constants (ECV's) contained within the report must be current. Discrete data values (DVVAL's) are only guaranteed to be valid upon the occurrence of a specific collection event. If DVVAL cannot be specified in equipment due to some restrictions depend on hardware and/or software conditions, the zero length item is reported.

4.2.2.4 Requirements

— Variable data items (V's) and associated units of measure must be provided by the equipment manufacturer.

¹⁷ Define Report Acknowledge Code, see SEMI E5 for full definition of this Data Item.

4.2.2.5 Scenario

Host Requests Report:

COMMENT	HOST	EQUIPMENT	COMMENT
Host requests data variables contained in report RPTID	S6,F19-->	<--S6,F20	Equipment responds with a list of variable data for the given RPTID.

4.2.3 Trace Data Collection

4.2.3.1 Purpose — Trace data collection provides a method of sampling data on a periodic basis. The time-based approach to data collection is useful in tracking trends or repeated applications within a time window, or monitoring of continuous data.

4.2.3.2 Definitions

Data Sample Period (DSPER) — The time delay between samples. See SEMI E5 for a full definition of this data item.

Reporting Group Size (REPGSZ) — The number of samples included per trace report transmitted to the host. See SEMI E5 for a full definition of this data item.

Status Variable (SV) — Status data item (included in trace report). See SEMI E5 for a full definition of this data item.

Status Variable ID (SVID) — A unique identifier of a status variable. See SEMI E5 for a full definition of this data item.

Total Samples (TOTSMP) — Number of samples to be taken during a complete trace period. See SEMI E5 for a full definition of this data item.

Trace Request ID (TRID) — An identifier associated with a trace request definition. See SEMI E5 for a full definition of this data item.

4.2.3.3 Detailed Description — The equipment shall establish a trace report as instructed by the host (S2,F23). For a trace report (S6,F1), the host shall designate a name for the trace report (TRID), a time interval for data sampling (DSPER), the total number of samples to be taken (TOTSMP), the number of samples per trace report (REPGSZ), and a listing of which data will be sent with the report (SVID's). The number of trace reports sent to the host is determined by total samples divided by reporting group size (TOTSMP/REPGSZ).

The equipment shall sample the specified data (SV's) at the interval designated by the host (DSPER) and shall send a predefined trace report to the host for the

specified reporting group size (REPGSZ). The trace report definition shall be automatically deleted from the equipment after the last trace report has been sent.

The host may modify or re-initiate a trace function currently in progress by specifying the same TRID in a trace request definition, at which point the old trace shall be terminated and the new trace shall be initiated, or the host can instruct the equipment to terminate a trace report prior to its completion by specifying TOTSMP = 0 for that TRID, at which point the trace report definition shall be deleted.

A detailed example is included as Application Note A.4.

4.2.3.4 Requirements

- The equipment must have a local mechanism (e.g., internal clock) for triggering the periodic sampling and transmission of trace reports to the host.
- A minimum of four (4) concurrent traces shall be supported by the equipment. The same SVID may be collected in multiple traces simultaneously.
- All SVID's available at the equipment shall be supported for trace data collection. The exception to this is any SV that will not fit into a single block.

NOTE 5: SEMI E5 provides for SV's to be of a list format. Since this may in practice be a variable list, there is a potential problem with such an SV supported by the Trace Data Collection capability. This is a problem with the SEMI E5 standard. Care should be exercised in the use of SV's using the list format.

4.2.3.5 Scenario

Host Initiates Trace Report:

COMMENT	HOST	EQUIPMENT	COMMENT
Trace Data initialization requested	S2, F23-->	<--S2, F24	Acknowledge, trace initiated [D0] TOTSMPS REPGSZ times [D0] REPGSZ many times: collect SVID ₁ ,...SVID _n data, delay time by DSPER. [END_D0]
Acknowledge receipt	S6, F2-->	<--S6, F1	Send SV ₁ ,...SV _n [END_D0]
Optional: Request trace termination prior to completion (TOTSMPS = 0)	S2, F23-->	<--S2, F24	Acknowledge premature termination

4.2.4 Limits Monitoring — This capability relates to the monitoring of selected equipment variables and has three primary aspects:

- Defines a standard set of monitoring zones and limits.
- Provides for reporting to the host when selected equipment variables transition between monitoring zones.
- Empowers the host to modify the values of the variable limit attributes for these same selected equipment variables.

4.2.4.1 Purpose — The limits monitoring capability provides the host a means of monitoring equipment conditions by a flexible, efficient, and asynchronous method which is consistent across equipment. It eliminates the need for constant polling of equipment by the host for current status values. Further, this capability allows the host to implement changes in the monitoring range as needed. This capability has application to both production operation and diagnostic/testing scenarios, and it also has applicability to statistical process control.

4.2.4.2 Definitions

LimitVariable — DVVAL containing the VID of a specific equipment variable for which a zone transition collection event has been generated.

EventLimit — DVVAL containing the LIMITID of the limit crossed by LimitVariable.

TransitionType — DVVAL which defines the direction of the zone transition which has occurred: 0 = transition

from lower to upper zone, 1 = transition from upper to lower zone.

Limit — Used in this section to represent the set of variable limit attributes that completely describe a variable monitoring “barrier.” The attributes include VID, Units, UPPERDB, LOWERDB, LIMITMAX, and LIMITMIN. In some contexts it may be interpreted more narrowly as the combination of UPPERDB and LOWERDB.

LIMITID_n — Refers to the identifier of a specific limit (as defined by UPPERDB and LOWERDB) among the set of limits for a monitored equipment variable. LIMITIDs are consecutively numbered, beginning at one through the number of limits possible (seven minimum).

Monitoring Zone — A subset of the possible range of values for a variable of interest to the host. A single limit divides the range into two zones. Multiple limits may be combined to divide the range even further.

Zone Transition — The movement of a variable value from one monitoring zone to another. This transition is a collection event and has a corresponding CEID.

Deadband — An overlap of two zones implemented to prevent constant zone transitions by a variable sitting on or near a limit (i.e., “chattering”).

UPPERDB — A variable limit attribute that defines the upper boundary of the deadband of a limit.¹⁸ The value applies to a single limit (LIMITID) for a specified VID.

¹⁸ The format and units must be the same as the format of the variable being monitored.

Thus, UPPERDB and LOWERDB as a pair define a limit.

LOWERDB — A variable limit attribute that defines the lower boundary of the deadband of a limit.¹⁸ The value applies to a single limit (LIMITID) for a specified VID. Thus, UPPERDB and LOWERDB as a pair define a limit.

UPPER ZONE — The range of values lying above a limit.

LOWERZONE — The range of values lying below a limit.

LIMITMAX — The maximum value for any limits of a specific equipment variable. This value is set by the equipment manufacturer and typically coincides with the maximum value allowed for the monitored variable.

LIMITMIN — The minimum value for any limits of a specific equipment variable.¹⁹ This value is set by the equipment manufacturer and typically coincides with the minimum value allowed for the monitored variable.

Undefined — When used in reference to variable limits, it indicates that monitoring/reporting of zone transitions involving that particular limit are disabled.

4.2.4.3 Description — The limits monitoring capability provides the host with a minimum of seven configurable limits or barriers that may be applied to selected equipment status variables (SV's) of the types floating point, integer, and boolean. When one of these barriers is crossed, a collection event is generated to alert the host to a change in monitoring zone or state of the monitored variable. These seven limits may be combined in a variety of ways to match the needs of the host system.¹⁹ An illustration of a combination of five of the limits to provide one type of variable monitoring is shown in Figure 4.2.1.²⁰ This section describes the key aspects of limits monitoring. Detailed implementation examples of limits monitoring are provided as Application Note A.7.

NOTE 6: While the SEMI E5 standard allows SV's to be lists, such variable lists are not allowed under this capability.

4.2.4.3.1 Monitoring Limit Characteristics — A limit is defined by a set of attributes that include the variable (VID) to which the limit corresponds, the units of that variable, the maximum and minimum possible values of

the limit (LIMITMAX and LIMITMIN) and the specific borders of the limit (UPPERDB and LOWERDB). See Figure 4.2.2. There is a limitation to the values of UPPERDB and LOWERDB which may be stated as:

$$\text{LIMITMAX} \geq \text{UPPERDB} \geq \text{LOWERDB} \geq \text{LIMITMIN}$$

A limit divides the possible range of variable values into two parts, the upper zone and the lower zone. At any time, the monitored variable is considered to be in one and only one of these zones. However, as Figure 4.2.2 shows, these two zones have an area of overlap. This is called the deadband.

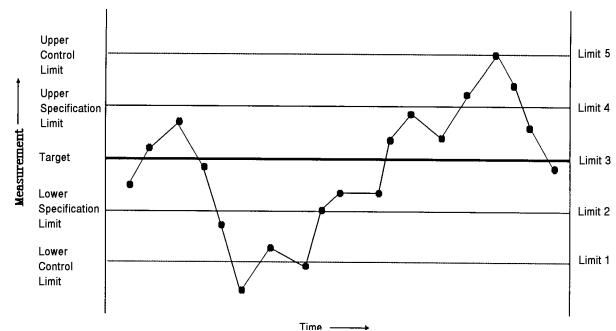


Figure 4.2.1
Limit Combination Illustration: Control Application

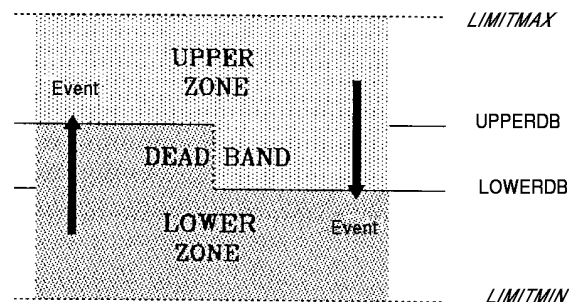


Figure 4.2.2
Elements of One Limit

The deadband is a key concept of limits monitoring, especially for floating point variables. Its purpose is to prevent a phenomenon known as chattering — the repeated changing of zones due to small, rapid fluctuations in variable value while near the zone boundary. In practice, the value of a variable must reach the opposite boundary of the deadband before a zone transition can occur. Thus, if a variable's value

¹⁹ Note that while at least seven limits per variable are available from the equipment, the host need not use all seven.

²⁰ This illustration shows the reading which might be available to the equipment, not the limit excursions reported to the host. Reporting is covered later in the section.

reaches the UPPERDB and transitions into the upper zone, it will not return to the lower zone until it falls back to the LOWERDB. The difference between UPPERDB and LOWERDB should always be greater than the typical amplitude of those fluctuations deemed to be insignificant. In some cases, the width of the deadband may set to zero (i.e., UPPERDB = LOWERDB). At first glance, this would seem to make indeterminate the current zone when an integer value sits on the limit. This is not the case, however, when movement of the value is considered. To illustrate, an example is given, assuming that UPPERDB = LOWERDB = 100. The list shows consecutive readings of the variable and the resultant zone:

```

99 Lower Zone (Initial Reading)
101 Upper Zone (Zone Transition)
100 Lower Zone (Zone Transition)
100 Lower Zone
99 Lower Zone
100 Upper Zone (Zone Transition)

```

Transition from one zone into another generates a collection event, as might be reported via S6,F11. The host has the option of receiving notification by enabling event reporting for the event. For each variable that has monitoring capability, one CEID is reserved to indicate zone transitions for that variable. To aid in the determination of the nature of a transition event, three DVVAL's have been defined:

LimitVariable — The VID of the monitored variable to which the collection event refers.

EventLimit — Contains the LIMITID of the limit reached or crossed by LimitVariable.

TransitionType — Defines the direction of the zone transition which has occurred: 0 = transition from lower to upper zone, 1 = transition from upper to lower zone.

Sampling frequency is an important element of limits monitoring and should be considered during equipment specification. If changes in variable value are relatively fast compared to sampling frequency, it is possible for some zone transitions to be missed or for multiple zone transitions to occur between readings. Since it is possible for zone transitions to occur “simultaneously” or for limits to be identically defined, the DVVAL EventLimit has been defined to allow for a list of multiple zone transitions of a variable to be reported with a single collection event.

It also should be emphasized that a single CEID is used to report transitions in both directions across a limit.

Thus, reporting for one direction but not the other cannot be configured.

The functionality of each limit for each variable can be described with the state model shown in Figure 4.2.3. Below, the three states are described more fully, followed by a table defining the transitions.

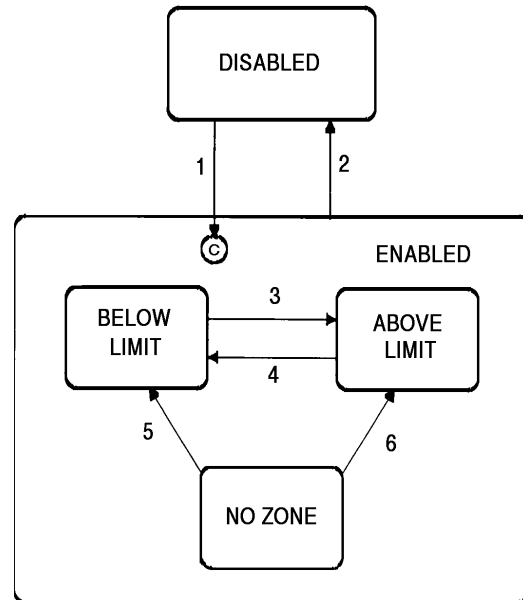


Figure 4.2.3
Limit State Model

ABOVE LIMIT

A variable is considered to be above a limit when its value increases to equal or exceed the upper boundary of the deadband, UPPERDB. The significance attached to this state is a function of the host's usage.

BELOW LIMIT

A variable is considered to be below a limit when its value decreases to equal or fall below the lower boundary of the deadband, LOWERDB. The significance attached to this state is a function of the host's usage.

NO ZONE

In some circumstances it is possible for the variable value to be in neither the upper zone nor the lower zone. This may occur upon definition of a new limit or upon equipment startup when the value of the variable lies in the deadband. In this case, the active state of the limit is considered to be NO ZONE. The limit shall remain in this state until the variable value reaches either boundary of the deadband.

4.2.4.3.2 *Modification of Limit Values* — Values for the monitoring limits on any monitored variable may be modified by the host using the message transaction S2,F45/F46 (Define Variable Limit Attributes). The equipment must reject any S2,F45 message which contains limit information which conflicts with the following rules:

- $LIMITMAX \geq UPPERDB \geq LOWERDB \geq LIMITMIN$;
- If either UPPERDB or LOWERDB is defined, both must be defined; if either UPPERDB or LOWERDB is undefined, both must be undefined.

The first rule is defined and graphically depicted in Figure 4.2.2. The second rule refers to the host's ability to turn any limit "on" or "off". While a minimum of seven limits must be available for each monitored variable, it will be common for the host application to require less than seven or even none of the limits be used. The limits not needed can be disabled by leaving the values for UPPERDB and LOWERDB "undefined". Limits may be disabled for a VID or for all monitored VIDs by using zero length lists in the S2,F45 message.

All monitored variables must be one of three types: integer, floating point, or Boolean. This may be accomplished by using the following formats: 11, 20, 3(), 4(), 5().

NOTE 7: The binary format is not allowed. If the ASCII format is used, the equipment shall perform a conversion into

one of the numeric types before performing any value comparisons, both for limit validations and zone transitions.

4.2.4.3.3 *Limit Values Request* — The host may request the current limit values for a specified VID using the message transaction S2,F47/F48 (Variable Limit Attribute Request).

4.2.4.4 *Requirements*

- A minimum of seven limits per monitored variable must be available.
- One CEID per monitored variable must be supplied for zone transition reporting.
- All limit definitions must be kept in non-volatile storage.
- The equipment must enforce the limit validation rules defined above.
- The specification and documentation of which variables may be monitored with this capability is the responsibility of the equipment manufacturer based on the specific instance of equipment. This subject also may be addressed by equipment models of classes of semiconductor equipment.

Table 4.2 Limit State Transition Table

#	Current State	Trigger	New State	Action	Comment
1	DISABLED	Limit attributed defined w/ S2,F45.	ENABLED	None	The substate of ENABLED is determined by the current value of the monitored variable.
2	ENABLED	Limit attributes set to undefined w/ S2,F45.	DISABLED	None	None
3	BELOW LIMIT	Variable Increased to be $\geq UPPERDB$	ABOVE LIMIT	None	Zone Transition.
4	ABOVE LIMIT	Variable decreases to be $\leq LOWERDB$	BELOW LIMIT	None	Zone Transition.
5	NO ZONE	Variable decreases to be $\leq LOWERDB$	BELOW LIMIT	None	Zone Transition.
6	NO ZONE	Variable increases to be $\geq UPPERDB$	ABOVE LIMIT	None	Zone Transition.



4.2.4.5 Scenarios — Zone Transition Event occurs in equipment:

COMMENTS	HOST	EQUIPMENT	COMMENTS
Send enabled event report to host as shown in Section 4.2.1.			

Host defines Limit Attributes:

COMMENTS	HOST	EQUIPMENT	COMMENTS
[IF] S2,F45 is Multi-block [THEN] Send Multi-block inquire S2,F39-->		<--S2,F40	Multi-block grant.
[END IF] Host defines new variable limit attributes.	S2,F45-->	<--S2,F46	Equipment acknowledges host request.

Host queries equipment for current Limits:

COMMENTS	HOST	EQUIPMENT	COMMENTS
Host queries equipment for current variable limit attribute definitions.	S2,F47-->	<--S2,F48	Equipment returns report containing requested variable limit attribute values.

4.2.5 Status Data Collection

4.2.5.1 *Purpose* — This capability allows the host to query the equipment for selected status information and is useful in synchronizing with equipment status.

4.2.5.2 Definitions

Status variable value (SV) — A data item containing the value of a status variable. See SEMI E5 for a full definition of this data item.

Status variable ID (SVID) — A unique identifier of a status variable. See SEMI E5 for a full definition of this data item.

4.2.5.3 *Detailed Description* — The host may query equipment status by specifying the desired SVID's. Upon such a request, the equipment sends the host the value of the selected status variables. The host also may request the description (name and units) of any or all available status variables.

4.2.5.4 Requirements

- The equipment manufacturer must provide unique SVID's for the various status variables (SV) available for data collection in the equipment.
- All status data is available for status data collection. See Section 5.2, Variable Item List, for a list of status variables.
- All SV's must contain valid data whenever provided to the host.

4.2.5.5 Scenario

Request Equipment Status Report:

COMMENT	HOST	EQUIPMENT	COMMENT
Host requests report of selected status variable values	S1,F3-->	<--S1,F4	Equipment responds with the requested status variable data.

Request Equipment Status Variable Name list:

COMMENT	HOST	EQUIPMENT	COMMENT
Host requests equipment to identify selected status variables.	S1,F11-->	<--S1,F12	Equipment responds with the requested status variable descriptions.

4.2.6 On-line Identification

4.2.6.1 *Purpose* — Implementation of SEMI E5 (a GEM Fundamental Requirement) requires the equipment to accept the S1,F1 from the Host at any time while it is ONLINE and COMMUNICATING, and respond with S1,F2. The On-line Identification capability describes the host-initiated scenario. The equipment-initiated scenario is used for a different purpose and is defined in sections 3.3 and 4.12 describing the GEM “Control Model”.

4.2.6.2 Definitions

Equipment Model Type (MDLN) — ASCII string containing the equipment model. See SEMI E5 for a full definition of this data item.

Equipment Software Revision Code (SOFTREV) — ASCII string containing the equipment software revision. See SEMI E5 for a full definition of this data item.

4.2.6.3 *Detailed Description* — On-line Identification allows the host to verify the presence and identity of the equipment.

4.2.6.4 *Requirements* — (Host-Initiated) An S1,F2 response from the equipment must provide MDLN and SOFTREV information which reflects the hardware and software configuration of the equipment.

SOFTREV must uniquely identify different releases of equipment software. Any change in equipment software must result in a corresponding change to SOFTREV.

The equipment-initiated S1,F1 is not required except as described in sections 3.3 and 4.12 describing the GEM “Control Model”.

4.2.6.5 Scenario:

Host Initiated

COMMENT	HOST	EQUIPMENT	COMMENT
Are you there?	S1,F1-->	<--S1,F2	Equipment replies with MDLN and SOFTREV.

4.3 Alarm Management — The alarm management capability provides for host notification and management of alarm conditions occurring on the equipment.

4.3.1 Purpose — Historically, a precise definition of an equipment alarm has been absent. Consequently, differing interpretations have resulted in inconsistent implementations. This is addressed by providing a more rigorous definition (see definition in Section 4.3.2 below) of an alarm.

In addition, it is often important for equipment to report more extensive information to the host than has been available in the S5,F1/F2 (Alarm Report Send/Acknowledge) transaction. The data required in such cases is very dependent on equipment type, host information requirements, and alarm situation. This issue is addressed by providing event reporting methods that are tied to alarm state changes.

Lastly, the alarm management capability provides mechanisms for:

- Reporting the time of an alarm state change,
- Uploading a list of alarm texts,
- Enabling and disabling the notification of specific alarms, and
- Host query of alarms set and enabled status on the equipment.

4.3.2 Definitions

Alarm — An alarm is related to any abnormal situation on the equipment that may endanger people, equipment, or material being processed. Such abnormal situations are defined by the equipment manufacturer based on physical safety limitations. Equipment activities potentially impacted by the presence of an alarm shall be inhibited.

Note that exceeding control limits associated with process tolerance does not constitute an alarm nor do normal equipment events such as the start or completion of processing.

AlarmsEnabled — Status value consisting of a list of the alarm ID's currently enabled for reporting to the host. See SEMI E5 for a full definition of this variable data item.

AlarmsSet — Status value consisting of a list of the alarm ID's currently in the ALARM SET (or unsafe) state. See SEMI E5 for a full definition of this variable data item.

ALCD — Alarm code data item used in the S5,F1 (Alarm Report Send) and S5,F6 (List Alarm Data) messages. This code is divided into two parts, the alarm set/cleared bit and the 7 bit alarm category code. Only the set/cleared bit is used—bit 8 = 1 means alarm set, = 0 means alarm cleared. The alarm category code is not used. See SEMI E5 for a full definition of this data item.

ALID — Alarm identifier. See SEMI E5 for a full definition of this data item.

ALTX — Data item contained in the S5,F1 and S5,F6 messages containing a brief textual description of an alarm. See SEMI E5 for a full definition of this data item.

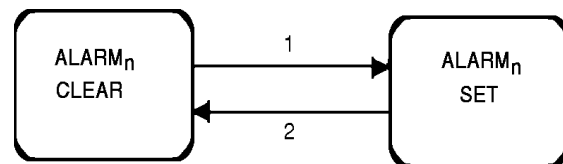


Figure 4.3
State Diagram for Alarm ALID_n

4.3.3 Detailed Description — Two alarm notification mechanisms are defined to achieve the flexibility necessary for the reporting required by host systems. First, stream 5 alarm reporting enables a brief, yet fixed, method for notification of alarm occurrences using the S5,F1/F2 transaction. Second, to address the host's potential need for more extensive and flexible data reporting, two collection events ("alarm set" and "alarm cleared") are defined for each possible alarm condition on the equipment to allow the use of event data collection mechanisms. In the latter case, reports are sent by the equipment using the Event Report/Acknowledge transaction (see Section 4.2 on event data collection).

The alarm_n detected and cleared events are derived from the state model for an alarm (see Figure 4.3 and Table 4.3.1). In this model each of n alarms can be in one of two possible states, either ALARM CLEAR or ALARM SET. The transition from the ALARM CLEAR to the ALARM SET state is defined as the collection event "Alarm_n detected" (transition 1). Conversely, the transition from ALARM SET to ALARM CLEAR is defined as the collection event "Alarm_n cleared" (transition 2).

NOTE 8: The alarm capability is intended as an addition to standard safety alarms (e.g., lights, horns). There is no intent to replace direct operator notification of such problems, nor is there the expectation that the host can prevent or directly address such alarms.

Table 4.3.1 Alarm State Transition Table

#	Current	Trigger	New State	Action	Comment
1	ALARM _n CLEAR	Alarm _n is detected on the equipment.	ALARM _n SET	Initiate local actions (if any) to ensure safety. Update AlarmsSet and ALCD _n values. Generate and issue alarm message if enabled.	Inhibited activities require operator or host intervention prior to resuming.
2	ALARM _n SET	Alarm _n is no longer detected on the equipment.	ALARM _n CLEAR	Update AlarmsSet and ALCD _n values. Generate and issue alarm message if enabled.	Inhibited activities require operator or host intervention prior to resuming.

The equipment manufacturer is responsible for identifying all alarms on their equipment by:

- Applying the above alarm definition,
- Consulting Application Note A.3 for examples of alarms for various equipment configurations,
- Noting that the presence of an alarm typically requires some action or intervention before resuming safe operation of the equipment, and by
- Referring to Table 4.3.2 below which delineates the differences between events and alarms.

Table 4.3.2

EVENT	ALARM
Any occurrence detectable by the equipment.	Related to only those occurrences that are abnormal, undesirable, AND endanger people, equipment, or physical material being processed.
Certain events may trigger a state transition(s).	Each alarm has an associated two-state state model: ALARM SET (or unsafe) and ALARM CLEAR (or safe).
Equipment activities are not necessarily inhibited by the occurrence of an event (unless it is associated with an alarm or intentional inhibit).	The presence of an alarm inhibits equipment activities to ensure safe operation until the alarm condition is cleared.
Certain events may occur in an expected sequence.	Alarms may occur at any time.

4.3.3.1 Enable/Disable Alarms — Upon request from the host, the equipment shall enable or disable reporting of certain alarms. Enabling or disabling a given alarm shall impact the communication of both the alarm set and clear messages equally (i.e., turn them both on or both off). This is not the case for enabling/disabling of the associated collection events, where the alarm-set and alarm-cleared events can be enabled and disabled separately. The current enable/disable settings must be stored in non-volatile memory. Changes to the enable/disable settings must be reflected in the AlarmsEnabled status value.

NOTE 9: The alarm itself is not being enabled or disabled, but the reporting of the alarm through SECS-II messages is being enabled or disabled.

4.3.3.2 Send Alarm Report — Upon detecting a change in the status of a given alarm, an associated alarm state model shall transition to the opposite state. Following initiation of local actions necessary to ensure safety, the equipment must update the AlarmsSet and associated ALCD values and send an alarm message and/or an event message to the host assuming one or both are enabled. Alarm messages must be sent before the corresponding event messages if both are enabled.

NOTE 10: The ALCD is divided into two parts, the alarm set/cleared bit and the 7 bit alarm category code. Only the set/cleared bit is used—bit 8 = 1 means alarm set, = 0 means alarm cleared. The alarm category code is not used.

4.3.3.3 List Alarm Text — Upon request from the host, the equipment sends values of alarm text associated

with a specified list of alarm ID's using the S5,F5/F6 (List Alarms Request/Data) transaction.

4.3.3.4 List Currently Set Alarms — The host may obtain a listing of currently set alarms by employing any data collection method specified by GEM and referencing the variable data item called "AlarmsSet" (e.g., including " AlarmsSet in a report definition). When reported, AlarmsSet must contain a list of all currently set alarms. Each alarm set or clear occurrence must cause a change to the AlarmsSet status value prior to reporting it to the host.

4.3.3.5 List Currently Enabled Alarms — The host may obtain a listing of currently enabled alarms by employing any data collection method specified by GEM and referencing the variable data item called "AlarmsEnabled" (e.g., including AlarmsEnabled in a report definition). When reported, AlarmsEnabled must contain a list of all alarm ID's currently enabled for reporting. The equipment must update the value of AlarmsEnabled upon corresponding change(s) to the enable/disable settings.

4.3.4 Requirements

4.3.5 Scenarios

NOTE 11: Consult event reporting sections of this document for descriptions of enabling, disabling, and sending collection event reports.

Enable/Disable Alarms:

COMMENTS	HOST	EQUIPMENT	COMMENT
Enable/disable Alarm	S5,F3-->	<-- S5,F4	Acknowledge

Upload Alarm Information:

COMMENTS	HOST	EQUIPMENT	COMMENT
Request alarm data/text	S5,F5-->	<-- S5,F6	Send alarm data/text

Send Alarm Report:

COMMENTS	HOST	EQUIPMENT	COMMENT
Alarm occurrence detected by the equipment.		<-- S5,F1	Send alarm report (if enabled).
Acknowledge	S5,F2-->	<-- S6,F11	Send event report (if enabled).
Acknowledge	S6,F12-->		

4.4 Remote Control

4.4.1 *Purpose* — This capability provides the host with a level of control over equipment operations.

4.4.2 Definitions

Host Command Parameter (CPNAME/CPVAL/CEPVAL) — A parameter name/value associated with a particular host command (S2,F41/S2,F49). The equipment manufacturer must provide unique names (CPNAMEs) for any supported command parameters. Command parameters are not specified in this document but are left to equipment manufacturers to define. Equipment models of specific classes of semiconductor equipment also may address this issue. Note that if there are no associated parameters a zero-length list is sent. The data item CEPVAL, which can be defined as a list, allows grouping of related parameters within a main parameter. If the CEPVAL is defined as a single (non-list) item, then it is the equivalent of a CPVAL.

The uses of OBJSPEC in the header structure of the S2,F49 Enhanced Remote Command allows the equipment supplier to define a set of unique identifiers for different objects within the equipment such as: equipment sub-systems, sub-system components, processing stations, ports, and exchange stations.

4.4.3 *Description* — The equipment responds to host commands that provide the following functions relative to individual equipment implementations:

- Start processing
- Select a process program or recipe
- Stop processing
- Temporarily suspend processing
- Resume processing
- Abort processing

Additional commands may be implemented by the equipment manufacturer (e.g., vent chamber, clear material, open door).

Remote commands shall be interpreted as “request action be initiated” rather than “do action.” The equipment may then respond via S2,F42/S2,F50 with HCACK = 4 if the command “is going to be performed.” This alleviates any transaction timeouts for commands that may take a long time to perform. The completion of the action initiated by the remote command (i.e., HCACK = 0 or 4) must result in either a state transition or other action that generates a collection event upon normal/abnormal completion.

The format for all remote commands is ASCII, with a maximum length of 20 characters. The character set is restricted to the printable characters (hexadecimal 21 through 7E). Note that spaces are not allowed.

The following remote commands (RCMDs), if implemented on the equipment, shall be supported as described below (see Section 3.4 for a description of Equipment Processing States).

NOTE 12: The terms “current cycle” and “safe break point” used below are to be defined by the supplier or within the models of classes of semiconductor equipment.

START — This command is available to the host when a process program or recipe has been selected and the equipment is in the “ready” processing state. The START command instructs the equipment to initiate processing. Variable parameter settings may be included as name/value command parameters CPNAME/CPVAL/CEPVAL.

PP-SELECT — This command instructs the equipment to make the requested process program(s) available in the execution area. The process programs (PPIDs) are specified via the command parameter list. A status variable (PPExecName) contains the PPID of the process program(s) currently selected.

RCP-SELECT — This command uses the Enhanced Remote Command S2,F49 to instruct the equipment to prepare the requested recipes for execution in the execution area. The recipes and variable parameters are specified via command parameter lists. Each recipe specification may be accompanied by new variable parameter settings, if any, in the command parameter list. A status variable RcpExecName contains the recipe specifiers or identifiers of the recipes currently selected.

STOP — Command to complete the current cycle, stop in a safe condition and return to the “idle” processing state. Stop has the intent of stopping the process. The equipment is not required to support the continuation of processing. Stop leaves material either fully processed or partially processed so that the processing can be later completed. For example, for a single wafer process tool, five wafers have been processed while the remaining wafers remain unprocessed.

PAUSE — Command to suspend processing temporarily at the next safe break point. Pause has the intent of resuming the process at the same point where it was paused. The process may be RESUMED, STOPPED, or ABORTED while in a PAUSED condition. RESUME shall be able to continue the process from the same point where it was paused.

RESUME — Command to resume processing from the point where the process was paused.

ABORT — Command to terminate the current cycle prior to its completion. Abort has the intent of immediately stopping the process and is used because of abnormal conditions. Abort makes no guarantee about the subsequent condition of material. In the above example, the wafers being processed at the time of the abort may not be completely processed. Other AbortLevels > 1 may be defined by the manufacturer or addressed by models of specific classes of semiconductor equipment.

CPNAME = AbortLevel, CPVAL = 1 means terminate current cycle at the next “safe break point,” retrieve all material, stop in a safe condition and return to the idle state in the processing state machine.

4.4.4 Requirements

- The following Remote Commands, as defined under Descriptions, must be implemented on equipment to satisfy minimum requirements for this capability:
 - START
 - STOP
- The RCMD value for all commands supported on the equipment must be recognized if sent with all upper-case characters (e.g., “STOP”, “START”, “PP-SELECT”, “PAUSE”, etc.). In addition to accepting strings with all upper-case characters, the equipment can optionally accept strings with all lower-case characters or mixed-case strings. The equipment documentation should describe whether or not the optional lower-case or mixed-case strings are supported.
- Stream 2 currently provides for Host Command Send and Enhanced Remote Command. The equipment shall support one or both methods, based on appropriateness.
- The Enhanced Remote Command is used to address size, complexity, or the need to target a specific sub-system within the equipment, (i.e., processing station, port, exchange station, material handler, chamber).

4.4.5 Scenarios

4.4.5.1 Host Command Send Scenario

COMMENTS	HOST	EQUIPMENT	COMMENT
Host Command	Send S2,F41-->	<-- S2,F42	Host Command Acknowledge
			[IF] Command Accepted (HCACK = 0 or 4)
		<-- S6,F11	[THEN] Event Report-state change or other collection event occurrence.
Event Report Acknowledge	S6,F12-->		

4.4.5.2 Enhanced Remote Command Scenario

COMMENTS	HOST	EQUIPMENT	COMMENT
Enhanced Remote Command	S2,F49-->	<--S2,F50	EnhancedRemote Command Acknowledge
			[IF] Command Accepted (HACK = 0 or 4)
		<--S6,F11	[THEN] Event Report-state change or other collection event occurrence.
Event Report Acknowledge	S6,F12-->		

4.5 Equipment Constants

4.5.1 *Purpose* — This capability provides a method for the host to read and to change the value of selected equipment constants on the equipment.

4.5.2 *Definitions* — None.

4.5.3 *Description* — This capability allows the host to reconfigure equipment constants to support a variety of situations. The following functions are included:

Host Sends Equipment Constants — Allows the host to change the value of one or more equipment constants.

Host Equipment Constant Request — Allows the host to determine the current value of equipment constants.

Host Equipment Constant Namelist Request — Allows the host to retrieve basic information about the equipment constants available at the equipment.

4.5.4 Requirements

- Equipment constants must be stored in non-volatile memory.
- The equipment must be in a “safe” condition to accept new constant(s) settings as defined by the equipment manufacturer.
- The equipment must provide a collection event to alert the host whenever an equipment constant is changed by the operator. Information indicating which constant was changed shall be available for the event report.

4.5.5 Scenarios

Host Sends Equipment Constants:

COMMENTS	HOST	EQUIPMENT	COMMENTS
Host sends equipment constants	S2,F15-->	<-- S2,F16	EAC = 0 equipment sets constants

Host Equipment Constants Request:

COMMENTS	HOST	EQUIPMENT	COMMENTS
Host constant request	S2,F13-->	<-- S2,F14	Equipment constant data

NOTE: This capability also can be accomplished using S6,F19 & S6,F20. See Section 4.2.

Host Equipment Constant Namelist Request:

COMMENTS	HOST	EQUIPMENT	COMMENTS
Host constant namelist request	S2,F29-->	<-- S2,F30	Equipment constant namelist

Operator Changes Equipment Constant

COMMENTS	HOST	EQUIPMENT	COMMENTS
			Operator changes equipment constant at equipment operator console.
		<-- S6,F11	Equipment reports equipment constant change.
Host acknowledges event	S6,F12-->		

4.6 Process Program Management — Process programs and recipes must be managed through interaction between the equipment and the host system.

4.6.1 Purpose — Process program management provides a means to transfer process programs or recipes, and to share the management of those process programs or recipes, between the host and equipment.

4.6.2 Definitions

PPError — A text data value with information about verification errors of a process program that failed verification. If the equipment provides an event for recipe verification and/or recipe verification failure, then PPErrror shall be a DVVAL. Otherwise, PPErrror shall be an SV.

PPFormat — A variable (SV) indicating the type or types of process programs and recipes that are supported.

- 1 = Unformatted process programs
- 2 = Formatted process programs
- 3 = Both unformatted and formatted process programs
- 4 = Execution recipes
- 5 = Large unformatted process programs
- 6 = Large formatted process programs
- 7 = Both large unformatted and large formatted process programs
- 8 = Large execution recipes

and a combination of these formats. See SEMI E5 for a complete list.

4.6.2.1 Definitions for Process Programs

Process Program — A process program is the pre-planned and reusable portion of the set of instructions, settings, and parameters under control of the equipment that determine the processing environment seen by the manufactured object and that may be subject to change between runs or processing cycles.

Process Program Identifier — A text string (PPID) used to identify a process program.

Formatted Process Program — A process program that is presented as an ordered sequence of command codes with their associated parameters as dictated by S7,F23, and S7,F26. Where formatted process programs are supported, equipment must also provide information sufficient to allow a user at the host to create, display, modify, and partially verify their contents (for example, that information provided in S7,F22).

Unformatted Process Program — An unformatted process program is transferred without structure as the single data item PPBODY (refer to SEMI E5 for a complete description of PPBODY).

Process Program Change Event — The collection event associated with the occurrence of the creation, modification, or deletion of a process program by the operator.

PPChangeName — A data value (DVVAL) containing the PPID of the process program affected by the event Process Program Change Event. See SEMI E5 for a full definition of this variable data item.

PPChangeStatus — The action taken on the process program named by PPChangeName. This variable is valid for the collection event Process Program Change Event. See SEMI E5 for a full definition of this variable data item.

PPExecName — The status variable containing the PPID(s) of the currently selected process program(s). See SEMI E5 for a full definition of this variable data item.

PP-SELECT — The remote command used to select one or more process programs for execution. The process programs are specified by PPID via the command parameter list.

Process Program Verification — Verification is syntax checking of a process program. Verification ensures only that a process program is structured correctly. It does not ensure that the program has the correct parameters to run a particular process or product (see Process Program Validation). Equipment supporting unformatted process programs should provide a variable DVVAL PPErrror that provides information to the user concerning the error or errors when an attempt to verify a process program fails.

NOTE 14: It may not be possible for the equipment to verify unformatted process programs other than to check the size of the program and internal program checksums. Equipment has no standard means of indicating the type of error encountered in an unformatted process program.

Process Program Validation — Validation is type-and-range checking of parameters in a process program, and is performed after verification.

4.6.2.2 Definitions for Recipes

Execution recipe — A type of recipe stored by the equipment for purposes of editing, verification, and execution.

For complete definitions of execution recipes and their standard attributes, see SEMI E42, Section 6.

Execution Recipe Change Event — The collection event associated with the occurrence of the modification or deletion of an execution recipe stored by the equipment.

Note 15: A recipe is modified whenever its body is changed.

New Execution Recipe Event — The collection event associated with the creation of a new execution recipe at the equipment.

Object form recipe — A recipe with body in a proprietary format that may be presented without structure.

RcpChangeName — A data value (DVVAL) containing the identifier of the recipe affected by the event Execution Recipe Change Event or New Execution Recipe Event. See the SEMI E5 Standard for a full definition of this variable data item.

RcpChangeStatus — The action taken on the recipe named by RcpChangeName. This variable is valid for the

collection event Execution Recipe Change Event or New Execution Recipe Event. See the SEMI E5 Standard for a full definition of this variable data item.

RcpExecName — The status variable containing the specifiers of the currently selected recipe(s). See the SEMI E5 Standard for a full definition of this variable data item.

RCP-SELECT — The remote command used to select one or more recipes for execution. See Section 4.4.3.

Recipe Attribute — Information about the recipe that is transferred with the recipe as a name/value pair. The value may be a single item or a list.

Recipe — A recipe contains both a set of instructions, settings, and parameters that the equipment uses to determine the processing environment (its body or process program) and a set of attributes that provide information about the recipe, such as the date and time the body was last changed.

SEMI E42 defines two types of recipes: *managed recipes* and *execution recipes*. For purposes of GEM, the term *recipe* refers to an *execution recipe* only.

Recipe identifier — A recipe identifier is a formatted text string (RCPID) used to identify the recipe.

Recipe specifier — A formatted text string (RCPSPEC) used in messages to indicate a specific recipe. A recipe specifier includes the recipe identifier. It may also include additional information, such as the name of the specific component of the equipment where the recipe is to be executed (e.g. a process chamber) and the name of a recipe repository on the host.

Recipe Verification — Verification is syntax checking of a recipe's body. Verification ensures that a recipe body is structured correctly and has the correct syntax. It may also provide a check of semantics. It does not ensure that the body has the correct parameters to run a particular process or product (see Recipe Validation).

NOTE 16: Unverified recipes shall be verified upon download.

Recipe Validation — Validation is type-and-range checking of parameters in a recipe, and is performed when the recipe is selected for execution. The recipe may be correct in its syntax and semantics but should fail validation if it can not be executed with the current equipment configuration.

Source form recipe — A recipe with a body that is presented as an ordered sequence of text. A source form recipe may be created and edited off-line to the equipment. Definition of syntax requirements shall be documented, in order to allow proper off-line editing.

Variable Parameters — Variable parameters are recipe parameters that are defined in the body of the recipe and whose run-time values may be set outside of the recipe when the recipe is selected for execution and/or when processing is started. Both the host and the operator may specify new settings as a parameter name/value pair.

Variable Parameter Definition — A variable parameter definition has three parts: the name of the variable parameter, its default setting, and restrictions on the run-time value selected. Variable parameter definitions are stored in the recipe attribute “Parameters”.

4.6.3 Description

4.6.3.1 Process Program Description

Process programs allow the equipment’s process, and/or the parameters used by that process, to be set and modified by the engineer to achieve different results. Different process programs may be required for different products, while often the same process program will be used for all lots of a given product. The engineer must be able to create such programs, to modify current programs, and to delete programs from equipment storage.

For the host to ensure that the proper process programs are in place at the equipment, there must be a means of transferring them from equipment to host and from host to equipment. The host also may need to delete process programs from the equipment’s storage to make room for a process program to be downloaded. In addition, the host must be kept informed whenever a local change occurs in the contents or status of a process program.

Both formatted and unformatted process programs may be uploaded and downloaded. This capability provides for both host- and equipment-initiated transfers. The equipment-initiated transfer may be used at the request of the process engineer or operator at the equipment.

If a process program exists with the same PPID as the one given in the SECS-II message, the old process program must be replaced. The PPID in the e process program in non-volatile storage.

4.6.3.2 Recipe Description

Specifications in Section 4.6.3.1 apply to recipes as well as process programs, with the following differences:

- A recipe contains a body corresponding to a process program. In addition, it contains attributes defined for execution recipes in SEMI E42, Section 6. Recipe attributes are transferred whenever the recipe is downloaded or uploaded.

- The same SECS-II messages are used for all execution recipes, regardless of the internal structure of the recipe body.
- If an execution recipe already exists with the same identifier as the one given in the SECS-II message, the downloaded recipe shall be rejected (not stored) unless the host has specified a “forced overwrite” in the data item RCPOWCODE.
- A recipe currently being edited shall be protected from inadvertent change or overwriting by a recipe with the same identifier that is downloaded during this time. If the downloaded recipe is accepted (stored), the equipment shall require the operator either to save the edited recipe to a new (unused) identifier or to discard it.
- For the equipment to initiate either an upload or download of a recipe, it shall request the host to initiate an upload or download procedure. In addition, it may be necessary to also specify the name of the repository (recipe namespace) at the host.

4.6.3.3 Large Process Programs and Recipes

4.6.3.3.1 Process programs and recipes for certain types of equipment, such as metrology or inspection equipment, contain images and are, therefore, very large. Such process programs and recipes cannot fit into a single multi-block SECS-II message that has a maximum theoretical size of 7,995,148 text bytes. Furthermore, process programs and recipes may also require some preparation prior to transfer.

4.6.3.3.2 The commonly used Stream 7 and Stream 15 transfer functions require that the entire process program be sent in a single message. Thus, it is not possible to send a large process program or recipe with such messages. However, there is an alternative set of Stream 7 and Stream 15 functions that supports large transfers. These messages invoke Stream 13 Data Set Transfer Protocol messages to transfer large process programs (or recipes) by using a sequence of read messages. The completion of such a read transaction is indicated by “ERROR: End of Data”. The Data Set Transfer Protocol does not set any limit on the size of the data set.

4.6.4 Requirements

- The equipment manufacturer shall provide a method to create, modify, and delete process programs or recipes. This method shall exist on either the equipment or on a separate computing system.

- A CEID shall be defined for a collection event for the creation, the deletion, or the modification (completion of an editing session) of a process program (Process Program Change Event). For recipes, there are two separate CEIDs and collection events, one for the creation of a new recipe (New Execution Recipe Event) and one when a recipe is changed or deleted (Execution Recipe Change Event). A New Execution Recipe Event shall occur whenever a new recipe identifier is created through download, edit, copy, or rename operations. A Execution Recipe Change Event shall occur whenever the body of an existing recipe is modified.
- The name (identifier) that the engineer or operator uses to refer to the process program or recipe is the same as the identifier used by the host.
- Upon request from the host or operator, the equipment shall perform the following actions with regard to process programs and recipes stored in non-volatile storage: upload, download, delete, and list current equipment process program or recipe directory.
- The equipment shall be able to store in non-volatile memory the number of process programs or recipes sufficient to execute three unique process cycles. For example, if a wire-bonder requires both an “ALIGN” process program and a “BOND” process program for a full process cycle, then it must provide non-volatile storage for at least three pairs of process programs. These stored process programs or recipes may not be modified in any way by the execution process, nor may the execution process be affected by the modification of any process program or recipes in storage, either by downloading or by local editing, while that process program or recipes is being executed.
- The equipment must provide verification and validation of all downloaded process programs and recipes.
- Stream 7 provides for formatted and unformatted process programs, while Stream 15 provides for recipes. The equipment must support at least one of these three methods.
- The equipment supplier shall document any restrictions on the length or test format of PPID. The maximum length allowed by equipment may be less than that allowed by SECS-II.
- Where recipes are supported, the following requirements also apply:
 - The variable PPFormat shall be provided to indicate to the user the messages supported by the equipment.
 - The recipe and its attributes shall comply to the requirements for execution recipes as defined in SEMI E42, Section 6.

4.6.5 Scenarios for Process Programs

4.6.5.1 Process Program Creation, Editing, or Deletion Process Program Created, Edited, or Deleted by Operator

COMMENTS	HOST	EQUIPMENT	COMMENTS
			New process program created, edited, or deleted by operator at equipment. PPChangeName = PPID PPChangeStatus = 1 (Created) = 2 (Edited) = 3 (Deleted) [IF] CEID for Process Program Change Event enabled [THEN] Send Event Report [END_IF]
Event Report Acknowledge	S6,F12-->	<-- S6,F11	



Process Program Deletion by the Host:

COMMENTS	HOST	EQUIPMENT	COMMENTS
Delete Process Program Send	S7,F17-->	<-- S7,F18	Delete Process Program Acknowledge. The process program is removed from non-volatile storage.

4.6.5.2 Process Program Directory Request

COMMENTS	HOST	EQUIPMENT	COMMENTS
Current EPPD Request	S7,F19-->	<-- S7,F20	Current EPPD Data



4.6.5.3 Process Program Upload

Host-Initiated Process Program Upload — Formatted:

COMMENTS	HOST	EQUIPMENT	COMMENTS
Formatted Process Program Request	S7,F25-->	<-- S7,F26	Formatted Process Program Data
Large Formatted Process Program Request	S7,F43-->	<-- S7,44	Large Formatted Process Program Acknowledge Acknowledge may be "accepted and will be performed later with completion signaled". When the equipment is ready to send:
Send Data Set Send Acknowledge	S13,F2-->	<-- S13,F1	Send Data Set Send
Open Data Set request	S13,F3-->	<-- S13,F4	Open Data Set Data
[START] Repeat until ACKC13 indicates error: Read Data Set Request	S13,F5-->		
[END]		<-- S13,F6	Read Data set Data
Event Report Acknowledge	S6,F12-->	<-- S6,F12	[IF] CEID for upload event enabled [THEN] [IF] ACKC13 is "ERROR: END OF DATA" [Then] Send Event Report "successful upload" [ELSE]
Event Report Acknowledge	S6,F12-->	<-- S6,F11	Send Event Report "bad upload" [END_IF] [END_IF]
Close Data Set Send	S13,F7-->		
		<-- S13,F8	Close Data Set Acknowledge



Host-Initiated Process Program Upload — Unformatted:

COMMENTS	HOST	EQUIPMENT	COMMENTS
Process Program Request	S7,F5-->	<-- S7,F6 ²¹	Process Program Data
Large Process Program Request	S7,F41-->	<-- S7,F42	Large Process Program Acknowledge
			Acknowledge may be "accepted and will be performed later with completion signaled". When the equipment is ready to send:
		<-- S13,F1	Send Data Set Send
Send Data Set Send Acknowledge	S13,F2-->		
Open Data Set request	S13,F3-->	<-- S13,F4	Open Data Set Data
[START] Repeat until ACKC13 indicates error: Read Data Set Request [END]	S13,F5-->	<-- S13,F6	Read Data set Data [IF] CEID for upload event enabled [THEN] [IF] ACKC13 is "ERROR: END OF DATA" [Then]
Event Report Acknowledge	S6,F12-->	<-- S6,F11	Send Event Report "successful upload"
Event Report Acknowledge	S6,F12-->	<-- S6,F11	[ELSE] Send Event Report "bad upload" [END_IF] [END_IF]
Close Data Set Send	S13,F7-->	<-- S13,F8	Close Data Set Acknowledge

²¹ If the process program does not exist, a zero-length list will be sent.



Equipment Initiated Process Program Upload — Formatted:

COMMENTS	HOST	EQUIPMENT	COMMENTS
			[IF] Process program is multi-block
			[THEN]
Process Program Load Grant	S7,F2-->	<-- S7,F1	Process Program Load Inquire
			[END_IF]
Formatted Process Program Acknowledge	S7,F24-->	<-- S7,F23	Formatted Process Program Send
Large Formatted Process Program Acknowledge	S7,F40-->	<-- S7,F39	Large Formatted Process Program Send
Send Data Set Send Acknowledge	S13,F2-->	<-- S13,F1	Send Data Set Send
Open Data Set request	S13,F3-->	<-- S13,F4	Open Data Set Data
[START] repeat until ACKC13 is error:			
Read Data Set Request	S13,F5-->	<-- S13,F6	Read Data set Data
[END]			
			[IF]
			CEID for upload event enabled
			[THEN]
			[IF]
			ACKC13 is "ERROR: END OF DATA"
			[Then]
Event Report Acknowledge	S6,F12-->	<-- S6,F11	Send Event Report "successful upload"
			[ELSE]
Event Report Acknowledge	S6, F12-->	<-- S6,F11	Send Event Report "bad upload"
			[END_IF]
			[END_IF]
Close Data Set Send	S13,F7-->	<-- S13,F8	Close Data Set Acknowledge



Equipment-Initiated Process Program Upload — Unformatted:

COMMENTS	HOST	EQUIPMENT	COMMENTS
			[IF] Process program is multi-block [THEN]
Process Program Load Grant	S7,F2-->	<-- S7,F1	Process Program Load Inquire
Process Program Acknowledge	S7,F4-->	<-- S7,F3	[END_IF] Process Program Send
Large Process Program Acknowledge	S7,F38-->	<-- S7,F37	Large Process Program Send
Send Data Set Send Acknowledge	S13,F2-->	<-- S13,F1	Send Data set Send
Open Data Set request	S13,F3-->	<-- S13,F4	Open Data Set Data
[START] Repeat until ACKC13 is error: Read Data Set Request [END]	S13,F5-->	<-- S13,F6	Read Data set Data
			[IF] CEID for upload event enabled [THEN] [IF] ACKC13 is "ERROR: END OF DATA" [Then]
Event Report Acknowledge	S6,F12-->	<-- S6,F11	Send Event Report "successful upload"
Event Report Acknowledge	S6,F12-->	<-- S6,F11	[ELSE] Send Event Report "bad upload"
Close Data Set Send	S13,F7-->		[END_IF] [END_IF]
		<-- S13,F8	Close Data Set Acknowledge



4.6.5.4 *Process Program Download*

NOTE 17: Formatted process programs must be verified immediately following any download.

While the Process Program Load Inquire/Grant transaction (S7,F1/F2) is required only for the transfer of multi-block process programs, its use is recommended prior to all host-initiated downloads. It provides a means of verifying process program size.



Host-Initiated Process Program Download — Formatted:

COMMENTS	HOST	EQUIPMENT	COMMENTS
[IF] Process program is multi-block [THEN] Process Program Load Inquire	S7,F1 ²² -->		
		<-- S7,F2	Process Program Load Grant
[ENDIF] Formatted Process Program Send	S7,F23-->		
		<--S7,F24	Formatted Process Program Acknowledge Verify process program
			[IF] S7,F27 is multi-block [THEN] Process Program Verification Inquire
	S7,F30-->		Process Program Verification Grant
		<--S7,F29	[END IF] Process Program Verification Send
Process Program Verification Acknowledge	S7,F28-->		
Large Formatted Process Program Send	S7,F39-->		
		<-- S7,F40	Large Formatted Process Program Acknowledge
		<-- S13,F3	Open Data Set Request
Open Data Set Data	S13,F4-->		[START] repeat until ACKC13 is error:
		<-- S13,F5	Read Data Set Request
Read Data Set Data	S13,F6-->		[END]
		<-- S13,F7	Close Data Set Send
Close Data Set Acknowledge	S13,F8-->		
			[IF] S7,F27 is multi-block [THEN] Process Program Verification Inquire
Process Program Verification Grant	S7,F30-->		[END_IF]
		<-- S7,F27	Process Program Verification Send
Process Program Verification Acknowledge	S7,F28-->		

²² S7,F1 should be used only to request permission to transfer a multi-block formatted or unformatted process program. It should not be used to select a process program. For selecting a process program for execution, the remote command PP-SELECT should be used.



Host-Initiated Process Program Download — Unformatted:

COMMENTS	HOST	EQUIPMENT	COMMENTS
[IF] Process program is multi-block [THEN] Process Program Load Inquire	S7,F1 ²¹ -->		
		<--S7,F2	Process Program Load Grant
[END_IF] Process Program Send	S7,F3-->		
		<--S7,F4	Process Program Acknowledge
Large Process Program Send	S7,F37-->		
		<--S7,F38	Large Process Program Acknowledge
		<--S13,F3	Open Data Set Request
Open Data Set Data	S13,F4-->		
		<--S13,F5	[START] repeat until ACKC13 is error: Read Data Set Request
Read Data set Data	S13,F6-->		
		<--S13,F7	[END]
Close Data Set Acknowledge	S13,F8-->		Close Data Set Send
			[IF] S7,F27 is multi-block [THEN] Process Program Verification Inquire
Process Program Verification Grant	S7,F30-->	<--S7,F29	
			[END_IF]
		<--S7,F27	Process Program Verification Send
Process Program Verification Acknowledge	S7,F28-->		



Equipment-Initiated Process Program Download — Formatted:

COMMENTS	HOST	EQUIPMENT	COMMENTS
		<--S7,F25	Formatted Process Program Request
Formatted Process Program Data	S7,F26-->		[IF] S7,F27 is multi-block [THEN]
		<--S7,F29	Process Program Verification Inquire
	S7,F30-->		Process Program Verification Grant
		<--S7,F27	[END IF] Process Program Verification Send
Process Program Verification Acknowledge	S7,F28-->		
		<--S7,F43	Large Formatted Process Program Request
Large Formatted Process Program Acknowledge	S7,F44-->		
		<--S13,F3	Open Data Set Request
	S13,F4-->		
Open Data Set Data			[START] repeat until ACK13 is error:
		<--S13,F5	Read Data Set Request
Read Data Set Data	S13,F6-->		[END]
		<--S13,F7	Close Data Set Send
Close Data Set Acknowledge	S13,F8-->		
			[IF] S7,F27 is multi-block [THEN]
		<--S7,F29	Process Program Verification Inquire
Process Program Verification Grant			[END_IF]
		<--S7,F27	Process Program Verification Send
Process Program Verification Acknowledge	S7,F28-->		



Equipment-Initiated Process Program Download — Unformatted:

COMMENTS	HOST	EQUIPMENT	COMMENTS
Process Program Send	S7,F6-->	<--S7,F5	Process Program Request
Large Process Program Program Acknowledge	S7,F42-->	<--S7,F41	Large Process Program Request
Open Data Set Data	S13,F4-->	<--S13,F3	Open Data Set Request
			[START] repeat until ACKC13 is error:
		<--S13,F5	Read Data Set Request
Open Data Set Data	S13,F6-->		[END]
		<--S13,F7	Close Data Set Send
Close Data Set Acknowledge	S13,F8-->		[IF] S7,F27 is multi-block [THEN] Process Program Verification Inquire [END_IF]
Process Program Verification Grant	S7,F30-->	<--S7,F29	
Process Program Verification Acknowledge	S7,F28-->	<--S7,F27	Process Program Verification Send

4.6.6 Scenarios for Recipes

4.6.6.1 Recipe Creation, Editing, or Deletion

Recipe Created by Operator:

COMMENTS	HOST	EQUIPMENT	COMMENTS
			New recipe created by operator at equipment RcpChangeName = RCPID RcpChangeStatus = 1 (Created) [IF] CEID for New Execution Recipe Event enabled [THEN] Send Event Report
Event Report Acknowledge	S6,F12-->	<--S6,F11	[END_IF]



Recipe Edited or Deleted by Operator:

COMMENTS	HOST	EQUIPMENT	COMMENTS
			New recipe edited, or deleted at equipment RcpChangeName = RCPID RcpChangeStatus = 2 (Modified) or 5 (Deleted) [IF] CEID for Execution Execution Recipe Change Event enabled [THEN] Send Event Report [END_IF]
Event Report Acknowledge	S6,F12-->	<--S6,F11	

Recipe Deletion by the Host:

COMMENTS	HOST	EQUIPMENT	COMMENTS
Recipe Delete Request	S15,F35-->	<--S15,F36	Recipe Delete Acknowledge. The recipe is removed from non-volatile storage.

4.6.6.2 Recipe Directory Request

Host requests a list of identifiers of currently stored recipes.

COMMENTS	HOST	EQUIPMENT	COMMENTS
GetAttr Request	S14,F1-->	<--S14,F2	GetAttr Data

4.6.6.3 Recipe Upload

Host-Initiated Recipe Upload:

COMMENTS	HOST	EQUIPMENT	COMMENTS
Recipe Upload Request	S15,F31-->		
		<--S15,F32	Recipe Upload Data
Large Recipe Upload Request	S15,F51-->		
		<--S15,F52	Large Recipe Upload Acknowledge
			Acknowledge may be "accepted and will be performed later with completion signaled". When the equipment is ready to send:
		<--S13,F1	Send Data Set Send
Send Data Set Send Acknowledge	S13,F2-->		
Open Data Set request	S13,F3-->		
		<--S13,F4	Open Data Set Data
[START] repeat until ACKC13 is error: Read Data Set Request [END]	S13,F5-->		
		<--S13,F6	Read Data Set Data
			[IF] CEID for upload event enabled [THEN] [IF] ACKC13 is "ERROR: END OF DATA" [Then] Send Event Report "successful upload" [ELSE] Send Event Report "bad upload" [END_IF] [END_IF]
Event Report Acknowledge	S6,F12-->	<--S6,F11	
Event Report Acknowledge	S6,F12-->	<--S6,F11	
Close Data Set Send	S13,F7-->		
		<--S13,F8	Close Data Set Acknowledge



Equipment-Initiated Recipe Upload:

COMMENTS	HOST	EQUIPMENT	COMMENTS
			RCPCMD = Upload
			[IF] multi-block request
		<--S15,F1	[THEN] Recipe Management Multi-block inquire
Recipe Management Multi-block Grant	S15,F2-->		
			[END_IF]
		<--S15,F21	Recipe Action Request
Recipe Action Acknowledge	S15,F22-->		
Host requests upload	S15,F31-->		
		<--S15,F32	Recipe Upload Data
			RCPCMD = Upload
			[IF] multi-block request
		<--S15,F1	[THEN]
Recipe Management Multi-block Grant	S15,F2-->		Recipe Management Multi-block Inquire
			[END_IF]
		<--S15,F21	Recipe Action Request
Recipe Action Acknowledge	S15,F22-->		
		S15,F51-->	
Large Recipe Upload Request			
		<--S15,F52	Large Recipe Upload Acknowledge
		<--S13,F1	Send Data Set Send
Send Data Set Send Acknowledge	S13,F2-->		
Open Data Set Request	S13,F3-->		
		<--S13,F4	Open Data Set Data
[START] repeat until ACKC13 is error:			
Read Data Set Request	S13,F5-->		
[END]		<--S13,F6	Read Data Set Data
			[IF]
			CEID for upload event enabled
			[THEN]
			[IF]
			ACKC13 is "ERROR: END OF DATA"
			[Then]
		<--S6,F11	Send Event Report "successful upload"
Event Report Acknowledge	S6,F12-->		[ELSE]
		<--S6,F11	Send Event Report "bad upload"
Event Report Acknowledge	S6,F12-->		
			[END_IF]
			[END_IF]
Close Data Set Send	S13,F7-->		
		<--S13,F8	Close Data Set Acknowledge

4.6.6.4 Recipe Download

The Recipe Management Multi-block Inquire/Grant transaction (S15,F1/F2) is required for the transfer of multi-block recipes. However, its use is recommended prior to all downloads, as it provides recipe size to the equipment.

NOTE 18: If the data item RCPOWCODE is TRUE in S15,F27, then a pre-existing recipe with the same identifier shall be overwritten.

Host-Initiated Recipe Download:

COMMENTS	HOST	EQUIPMENT	COMMENTS
[IF] Recipe is multi-block [THEN]			
Recipe Management Multi-block Inquire	S15,F1-->		
		<--S15,F2	Recipe Management Multi-block Grant
[END IF] Recipe Download Request	S15,F27-->		
			[IF] RCPOWCODE = TRUE delete any pre-existing recipe with the same identifier before storing new recipe [END_IF]
Large Recipe Download Request	S15,F49-->	<--S15,F28	Recipe Download Acknowledge
			[IF] RCPOWCODE = TRUE delete any pre-existing recipe with the same identifier before storing new recipe [END_IF]
		<--S15,F50	Large Recipe Download Acknowledge
		<--S13,F3	Open Data Set Request
Open Data Set Data	S13,F4-->		
			[START] repeat until ACKC13 is error:
Read Data Set Data	S13,F6-->	<--S13,F5	Read Data Set Request
			[END]
Close Data Set Acknowledge	S13,F8-->	<--S13,F7	Close Data Set Send
Recipe Management Multi-block Grant	S15,F2-->	<--S15,F1	[IF] multi-block request [THEN] Recipe Management Multi-block Inquire [END_IF]
		<--S15,F53	
Recipe Verification Acknowledge	S15,F54-->		Recipe Verification Send



Equipment-Initiated Recipe Download:

COMMENTS	HOST	EQUIPMENT	COMMENTS
			RCPCMD = Download [IF] multi-block request [THEN] Recipe Management Multi-block inquire
Recipe Management Multi-block Grant	S15,F2-->	<--S15,F1	
		<--S15,F21	[END_IF] Recipe Action Request
Recipe Action Acknowledge [IF] Recipe is multi-block [THEN]	S15,F22-->		
Recipe Management Multi-block Inquire	S15,F1-->	<--S15,F2	Recipe Management Multi-block Grant
[END IF] Recipe Download Request	S15,F27-->		[IF] RCPOWCODE = TRUE delete any pre-existing recipe with the same identifier before storing new recipe [END_IF] Recipe Download Acknowledge RCPCMD = Download [IF] multi-block request [THEN] Recipe Management Multi-block Inquire [END_IF]
		<--S15,F28	
Recipe Management Multi-block Grant	S15,F2-->	<--S15,F1	
		<--S15,F21	Recipe Action Request
Recipe Action Acknowledge	S15,F22-->		
Large Recipe Download Requests	S15,F49-->		[IF] RCPOWCODE = TRUE delete any pre-existing recipe with the same identifier before storing new recipe [END_IF]
		<--S15,F50	Large Recipe Download Acknowledge
Open Data Set Data	S13,F4-->	<--S13,F3	Open Data Set Request
		<--S13,F5	START] repeat until ACKC13 is error: Read Data Set Request
Read Data Set Data	S13,F6-->		[END]
		<--S13,F7	Close Data Set Send

Close Data Set Acknowledge	S13,F8-->	[IF] multi-block request [THEN]
	<--S15,F1	Recipe Management
Recipe Management	S15,F2-->	Multi-block Inquire
Multi-block Grant		[END_IF]
	<--S15,F53	Recipe Verification Send
Recipe Verification Acknowledge	S15,F54-->	

4.6.6.5 Recipe Verification

Host requests equipment to verify a recipe that it has stored:

COMMENTS	HOST	EQUIPMENT	COMMENTS
Recipe Verify Request	S15,F29-->		
		<--S15,F30	Equipment verifies recipe Recipe Verify Data

4.7 Material Movement — The material movement capability includes the physical transfer of material (WIP, tools, expendable materials, etc.) between equipment, buffers, and storage facilities. The transfer of material can be performed by operators, AGV robots, tracks, or dedicated fixed material handling equipment.

4.7.1 Purpose — This capability is limited in implementation, serving to notify the host of the appearance or removal of material at the equipment's ports.

4.7.2 Definitions

Port — A point or area on the equipment at which a change in equipment ownership of material may occur.

4.7.3 Description — This capability consists of alerting the host whenever material is sent or received from any of the ports on the equipment. Event-specific information, such as port identification and material identification, also may be useful, but definition of these and other related DVVAL's are left to the implementation.

4.7.4 Requirements — The equipment must supply two CEIDs, one to report when material is sent from any port and the other to report when material is received at any port.

4.7.5 Scenarios

COMMENTS	HOST	EQUIPMENT	COMMENTS
			Material is sent or received at an equipment port.
		<--S6,F11	Send collection event to host.
Host acknowledges.	S6,F12-->		

4.8 Equipment Terminal Services — Equipment Terminal Services allows the host to display information on the equipment's display device or the operator of the equipment to send information to the host.

4.8.1 Purpose — Equipment Terminal Services allows the factory operators to exchange information with the host from their equipment workstations.

4.8.2 Definitions — **Message Recognition:** a positive action by the equipment operator indicating the operator has viewed the text of a host initiated message.

4.8.3 Detailed Description — The equipment must be capable of displaying information passed to it by the host for the operator's attention. The information, or an indication of a message, must remain on the equipment's display until the operator indicates message recognition. Message recognition results in a collection event that informs the host that the operator has actually viewed the information.



The equipment must be capable of passing information to the host that has been entered from the operator's equipment console. This information is intended for host applications and is not processed by the equipment.

The equipment has no responsibility for interpreting any of the data passed to or from the host using this method.

4.8.4 Requirements

- Any new Terminal Display message sent by the host shall overwrite an unrecognized message at the same equipment terminal.
- The equipment must provide a display device capable of displaying at least 160 characters to the operator.
- The equipment must provide a mechanism for displaying information sent to it by the host.
- The equipment must provide an indicator to notify the operator when an unrecognized message is present.
- The equipment must support single-block messages as a minimum. Support of multi-block messages is optional.
- The equipment must provide a mechanism for the operator to indicate message recognition (e.g., push button, terminal function).
- A Terminal Display message received by the equipment with a zero length TEXT data item shall be accepted and replace any previous unrecognized message, but shall not itself be considered an unrecognized message. This provides a method of clearing an unrecognized message and turning off the unrecognized message indicator.
- The equipment must provide a means for alpha numeric data entry that can be used by the operator.
- The equipment must support operator entry of at least 160 characters per message.
- The equipment must have a mechanism to send operator-entered messages to the host.

4.8.5 Scenarios

Host sends information to an equipment's display device:

COMMENTS	HOST	EQUIPMENT	COMMENTS
Host sends textual information to equipment for display to the operator on terminal x.	S10,F3-->	<--S10,F4	Equipment acknowledges request to display text. (Equipment sets unrecognized message indicator.) Operator indicates message recognition. (Equipment clears unrecognized message indicator.)
		<--S6,F11	Message recognition event. (See Section 4.2.1, Event Data Collection, for details.)
Host acknowledges Optional:	S6,F12-->	<--S10,F1	Operator responds with text via terminal x.
Host acknowledges receipt of operator text.	S10,F2-->		



Host sends information to an equipment's display device and then overwrites the information before operator recognizes message:

COMMENTS	HOST	EQUIPMENT	COMMENTS
Host sends textual information to equipment for display to the operator on terminal x.	S10,F3-->		
		<--S10,F4	Equipment acknowledges request to display text. (Equipment sets unrecognized message indicator.)
Host sends textual information to equipment for display to the operator on terminal x. This message overwrites the first one sent by the host since it is still unrecognized.	S10,F3-->		
		<--S10,F4	Equipment acknowledges request to display text. (Equipment sets unrecognized message indicator.)
			Operator indicates message recognition. (Equipment clears unrecognized message indicator.)
		<--S6,F11	Message recognition event. (See Section 4.2.1, Event Data Collection, for details.)
Host acknowledges	S6,F12-->		

Operator sends information to the host:

COMMENTS	HOST	EQUIPMENT	COMMENTS
		<--S10,F1	Operator sends textual information via equipment terminal x.
Host acknowledges receipt of operator initiated message.	S10,F2-->		
Optional: Host responds with information for display to the operator on terminal x.	S10,F3-->		
		<--S10,F4	Equipment acknowledges receipt of request to display text. (Equipment sets unrecognized message indicator.)
			Operator indicates message recognition.
		<--S6,F11	Message recognition event. (See Event data collection for details.)
Host acknowledges	S6,F12-->		



Host sends a multi-block display message:

COMMENTS	HOST	EQUIPMENT	COMMENTS
Send information	S10,F5-->		
		<--S10,F6	Accepted or denied. [IF] Message from host is multi-block and multi-block is not supported by the equipment, [THEN] Send Multi-block Not Allowed. [END_IF]
		<--S10,F7	

4.9 Error Messages

4.9.1 *Purpose* — Error messages provide the host with information describing the reason for a particular message or communication fault detected by the equipment.

4.9.2 Definitions

Communication Fault — Refer to Section 2 for the definition.

Message Fault — Refer to Section 2 for the definition.

4.9.3 *Detailed Description* — The equipment must inform the host that it cannot process a message due to an incorrect:

- device ID,
- message stream type,
- message function,
- message format, or
- data format.

The equipment must inform the host if the message has more data than it can handle.

The equipment must inform the host if the equipment's transaction timer expires.

The equipment shall treat the above conditions as application-level errors and shall not take any further action on any message in error.

Error messages are invoked whenever the equipment detects communication or message faults.

4.9.4 *Requirements* — Support of all Stream 9 messages.

4.9.5 Scenario

Message Fault Due to Unrecognized Device ID:

COMMENTS	HOST	EQUIPMENT	COMMENTS
Host sends a message.	Sx,Fy-->		
		<--S9,F1	Equipment detects an unrecognized device ID within the message from the host. Equipment reports to the host that an "unrecognized device ID" was detected in the received message.



Message Fault Due to Unrecognized Stream Type:

COMMENTS	HOST	EQUIPMENT	COMMENTS
Host sends a message.	Sx,Fy-->		
		<--S9,F3	Equipment detects an unrecognized stream type within the message from the host. Equipment reports to the host that an "unrecognized stream type" was detected in the received message.

Message Fault Due to Unrecognized Function Type:

COMMENTS	HOST	EQUIPMENT	COMMENTS
Host sends a message.	Sx,Fy-->		
		<--S9,F5	Equipment detects an unrecognized function type within the message from the host. Equipment reports to the host that an "unrecognized function type" was detected in the received message.

Message Fault Due to Illegal Data Format:

COMMENTS	HOST	EQUIPMENT	COMMENTS
Host sends a message.	Sx,Fy-->		
		<--S9,F7	Equipment detects illegal data format within the message from the host. Equipment reports to the host that "illegal data format" was detected in the received message.

Communication Fault Due to Transaction Timer Timeout:

COMMENTS	HOST	EQUIPMENT	COMMENTS
			Equipment does not receive an expected reply message from the host and a transaction timer timeout occurs.
		<--S9,F9	Equipment reports to the host that a transaction timer timeout occurred.

Message fault due to data too long:

COMMENTS	HOST	EQUIPMENT	COMMENTS
Host sends a message.	Sx,Fy-->		
		<--S9,F11	Equipment detects that the message from the host contains more data than it can handle. Equipment reports to the host that "data too long" was detected in the received message.

Communication Fault Due to Conversation Timeout:

COMMENTS	HOST	EQUIPMENT	COMMENTS
Host sends a message.	Sx,Fy-->		
		<--Sx,Fy+1	Equipment sends reply. Equipment is now expecting a specific message from the host as a result of the previous transaction. Equipment has not received the expected message from the host and a conversation timeout occurs.
		<--S9,F13	Equipment reports to the host that a conversation timeout occurred.

4.10 Clock — The clock capability enables host management of time-related activities and occurrences associated with the equipment and across multiple pieces of equipment.

4.10.1 Purpose — The purpose of the clock capability is to enable time stamping of collection event and alarm reports. Time stamping is useful for resolving relative order of event/alarm occurrences and scheduling of equipment activities by the host.

The ability for the host to instruct the equipment to set an internal clock to a specified time value, and for the equipment to request the current date and time, is needed for effective time management and synchronization between host and equipment.

4.10.2 Definitions

Clock — Clock is a status variable containing the current value of time at the equipment. Clock may be included in report definitions and/or queried separately by the host. See SEMI E5 for a full definition of this variable data item and its required formatting.

TIME — TIME is a data item contained in messages used by the host to set time at the equipment and by the equipment or host to request the current time from the

other. (See SEMI E5 for a full definition of this data item.)

4.10.3 Detailed Description — The clock capability assumes the existence of a relative time reference on the equipment. This time reference is used as a basis for updating the time value of an equipment status variable called "Clock." The time reference must reflect the current time to within a resolution range of seconds to centiseconds (refer to the format for Clock in the SEMI E5 Standard). The purpose of time stamping with centiseconds is to resolve the order in which nearly simultaneous events occur rather than to provide a more precise record of the time of day at which they occurred. Where more than one event occurs within a given period of clock resolution, the centiseconds reported in the time stamps for these events must reflect the actual order in which the events were detected. Equipment with a clock resolution of less than a second should report centiseconds. Otherwise, centiseconds should be assigned to reflect the relative order in which events were detected. Equipment unable to resolve time to less than a second and unable to reflect the relative order in which events were detected may report centiseconds as "00".

The host employs the “Date and Time Set Request” message (S2,F31) to initialize the value of Clock to the value contained in the TIME data item. Similarly, the equipment may employ the “Date and Time Request” message (S2,F17) to obtain a new initialization time for Clock. As before, the value of TIME returned by the host is used to set Clock. Note, in the event that the precision of TIME is seconds and that for Clock is centiseconds, in both cases the initial value of Clock shall contain “00” for its Centisecond digits upon initialization. Additionally, for any field in TIME that is not supported by the equipment, the local value of this field is equipment dependent. For example, Equipment that cannot resolve time to less than a second might round or ignore centiseconds and always set the Centisecond field to “00”.

4.10.4 Requirements

- The resolution and update rate of the internal time reference must be sufficient to distinguish between two nearly simultaneous collection events and/or alarms.
- The equipment supplier shall provide documentation describing the resolution of the internal time reference.
- The equipment supplier shall provide documentation describing how centisecond values are assigned, including the case of unresolvable simultaneous events.

4.10.5 Scenarios

Equipment Requests TIME (Optional Scenario):

COMMENTS	HOST	EQUIPMENT	COMMENTS
		<--S2,F17	Equipment requests a time value from the host.
Host responds with a TIME value S2,F18-->			Equipment sets its internal time reference to the value of TIME received from the host.

Host Instructs Equipment to Set Time:

COMMENTS	HOST	EQUIPMENT	COMMENTS
Host instructs equipment to set its time.	S2,F31-->		
		<--S2,F32	Equipment sets its internal time reference to the value of TIME received from the host and acknowledges completion.

Host Requests Equipment’s Current Time Value:

COMMENTS	HOST	EQUIPMENT	COMMENTS
Host requests equipment time.	S2,F17-->		
		<--S2,F18	Equipment returns its internal time reference value to the host.

4.11 Spooling — Spooling is a capability whereby the equipment can queue messages intended for the host during times of communication failure and subsequently deliver these messages when communication is restored. Spooling is limited to primary messages of user-selected streams.

4.11.1 Purpose — The purpose of spooling is to provide a method for retaining equipment message data that might otherwise be lost due to communication failure. The motive for producing this functionality is to retain valuable data used to track material and to improve product quality. The spooling capability fills a gap in the SEMI E5 standard. In the past, without a spooling capability, the equipment has typically discarded messages that could not be delivered, or turned messaging off altogether. It is intended that the host initiate the spool unload process immediately following the reestablishment of communications.

4.11.2 Definitions

MaxSpoolTransmit — An equipment constant containing the maximum number of messages that the equipment shall transmit from the spool in response to an S6,F23 “Transmit Spooled Messages” request. If MaxSpoolTransmit is set to zero, no limit is placed on the messages sent from the spool. Multi-block inquire/grant messages are not counted in this total.

OverWriteSpool — A boolean equipment constant used to indicate to the equipment whether to overwrite data in the spool area or to discard further messages whenever the spool area limits are exceeded.

Send Queue — Refers to the queue into which equipment generated SECS messages are placed in preparation for transmission to the host.

Spool — The spool is an area of non-volatile storage in which the equipment stores certain messages that cannot be delivered to the host (when the equipment is in the NOT COMMUNICATING substate of COMMUNICATIONS ENABLED). The spool area can be thought of as a sequential “ring” buffer. The term spool is also used to denote the action of placing messages into the spool area.

SpoolCountActual — A status variable used to keep a count of the messages actually stored in the equipment’s spool area. Multi-block inquire/grant messages are not spooled and not included in this count.

SpoolCountTotal — A status variable used to keep a count of the total number of primary messages directed to the spool, regardless of whether placed or currently retained in the spool. Multi-block inquire/grant

messages are not spooled and not included in this count.

SpoolFullTime — A status variable containing the timestamp when the spool last became full. If the spool was not filled during the last spooling period, this will contain a time value prior to the current SpoolStartTime.

SpoolStartTime — A status variable containing the timestamp from when spooling was last activated.

4.11.3 Description

4.11.3.1 Spooling State Model — There are two major states of spooling: SPOOL INACTIVE and SPOOL ACTIVE. SPOOL ACTIVE has two components: SPOOL UNLOAD and SPOOL LOAD. These are each broken into substates. A description of all spooling states, substates, and applicable state transitions follows. The POWER OFF and POWER ON parent states are common to all equipment subsystems. They are shown here to illustrate the retention of spooling context during a power down situation.

NOTE 20: Disabling SECS communications does not affect the current spooling state since no messages are generated until communications are subsequently enabled. Spooling is effectively frozen in this case.

POWER OFF

The equipment has lost power for any reason (e.g., power failure, power switch set to off).

POWER ON

The equipment is powered up.

SPOOL INACTIVE

This is the normal operating mode. No spooling occurs. The spool area is empty. Primary SECS-II messages are transmitted normally.

SPOOL ACTIVE

All primary SECS-II messages ready for sending and for which spooling is enabled (see S2,F43) are directed to the spool area. All other primary messages, except Stream 1, are discarded. The equipment shall attempt to send any secondary messages that are generated and discard these messages should the attempt to send fail.

Spool state transitions from SPOOL INACTIVE to SPOOL ACTIVE if the communications state changes from COMMUNICATING to NOT COMMUNICATING (Communication State Transition Table, #14) or from WAIT CRA to WAIT DELAY (Communication STATE Transition Table, #6).

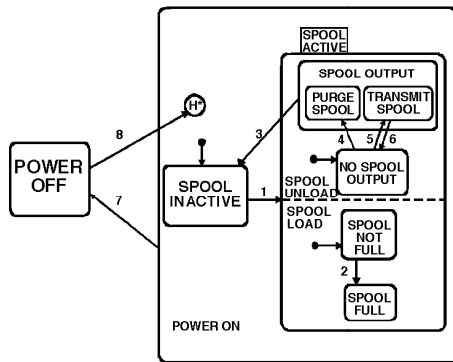


Figure 4.11
Spooling State Diagram

Once communications are established, the host must initiate the spool unload sequence to restore full functionality (see below). Since the equipment will deliver secondary messages, the host may inquire for information or send commands as needed.

The SPOOL ACTIVE state has two AND states: SPOOL LOAD and SPOOL UNLOAD. This means that they operate independently, though sharing data and some state change stimuli. See Section 3.1 for explanation of state model terms and notation.

SPOOL LOAD

The SPOOL LOAD component enters messages into the spool area. It is divided into two substates: SPOOL NOT FULL and SPOOL FULL. SPOOL NOT FULL is the default entry substate of the parent state SPOOL LOAD.

SPOOL NOT FULL

As primary SECS-II messages are directed to the spool area, the equipment shall “write” the SECS-II message to the end of the spool. Status variables SpoolCountTotal and SpoolCountActual shall be incremented each time a message is placed in the spool area.

SPOOL FULL

In this state, all of the allocated spooling area is filled. Choice of the following options shall be controlled by the setting of a Boolean equipment constant called “OverWriteSpool.” The first message to be dealt with is that which could not be fit into the spool prior to transition from SPOOL NOT FULL (see transition table below).

OverWriteSpool is True: The equipment deletes as many of the “oldest” records (e.g., SECS-II messages) contained in the spool area necessary to make space for the new message and then adds the message. Status variable SpoolCountTotal shall be incremented whenever a message is submitted to the spool area. Status variable SpoolCountActual shall be manipulated to keep an accurate count of the number of messages contained in the spool area. For example, if it is necessary to delete three messages in the spool area to spool one message, SpoolCountActual would have three subtracted and then one added to the total.

OverWriteSpool is False: Any subsequent primary messages shall be discarded. When such a message is discarded, SpoolCountTotal is incremented.

SPOOL UNLOAD

The SPOOL UNLOAD component of SPOOLACTIVE deals with movement of messages out of the spool. It has an active substate (SPOOL OUTPUT) and a passive substate (NO SPOOL OUTPUT). NO SPOOL OUTPUT is the default entry substate, since the equipment is NOT COMMUNICATING at the time spooling is initiated. When communications between equipment and host are restored, there is an opportunity for the host to recover spooled messages. No action is taken until the host initiates the spool output process via the S6,F23 (Request Spooled Data). The host has the option to either receive the spooled messages (see substate TRANSMIT SPOOL) or discard all messages in the spool (see substate PURGE SPOOL).

NO SPOOL OUTPUT

In this state, no messages are removed from the spool.

SPOOL OUTPUT

The SPOOL OUTPUT state encompasses the removal of messages from the spool. Its substates are TRANSMIT SPOOL and PURGE SPOOL.

TRANSMIT SPOOL

The host elects to receive all messages contained in the spool area. The equipment is expected to keep track of the oldest record (i.e., message) within the spool area. When communications are re-established with the host and transmission of the spool area is started, the oldest record must be the first record transmitted, then the next oldest record, etc. There is no prioritization of messages to be sent from the spool.

As each spooled message is successfully transmitted to the host, it is removed from the spool area upon successful completion of the transaction. Spool-CountActual is decremented as each message is removed from the spool. The equipment shall transmit

messages only from the spool area until all spooled messages have been completely transmitted to the host.

Flow control of the spool transmit process is achieved in two ways. First, only one open transaction on the equipment is allowed during spool unload. Thus, if a message requires a reply, the equipment shall wait for that reply before transmitting the next spooled message. Messages which require no reply may be transmitted sequentially as rapidly as the message transfer mechanism will allow.

The second flow control method is to allow the host to limit the maximum number of messages sent from the spool in response to the S6,F23 request. An equipment constant named MaxSpoolTransmit may be set by the host to achieve this behavior. If MaxSpoolTransmit is set to five, for example, the equipment will send the first five messages from the spool and then transition to the NO SPOOL OUTPUT state, awaiting the next S6,F23 request. There is no event report generated when MaxSpoolTransmit is reached. The host is responsible for determining this situation by a) counting the messages received, b) timing out waiting for the next message, c) inquiring to the equipment for the current value of the SpoolCountActual status variable, or d) some combination of the above. If MaxSpool-Transmit is set to zero, the spool shall be transmitted completely in response to S6,F23.

Normal spooling continues during the spool transmit process. If the SPOOL LOAD component should transition to SPOOL FULL, it shall not have any effect on the SPOOL UNLOAD component. Once full, the spool cannot make the transition back to SPOOL NOT FULL except via the SPOOL INACTIVE state. Space made available due to the spool unload process shall not be used in this case.

When a multi-block message is to be transmitted from the spool, any required inquire/grant transaction shall be initiated. If the host's response denies permission to send the multi-block message, the equipment shall discard that message and continue with the transmit process. This sequence shall count as one message in the MaxSpoolTransmit count.

There is one area where SPOOL LOAD and SPOOL UNLOAD may interact: When the spool is full and OverWriteSpool is True. During the spool transmit process, spooled messages are being removed and new primary messages are being written to the spool. These new messages are overwriting the oldest messages available, unless the unload process has freed sufficient spool space. There is a possibility that the unload and overwrite processes may compete for control of the same message area. For example, if the spool holds messages ABCDE, with A oldest and E newest, A might be sent to the host, B (and the space from A) overwritten by the new message F, C sent to the host, D and E (and the space from message C) overwritten by G, etc. The loss of continuity may be "disorienting" to the host program receiving the messages. It is expected that the unload process will be fast relative to the generation of new messages, so that this occurrence will be rare.

Should a communication failure occur during the spool transmit process, spooling shall continue as before the transmit process began. However, the spool unload sequence shall terminate (i.e., transition to NO SPOOL OUTPUT will occur — see transition table below).

PURGE SPOOL

The equipment shall discard all messages in the spool and, when the spool is empty, zero SpoolCountActual.



Spooling State Transitions

A table follows detailing all spooling state transitions as presented in the state transition diagram.

Table 4.11 Spooling State Transition

#	Current State	Trigger	New State	Action	Comment
1	SPOOL INACTIVE	Communication state changes from COMMUNICATING to NOT COMMUNICATING or from WAIT CRA to WAIT DELAY and Enable Spool is true.	SPOOL ACTIVE	SpoolCountActual and SpoolCountTotal are initialized to zero. Any open transactions with the host are aborted. SpoolStartTime (SV) is set to current time. Alert the operator that spooling is active.	The default state in each AND substate is entered. The message which could not be sent remains in the send queue and is dealt with in Spool Active state. The collection event Spooling Activated has occurred.
2	SPOOL NOT FULL	Message generate which will not fit into spool area.	SPOOL FULL	SpoolFullTime (SV) is set to current time. Alert the operator that spool is full.	The message which would not fit into the spooling area is dealt with after the transition. No collection event is generated.
3	SPOOL OUTPUT	Spool area emptied.	SPOOL INACTIVE	Spooling process disabled. Alert the operator that spooling has been terminated.	The collection event Spooling Deactivated has occurred. Transition from the AND substate Spool Unload component occurs.
4	NO SPOOL OUTPUT	S6,F23 received w/RSDC = 1.	PURGE SPOOL	No action.	Initiates purging process. No collection event is generated since this is based on host request.
5	NO SPOOL OUTPUT	S6,F23 received w/RSDC = 0.	TRANSMIT SPOOL	No action.	Initiates message transmission from spool. No collection event is generated since this is based on host request.
6	TRANSMIT SPOOL	Communication failure or MaxSpoolTransmit reached.	NO SPOOL OUTPUT	Spool transmission process suspended.	If communications failure, the event Spool Transmit Failure has occurred. No collection event generated for MaxSpoolTransmit reached.
7	POWER ON	Equipment power source discontinued.	POWER OFF	No action.	Spooling context has been maintained in non-volatile storage prior to this transition.
8	POWER OFF	Equipment power source restored.	POWER ON	Spooling context restored from non-volatile memory.	If spooling were active prior to power down, it shall continue. If TRANSMIT SPOOL were active at powerdown, transition #6 is expected to follow since communications state is initially NOT COMMUNICATING.

4.11.3.2 Enabling Spooling — The equipment shall provide the host with the ability to enable and disable Spooling for any message (except Stream 1 messages (i.e., S1,F1, S1,F13)) via the S2,F43/F44 transaction. Spooling may be enabled for an entire Stream, for individual messages within a stream, or for any combination of the two. Streams and Functions not referenced in this message are not spooled. Spooling can be totally disabled by sending an S2,F43 with a

zero length list for the first item (see S2,F43 definition). In addition, the equipment shall provide an equipment constant, EnableSpooling, to allow setting the enable or disable of spooling. NOTE: When EnableSpooling is false, the SPOOL state cannot transition from SPOOLINACTIVE to SPOOLACTIVE. Changing EnableSpooling does not change the spool state, purge the spool or change streams and functions enabled for spooling. Once the equipment is “SPOOL ACTIVE”, the host must initiate the spool unload sequence to



restore full functionality even though “Enable Spooling” has changed to false.

4.11.4 Requirements — The following items are required to support the spooling capability:

- At a minimum, the equipment shall reserve for spooling non-volatile storage with sufficient capacity to store all of the primary SECS-II messages that would occur during a normal processing cycle.
- While spooling is enabled, the equipment shall reply to primary messages sent by the host with the appropriate secondary message.
- Secondary messages which cannot be delivered shall be discarded, never spooled.
- All spooling-related status variables and setup information (as per S2,F43) must be stored in non-volatile memory along with any other information

required for the potential unloading of the spool area after a power loss.

- Upon powerup, the equipment shall retain all spooling context from the time the equipment was last shutdown or reset. This means that spooling, if previously active, continues upon system powerup.
- The Equipment must reject any message that attempts to set “Spooling” for Stream 1.
- If a multi-block primary message need for the inquire/grant is to be sent during SPOOL ACTIVE, the message should be placed in the spool and the grant resolved during spool transmit.

4.11.5 Scenarios

Define the Set of Messages to be Spooled:

This Scenario is used to set up the list of messages that the equipment should spool (or by defining none, to disable spooling).

COMMENTS	HOST	EQUIPMENT	COMMENTS
Host defines messages to be spooled in case of communications failure.	S2,F43-->		
		<--S2,F44	Equipment acknowledges setup.

Define the Maximum Number of Messages to Send in Response to S6,F23:

This Scenario sets the value of the equipment constant MaxSpoolTransmit.

COMMENTS	HOST	EQUIPMENT	COMMENTS
Host sends value for equipment constant MaxSpoolTransmit.	S2,F15-->		
		<--S2,F16	Equipment acknowledges equipment constant change.



Request or Delete Spooled Data (MaxSpoolTransmit = 0):

This Scenario is used to initiate the transfer of the spooled data from the equipment to the host or to purge the spool.

COMMENTS	HOST	EQUIPMENT	COMMENTS
			Communications were lost and then re-established.
Host requests data that includes spool-related status variables.	S1,F3-->		
		<--S1,F4	Send status data.
NOTE: S1,F3 is one of various methods that could be used. Request or delete spooled data.	S6,F23-->		
		<--S6,F24	Request spooled data acknowledgement. [IF] RSDC = 0 (Spool data requested.) [THEN] The appropriate Streams and Functions are used to transmit the spooled data to the host. [ELSE_IF] RSDC = 1 [THEN] Spool data discarded. [END_IF]
		<--S6,F11	Spooling Deactivated event report sent.
Acknowledge	S6,F12-->		



Request or Delete Spooled Data (MaxSpoolTransmit > 0):

This Scenario shows the affect of MaxSpoolTransmit < SpoolCountActual on the Spool Transmit process. For the purpose of illustration, the value of MaxSpoolTransmit is 5 and the SpoolCountActual is 8 (at the time communications are re-established). No messages are added to the Spool during the transmit process.

COMMENTS	HOST	EQUIPMENT	COMMENTS
Communications were lost and then re-established.			
Host requests data that includes spool-related status variables.	S1,F3-->	<--S1,F4	Send status data (e.g. SpoolCountActual = 8, MaxSpoolTransmit = 5).
Host requests spooled data (RSDC=0).	S6,F23-->	<--S6,F24	Request spooled data acknowledgement. The five oldest messages in the Spool are transmitted to the host. Spooling remains active.
Host recognizes that MaxSpoolTransmit is reached.			
Host requests additional spooled data (RSDC = 0).	S6,F23-->	<--S6,F24	Request spooled data acknowledgement. The three remaining messages are transmitted from the spool.
		<--S6,F11	Spooling Deactivated event report sent.
Acknowledge	S6,F12-->		

4.12 *Control* — The control-related capabilities allow for configuration and manipulation of the control state model. In this way the host and/or user may modify the equipment's control-related behavior.

4.12.1 *Purpose* — This section complements the CONTROL state model description found in Section 3.3. It defines the requirements for implementation of this model.

4.12.2 *Definitions* — None.

4.12.3 *Description*

4.12.3.1 *Control Configuration* — The control state model has two areas of configuration. The first area is related to default entry states of the state model. Upon system initialization, the system must activate either the ON-LINE or OFF-LINE state. Upon entry to OFF-LINE, the system must in turn activate one of the substates of OFF-LINE (EQUIPMENT OFF-LINE, ATTEMPT ON-LINE, or HOST OFF-LINE). In both these cases, the user shall configure the equipment to make the choices appropriate to that factory. Entry to the ON-LINE state also involves a choice of substates.

In this case, the equipment reads the front panel REMOTE/LOCAL switch to determine the appropriate state.

The second area of configuration involves the transition to be made if the ON-LINE attempt should fail. The model may be set to transition to either HOST OFF-LINE or to EQUIPMENT OFF-LINE should the S1,F1 transaction be terminated unsuccessfully. Choosing HOST OFF-LINE allows the host to cause the equipment to transition to ON-LINE when the host becomes ready. This is accomplished via the message S1,F17 (see below).

4.12.3.2 *Changing Control State* — In the control state model, both the operator and the host can affect the control state. The operator retains ultimate authority to set the equipment OFF-LINE by means of an OFF-LINE switch mechanism. The operator also can cause the equipment to attempt to go ON-LINE. Under some circumstances, the host can initiate the transition to ON-LINE.

If the operator requests ON-LINE, the equipment will send an S1,F1 to the host. The host may confirm ON-



LINE with an S1,F2 or deny ON-LINE by sending an S1,F0.²³

When the equipment is ON-LINE, the host may request that it transition to OFF-LINE. It will transition into the HOST OFF-LINE substate. When the equipment HOST OFF-LINE state is active, the host may request that it transition to ON-LINE. The combination of these two allow the host to cycle the equipment between ON-LINE and OFF-LINE.

Only the operator may change the ON-LINE substate (REMOTE or LOCAL).

via the operator console display. It is recommended that this indicator be visible at all times.

- The equipment shall supply a status variable that contains the current state/substate of the CONTROL state model.
- Whenever the ON-LINE/REMOTE state is active and the operator issues a command to the equipment, the equipment shall cause an “operator command issued” event.

4.12.4 Requirements

- The equipment shall supply a method for configuring the default CONTROL state to be activated upon system initialization. The choice of states must be among ATTEMPT ON-LINE, EQUIPMENT OFF-LINE, HOST OFF-LINE, and ON-LINE.
- The equipment shall supply a method for configuring which state should be activated when the attempt to go ON-LINE fails. The option is a transition to either the HOST OFF-LINE state or the EQUIPMENT OFF-LINE state.
- The equipment shall supply a momentary switch which will initiate the transition to OFF-LINE and another which will begin the process to go ON-LINE. Discrete position switches shall not be used. These should be designed so that they may not be actuated simultaneously. The switch may be mounted on the front panel or be available via keyboard input at the operator console.
- The equipment shall supply a discrete two-position switch which the operator may use to indicate the desired substate for ON-LINE (i.e., REMOTE or LOCAL). The switch may be mounted on the front panel or be available via keyboard input at the operator console. If implemented in software, this setting shall be retained in non-volatile storage.
- The equipment shall supply an indicator on the front panel which displays the full identification of the current CONTROL state/substate (e.g., OFF-LINE/ATTEMPT ON-LINE). This may be accomplished either with labelled display lights or

²³ If there is no host response (i.e. reply timeout), the equipment shall treat it as a denial.



4.12.5 Scenarios

4.12.5.1 Operator-Initiated Scenarios

Host Accepts ON-LINE:

COMMENTS	HOST	EQUIPMENT	COMMENTS
Operator actuates ON-LINE switch when equipment OFF-LINE state is active.			
Host grants ON-LINE.	S1, F2-->	<--S1, F1	Equipment requests ON-LINE.
Acknowledge.	S6, F12-->	<--S6, F11	"Control State LOCAL (or REMOTE)" event.

Host Denies ON-LINE:

COMMENTS	HOST	EQUIPMENT	COMMENTS
Operator actuates ON-LINE switch when equipment OFF-LINE state is active.			
Host denies ON-LINE.	S1, F0-->	<--S1, F1	Equipment requests ON-LINE.

Operator Sets OFF-LINE:

COMMENTS	HOST	EQUIPMENT	COMMENTS
Operator actuates OFF-LINE switch when equipment ON-LINE state is active.			
Acknowledge.	S6, F12-->	<--S6, F11	"Equipment requests OFF-LINE" event.

Operator Sets REMOTE:

COMMENTS	HOST	EQUIPMENT	COMMENTS
Operator sets switch from LOCAL to REMOTE.			
Acknowledge.	S6, F12-->	<--S6, F11	"Control State REMOTE" event.

Operator Sets LOCAL:

COMMENTS	HOST	EQUIPMENT	COMMENTS
Operator sets switch from REMOTE to LOCAL.			
Acknowledge.	S6, F12-->	<--S6, F11	"Control State LOCAL" event.



4.12.5.2 Host-Initiated Scenarios

Host Sets OFF-LINE:

COMMENTS	HOST	EQUIPMENT	COMMENTS
Host requests OFF-LINE.	S1,F15-->		[IF] Equipment is OFF-LINE [THEN]:
		<--S1,F0	Equipment does not process requests.
		<--S1,F16	[ELSE] Equipment ON-LINE Equipment acknowledges request and transitions to OFF-LINE.
		<--S6,F11	"Equipment OFF-LINE" event.
Acknowledge	S6,F12-->		[END_IF]

Host Sets ON-LINE:

COMMENTS	HOST	EQUIPMENT	COMMENTS
Host requests ON-LINE.	S1,F17-->		[IF] Equipment is HOST OFF-LINE state not active.
		<--S1,F18	[THEN] Equipment denies request (ONLACK = 1).
		<--S1,F18	[ELSE] Equipment HOST OFF-LINE state is active.
		<--S6,F11	Equipment acknowledges request (ONLACK != 1). "Control state LOCAL (or REMOTE)" event. (only if ONLACK = 0)
Acknowledge	S6,F12-->		[END_IF]



5 Data Items

The following sections specify which data items and variable data items are required.

Except for the specified format restrictions, all data items and variable data items follow the definitions contained in SEMI E5.

5.1 Data Item Restrictions — The following is a subset of the data items used by SECS-II messages specified in this standard. Each data item listed in this section is restricted in its SEMI E5-defined usage. Most are limited to a single format from their standard list of formats. Data items used by SECS-II messages contained in this document, but which have no restrictions are not duplicated here.

NOTE 21: One data item, ALCD, is restricted in other than its format. Take note of this restriction as described below.

NOTE 22: The equipment supplier shall document any restrictions on length or format of CPNAME. Suppliers shall document the behavior of spaces in a CPNAME. The maximum length of CPNAME shall be 40. This change became effective in September 1995.

ACKC7A	Format: 51
ALCD	Only bit 8 (alarm set/cleared) of the binary byte is used. Bits 1–7, denoting alarm category, are not used.
ALID	Format: 5()
CCODE	Format: 20, 32, 34, 52, 54
CEID	Format: 5()
CPNAME	Format: 20
DATAID	Format: 5()
DATALLENGTH	Format: 5()
ECID	Format: 5()
LENGTH	Format: 5()
PPID	Format: 20
RCMD	Format: 20
REPGSZ	Format: 5()
RPTID	Format: 5()
SEQNUM	Format: 52
SMPLN	Format: 5()
SVID	Format: 5()
TEXT	Format: 20
TOTSMP	Format: 5()
TRID	Format: 5()
VID	Format: 5()

5.2 Variable Item List — The following variable data items from the Variable Item Dictionary in SEMI E5 are required. Format restrictions are noted.

DVVAL's:

AlarmID Format: 5()
 EventLimit
 LimitVariable
 PPChangeName Format: 20
 PPChangeStatus
 PPErrors
 RcpChangeName
 RcpChangeStatus
 TransitionType

ECV's:

EstablishCommunicationsTimeout
 MaxSpoolTransmit
 OverWriteSpool
 TimeFormat

SV's:

AlarmsEnabled
 AlarmsSet
 Clock
 ControlState
 EventsEnabled
 PPErrors
 PPExecName Format: 0,20
 PPFormat
 PreviousProcessState
 ProcessState
 RcpExecName
 SpoolCountActual
 SpoolCountTotal
 SpoolFullTime
 SpoolStartTime

6 Collection Events

Table 6.1 provides the list of collection events required to support the capabilities addressed within this standard. Also shown are typical variable data that would most likely be included in the associated collection event report and a reference to the event trigger and to the appropriate section of the standard.

This list does not represent all events that might be needed to properly monitor/control equipment. Many events are unique to the specific equipment characteristics. The needed additions are a matter for other standards and for collaboration between equipment supplier and user.

See Section 5.2 for further detail of variable data items.

Table 6.1 GEM-Defined Collection Events

<i>Event Designation</i>	<i>Typical Variable Data</i>	<i>Reference</i>
Control-Related Events:		Section 3.3
Equipment OFF-LINE	ControlState, Clock	ON-LINE->OFF-LINE
Control State LOCAL	ControlState, Clock	REMOTE->LOCAL or OFF-LINE->LOCAL
Control State REMOTE	ControlState, Clock	LOCAL->REMOTE or OFF-LINE->REMOTE
Operator Command Issued	OperatorCommand	Operator Activity while REMOTE state is active.
Processing-Related Events:	Note: Any transition in the implemented processing state model must have a corresponding collection event.	Section 3.4
Processing Started	Clock, PreviousProcessState	Entry into EXECUTING state.
Processing Completed	Clock, PreviousProcessState	Normal exit of EXECUTING state.
Processing Stopped	Clock, PreviousProcessState	Result of STOP command from host or operator.
Processing State Change	Clock, ProcessState, PreviousProcessState	Any processing state transition.
Alarm Management Events:		Section 4.3
Alarm _n Detected	Clock, AlarmID, AlarmsSet, Associated variable data	ALARM _n CLEAR->ALARM _n SET
Alarm _n Cleared	Clock, AlarmID, AlarmsSet	ALARM _n SET->ALARM _n CLEAR
Equipment Constant Events:		Section 4.5
Operator Equipment Constant Change	ECID	Operator activity
Limits Monitoring:		Section 4.2.4
Limit Zone Transition _n (separate CEID per variable)	Clock, LimitVariable, EventLimit, Transition Type	Entry into BELOW LIMIT or ABOVE LIMIT states.
Process Program Management Events:		Section 4.6
Process Program Change	PPChangeName, PPChangeStatus	Operator activity
Process Program(s) Selected	PPExecName	Operator/Host activity
Material Movement Events:		Section 4.7
Material Received	Clock	
Material Removed	Clock	
Spooling Events:		Section 4.11
Spooling Activated	SpoolStartTime	SPOOL INACTIVE->SPOOL ACTIVE
Spooling Deactivated	SpoolCountTotal	SPOOL OUTPUT->SPOOL INACTIVE
Spool Transmit Failure	Clock, SpoolCountActual SpoolCountTotal	TRANSMIT SPOOL->NO SPOOL OUTPUT
Terminal Services Events:		Section 4.8
Message Recognition	Clock	Operator
New Execution Recipe Event	RcpChangeName, RcpChangeStatus	Section 4.6.2.2
Execution Recipe Change Event	RcpChangeName, RcpChangeStatus	Section 4.6.2.2
Successful Upload	DataSetName	Sections 4.6.5.3, 4.6.6.3
Bad Upload	DataSetName	Sections 4.6.5.3, 4.6.6.3



7 SECS-II Message Subset

This section lists the required set of SECS-II messages as referenced in this document. Definitions for these messages can be found in SEMI E5. All primary messages (for which SEMI E5 defines replies) should have replies available. Replies are required or optional as specified in SEMI E5.

STREAM 1: Equipment Status

S1,F1 Are You There Request (R)	S,H<->E
S1,F2 On-Line Data (D)	S,H<->E
S1,F3 Selected Equipment Status Request (SSR)	S,H->E
S1,F4 Selected Equipment Status Data (SSD)	M,H<-E
S1,F11 Status Variable Namelist Request (SVNR)	S,H->E
S1,F12 Status Variable Namelist Reply (SVNRR)	M,H<-E
S1,F13 Establish Communications Request (CR)	S,H<->E
S1,F14 Establish Communications Request Acknowledge (CRA)	S,H<->E
S1,F15 Request OFF-LINE (ROFL)	S,H->E,reply
S1,F16 OFF-LINE Acknowledge (OFLA)	S,H<-E
S1,F17 Request ON-LINE (RONL)	S,H->E,reply
S1,F18 ON-LINE Acknowledge (ONLA)	S,H<-E

STREAM 2: Equipment Control and Diagnostics

S2,F13 Equipment Constant Request (ECR)	S,H->E
S2,F14 Equipment Constant Data (ECD)	M,H<-E
S2,F15 New Equipment Constant Send (ECS)	S,H->E
S2,F16 New Equipment Constant Acknowledge (ECA)	S,H<-E
S2,F17 Date and Time Request (DTR)	S,H<->E
S2,F18 Date and Time Data (DTD)	S,H<->E
S2,F23 Trace Initialize Send (TIS)	S,H->E
S2,F24 Trace Initialize Acknowledge (TIA)	S,H<-E
S2,F29 Equipment Constant Namelist Request (ECNR)	S,H->E
S2,F30 Equipment Constant Namelist (ECN)	M,H<-E
S2,F31 Date and Time Send (DTS)	S,H->E
S2,F32 Date and Time Acknowledge (DTA)	S,H<-E
S2,F33 Define Report (DR)	M,H->E
S2,F34 Define-Report Acknowledge (DRA)	S,H<-E
S2,F35 Link Event Report (LER)	M,H->E
S2,F36 Link Event Report Acknowledge (LERA)	S,H<-E



S2,F37 Enable/Disable Event Report (EDER)	S,H->E,reply
S2,F38 Enable/Disable Event Report Acknowledge (EDEA)	S,H<-E
S2,F39 Multi-Block Inquire (DMBI)	S,H->E
S2,F40 Multi-Block Grant (DMBG)	S,H<-E
S2,F41 Host Command Send (HCS)	S,H->E
S2,F42 Host Command Acknowledge (HCA)	S,H<-E
S2,F43 Reset Spooling Streams and Functions (RSSF)	S,H->E
S2,F44 Reset Spooling Acknowledge (RSA)	M,H<-E
S2,F45 Define Variable Limit Attributes (DVLA)	M,H->E
S2,F46 Variable Limit Attribute Acknowledge (VLAA)	M,H<-E
S2,F47 Variable Limit Attribute Request (VLAR)	S,H->E
S2,F48 Variable Limit Attributes Send (VLAS)	M,H<-E
S2,F49 Enhanced Remote Command	M,H->E
S2,F50 Enhanced Remote Command Acknowledge	M,H<-E

STREAM 5: Exception (Alarm) Reporting

S5,F1 Alarm Report Send (ARS)	S,H<-E
S5,F2 Alarm Report Acknowledge (ARA)	S,H->E
S5,F3 Enable/Disable Alarm Send (EAS)	S,H->E
S5,F4 Enable/Disable Alarm Acknowledge (EAA)	S,H<-E
S5,F5 List Alarms Request (LAR)	S,H->E
S5,F6 List Alarm Data (LAD)	M,H<-E

STREAM 6: Data Collection

S6,F1 Trace Data Send (TDS)	S,H<-E
S6,F2 Trace Data Acknowledge (TDA)	S,H->E
S6,F5 Multi-block Data Send Inquire (MBI)	S,H<-E
S6,F6 Multi-block Grant (MBG)	S,H->E
S6,F11 Event Report Send (ERS)	M,H<-E
S6,F12 Event Report Acknowledge (ERA)	S,H->E
S6,F15 Event Report Request (ERR)	S,H->E
S6,F16 Event Report Data (ERD)	M,H<-E
S6,F19 Individual Report Request (IRR)	S,H->E
S6,F20 Individual Report Data (IRD)	M,H<-E
S6,F23 Request Spooled Data (RSD)	S,H->E
S6,F24 Request Spooled Data Acknowledgement Send (RSDAS)	S,H<-E



STREAM 7: Process Program Load

S7,F1 Process Program Load Inquire (PPI)	S,H<->E, reply
S7,F2 Process Program Load Grant (PPG)	S,H<->E
S7,F3 Process Program Send (PPS)	M,H<->E
S7,F4 Process Program Acknowledge (PPA)	S,H<->E
S7,F5 Process Program Request (PPR)	S,H<->E
S7,F6 Process Program Data (PPD)	M,H<->E
S7,F17 Delete Process Program Send (DPS)	S,H->E
S7,F18 Delete Process Program Acknowledge (DPA)	S,H<-E
S7,F19 Current EPPD Request (RER)	S,H->E
S7,F20 Current EPPD Data (RED)	M,H<-E
S7,F23 Formatted Process Program Send (FPS)	M,H<->E
S7,F24 Formatted Process Program Acknowledge (FPA)	S,H<->E
S7,F25 Formatted Process Program Request (FPR)	S,H<->E
S7,F26 Formatted Process Program Data (FPD)	M,H<->E
S7,F27 Process Program Verification Send (PVS)	S,H<-E
S7,F28 Process Program Verification Acknowledge (PVA)	S,H->E
S7,F29 Process Program Verification Inquire (PVA)	
S7,F30 Process Program Verification Grant (PVG)	
S7,F37 Large Process Program Send	S,H <-> E,reply
S7,F38 Large Process Program Acknowledge	S,H <-> E
S7,F39 Large Formatted Process Program Send	S,H <-> E,reply
S7,F40 Large Formatted Process Program Acknowledge	S,H <-> E
S7,F41 Large Process Program Request	S,H <-> E,reply
S7,F42 Large Process Program Acknowledge	S,H <-> E
S7,F43 Large Formatted Process Program Request	S,H <-> E,reply
S7,F44 Large Formatted Process Program Acknowledge	S,H <-> E

STREAM 9: System Errors

S9,F1 Unrecognized Device ID (UDN)	S,H<-E
S9,F3 Unrecognized Stream Type (USN)	S,H<-E
S9,F5 Unrecognized Function Type (UFN)	S,H<-E
S9,F7 Illegal Data (IDN)	S,H<-E
S9,F9 Transaction Timer Timeout (TTN)	S,H<-E
S9,F11 Data Too Long (DLN)	S,H<-E
S9,F13 Conversation Timeout (CTN)	S,H<-E



STREAM 10: Terminal Services

S10,F1 Terminal Request (TRN)	S,H<-E
S10,F2 Terminal Request Acknowledge (TRA)	S,H->E
S10,F3 Terminal Display, Single (VTN)	S,H->E
S10,F4 Terminal Display, Single Acknowledge (VTA)	S,H<-E
S10,F5 Terminal Display, Multi-block (VMN)	M,H->E
S10,F6 Terminal Display, Multi-block Acknowledge (VMA)	S,H<-E
S10,F7 Multi-block Not Allowed (MNN)	S,H<-E

STREAM 14: Object Services

S14,F1 GetAttr Request	S,H <-> E
S14,F2 GetAttr Data	M,H <-> E

STREAM 15: Recipe Management

S15,F1 Recipe Management Multi-block Inquire	S,H <-> E
S15,F2 Recipe Management Multi-block Grant	S,H <-> E
S15,F21 Recipe Action Request	M,H <-> E
S15,F22 Recipe Action Acknowledge	M,H <-> E
S15,F27 Recipe Download Request	M,H -> E
S15,F28 Recipe Download Acknowledge	M,H <- E
S15,F29 Recipe Verify Request	M,H -> E
S15,F30 Recipe Verify Data	M,H <- E
S15,F31 Recipe Upload Request	S,H -> E
S15,F32 Recipe Upload Data	M,H <- E
S15,F35 Recipe Delete Request	M,H -> E
S15,F36 Recipe Delete Acknowledge	M,H <- E
S15,F49 Large Recipe Download Request	S,H -> E,reply
S15,F50 Large Recipe Download Acknowledge	S,H <- E
S15,F51 Large Recipe Upload Request	S,H -> E,reply
S15,F52 Large Recipe Upload Acknowledge	S,H <- E
S15,F53 Recipe Verification Send	M,H <- E,reply
S15,F54 Recipe Verification Acknowledge	S,H -> E

8 GEM Compliance

This section defines compliance to the GEM standard. It describes the fundamental GEM requirements and additional GEM capabilities. It provides references to other sections of the standard where detailed requirements are located. This section also defines standard terminology and documentation that can be used by equipment suppliers and device manufacturers to describe compliance with this standard.

The GEM standard contains two types of specifications:

- fundamental GEM requirements and
- requirements pertaining to additional GEM capabilities.

The fundamental GEM requirements form the foundation of the GEM standard. The additional GEM capabilities provide functionality required for some types of factory automation or functionality applicable to specific types of equipment. Figure 8.1 illustrates the relationship of the fundamental GEM requirements and the additional GEM capabilities.

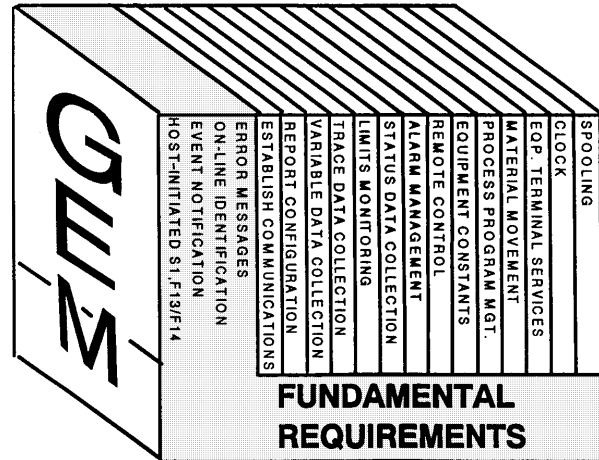
8.1 Fundamental GEM Requirements — All equipment shall comply with the fundamental GEM requirements listed in Table 8.1. Compliance to these requirements involves precise and complete adherence to all sections of the GEM standard referenced in Table 8.1.

Table 8.1 Fundamental GEM Requirements

Requirement	Section References
State Models	3.0, 3.1, 3.3
Equipment Processing States	3.4
Host-Initiated S1,F13/F14 Scenario	4.1.5.1
Event Notification	4.2.1.1
On-line Identification	4.2.6
Error Messages	4.9
Control (Operator-Initiated)	4.12 (except 4.12.5.2)
Documentation	8.4

In addition, compliance requires adherence to the portions of the following sections that are applicable to the fundamental GEM requirements:

- Variable data items (GEM, Section 5)
- SECS-II data item restrictions (GEM, Section 5)
- Collection events (GEM, Section 6)



Vertical text represents capabilities.

Some capabilities are also fundamental requirements.

Figure 8.1
GEM Requirements and Capabilities

8.2 GEM Capabilities — The following table lists all GEM capabilities and the sections of the GEM standard where they are specified. These sections contain the detailed requirements for implementing a GEM capability. Requirements for an individual capability include any referenced portions of the document. As an example, the Alarm Management capability requires implementation of the status variables “AlarmsEnabled” and “AlarmsSet” as defined in Section 5.

Table 8.2 Section References for GEM Capabilities

Capability	Section References
Establish Communications	4.1, 3.2
Event Notification	4.2.1.1
Dynamic Event Report Configuration	4.2.1.2
Variable Data Collection	4.2.2
Trace Data Collection	4.2.3
Limits Monitoring	4.2.4
Status Data Collection	4.2.5
On-line Identification	4.2.6
Alarm Management	4.3
Remote Control	4.4
Equipment Constants	4.5
Process Program Management	4.6
Material Movement	4.7
Equipment Terminal Services	4.8

Capability	Section References
Error Messages	4.9
Clock	4.10
Spooling	4.11
Control (Operator-Initiated)	4.12 (except 4.12.5.1)
Control (Host-Initiated)	4.12.5.1

8.3 Definition of GEM Compliance — The term “GEM Compliance” is defined with respect to individual GEM capabilities to indicate adherence to the GEM standard for a specific capability. Equipment is GEM-compliant for a specific GEM capability if, and only if, the following three criteria are met:

- The fundamental GEM requirements are satisfied.
- The capability is implemented to conform with all applicable definitions, descriptions, and requirements defined for the capability in this standard.
- The equipment does not exhibit behavior related to this capability that conflicts with the GEM behavior defined for the capability.

For example, equipment that provides SECS-II messages for management of process programs must precisely implement the GEM Process Program Management capability to be “GEM-Compliant for Process Program Management.”

Equipment may supply additional functionality not specified in the GEM standard by using any messages defined in the SECS-II standard as long as the additional functionality does not conflict with compliance to GEM capabilities.

Figure 8.2 illustrates the host view of equipment communications in relationship to the components of the GEM standard. The GEM capabilities are built upon the fundamental GEM requirements and present GEM-compliant behavior to the host when they are not obstructed by conflicting functionality. Additional non-GEM capabilities and non-obstructing extensions to GEM capabilities provide additional functionality while maintaining GEM behavior from the host view.

One additional term is defined to facilitate discussion of GEM capability. Equipment is “Fully GEM Capable” if and only if it meets the following two criteria:

- The equipment supplies all the GEM capabilities listed in Section 8.2.

— Every implemented GEM capability is GEM-Compliant.

8.4 Documentation — This section describes documentation requirements in addition to those specified in Sections 3 and 4 of this standard. All documentation of the SECS-II interface shall be supplied as a single volume, including Message Documentation, a Compliance Statement and the documentation required by Sections 3 and 4.

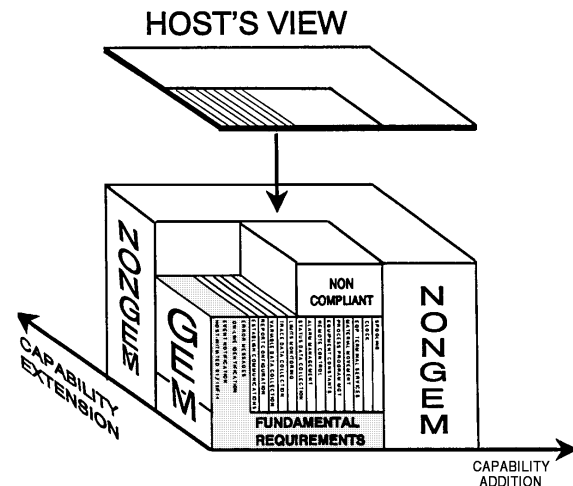


Figure 8.2
Host View of GEM

8.4.1 Message Documentation — The equipment supplier shall provide message documentation in conformance with Chapter 8 (Message Documentation) of SEMI E5.

8.4.2 GEM Compliance Statement — The SECS-II interface documentation provided by an equipment supplier shall address GEM compliance. This documentation shall include a GEM Compliance Statement that accurately indicates for each capability whether it has been implemented and whether it has been implemented in a GEM-compliant manner. The format for this statement is supplied as Table 8.3.

The table consists of three columns. The first column lists the requirements and capabilities. The other two columns pose questions to the supplier:

Implemented: Does the equipment provide functionality that is similar to that defined for the GEM requirement or capability?

GEM-Compliant: Has that requirement or capability been implemented in a GEM-compliant manner?

8.4.3 The equipment supplier may provide documentation on the format of required data items (see Section 5) using SECS Message Language™ Notation (SML®). The SML formats are provided in Table 8.4.

Table 8.3 GEM Compliance Statement

GEM COMPLIANCE STATEMENT		
FUNDAMENTAL GEM REQUIREMENTS	IMPLEMENTED	GEM-COMPLIANT
State Models	☐ Yes ☐ No	☐ Yes (See NOTE 1.) ☐ No
Equipment Processing States	☐ Yes ☐ No	
Host-Initiated S1= F13/F14 Scenario	☐ Yes ☐ No	
Event Notification	☐ Yes ☐ No	
On-Line Identification	☐ Yes ☐ No	
Error Messages	☐ Yes ☐ No	
Documentation	☐ Yes ☐ No	
Control (Operator Initiated)	☐ Yes ☐ No	
ADDITIONAL CAPABILITIES	IMPLEMENTED	GEM-COMPLIANT (See NOTE 2.)
Establish Communications	☐ Yes ☐ No	☐ Yes ☐ No
Dynamic Event Report Configuration	☐ Yes ☐ No	☐ Yes ☐ No
Variable Data Collection	☐ Yes ☐ No	☐ Yes ☐ No
Trace Data Collection	☐ Yes ☐ No	☐ Yes ☐ No
Status Data Collection	☐ Yes ☐ No	☐ Yes ☐ No
Alarm Management	☐ Yes ☐ No	☐ Yes ☐ No
Remote Control	☐ Yes ☐ No	☐ Yes ☐ No
Equipment Constants	☐ Yes ☐ No	☐ Yes ☐ No
Process Program Management	☐ Yes ☐ No	☐ Yes ☐ No
Material Movement	☐ Yes ☐ No	☐ Yes ☐ No
Equipment Terminal Services	☐ Yes ☐ No	☐ Yes ☐ No
Clock	☐ Yes ☐ No	☐ Yes ☐ No
Limits Monitoring	☐ Yes ☐ No	☐ Yes ☐ No
Spooling	☐ Yes ☐ No	☐ Yes ☐ No
Control (Host-Initiated)	☐ Yes ☐ No	☐ Yes ☐ No

NOTE 1: Do not mark YES unless all fundamental GEM requirements are implemented and GEM-compliant.

NOTE 2: Additional capabilities may not be marked GEM-compliant unless the fundamental GEM requirements are GEM-compliant.

Table 8.4 SML Notation

<i>Item Format</i>	<i>SECS-II Format Code</i>		<i>SML Item Format Mnemonic</i>
	<i>Binary</i>	<i>Octal</i>	
LIST	000000	00	L [length]
Binary	001000	10	B
Boolean	001001	11	BOOLEAN
ASCII	010000	20	A [length] or A [min., max.]
JIS-8	010001	21	J [length] or J [min., max.]
8-byte integer (signed)	011000	30	I8
1-byte integer (signed)	011001	31	I1
2-byte integer (signed)	011010	32	I2
4-byte integer (signed)	011100	34	I4
8-byte floating point	100000	40	F8
4-byte floating point	100100	44	F4
8-byte integer (unsigned)	101000	50	U8
1-byte integer (unsigned)	101001	51	U1
2-byte integer (unsigned)	101010	52	U2
4-byte integer (unsigned)	101100	54	U4

A. Application Notes

NOTE: The material contained in these Application Notes is not an official part of this SEMI standard and is not intended to modify or supersede the official standard. Rather, these notes are auxiliary information describing possible methods for implementing the protocol described by the standard and are included as reference material. The standard should be referred to in all cases. SEMI makes no warranties or representations as to the suitability of the material set forth herein for any particular application. The determination of the suitability of the material is solely the responsibility of the user.

A.1 Factory Operational Script

An Operational Script is a series of capabilities arranged in a typical factory operation sequence. The intent of having an Operational Script is to help put the SECS-II message Scenarios into a context. Although this context will vary, it represents a typical operational sequence found in most semiconductor device manufacturers' applications.

- System Initialization
- Synchronization
- Machine Setup
- Production Setup
- Processing
- Post-Processing
- Shutdown

The following script is not intended to be complete, but to serve as an example to be further developed on an implementation basis.

A1.1 Anytime Capabilities — All capabilities can generally occur at anytime during the operational script sequence.

A1.2 System Initialization and Synchronization — Upon system initialization, the default setting for communication (enabled or disabled) becomes effective, as well as any equipment constants or other information retained in non-volatile storage. The initial communication status is displayed at the equipment.

Assuming the communication state is enabled, the equipment will attempt to establish communication with a host computer. See Section 4.1 for a description of the scenario for establishing communications.

Upon receiving an indication that the equipment was previously not communicating, the host would typically

perform synchronization activities including setting the equipment's clock and requesting selected status information. Note that synchronization activity is host application-dependent and may be implemented using various scenarios.

A1.3 Production Set-Up — The host typically has the following information:

- what material
- what process step
- what process program to use (PPID)
- current equipment status, VID's, SVID's
- data collection requirements (trace data & event data)
- VID's needed
- Equipment constants (ECID's)

Based upon the above information, the host will perform setup activities as required. It must be verified that the correct process program is available and selected at the equipment.

A1.3.1 Auxiliary Material and Manual Set-Up — Auxiliary material can be checked and verified at this point. If status variables exist for auxiliary material, they may be requested by the host.

Any other manual, non-process, and/or non-product specific set-up also may take place at this point. The operator may interact with the equipment and the host. If the operator interacts with the equipment, the equipment communications link with the host should stay operational.

The operator and the host may exchange information via equipment terminal services.

A1.3.2 Product/Process Set-Up — Specific product and/or process information is communicated to the equipment prior to processing material.

A1.3.3 Material Load — The host may instruct an operator or a material handling system to deliver material to the equipment.

Once the material has arrived at the equipment, the equipment or the operator will notify the host.

A1.3.4 Production Data Collection Set-Up — The host instructs the equipment to collect event-based data. Reports are defined and linked to events. Event reports can be enabled or disabled.

The host instructs the equipment to collect data from the equipment based on time intervals.

The host configures the equipment to monitor specific variables and to send event reports when variables transition between monitoring zones.

A1.4 Processing

A1.4.1 Start Process Executing — The host or operator issues a command to start.

A1.4.2 Equipment Signals End of Run — When process execution is completed, the equipment generates events. If any of the events are enabled, they will be sent as event reports.

A1.5 Post-Processing — The equipment has completed processing material. It now makes the material available to the operator or material handling system for removal. The equipment signals the host that it is available for more work.

A1.5.1 Material Unload — Material is unloaded from the equipment by an operator or material handling system.

A.2 Equipment Front Panel

In the GEM standard, several requirements are stated that involve the display or input of information at the equipment front panel. The “equipment front panel” refers to an area on the equipment that is available to the operator under normal use (i.e., without removing maintenance access panels). This may include a CRT display, keyboard, switches, and lights.

This application note provides some guidance for implementation of the GEM front panel capabilities. All of these requirements map directly to state models and capabilities defined in Sections 3 and 4. All capabilities may be implemented in either hardware (buttons, switches, lights) or in a software/CRT equivalent.

A2.1 Displays and Indicators — The intent of various displays is to inform the operator of either the current state of the equipment or of a recent change of state (or both). Therefore, it is most useful if these displays are continuously visible and easily recognized at a distance. Required displays/indicators include:

Communications State: This means that three distinct states must be represented: DISABLED, ENABLED/NOT COMMUNICATING, and ENABLED/COMMUNICATING.

Terminal Services: An “New Host Message” indicator must be supplied.

A2.2 Switches/Buttons — Note that discrete switches also contain information for the user. However, these

tend to represent the desired states of the operator/user. The equipment’s response to a change of a switch may not be instantaneous. Still, the current position of switches should be available to the operator.

It may be appropriate to limit the access to some switches and buttons. This might be done via any of the standard methods, keys, passwords, combinations, etc. This is especially true for system default switches that would not often be changed. Required switches/buttons include:

Communications State System Default: In what communications state should the equipment be when system initialization is complete? The choices are DISABLED and ENABLED.

Communications State Selector: This is a toggle or button that will initiate a transition from ENABLED to DISABLED or vice versa.

Message Recognition Button: This button is used to initiate an event message to the host which indicates that the “New Host Message” has been read. This button should function only when the New Host Message Indicator is activated and when the received message is displayed in the terminal display.

A.3 Examples of Equipment Alarms

Table A.3 provides alarm examples pertaining to various configurational aspects of equipment.

NOTE: It is important to stress that these are just examples intended to illustrate that alarms pertain to situations in which there exists a potential for exceeding physical safety limits associated with people, equipment, and material being processed as per the GEM definition of an alarm.

NOTE: The alarm capability is intended as an addition to standard safety alarms (e.g., lights, horns). There is no intent to replace direct operator reaction to such problems. Nor is there the expectation that the host can necessarily prevent or directly address such alarms.

An actual machine shall have an associated set of alarms defined by the manufacturer that pertains to its specific configuration and design. The equipment manufacturer is responsible for supplying documentation associated with these alarm definitions.

Table A.3 Alarm Examples Per Equipment Configuration

<i>Subsystem</i>	<i>Alarm Description</i>	<i>ALID</i>	<i>Trigger</i>	<i>Reset</i>	<i>Operator</i>	<i>Equipment</i>	<i>Material</i>
Mainframe Power Supply	Overvoltage		Voltage supply over maximum limit			X	
	Undervoltage		Voltage supply under minimum limit			X	
Internal Power Distribution Bus	AC Low		AC under minimum limit			X	X
Cooling System	Overtemp		Temperature over maximum		X		X
	Pressure Low		Pressure below minimum				X

Subsystem	The subsystem of the equipment to which the alarm is related
Alarm Description	Description of the alarm
ALID	The Alarm ID as specified by SECS-II
Trigger	Text description of what caused the alarm
Reset	Description of how to resolve the alarm condition
Affected	Who or what is affected by the alarm trigger: Operator, Equipment, and Material

A.4 Trace Data Collection Example

This example shows an implementation of the Trace Data Collection capability defined in Section 4.2.3.

S2,F23 sent by host:

```

TRID = ABCD
DSPER = 000100 (One minute per
period)
TOTSMP = 9
REPGSZ = 3
SVID1 = Temperature
SVID2 = Relative humidity

```

S6,F1 looks like this (starting at time 1 a.m.):

1st transmission <L,4>

1. ABCD (trace ID)
2. 3 (last sample of the transmission)
3.

88	5	01	01	03	00
Year	Month	Day	Hour	Min	Sec
4. <L, n> n = 2 SVID's x REPGSZ of
3 = 2 x 3 = 6
72 (temperature)
0.29 (relative humidity)
73 (temp.)
0.30 (r.h.)
71 (temp.)
0.30 (r.h.)

2nd transmission <L,4>

1. ABCD
2. 6
3.

88	05	01	01	06	00
			hr	min	
4. <L, 6>
73
0.31
71
0.32
71
0.31

3rd and last transmission <L,4>

1. ABCD
2. 9
3. 88 05 01 01 09 00
hr min
4. <L,6>
71
0.30
72
0.30
71
0.31

A.5 Harel Notation

Harel's statecharts extend traditional state-transition diagrams with several additional concepts, most important of which are hierarchy and concurrence. Statecharts depict the behavior of a system by showing states it may take, events that prompt a change of state, and the composition of states. What follows is a very brief description of the symbols defined for use and how these are useful to describe a system. See Figure A.5.1 for the basic notational symbols.

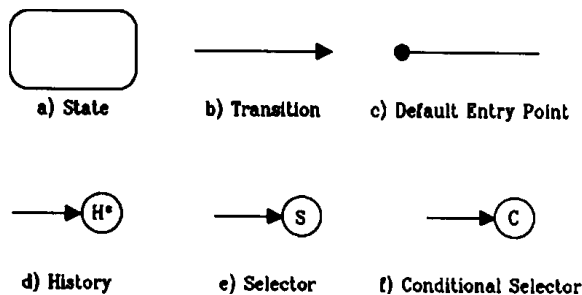


Figure A.5.1
Harel Statechart Symbols

States are represented by rounded boxes. A state transition is shown graphically with a line from the old state terminating with the arrow symbol at the new state. Transitions are unidirectional-while the reverse transition may be possible, it is considered a different transition with different conditions for initiation and different resultant actions.

States may be subdivided into substates to facilitate more concise definition of behavior. Thus, a hierarchy is defined whereby any state may be a substate of some parent state and in turn be the parent of its own substates. Substates must be one of two types, termed AND substates and OR substates.

A parent maybe divided into two or more OR substates of which one and only one is the active substate at any time. The accepted term for this exclusivity is XOR. Figure A.5.2 gives an example of a simple case of OR substates. In this example, some system (perhaps a motor) has a state named FUNCTIONAL. When the motor is FUNCTIONAL, it may be either ON or OFF, but never both.

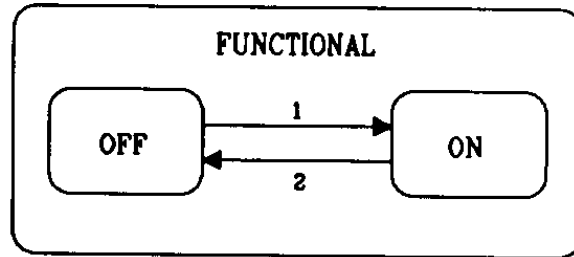


Figure A.5.2
Example of OR Substates

Another way of dividing a parent state corresponds roughly to subsystems. These AND substates represent parallelism, such that every AND substate of an active parent state is considered active. Harel also uses the term "Orthogonal Component" to refer to AND substates. However, these parallel substates tend to be highly interactive and interdependent. For this reason, the word orthogonal is considered confusing and has been excluded from use in this document. Figure A.5.3 shows an example of AND substates representing (in part) an automobile. Note the convention of attaching the name of the parent state AUTOMOBILE to the outside of the state in a small box. The substates shown are independent components and may have their own substates (of either the AND or OR type):

- LIGHTS may be ON or OFF;
- DOOR may be OPEN or CLOSED;
- ENGINE is constructed of components such as pumps, pistons, carburetor, etc.

Exiting one of a set of AND substates requires the exit of all others. In some cases, a transition arrow will be shown from only one of the substates with the others implied.

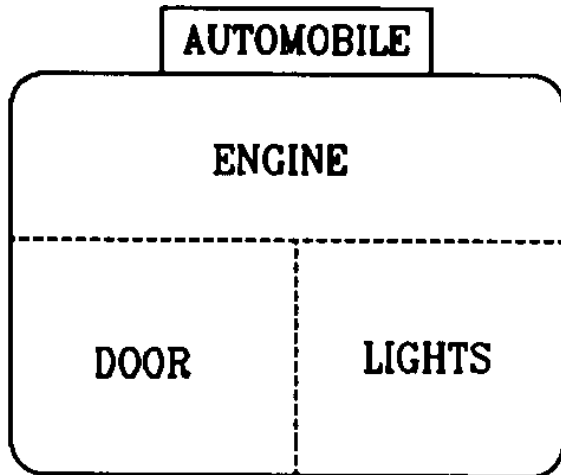


Figure A.5.3
Example of AND Substates

A simplification that also helps to prevent indeterminacy is implemented with the symbol for default entry point. This symbol will indicate which OR substate is initially active when there is not an explicit choice. This lack of specification is indicated by a transition arrow from one state to another that does not cross the boundary of the parent to point specifically to a substate.

An entrance to a state terminating in a history symbol (see Figure A.5.1) indicates that the OR substate to be entered should be that which was active the last time the parent state was active (i.e., last time the car was running, the radio was on). The history symbol H refers to the choice of substates of the parent. The symbol H* extends further to the lowest level substates defined. In the absence of memory of a “last time”, the default entry is used.

The selector and conditional selector symbols serve to abbreviate complex entrances to states. Their meaning is similar and indicate that the choice of OR substate upon entry of a parent state depends on some condition that is not shown. The selector is usually used to combine several similar transition events, while the conditional selector will typically require some computation or test of conditions external to the stimulus for state transition. Please examine the referenced article for more detail.

NOTE: Within the body of this document, the term statechart is not used in favor of the more traditional term state diagram.

A5.1 State Definitions — The state diagram provides a concise description of the function of a system. However, a full definition requires detail that cannot be

included on the diagram. A description of each state is required that covers the boundaries of the state and any responses that occur within that state to the environment. The convention in this document is to provide state names in ALL CAPS to help the reader identify where these are used. A sample state description of the ON state depicted in the Figure A.5.2 might be:

ON

The switch is in the on position. Power is available to the motor. Speed of the motor will change in proportion to the speed knob adjustment.

A5.2 Transition Table — The last piece of the state model is the transition table. It consists of several columns that list the transition number from the diagram, the starting and ending state for the transition, and three columns titled trigger, action, and comment. The trigger column describes the combination of events and conditions that initiates the transition (e.g., message Sx,Fy received). The trigger should be related to a single clearly defined event at the equipment. The action column identifies the activities associated directly with the transition. These activities may be of three types: a) actions taken upon exit of the old state, b) actions taken upon entry to the new state, and c) actions not associated with either state. These are not differentiated in this document. The final column allows for additional comments that help to clarify the transition. Table A.5, an example of transition table, illustrates the motor example in Figure A.5.2.

Table A.5 Transition Table for Motor Example

#	Current State	Trigger	New State	Action	Comment
1	OFF	Switch turned to on position.	ON	Power supplied to motor.	Power supply assumed available. Motor begins to turn.
2	ON	Switch turned to off position.	OFF	Power supply to motor disconnected.	Motor begins deceleration.

A.6 Example Control Model Application

This section provides one example of a host’s interaction with an equipment’s control model. A host system must have a view of the control model to understand and predict equipment behavior. However, the implementor may simplify the host’s view by assuming that some configuration settings are fixed and that the host-initiated features are not implemented. Applying these assumptions simplifies the behavior the host expects to see.

Figure A.6.1 shows the effective control model²⁴ based on the following host assumptions:

- The fundamental requirements are met, but the additional host-initiated control capability is not implemented.
- The configuration for the default entry to CONTROL is set to an OFF-LINE substate (either ATTEMPT ON-LINE or EQUIPMENT OFF-LINE).
- The destination state for transition 4 (failure of S1,F1 transaction) is configured to EQUIPMENT OFF-LINE.

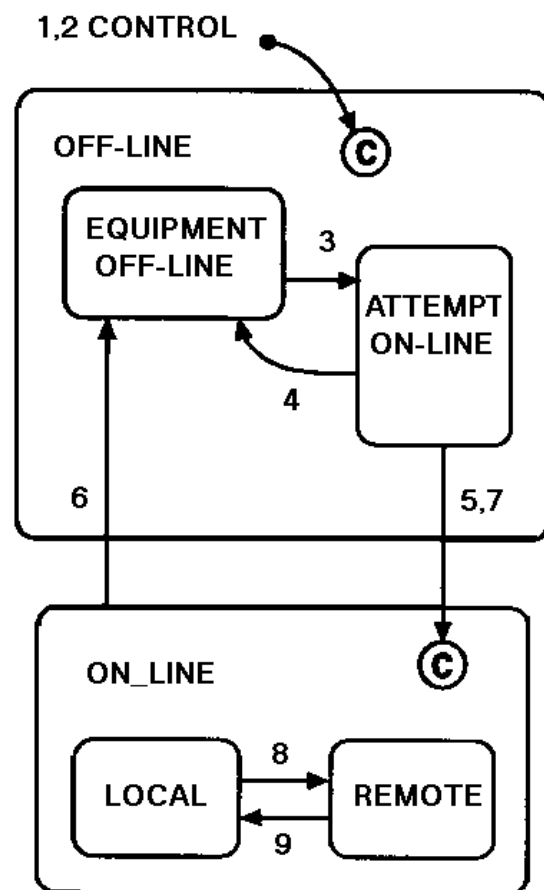


Figure A.6.1
Example of the Simplified "Effective" Control Model

This view of the model has two further settings that the host recognizes as changeable at the equipment. The first is the configuration of which substate of OFF-

LINE to be activated upon system initialization. The second is the front panel switch that determines whether the active system substate is LOCAL or REMOTE when ON-LINE.

This application has the following implications:

- This application requires that the equipment begin with the OFF-LINE state active. Thus, an equipment initiated S1,F1/F2 transaction must be completed before the equipment will begin sending all messages to the host.
- If a failed attempt to go ON-LINE is made by the equipment, it will not allow the host to complete the transition at a later time. An operator will be required to re-initiate the transition to ON-LINE when the host becomes ready.
- Once ON-LINE, the equipment will remain ON-LINE until an operator sets the equipment OFF-LINE at the equipment front panel.
- Since all transitions into the HOST OFF-LINE state are eliminated, this state is effectively eliminated from the host view of the control model.

This application retains the following features:

- The ON-LINE state is achieved only after the host acknowledges the equipment by replying to the S1,F1 with and S1,F2. This confirms to the operator attempting to put the equipment ON-LINE that the host application is ready for work to begin.
- It provides the operator the means to set the equipment OFF-LINE for non-host-related activities²⁵ (e.g., maintenance, test lots).
- The operator has the ability to operate the equipment with either the REMOTE or LOCAL state active. As the equipment transitions to ON-LINE, the preferred substate is automatically chosen (based on a front panel switch).
- The user may configure which substate of OFF-LINE the equipment will initially activate at system initialization. If ATTEMPT ON-LINE is chosen, the equipment will automatically attempt the transition to the ON-LINE state as system initialization.

²⁴ See Section 3.3 for details of the control model.

²⁵ Which activities are "non-host-related" varies from factory to factory. In general, fewer activities are "non-host-related" as a factory's automation level increases.

A.7 Examples of Limits Monitoring

A7.1 Introduction

A7.1.1 Four limits monitoring examples are included below to help clarify the use of limits and to illustrate typical applications. The first example shows how to apply limits to boolean values. The second illustrates application of several limits to a floating point variable in a classical control zone style. The third example shows an integer counter variable used to prompt for equipment maintenance.

A7.2 Examples

A7.2.1 Example 1 — Valve Monitoring

A7.2.1.1 The ACME Shine-Um-Rite Model 13 includes a sump which contains the chemical agent used to clean bare wafers. A chemical feeder system serves to refill the sump when the level drops below a certain level. The fill is accomplished via an on-off valve driven by sensors in the sump. Facilities must be informed of the proportion of the time the valve is open (approximates usage) and any time the value remains open for more than 5 minutes (valve likely broken).

A7.2.1.2 To implement this requirement, a limit was defined for the Boolean status variable which contains the current state of the valve (i.e., 0 = Closed, 1 = Open). See Figure A7.1 for illustration. LIMITID1 was defined with UPPERDB = LIMITMAX = 1 (Open) and LOWERDB = LIMITMIN = 0 (Closed). As a result, any time the valve opens, a collection event is generated with TransitionType = 0 and when the valve closes, a collection event is generated with TransitionType = 1. An event report containing the DVVAL LimitVariable was attached to each collection event and reporting for the event was enabled.

NOTE: Boolean values are defined as 0 = False/Closed/Off and any value > 0 = True/Open/On — never depend on a value of 1.

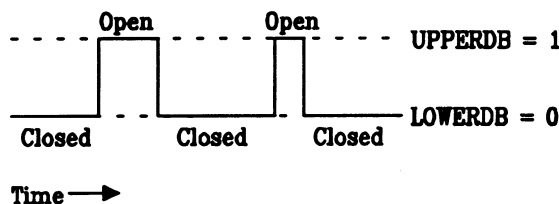


Figure A7.1
Valve Monitoring Example

A7.2.2 Example 2 — Environment Monitoring

A7.2.2.1 ACME also makes a Model 2 Stepper. The environmental control system of this equipment is

designed to hold the internal temperature relatively constant, but is sensitive to large changes in the external environment, opening of access doors, etc. To ensure that processing conditions are appropriate, the internal stepper temperature is monitored to ensure it remains in a safe operating zone (within “Shutdown” limits). In addition, a second set of limits are used within the Shutdown limits to bound the “Normal” operating range. Frequent excursions from the normal range into the “warning” range will prompt service on the environmental control system. The target temperature range is specified as 98–100°, the shutdown limits as 95–103°.

A7.2.2.2 Event reports are desired when the internal temperature moves outside of the normal operating zone into a warning zone (above or below), when the temperature moves back into the normal operating zone from the warning zones, and when the temperature moves out of the warning zones into the shutdown zones. Furthermore, temperature fluctuations of 0.5° should not trigger multiple event reports.

A7.2.2.3 Probably the most intuitive use of the limits monitoring capability is in establishing normal, warning, and shutdown zones for a particular equipment variable. Limits may be combined to provide such a scenario. The method is described below and illustrated in Figure A7.2. Please note that in the figure, limits are denoted as solid lines for simplicity, with deadbands indicated using the ± notation.

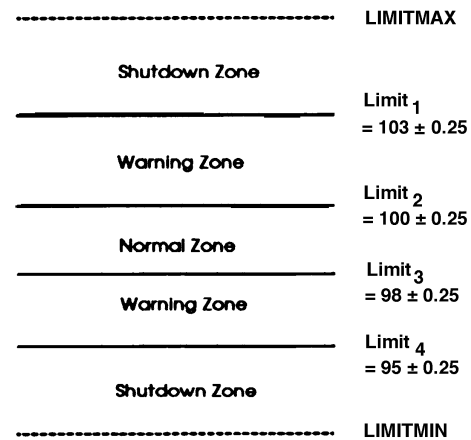


Figure A7.2
Environment Monitoring Example

A7.2.3 Example 3 — Calibration Counter

A7.2.3.1 Another ACME equipment is the multi-chamber Duz-It-All Model 7. This machine includes redundant chambers to increase throughput. One particular chamber on this equipment requires periodic

calibration. The need for calibration is a non-linear function of the number of wafers processed in that chamber. A status variable exists which contains the number of wafers processed since the last calibration was performed. Maintenance is definitely required after every 8 cycles, but the machine must be checked after 5 and 7 cycles to determine whether early calibration is necessary. This checking may be done by examining certain other equipment status variables.

A7.2.3.2 To meet this need, three limits are defined for the counter variable. Three limits were set, at 5, 7, and 8. Deadbands are set to zero, since chattering is not a problem. All the pertinent information is placed in an event report which is attached to the CEID for the limits of the counter to negate the need for further message exchange. Event reports are generated as each limit is reached (one zone transition each), and when the counter is reset following calibration (one, two, or three zone transitions referenced in one report). Figure A7.3 illustrates this example. Note that disabling the report upon counter reset (downward transitions) is not possible.

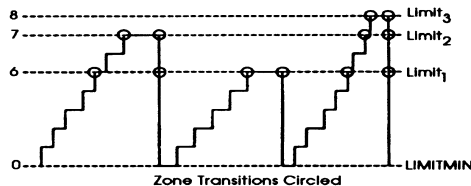


Figure A7.3
Calibration Counter Example

A7.2.4 Example 4 — Derived Variables

A7.2.4.1 The flagship of the ACME line is the new HotDog Furnace. This is a vertical furnace which exposes wafers to a variety of temperatures during processing. The temperature profile during the run is critical to the process and is typically contains a number of plateau's at different levels during the run. The owner wishes to monitor the temperature and be alerted whenever the actual temperature profile differs from the ideal by $> 0.5^\circ$. The derived variable was created to provide a steady target range during the run, no matter what the desired temperature range happened to be. Deviation from "ideal temperature profile" was chosen as the new variable to be monitored. The equipment already had access to the profile for the run, which described the desired temperature at a given time into the process. The manufacturer added a calculation each time the actual temperature was sampled, subtracting the ideal temperature from the actual. They provided as status variables the actual temperature, the ideal temperature, and the new "deviation from profile" variable. One limit was activated and set to 0.5 degrees

and a second set to -0.5 degrees (each with a deadband width of 0.05). Thus, when the temperature deviation from setpoint exceeds ± 0.5 , an event is generated containing the current desired temperature and the actual temperature. For good measure, additional data was added, providing time since start of run to document the precise point in the process that the problem occurred.

A7.2.4.2 In order to achieve the desired behavior, the host defines four monitoring limits. Two of the limits establish the target zone. These are responsible for reporting transitions from normal to warning zones in either direction. The other two limits establish the transitions between the warning zones and the error zones. The difference between UPPERDB and LOWERDB for each limit is 0.5. This may also be expressed as limit ± 0.25 . Combining limits does not change the way the equipment treats limits monitoring, but rather builds a method of interpreting limits from the host's point of view.

A.8 Process Parameter Modification for Process and Equipment Control

A8.1 Introduction

A8.1.1 In many equipment control applications there is a need for a GEM host to modify one or a small set of process parameters associated with a recipe. The number of parameters modified, frequency of modification (e.g., wafer-to-wafer, batch-to-batch, etc.), range of modification, etc., is largely a function of the equipment control application. Utilizing GEM, at least two methods are envisioned for modifying process parameters on a tool. With the first method, "Equipment Constants" can be used to relate process parameters of the updated recipe. Equipment Constants can also be used in a mode where they relate suggested modifications to process parameters from the stored recipe, i.e., the constants contain only the $\pm$ differential from a nominal value. The former mode is preferred because it better ensures data integrity between the controller and tool. With the second method the entire recipe could be downloaded, but this results in an enormous amount of communication overhead. Note that, in all cases, the Equipment Constants do not replace the process parameters inside a recipe, but are associated with (e.g., linked to) these parameters to relate modifications. The remainder of this application note provides a description of how process parameter modification can be implemented using existing GEM capabilities. The method may be used in a GEM compliant system provided that the specific GEM capabilities described are supported.

A8.2 *Equipment Constants*

A8.2.1 Incremental process parameter modification for process and equipment control can be supported over a GEM interface by using the *Equipment Constants* GEM capability (see Section 4.5). With this capability, each process parameter (or process parameter at a step) that can be modified, e.g., for purposes of process control, is associated with an equipment constant. Using the Equipment Constant GEM scenarios (see Section 4.5.5) the host can (1), send process parameters or parameter modifications, (2), request current values of modifiable recipe parameters, (3), retrieve name lists of equipment constants associated with modifiable parameters, and (4), be informed by the equipment when an operator changes one of the modifiable process parameters.

A8.2.2 The equipment constants should represent the actual values of the process parameters with which they are associated. Depending on the equipment operation and control application, the equipment constant could represent the actual value of a process parameter at a recipe step, or over the entire recipe. The equipment constants could also be utilized to represent the differentials of process parameters from nominal values. However it is important to note that, when using differential values to relate process parameter modifications, any loss of synchronization between equipment and host could result in an incorrect assessment of the value of a process setpoint by the host. Note also that, upon system startup, and whenever the appropriate process parameters are modified, the equipment constants should also be modified as necessary to always reflect the (absolute or relative) values of the associated process parameters.

A8.2.3 In order to maintain synchronization between equipment and host, it is recommended that equipment constants associated with recipe parameters be applied to only override the currently selected and active recipe. A selected recipe is considered to be active whenever the equipment is in the "PROCESSING" state and the recipe is the currently selected recipe (process program). If multiple recipes are utilized during one process event, e.g., cluster tool scenario, it is recommended that separate equipment constants be utilized for each recipe/process parameter pair.

A8.2.4 Note that timing and traceability issues associated with utilizing the equipment constants capability (for process control) are application specific and beyond the scope of this application note. Equipment that provides for recipe parameter overrides though setting of Equipment Constants should also provide additional Equipment Constants for the set that includes the name of the associated process program and a Boolean variable to enable and disable the override feature. In addition, the supplier should document for each parameter: (1), the associated Equipment Constant, and (2), any restrictions on the state (active or not) or the recipe in which the parameter may be modified. Also, since override of the process setpoints may be provided by a Host controller application element, it is recommended that the equipment provide an event report whenever the associated recipe has been modified.

A8.3 *Example*

A8.3.1 In the example of Figure A.8.1, a Chemical-Mechanical Planarizer (CMP) single wafer "polishing" system includes a GEM compliant planarizer (equipment), a thickness metrology unit and a (host) controller. The tool polishes a wafer to a target thickness. The post-process thickness is measured by the metrology unit and reported to the host controller. The controller utilizes a feedback control algorithm to determine the appropriate polish "time" recipe parameter for the next wafer. This time should be reported to the CMP equipment utilizing the equipment's GEM interface so that the information can be utilized for the next wafer processing event.

A8.3.2 The mechanism described in this application note could be utilized to implement process control as follows. A settable equipment constant is associated or "linked" with the "time" parameter on the equipment. Equipment system documentation indicates the linkage and the conditions under which the linkage is valid. When the controller determines an appropriate "time" parameter value for the next wafer to be polished, it sets the equipment constant to this value (see Section 4.5). The equipment is configured to accept this equipment constant change and, if the equipment is in (or possibly when the equipment reaches) the appropriate state, the recipe parameter is modified.

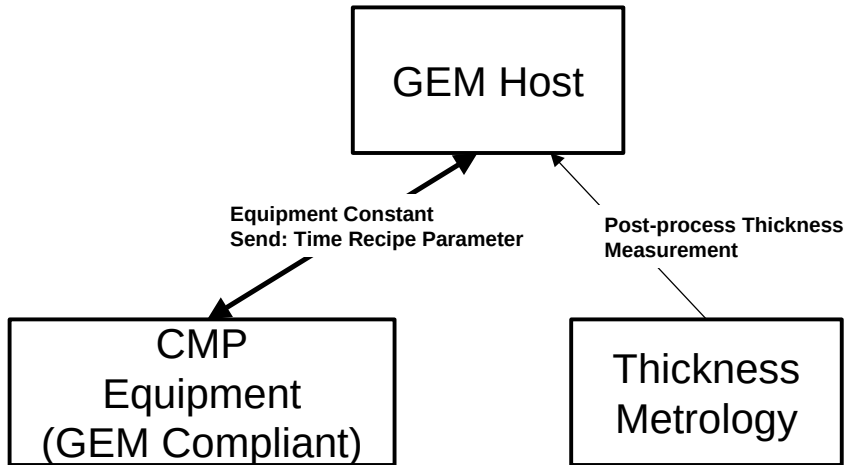


Figure A.8.1

CMP Single Wafer “Polishing” System with Host Recipe Parameter Modification Capability

NOTICE: These standards do not purport to address safety issues, if any, associated with their use. It is the responsibility of the user of these standards to establish appropriate safety and health practices and determine the applicability of regulatory limitations prior to use. SEMI makes no warranties or representations as to the suitability of the standards set forth herein for any particular application. The determination of the suitability of the standard is solely the responsibility of the user. Users are cautioned to refer to manufacturer’s instructions, product labels, product data sheets, and other relevant literature respecting any materials mentioned herein. These standards are subject to change without notice.

The user’s attention is called to the possibility that compliance with this standard may require use of copyrighted material or of an invention covered by patent rights. By publication of this standard, SEMI takes no position respecting the validity of any patent rights or copyrights asserted in connection with any item mentioned in this standard. Users of this standard are expressly advised that determination of any such patent rights or copyrights, and the risk of infringement of such rights, are entirely their own responsibility.

Index

A

ABORT	14, 17, 36
AbortLevel.....	36
alarm	5, 17, 21, 31, 32, 33, 34, 35, 50, 62, 73
ALARM CLEAR.....	32, 33
Alarm Management	1, 17, 31, 63, 67, 69
Alarm Report	31, 34, 35, 65
ALARM SET.....	31, 32, 33
alarm text.....	34
Alarm Text.....	34
AlarmID.....	62, 63
AlarmsEnabled	31, 34, 62, 67
AlarmsSet	31, 33, 34, 62, 63, 67
ALCD	31, 34, 62
ALCD _n	33
ALID	32, 34, 62, 73
ALID _n	1, 32
ALTX	32, 34
Anytime Capabilities	2, 71

B

button.....	45, 72
Button	72
buttons	5, 72
Buttons.....	2, 72

C

CEID.....	21, 26, 27, 28, 39, 41, 62, 63, 78
clock	24, 50, 71
Clock	1, 18, 50, 62, 63, 67, 69
COMMACK	8, 10, 11, 18, 19, 20
CommDelay.....	8, 10, 11
COMMUNICATING	9, 10, 11, 12, 18, 19, 20, 52, 53, 55, 72
communication failure	5, 8, 10, 11, 15, 47, 52, 54
Communication failure	11
communication fault.....	5, 10, 47
Communication Fault	47, 49
communication link	8
communications failure	8, 55
compliance.....	5, 6, 67, 68, 78
Compliance.....	1, 2, 4, 5, 6, 67, 68, 69
Compliance Statement	68
COMPLIANCE STATEMENT	69
compliant	68, 69
Compliant	68
COMPLIANT.....	69
connection transaction failure.....	8, 9, 10, 18
Connection transaction failure.....	11
control.....	4, 5, 6, 7, 12, 14, 25, 31, 35, 39, 54, 58, 59, 62, 76, 77
Control.....	1, 2, 3, 5, 12, 15, 18, 26, 58, 59, 60, 61, 63, 64, 67, 68, 69, 76
CONTROL	1, 3, 12, 13, 14, 15, 58, 59, 76

CPNAME	35, 36, 62
CPVAL	35, 36

D

data collection.....	4, 20, 22, 30, 32, 34, 47, 71
Data Collection.....	1, 17, 20, 23, 65, 67, 69, 72
Data Item Dictionary	4
data item restrictions.....	67
Data Item Restrictions	1, 62
data too long	49
Data Too Long.....	66
DATAID.....	62
DATALENGTH.....	62
Date and Time Data.....	64
Date and Time Request.....	50
Date and Time Send	64
Define Report	23, 64
disable.....	22, 23, 34, 35, 55, 56, 65
Disable	34, 35, 64, 65
disabled.....	10, 21, 26, 28, 34, 55, 71, 72
DISABLED	8, 9, 10, 11, 29, 72
display	9, 39, 45, 46, 47, 59, 72
Display.....	45, 66
document	4, 5, 6, 7, 8, 14, 22, 35, 40, 62, 64, 67, 75, 78
documentation	5, 16, 21, 22, 28, 36, 50, 67, 68, 69, 73
Documentation	2, 6, 67, 68, 69
DRACK	23
DSPER.....	24, 25, 74
DVNAME	22
DVVAL	22, 23, 25, 27, 39, 44, 62, 77

E

EAC.....	38
ECID.....	22, 62, 63, 71
ECV	22, 23, 62
enabled.....	8, 9, 11, 21, 22, 29, 31, 33, 34, 35, 41, 52, 55, 56, 71, 72, 77
ENABLED	9, 10, 11, 29, 52, 72
Equipment constant data.....	38
Equipment Constant Data.....	64
Equipment Constant Namelist	37, 38, 64
equipment model	6, 30
Equipment Model	6, 30
equipment status	26, 30, 50, 71, 77
Equipment Status	1, 30, 64
Equipment-Initiated Process Program Download	44
Equipment-Initiated Process Program Upload	42
ERACK	23
Error messages.....	47, 48
Error Messages	1, 10, 18, 47, 67, 69
Establish Communications Request	9, 10, 18, 19, 20, 64
Establish Communications Request Acknowledge	8, 64
EstablishCommunicationsTimeout.....	8, 18, 62
Event.....	15, 17, 20, 21, 23, 29, 32, 37, 39, 41, 44, 46, 47, 63, 64, 65, 67, 69, 72, 77, 78
event data collection	32
Event data collection	20, 47

event report.....	21, 22, 29, 35, 37, 54, 62, 77, 78
Event Report.....	17, 21, 22
event reporting.....	13, 20, 22, 27, 31, 35
Event Reporting.....	23
EventsEnabled	22, 62
executing	13, 16
Executing.....	16, 72
EXECUTING	16, 17, 63

F

Formatted Process Program.....	39, 41, 42, 43, 44, 66
Formatted Process Program Acknowledge.....	66
Formatted Process Program Request.....	66
Formatted Process Program Send.....	42, 43, 66

G

GEM-compliant.....	69
--------------------	----

H

HCACK.....	36, 37
------------	--------

I

idle.....	36
IDLE	16, 17
Illegal Data	49, 66
indicator.....	45, 46, 47, 59, 72
Indicator.....	72
INIT	16, 17

L

length.....	8, 23, 28, 35, 36, 40, 41, 45, 55, 62, 70
LENGTH.....	62
limits monitoring	5, 20, 25, 26, 27, 77, 78
Limits Monitoring	2, 5, 17, 25, 63, 67, 69, 77
List Alarm Data	31, 65
LOCAL.....	12, 13, 14, 15, 58, 59, 60, 61, 63, 76
LRACK	23

M

Material Movement	1, 17, 44, 63, 67, 69
MDLN	30, 31
Message Documentation.....	68
message fault	6, 47
Message fault.....	49
Message Fault.....	47, 48, 49
message recognition	45
Message recognition.....	45, 46, 47
Message Recognition.....	45, 63, 72
messages.....	3, 5, 6, 7, 8, 9, 10, 11, 12, 13, 17, 21, 22, 31, 32, 34, 45, 48, 50, 52, 53, 54, 55, 56, 58, 62, 64, 68, 76
Messages	52, 54, 56
Multi-block Data Send Inquire	65
Multi-block grant.....	21, 23, 29
Multi-Block Grant	65
Multi-block inquire.....	21, 23, 29, 52
Multi-Block Inquire.....	65

Multi-block Not Allowed	47, 66
must	6, 7, 9, 10, 14, 15, 16, 18, 21, 22, 23, 24, 26, 27, 28, 30, 31, 34, 35, 36, 37, 39, 40, 42, 44, 45, 47, 48, 50, 53, 54, 55, 56, 58, 59, 63, 68, 71, 72, 74, 76, 77

N

New Equipment Constant Acknowledge.....	64
New Equipment Constant Send.....	64
non-volatile memory.....	9, 34, 37, 40, 55, 56
non-volatile storage	22, 28, 39, 40, 41, 52, 55, 56, 59, 71
NOT.....	3, 9, 10, 11, 12, 19, 20, 52, 53, 54, 55, 72
NOT COMMUNICATING.....	9, 10, 11, 12, 19, 20, 52, 53, 55, 72

O

On-Line Data.....	64
On-line Identification	17, 30, 67
On-Line Identification	69

P

PAUSE	14, 16, 17, 36
PAUSED	36
Post-Processing	2, 71, 72
PPChangeName.....	39, 41, 62, 63
PPChangeStatus.....	39, 41, 62, 63
PPExecName	36, 39, 62, 63
PPID	36, 39, 40, 41, 62, 71
PP-SELECT.....	36, 39, 43
PreviousProcessState.....	16, 62, 63
PROCESS.....	16, 17
process program.....	6, 13, 14, 16, 35, 36, 39, 40, 41, 42, 43, 71
Process Program	1, 17, 39, 41, 42, 43, 44, 63, 65, 66, 67, 68, 69
Process Program Change	39, 63
Process Program Data.....	41
Process Program Download	42, 43
Process Program Load.....	1, 42, 43, 65
Process Program Load Grant.....	42, 43
Process Program Load Inquire	42, 43, 65
Process program management	39
Process Program Management	1, 17, 39, 63, 67, 68, 69
Process Program Request	41, 44, 66
Process Program Send	41, 42, 43, 44, 66
Process Program Validation	39
Process Program Verification.....	39, 43, 44, 66
Process Program Verification Send.....	66
Process Set-Up	71
processing.....	3, 5, 6, 7, 13, 14, 16, 17, 18, 21, 31, 35, 36, 39, 56, 63, 71, 72, 77, 78
Processing.....	1, 2, 16, 17, 36, 63, 67, 69, 71, 72
PROCESSING ACTIVE	16, 17
ProcessState.....	16, 62, 63

R

RCMD	36, 62
READY	16, 17
REMOTE.....	12, 13, 14, 15, 58, 59, 60, 61, 63, 76
remote command	3, 13, 36, 39, 43

Remote Command	35, 36, 37
Remote Control	1, 17, 35, 67, 69
REPGSZ	24, 25, 62, 74
require.....	4, 28, 33, 54, 75, 78
required.....	3, 4, 5, 10, 13, 14, 17, 21, 22, 31, 32, 36, 39, 42, 50, 54, 56, 62, 64, 67, 68, 69, 71, 75, 76, 77
Required	72
requirement.....	10, 16, 30, 68, 77
requirements	4, 5, 6, 17, 31, 36, 58, 67, 68, 69, 71, 72, 76
Requirements	1, 2, 18, 21, 22, 23, 24, 28, 30, 31, 34, 36, 37, 39, 44, 45, 48, 50, 56, 59, 67
REQUIREMENTS	69
resume	14, 36
Resume	35
RESUME.....	17, 36
RPTID	22, 23, 24, 62
RTY	5, 15

S

S1,F1.....	12, 15, 18, 31, 55, 58, 59, 60, 64, 76
S1,F2.....	12, 31, 59, 60, 64, 76
S1,F3.....	30, 57, 58, 64
S1,F4.....	30, 57, 58, 64
S10,F1.....	46, 47, 66
S10,F2.....	46, 47, 66
S10,F3.....	46, 47, 66
S10,F4.....	46, 47, 66
S10,F5.....	47, 66
S10,F6.....	47, 66
S10,F7.....	47, 66
S2,F13.....	38, 64
S2,F14.....	38, 64
S2,F15.....	38, 56, 64
S2,F16.....	38, 56, 64
S2,F17.....	50, 51, 64
S2,F18.....	51, 64
S2,F23.....	24, 25, 64, 74
S2,F24.....	25, 64
S2,F29.....	38, 64
S2,F30.....	38, 64
S2,F31.....	50, 51, 64
S2,F32.....	51, 64
S2,F33.....	21, 23, 64
S2,F34.....	23, 64
S2,F35.....	23, 64
S2,F36.....	23, 64
S2,F37.....	23, 64
S2,F38.....	23, 65
S2,F39.....	23, 29, 65
S2,F40.....	23, 29, 65
S2,F41.....	35, 37, 65
S2,F42.....	36, 37, 65
S2,F43.....	52, 55, 56, 65
S2,F44.....	56, 65
S2,F45.....	28, 29, 65
S2,F46.....	29, 65
S2,F47.....	28, 29, 65

S2,F48.....	29, 65
S5,F1.....	31, 32, 35, 65
S5,F2.....	35, 65
S5,F3.....	35, 65
S5,F4.....	35, 65
S5,F5.....	34, 35, 65
S5,F6.....	31, 32, 35, 65
S6,F1.....	24, 25, 65, 74
S6,F11.....	21, 22, 27, 35, 37, 38, 41, 44, 46, 47, 57, 58, 60, 61, 65
S6,F12.....	21, 35, 37, 38, 41, 44, 46, 47, 57, 58, 60, 61, 65
S6,F15.....	21, 22, 65
S6,F16.....	22, 65
S6,F19.....	24, 38, 65
S6,F2.....	25, 65
S6,F20.....	24, 38, 65
S6,F23.....	52, 53, 54, 55, 56, 57, 58, 65
S6,F24.....	57, 58, 65
S6,F5.....	21, 65
S6,F6.....	21, 65
S7,F1.....	42, 43, 65
S7,F17.....	41, 66
S7,F18.....	41, 66
S7,F19.....	41, 66
S7,F2.....	42, 43, 65
S7,F20.....	41, 66
S7,F23.....	39, 42, 43, 55, 66
S7,F24.....	42, 43, 66
S7,F25.....	41, 44, 66
S7,F26.....	39, 41, 44, 66
S7,F27.....	43, 44, 66
S7,F28.....	43, 44, 66
S7,F29.....	43, 44, 66
S7,F3.....	39, 42, 43, 66
S7,F30.....	43, 44, 66
S7,F4.....	42, 43, 66
S7,F5.....	41, 44, 66
S7,F6.....	39, 41, 44, 66
S9,F1.....	48, 66
S9,F11.....	49, 66
S9,F13.....	49, 66
S9,F3.....	48, 66
S9,F5.....	48, 66
S9,F7.....	49, 66
S9,F9.....	10, 49, 66
safety	5, 7, 31, 32, 33, 34, 73, 78
SECS-I.....	3, 5, 6, 8, 9, 15, 47
SECS-II	1, 3, 4, 5, 6, 9, 11, 13, 17, 22, 34, 39, 40, 52, 53, 56, 62, 64, 67, 68, 70, 71, 73
Selected Equipment Status Data.....	64
Selected Equipment Status Request.....	64
selector.....	75
Selector.....	72
SEQNUM	62
SETUP.....	16, 17

shall	3, 4, 5, 6, 8, 9, 10, 11, 12, 13, 14, 18, 21, 22, 24, 28, 31, 34, 36, 37, 40, 45, 48, 50, 52, 53, 54, 55, 56, 58, 59, 62, 67, 68, 73
shutdown	56, 77
Shutdown.....	71, 77
SMPLN.....	62
SMS	3, 5, 6, 10
SOFTREV	30, 31
Spooling.....	1, 12, 18, 52, 55, 56, 57, 58, 63, 65, 68, 69
START	16, 17, 36
State Model.....	1, 6, 7, 8, 9, 10, 12, 13, 18, 27, 52
Status Data Collection	17, 30, 67, 69
Status Variable Namelist Reply.....	64
Status Variable Namelist Request	64
STOP	14, 17, 36, 63
supplier	14, 21, 22, 35, 36, 40, 50, 62, 68, 69
SV	22, 23, 24, 25, 26, 30, 55, 62
SVID.....	22, 24, 30, 62, 71, 74
switch.....	7, 9, 11, 12, 15, 52, 58, 59, 60, 72, 75, 76
Switch.....	76
synchronization.....	18, 20, 23, 50, 71
Synchronization.....	2, 71
System Errors	1, 66

T

terminal services.....	71
Terminal Services.....	1, 13, 18, 45, 63, 66, 67, 69, 72
TEXT	45, 62
TIME	50, 51
toggle	72
TOTSMP	24, 25, 62, 74
Trace Data Acknowledge	65
trace data collection.....	24
Trace data collection.....	24
Trace Data Collection.....	2, 5, 17, 24, 25, 67, 69, 74
Trace Data Send	65
trace report.....	24
Trace Report	25
Transition Type	63
TRID.....	24, 62, 74

U

units	6, 22, 23, 26, 30
Units	25

V

Variable Item Dictionary	4, 62
Variable Item List.....	1, 22, 30, 62



SEMI E30.1-0200

INSPECTION AND REVIEW SPECIFIC EQUIPMENT MODEL (ISEM)

This standard was technically approved by the Global Information and Control Committee and is the direct responsibility of the North American Information and Control Committee. Current edition approved by the North American Regional Standards Committee on September 3, 1999. Initially available at www.semi.org November 1999; to be published February 2000. Originally published June 1998.

CONTENTS

1 Purpose

2 Scope

3 Limitations

4 Referenced Documents

4.1 SEMI Standards

4.2 Other References

5 Terminology

5.1 Abbreviations and Acronyms

5.2 Definitions

6 Communication Requirements

7 State Models

7.1 Processing State Model Requirements

7.2 Processing State Model Diagram

7.2.1 Working State for Inspection Equipment Model

7.2.2 Working State for Review Equipment

7.2.3 Working State for Inspection/Review Equipment

7.3 Processing State Definitions

7.4 Processing State Transition Table

8 Collection Event List

8.1 Requirements

9 Variable Items

9.1 Requirements

9.2 Variable Items and Reporting Levels

9.3 Variable Item Types

9.4 Variable Item Dictionary

10 Alarm List

10.1 Alarm List Table

10.2 Alarm ID, Alarm Set/Cleared Event Table

11 ISEM Tables

11.1 ISEM Table Data

11.2 TABLE-DEFs

11.3 Required ISEM Tables

11.3.1 "TABLE-AREA-DEF"

11.3.2 "TABLE-ALIGN-DEF"

11.3.3 "TABLE-ANOMALY-DEF"

11.3.4 "TABLE-M21-ANOMALY-DEF"

11.3.5 "TABLE-STANDARD-DEFECT-DATA-SET-DEF"

11.4 TABLE-DEF Column Header Descriptions and Formats

12 Process Program Management

12.1 Definition and Rules for ISEM Process Programs

12.2 Requirements

12.3 Process Program Structure

12.4 Process Program Variable Parameters

12.4.1 Overriding Process Program Variable Parameters Default Values

12.4.2 Requirements and Rules

12.4.3 Modifying Process Program Variable Parameters

12.5 Use of Process Programs, Remote Commands, and "PROCESS-BLD-GROUP"

13 Remote Commands

13.1 Requirements

13.2 Remote Commands Description

13.2.1 Remote Commands and Associated Host Command Parameters

13.2.2 Host Command Parameters Name and Values

13.2.3 Remote Commands vs. Processing States



14 Scenarios

14.1 Run Level Reporting Scenario

14.2 PP-UPDATE Remote Command Scenario

14.3 PP-SELECT Remote Command Scenario

14.4 Event Report and ISEM Table Transfer Command Scenario

15 GEM Capabilities

15.1 Requirements

Related Information 1

R1-1 Defect Classification Code Management

R1-1.1 Classification Codes and Defect Classification

R1-1.2 Requirements

R1-2 Reporting Coordinates and Coordinate Systems

R1-2.1 Site Location Accuracy

R1-2.2 Expected or Designed Locations vs. Actual Locations

R1-2.3 Substrate Coordinate Systems (Unpatterned)

R1-2.4 Substrate Pattern Coordinate System

R1-2.5 Pattern Element Coordinate System

R1-2.6 Parallel Coordinate Systems

R1-2.7 Requirements

R1-3 Coordinate System for a Substrate

R1-3.1 SEMI M20 Coordinate System

R1-3.2 M20P Coordinate System

R1-3.3 Establishing an M20P Coordinate System

R1-4 Layout of Rectangular Pattern Elements on a Substrate using SEMI M20 Coordinate System

R1-4.1 ISEM “M21” Layouts

R1-5 How an M20P Coordinate System is Established on a Substrate

Related Information 2 — Application Notes

R2-1 Using ISEM Table Attributes to Specify Process Related Data Item Variable Values



SEMI E30.1-0200

INSPECTION AND REVIEW SPECIFIC EQUIPMENT MODEL (ISEM)

This standard was technically approved by the Global Information and Control Committee and is the direct responsibility of the North American Information and Control Committee. Current edition approved by the North American Regional Standards Committee on September 3, 1999. Initially available at www.semi.org November 1999; to be published February 2000. Originally published June 1998.

1 Purpose

1.1 This standard establishes a Specific Equipment Model (SEM) for Inspection and Review Equipment (ISEM). The model consists of equipment characteristics and behavior that are to be implemented in addition to the SEMI E30 fundamental requirements and additional capabilities. The intent of this standard is to facilitate the integration of ISEM equipment into an automated (semiconductor fabrication) factory. This document accomplishes this by defining an operational model for ISEM equipment as viewed by a factory automation controller. This definition provides a standard host interface and equipment operational behavior (e.g., control, state models, data reports, and reporting levels). Several topics require additional activity that are within the scope of this standard: substrate pattern maps; defect classification code management; and review data management.

2 Scope

2.1 The scope of this standard is limited to the definition of Inspection, Review, and Inspection/Review equipment behavior as perceived by a SEMI Equipment Communications Standard II (SEMI E5/SECS-II) host that complies with SEMI E30. It defines the external view of the equipment through the SECS link; it does not define the internal operation of the equipment. This standard expands the SEMI E30 requirements and capabilities in the areas of the processing state model, remote commands, variable items, alarms, and data collection.

2.2 This standard is intended for ISEM equipment that generates data and information about anomalies and defects found on substrates. Inspection equipment finds anomalies. Anomalies are occurrences on a substrate that have been judged to be unexpected, abnormal, incongruous, or inconsistent. Anomalies may be examined using review equipment, at which time they may be classified as defects or non-defects. Some inspection equipment may generate, and some review equipment may use, coordinate data to locate anomalies on a substrate. The accuracy of the coordinate data generated or used is equipment-dependent.

2.3 This standard does not purport to address safety issues, if any, associated with its use. It is the responsibility of the users of this standard to establish appropriate safety and health practices and determine the applicability of regulatory limitations prior to use.

3 Limitations

3.1 This document addresses three distinct types of equipment: inspection, review, and inspection/review. The term ISEM equipment refers to all three types of equipment. These three equipment types are differentiated by the basic functions they perform:

3.1.1 Inspection Equipment that looks for anomalies on a substrate and reports information regarding those anomalies. Inspection equipment may determine the location of anomalies relative to a coordinate system. Inspection equipment may also provide other types of data related to the anomaly.

3.1.2 Review Equipment that accepts information about anomalies on a substrate, gathers information on those anomalies, and reports that data.

3.1.3 Inspection/Review Equipment having the characteristics of both inspection and review equipment.

4 Referenced Standards

4.1 SEMI Standards

SEMI E5 — SEMI Equipment Communications Standard 2 Message Content (SECS-II)

SEMI E30 — Generic Model for Communications and Control of SEMI Equipment (GEM)

SEMI E37 — High-Speed SECS Message Services (HSMS) Generic Services

SEMI E37.1 — High-Speed SECS Message Services Single-Session Mode (HSMS-SS)

SEMI E58 — Automated Reliability, Availability, and Maintainability Standard (ARAMS): Concepts, Behavior, and Services

SEMI M20 — Specification for Establishing a Wafer Coordinate System

SEMI M21 — Specification for Assigning Addresses to Rectangular Elements in a Cartesian Array

4.2 Other References

Harel, D., "Statecharts: A Visual Formalism for Complex Systems," Science of Computer Programming 8, (1987), 231-274

NOTE 1: As listed or revised, all documents cited shall be the latest publications of adopted standards.

5 Terminology

5.1 Abbreviations and Acronyms

5.1.1 *GEM* — generic equipment model

5.1.2 *PE* — pattern element

5.1.3 *TCP/IP* — Transmission Communication Protocol/Internet Protocol

5.2 Definitions

5.2.1 *align* — to put into proper relative position or orientation.

5.2.2 *alignment* — a procedure in which a coordinate system is established on a substrate.

5.2.3 *alignment mark* — a feature on the substrate selectively used for alignment.

5.2.4 *anomaly* — an occurrence on a substrate that has been judged to be unexpected. Something abnormal, incongruous, or inconsistent.

NOTE 2: After an anomaly is reviewed, it may be classified as a defect.

5.2.5 *batch* — a group of substrates or lots intended for a process sequence versus single substrate processing.

5.2.6 *carrier* — a container with one or more fixed positions at which material can be held.

5.2.7 *carrier location* — a physical place within the equipment capable of holding a carrier.

5.2.8 *cassette* — a container with one or more substrate locations (see *slot*).

5.2.9 *defect* — 1) A physical, optical, chemical, or structural irregularity that degrades the ideal substrate structure or the thin films built over the substrate. 2) An undesirable classified anomaly.

5.2.10 *defect classification* — the categorization of defects according to some systematic division based on their physical, optical, chemical, or structural properties.

5.2.11 *die* — 1) A field sub-unit. 2) An area of substrate that contains the device being manufactured.

5.2.12 *ended* — the end of a state that may be when it is normally completed, or its early end due to an

allowed or atypical condition (e.g., a STOP command, or an error or alarm condition).

5.2.13 *factory automation controller* — a computer system that provides integration of factory shop control and business systems with semiconductor equipment.

5.2.14 *feature* — 1) A line or a point (as a feature within a pattern). 2) A physical characteristic of the substrate (e.g., a substrate flat).

5.2.15 *field* — the printed pattern from a reticle.

5.2.16 *field of view* — the imaging area as seen at magnification of the inspection or review equipment.

5.2.17 *global alignment* — a procedure which establishes a coordinate system for the entire substrate (see *alignment*).

5.2.18 *group* — a logical collection of regions.

5.2.19 *group alignment* — a procedure which establishes a coordinate system for an area, which is a contiguous group (see *alignment*).

5.2.20 *inspect* — to detect anomalies and/or information about anomalies.

5.2.21 *inspection* — an examination to detect anomalies.

5.2.22 *inspection equipment* — equipment that looks for anomalies on a substrate and reports information regarding those anomalies. Inspection equipment may determine the location of anomalies relative to a coordinate system. Inspection equipment may also provide other types of data related to the anomaly.

5.2.23 *inspection/review equipment* — equipment having the characteristics of both inspection and review equipment.

5.2.24 *ISEM job* — the information required to specify an inspection or review that may include material identification and location and process program identifications as well as any other parameters required to obtain a desired result.

5.2.25 *layer* — one of a sequential series of overlaying photomasks that make up a device series.

5.2.26 *lot* — a group of one or more substrates of the same type (e.g., wafers, masks, CDs).

5.2.27 *major flat* — the flat of longest length that is commonly located with respect to a specific crystal plane (ASTM F 1241-89).

5.2.28 *mask* — a selective barrier to the passage of radiation. For example, a transparent plate containing an opaque pattern that is used to transfer that pattern to another substrate.

5.2.29 *material* — a piece or pieces of substrate, one or more substrates, a lot, a batch, or a run.

5.2.30 *metrology equipment* — any equipment that collects and reports information on specific predetermined locations or features on a substrate with consistent data structure or that reports general information about the entire substrate.

5.2.31 *notch* — a U-shaped cut on the edge of a substrate that is commonly located with respect to a specific crystal plane.

5.2.32 *overlay* — the actual distance between two features on different layers of a substrate, compared to the expected distance.

5.2.33 *pattern* — 1) The physical features on a substrate surface. 2) An ideal pattern is the arrangement of features expressed in a calculated or mathematical manner.

5.2.34 *pattern element* — 1) Any recognizable set of features. 2) A rectangular sub-unit of a pattern or a pattern element. There may be multiple levels of pattern elements.

5.2.35 *primary fiducial* — a key characteristic of a substrate used to align the substrate during processing (such as a *notch* or *major flat*).

5.2.36 *region* — a single field of view which may be a collection of sites.

5.2.37 *registration* — the actual distance between two features on the same layer of a substrate, compared to the expected distance.

5.2.38 *reticle* — a mask that contains the patterns to be reproduced on a substrate; the image may be equal to or larger than the final projected image.

5.2.39 *review* — the process of classification of anomalies which may result in the appending of additional data to inspection data. Used to create a field on a substrate.

5.2.40 *review equipment* — equipment that accepts information about anomalies on a substrate, gathers information on those anomalies, and reports that data.

5.2.41 *run (noun)* — the material processed during the EXECUTING state.

5.2.42 *run (verb)* — the actions of a process between the READY state and the STOPPING state.

5.2.43 *safe state* — a state in which the equipment presents no danger to the product or user. This implies that safety interlocks are in place such that the equipment can be serviced without harm to the operator and that the material being processed has been removed from the processing station into an accessible location.

5.2.44 *site* — a single x,y coordinate where an action can be performed (e.g., *alignment* or *review*). The area associated with a site is determined by the equipment accuracy (e.g., optics, stage algorithms).

5.2.45 *slot* — a physical location within a Carrier capable of containing a substrate. Also referred to as a carrier location.

5.2.46 *substrate* — the basic unit of material, processed by semiconductor equipment, such as wafers, CDs, flat panels, or masks.

6 Communication Requirements

6.1 It is required that any ISEM-compliant equipment follow the Communications State Model in SEMI E30. In addition, ISEM-compliant equipment shall support the High Speed Messaging Service Standard (SEMI E37/HSMS). It is a minimum requirement to support Single Session (SEMI E37.1/HSMS-SS) sending SECS-II messages over TCP/IP. The reason behind this requirement is the size of the process programs used by this class of equipment and the amount of data produced.

7 State Models

7.1 Processing State Model Requirements

7.1.1 The processing state model included in this standard is a requirement for ISEM equipment. This standard requires implementation of all SEMI E30 state models (such as control, communication, and on-line/off-line). A state model consists of state model diagrams, state definitions, and a state transition table. All state transitions in this standard, unless otherwise specified, shall correspond to collection events.

7.1.2 A state model is the host's view of the equipment and does not necessarily describe the internal equipment operation. All ISEM state model transitions shall be mapped sequentially into the appropriate internal equipment events that satisfy the requirements of those transitions. In certain implementations, the equipment may enter a state and have already satisfied all of the conditions required by the ISEM state model for transition to another state. The equipment makes the required transition without any additional actions in this situation.

7.1.3 Some equipment may need to include additional states other than those in this standard. Additional states may be added but shall not change the ISEM-defined state transitions. All expected transitions between ISEM states shall occur.

7.2 *Processing State Model Diagram* — Processing state models are detailed for ISEM equipment in Figure 1. This diagram contains all states and transitions that

are common to all three types of ISEM equipment. The WORKING state is different for each type of equipment. The same state names and transition identifiers are used to identify common states and transitions of the three types of equipment. The working states for the three types of equipment are presented in the following sections. All states and transitions are described in the section following the diagrams.

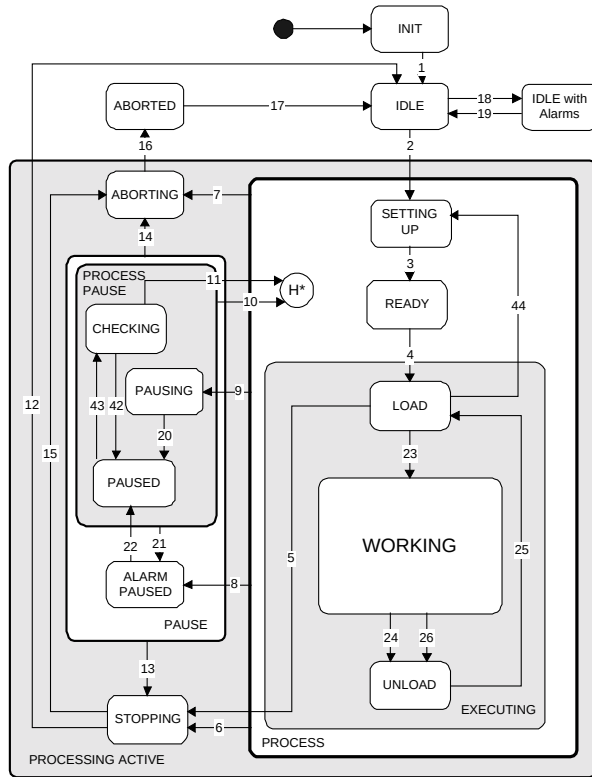


Figure 1
Generic ISEM Processing State Model Diagram

7.2.1 Working State for Inspection Equipment Model — The processing state model for inspection equipment is identical to the Generic ISEM Processing State Model (Figure 1). Only the WORKING state is unique to the inspection equipment processing state model. This is shown in Figure 2.

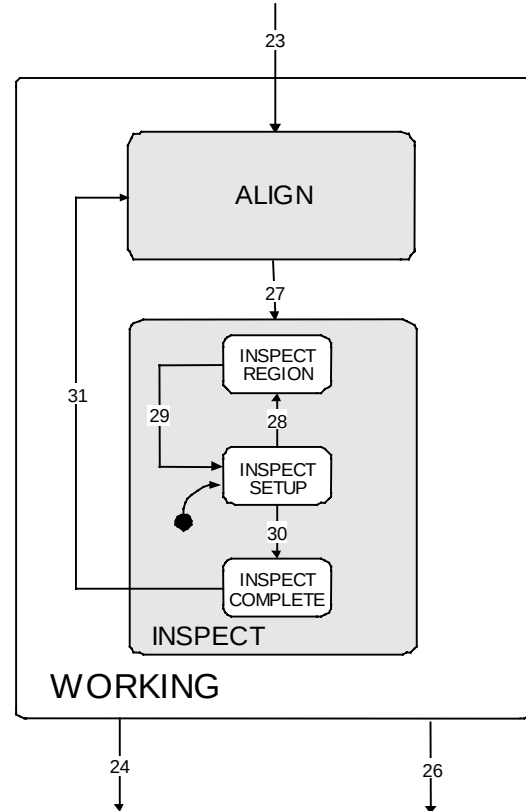


Figure 2
Working State for Inspection Equipment

7.2.2 Working State for Review Equipment — The processing state model for review equipment is identical to the generic ISEM Processing State Model (Figure 1). Only the WORKING state is unique to the review equipment processing state model. This is shown in Figure 3.

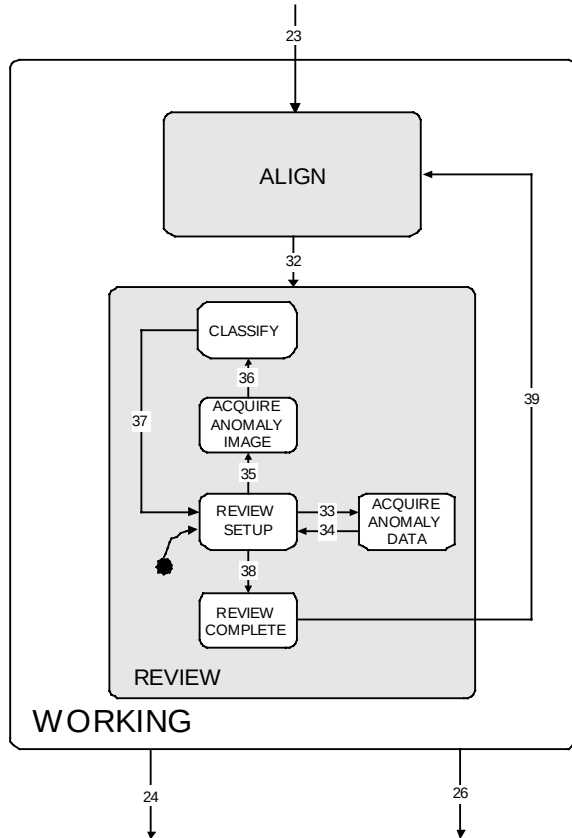


Figure 3
Working State for Review Equipment

7.2.3 Working State for Inspection/Review Equipment — The processing state model for inspection/review equipment is identical to the generic ISEM Processing State Model (Figure 1). Only the WORKING state is unique to the inspection/review equipment processing state model. This is shown in Figure 4.

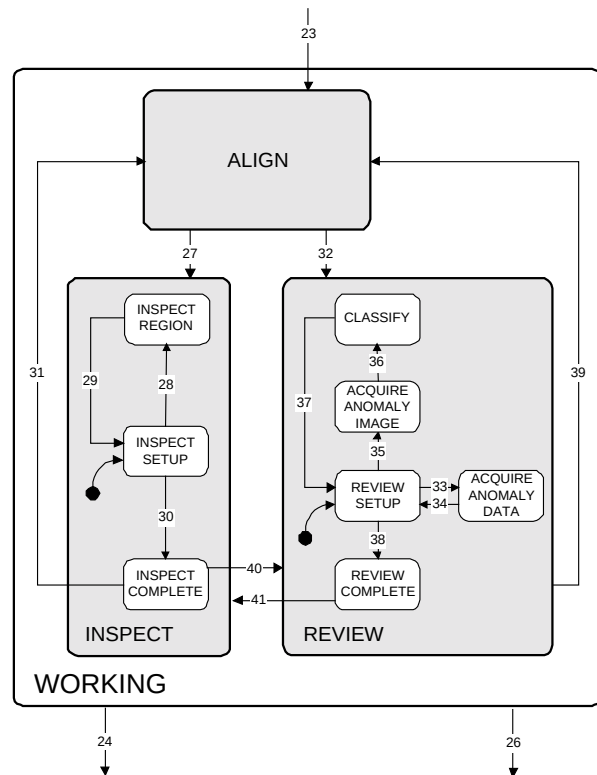


Figure 4
Working State for Inspection and Review Equipment

7.3 Processing State Definitions

7.3.1 ABORTED — All activity is suspended as a result of an ABORT command. Any alarm and abort conditions shall be cleared before exit from this state. The CLEANUP command is available to the operator or host to transition the equipment from the ABORTED state to IDLE state.

7.3.2 ABORTING (PROCESSING ACTIVE Sub-State) — The equipment has received an ABORT command. All normal activity is suspended. The equipment is taking appropriate action to put the equipment and material in a “safe state” where possible. Data may be invalid or not available.

7.3.3 ACQUIRE ANOMALY DATA (REVIEW Sub-State) — Data is being acquired about anomaly locations.

7.3.4 ACQUIRE ANOMALY IMAGE (REVIEW Sub-State) — The equipment is obtaining a view of the anomaly.

7.3.5 ALARM PAUSED (PAUSE Sub-State) — An alarm has occurred in the PROCESS or PROCESS PAUSE states, and the equipment is waiting for the alarm to be cleared or a command (STOP or ABORT).

7.3.6 ALIGN (WORKING Sub-State) — The equipment or operator is performing an alignment of the material to the equipment. If needed, within this state, the equipment shall refine or establish its SEMI M20 coordinate system and establish any secondary coordinate systems.

7.3.7 CHECKING (PROCESS PAUSE Sub-State) — The equipment verifies that the process program update request is valid. No process program parameters are changed unless “all” reported updates are valid. This is a similar procedure to that which is done in SETTING UP before the equipment is ready to transition to the READY state. Valid commands in this state are STOP, ABORT, and RESUME.

7.3.8 CLASSIFY (REVIEW Sub-State) — The operator or equipment is determining the classification of an anomaly.

7.3.9 EXECUTING (PROCESS Sub-State) — The equipment is processing material automatically and can continue to do so without external intervention but normally may include interaction with the host or operator.

7.3.10 IDLE — Checks for queued ISEM jobs or awaits a PP-SELECT, MAP-CARRIER, or PP-ASSIGN command. IDLE is free of alarm and error conditions. Any transition into this state shall deselect any selected Process program(s).

7.3.11 IDLE with ALARMS — An alarm has occurred in the IDLE state, and the equipment is waiting for all alarms to be cleared.

7.3.12 INIT — Equipment initialization is occurring. Equipment remains in this state unless initialization is successful.

7.3.13 INSPECT (WORKING Sub-State) — The current alignment area of the substrate is being inspected for anomalies.

7.3.14 INSPECT COMPLETE (INSPECT Sub-State) — The equipment has completed inspection of the current alignment area. Based on the recipe, the equipment determines if (a) additional alignment areas are required to do more inspections, (b) the recipe on this material is complete, or (c) a review of the current inspection area is required.

7.3.15 INSPECT REGION — A region on a substrate is being inspected for anomalies.

7.3.16 INSPECT SETUP (INSPECT Sub-State) — The equipment is in this sub-state immediately upon entering the INSPECT state. The equipment is determining if all conditions are satisfied to begin inspecting the regions in the current alignment as defined by the recipe and any commands.

7.3.17 LOAD (EXECUTING Sub-State) — The equipment is determining if the process program has completed. When additional processing is required, then the next unprocessed substrate shall be transferred to the equipment processing location, such as the stage. If equipment determines that there are more process programs in the “CARRIERBLD” ISEM job, the equipment makes the transition to setup for the next process program specified. Otherwise, the equipment transitions to IDLE through STOPPING.

7.3.18 PAUSE (PROCESS ACTIVE Sub-State) — PROCESS shall be suspended at the next opportunity. Actions to put the equipment in a “safe state” shall be performed. The equipment is awaiting a command (STOP or ABORT).

7.3.19 PAUSED (PROCESS PAUSE Sub-State) — PROCESS has been suspended, and the equipment is waiting for a command (RESUME, PP-UPDATE, STOP, or ABORT).

7.3.20 PAUSING (PROCESS PAUSE Sub-State) — PROCESS shall be suspended at the next opportunity, and the equipment shall be put in a “safe state.” ABORT, STOP, and RESUME commands are valid in this state.

7.3.21 PROCESS (PROCESSING ACTIVE Sub-State) — This state is the parent of those sub-states which refer to the active preparation and execution of a process program.

7.3.22 PROCESSING ACTIVE — This state is the parent of all sub-states where the context of a process program execution exists.

7.3.23 PROCESS PAUSE (PAUSE Sub-State) — The equipment is free of alarm conditions in the PAUSE state. The equipment is awaiting for a command (ABORT, RESUME, or STOP).

7.3.24 READY (PROCESS Sub-State) — The equipment is ready to begin processing and is awaiting a START command from the operator or host.

7.3.25 REVIEW (WORKING Sub-State) — Classification is being done on anomalies previously found in the current alignment area of the substrate.

7.3.26 REVIEW COMPLETE (REVIEW Sub-State) — The equipment has completed review of the current alignment area. Based on the recipe, the equipment determines if (a) additional alignment areas are required to do more classifications, (b) the recipe on this material is complete, or (c) an inspection is required.

7.3.27 REVIEW SETUP (REVIEW Sub-State) — The equipment is in this sub-state immediately upon entering the REVIEW state. The equipment is determining if all conditions are satisfied to begin

reviewing the anomalies in the current alignment as defined by the recipe and any commands.

7.3.28 SETTING UP (PROCESS Sub-State) — The equipment is being setup so that external conditions are satisfied to start processing the material. This includes the receipt of any process programs and material to be processed and their validation. Any of these conditions may be satisfied on entry to SETTING UP. For example, the selected process program may have already been loaded (e.g., if it was the default process program), or the specified material may have already been placed on the equipment material port. Additional information may come from the host during the execution of this state.

7.3.29 STOPPING (PROCESSING ACTIVE Sub-State) — The equipment has completed all process programs within a “CARRIERBLD” ISEM job or has been instructed to stop processing and shall do so gracefully at the next opportunity. All cleanup necessary is being completed within this state with regard to material, data, control system, etc. Data is normally preserved. Any alarm or error condition in this state causes a transition to ABORTING.

7.3.30 UNLOAD (EXECUTING Sub-State) — The substrate is being removed from the processing location.

7.3.31 WORKING (EXECUTING Sub-State) — The equipment is processing a specific material.

7.4 Processing State Transition Table

Table 1 Processing State Transition Table

Transition #	Previous State	Trigger	New State	Actions	Comments
1	INIT	All equipment initialization is complete with no alarms or error conditions.	IDLE	Equipment awaits for a PP-SELECT, PP-ASSIGN, or MAP-CARRIER command.	All equipment requires INIT to be free of errors and alarms when exited. IDLE state entry requires that no process program is selected.
2	IDLE	A ISEM job has been or is queued (PP-ASSIGN) or selected (PP-SELECT).	SETTING UP	The setup procedure is equipment-dependent.	Commit has been made to setup. Material may have been placed on the equipment before SETUP is entered. When the job becomes active, the process program gets selected.
3	SETTING UP	All setup activity has completed, and the equipment is ready to receive a START command.	READY	The equipment is waiting for a START command. START may be initiated by an operator or may be included in the process program.	The selected process program is available for execution. When running multiple process programs within a ISEM job, the equipment makes the next process program available for execution.
4	READY	The equipment receives a START command from the user or from within the body of the process program selected for execution.	LOAD	The equipment determines if processing is completed. If not, it transfers the next substrate to the processing location.	Equipment transitions to STOPPING when all process programs in the selected ISEM job are executed. If a new process program within the ISEM job needs to be executed, the equipment transitions to SETTING UP.
5	LOAD	The processing job is complete, and there are no more substrates to load or process programs to run.	STOPPING	Equipment initiates a cleanup to remove the completed ISEM job and process program.	Normal completion of the run.
6	PROCESS	The equipment has received a STOP command.	STOPPING	The equipment unloads the material and brings the equipment to a “safe state.”	Data is typically preserved and is valid.

<i>Transition #</i>	<i>Previous State</i>	<i>Trigger</i>	<i>New State</i>	<i>Actions</i>	<i>Comments</i>
7	PROCESS	The equipment has received an ABORT command.	ABORTING	The equipment is put in a “safe state” if necessary.	Data may be invalid or not available.
8	PROCESS	An alarm occurs.	ALARM PAUSED	PROCESS activity is suspended, and the equipment is waiting for all alarms to be cleared.	ALARM PAUSED is a PAUSE Sub-State.
9	PROCESS	The equipment has received a PAUSE command.	PAUSING	PROCESS shall be suspended at the next opportunity. Actions to put the equipment in a “safe state” will be performed.	This transition is required if the user wants to make changes to the current process program being executed.
10	PROCESS PAUSE	The equipment has received a RESUME command.	Previous PROCESS State	Proceed with the suspended process Sub-State.	If a RESUME command is received in the CHECKING state, then the PP-UPDATE command is canceled. Some equipment may only allow RESUME remote command from the PAUSE state.
11	CHECKING	The equipment has validated “all” requested updates to the current process program being executed; changes are done before entering into the next state.	Previous PROCESS State	Verification is appropriate in this state to check the changes made to the process program updated.	If an alarm occurs in the CHECKING state, then the PP-UPDATE command is canceled.
12	STOPPING	The equipment cleanup is complete, and the equipment is free of alarms.	IDLE	Equipment waits for a command/or determines if there is a ISEM job queued.	IDLE state entry requires that no process program is selected.
13	PAUSE	The equipment has received a STOP command.	STOPPING	The equipment proceeds with cleanup.	Normally, data is preserved and is valid.
14	PAUSE	The equipment has received an ABORT command.	ABORTING	Any unsafe condition is resolved if possible.	Data may be invalid or not available.
15	STOPPING	The equipment has received an ABORT command or an alarm was received while STOPPING.	ABORTING	Any unsafe condition is resolved if possible.	Data may be invalid or not available.
16	ABORTING	Unsafe conditions have been resolved where possible.	ABORTED	The equipment is waiting for alarm and ABORT conditions to be cleared.	The only state change allowed is to IDLE.
17	ABORTED	All alarms and abort conditions have been cleared.	IDLE	Equipment is waiting for a command (PP-SELECT, PP-ASSIGN, or MAP-CARRIER).	If needed, the CLEANUP command clears the abort conditions. IDLE state entry requires that no ISEM job or process program be selected.
18	IDLE	An alarm is set.	IDLE with ALARMS	The equipment waits for all alarms to be cleared.	None
19	IDLE with ALARMS	All alarms have been cleared.	IDLE	None	IDLE is free of alarms.

<i>Transition #</i>	<i>Previous State</i>	<i>Trigger</i>	<i>New State</i>	<i>Actions</i>	<i>Comments</i>
20	PAUSING	The equipment has achieved a “safe state.”	PAUSED	The equipment is waiting for a command (PP-UPDATE, RESUME, STOP, or ABORT).	None
21	PROCESS PAUSED	An alarm is set.	ALARM PAUSED	The equipment waits for all alarms to be cleared or for a STOP or ABORT command.	None
22	ALARM PAUSED	All alarms are cleared.	PAUSED	The equipment is waiting for a command (RESUME, PP-UPDATE, STOP, or ABORT).	None
23	LOAD	Material transfer to processing location is complete.	WORKING	The substrate is being processed.	None
24	WORKING	The processing of the specific material being processed successfully completed.	UNLOAD	This material is transferred from the processing location.	Normal completion of the substrate.
25	UNLOAD	The material unload is complete.	LOAD	The equipment returns to LOAD and determines if processing is complete, if not, transfers the next substrate to the processing location.	None
26	WORKING	The processing of the specific material being processed abnormally ended.	UNLOAD	This material is transferred from the processing location.	Error exit from WORKING. Data may be invalid.
27	ALIGN	The material alignment is complete, and inspection is required.	INSPECT	The equipment determines if another region needs to be inspected.	This transition is to the INSPECT SETUP Sub-State of INSPECT.
28	INSPECT SETUP	All inspect setup activity is complete, and the inspection is not complete.	INSPECT REGION	The equipment inspects the current alignment region.	None
29	INSPECT REGION	The region inspection has ended.	INSPECT SETUP	The equipment determines if another region needs to be inspected.	None
30	INSPECT SETUP	Inspection of this alignment group is complete.	INSPECT COMPLETE	The equipment determines if (a) additional alignment areas are required to do more inspections, (b) the recipe on this material is complete, or (c) a review of the current alignment area is required.	The next transition is conditional.
31	INSPECT COMPLETE	The inspection of this alignment area ended, and additional inspections may be required.	ALIGN	An inspection group is complete, and additional inspections may be required.	None
32	ALIGN	The material alignment is complete, and review is required.	REVIEW	The material is reviewed.	This transition is to the REVIEW SETUP Sub-State of REVIEW.
33	REVIEW SETUP	Anomaly data is needed to perform the review.	ACQUIRE ANOMALY DATA	Anomaly data is being acquired.	Anomaly data may come from the host or equipment.

<i>Transition #</i>	<i>Previous State</i>	<i>Trigger</i>	<i>New State</i>	<i>Actions</i>	<i>Comments</i>
34	ACQUIRE ANOMALY DATA	Anomaly data has been acquired for the review, or no more anomaly data is available.	REVIEW SETUP	The equipment determines what to do.	
35	REVIEW SETUP	The equipment has anomaly data, and the review is not complete.	ACQUIRE ANOMALY IMAGE	The equipment acquires the anomaly image at the specified site.	The image may be a stored image or from an imaging device.
36	ACQUIRE ANOMALY IMAGE	The equipment has acquired the anomaly image for the specified site.	CLASSIFY	The operator or equipment classifies the anomaly.	None
37	CLASSIFY	All anomalies have been classified for the site.	REVIEW SETUP	The equipment determines what to do.	None
38	REVIEW SETUP	The review of the alignment area is complete.	REVIEW COMPLETE	Transition to next state is to be determined.	None
39	REVIEW COMPLETE	The review of this alignment area ended, and additional review is required.	ALIGN	A review group is complete, and additional alignment is required.	None
40	INSPECT	The alignment area inspection is complete, and review is required.	REVIEW	The material is reviewed.	This transition is to the REVIEW SETUP Sub-State of REVIEW.
41	REVIEW COMPLETE	The review is complete, and inspection is required.	INSPECT	The material is inspected.	This transition is to the INSPECT SETUP Sub-State of INSPECT.
42	CHECKING	Validation of requested process program changes failed.	PAUSED	The equipment is waiting for a new command.	No process program parameters have been changed.
43	PAUSED	The equipment receives a PP-UPDATE command.	CHECKING	The equipment begins validating requested changes to the process program.	No process program parameters are updated or changed before "all" requested changes are validated.
44	LOAD	Previous process program has completed, and there are additional process programs assigned to the "CARRIERBLD". See Section 13.	SETTING UP	The equipment performs setup according to specifications of the next process program.	PROCESS-BLD-GROUP may include an AUTOSTART command within its body. Otherwise, the equipment waits for a START command.

8 Collection Event List

This section identifies data collection events and defines (Stream 6) reporting levels for variable items. The host can use the report definition scenario defined in SEMI E30 to define reports at ISEM-defined levels. The intent of this section is to ensure data is available at specific events and to optimize data reporting to the SECS-II host by allowing data to be grouped at reporting levels.

8.1 Requirements

8.1.1 This standard requires all collection events listed in the SEMI E30 standard. This standard requires the ISEM events in Table 2 for data collection (RunDataComplete, SubstrateDataComplete, GroupDataComplete, RegionDataComplete, and AnomalyDataComplete). These events are separate from the processing state transitions. These collection events shall occur before or on the processing state transition specified in Table 1. This was done to



ensure that the data and the material remain synchronous. As a result, in some cases material processing may be delayed due to extended data processing time.

8.1.2 The most fundamental level of data defined for ISEM equipment is the anomaly level for review equipment and region level for inspection equipment. For example, review equipment has data available for individual anomalies at the AnomalyDataComplete event. Anomaly data may be grouped for level reporting. For example, data for anomalies found within a region on a substrate would be available at the RegionDataComplete event. This data would be available as a list variable item for Region Anomalies. All anomalies found on a substrate would be available at the SubstrateDataComplete event. This could either be 1) a list of list variable items for Region Anomaly, or 2) a single list variable item of all Substrate Anomalies. In this way, data can be reported with less high-level event reports, rather than as more low-level event reports.

8.1.3 Data produced by ISEM equipment is customarily grouped for reporting by processing, material, and equipment constraints which are called reporting levels (i.e., run, substrate, group, site, and anomaly data). Level data is grouped by these constraints for a reporting level. Data shall be grouped within a reporting level according to other constraints by degree of processing (e.g., raw sensor, basic, or analyzed data), or statistically (e.g., summary, correlation, or comparison).

Table 2 Collection Events for ISEM Data Reporting

<i>Reporting Level</i>	<i>Data Collection Event</i>	<i>Inspection Equipment</i>	<i>Review Equipment</i>
Run	RunDataComplete	STOPPING → IDLE and LOAD → SETTING UP	STOPPING → IDLE and LOAD → SETTING UP
ProcessGroup	ProcessBuildGroup-Complete	LOAD → SETTING UP	LOAD → SETTING UP
Substrate	SubstrateDataComplete	UNLOAD → LOAD	UNLOAD → LOAD
Group	GroupDataComplete	INSPECT COMPLETE → ALIGN or WORKING → UNLOAD	REVIEW COMPLETE → ALIGN or WORKING → UNLOAD
Region	RegionDataComplete	INSPECT REGION → INSPECT SETUP	<i>Not Defined</i>
Anomaly	AnomalyDataComplete	<i>Not Defined</i>	CLASSIFY → REVIEW SETUP

NOTE 1: The data collection event shall occur before or on the processing state transition.

9 Variable Items

The purpose of this section is to define the list of variable item requirements for inspection and review equipment. Values of these variables shall be available to the host via collection event reports and host status queries. These variable items are separated into three categories: (a) common to all ISEM equipment; (b) specific to inspection equipment; (c) and specific to review equipment.

If equipment supports the data item functionality defined by ISEM, then it is required and shall be implemented as specified in Table 4 “Variable Item Dictionary”. That is, a variable item is only required if the equipment supports the functionality necessary to support it. For example, if an inspection instrument only has the hardware to count detected anomalies and lacks the hardware to determine their size, then the ISEM requires it to report anomaly count (e.g., as SubstrateAnomalyCount), but reporting anomaly size (as AnomalySize) is not required by the ISEM.

9.1 Requirements

- All variable items and data item restrictions defined in SEMI E30 are required on ISEM equipment.
- All variable items in the ISEM Variable Item Dictionary for specific equipment classifications are required for ISEM equipment. The data item restrictions are also required.

9.1.1 Variable items are categorized in the Variable Item Dictionary as follows:

- *Common Variables (CV)* — variables common to all ISEM equipment.
- *Inspection-Specific Variables (ISV)* — variables required only for inspection equipment.
- *Review-Specific Variables (RSV)* — variables required only for review equipment.



9.2 *Variable Items and Reporting Levels* — Table 3 defines reporting levels and associated Data Collection Events for which Variable Items are valid for.

Table 3 Variable Items and Reporting Levels

<i>Level</i>	<i>Reporting Level</i>	<i>Data Collection Event</i>
R	Run	RunDataComplete
P	ProcessGroup	ProcessGroupComplete
S	Substrate	SubstrateDataComplete
G	Group	GroupDataComplete
X	Region	RegionDataComplete
A	Anomaly	AnomalyDataComplete
ALL	Run, Substrate, Group, Region, and Anomaly	All of the above.

9.2.1 Variable items are documented in the ISEM Variable Item Dictionary using the following format:

<i>Variable Name</i>	<i>Type</i>	<i>Description</i>	<i>Level</i>	<i>Class</i>	<i>Format</i>	<i>Comments</i>
----------------------	-------------	--------------------	--------------	--------------	---------------	-----------------

Where:

Variable Name: A unique name for the variable item (this name is for reference only).

Type: Defined as Common (CV), Inspection (ISV), Review (RSV), or Inspection/Review specific variables (IRSV).

Description: If class is DVVAL, then the description shall contain a statement of when data is valid in terms of ISEM events.

Level: The report level at which this variable is used <R| S| G| X| A| ALL> as defined in Table 3. It also indicates when the variable item is valid.

Class: The data type of the item.

Format: <SECS Message Language (SML) mnemonic> acceptable formats are SEMI E5 lists, ASCII, floating point, unsigned integer, or signed integer. A description of “ANY” indicates that only the above formats are acceptable and is left to the tool vendor to decide.

Comments: Any additional information pertinent to the variable name.

9.3 Variable Item Types

9.3.1 *Equipment Constants (ECV)* — The value can be changed by the host using S2,F15. The operator may have the ability to change some or all of the values. The value of an equipment constant may be queried at any time by the host using the S2,F13/14 transaction or Stream 6 reports.

9.3.2 *Status Variables (SV)* — The values are valid at all times. A SV may not be changed by the host or operator but may be changed by the equipment. A host or operator command may change an equipment status, thus changing an SV. The value of status variables may be queried by the host at any time using the S1,F3/4 or Stream 6 reports.

9.3.3 *Data Variables (DVVAL)* — These are variables which are valid upon the occurrence of a specific collection event and which may or may not be valid at other times, depending upon the equipment. An attempt to read a variable item when it is invalid will not result in an error, but the data reported may not have relevant meaning.

9.4 Variable Item Dictionary

9.4.1 *Data Validity* — The “Level” column in Table 4 defines when the variable item is valid. The entry in this column corresponds to a reporting level defined in Section 9.2 “Variable Items and Reporting Levels”. For example, “RunAnomalyCount” is valid at the “RunDataComplete” event, and “AnomalySize” is valid at all reporting level data collection events.

Table 4 Variable Item Dictionary

<i>Variable Name</i>	<i>Type</i>	<i>Description</i>	<i>Level</i>	<i>Class</i>	<i>Format</i>	<i>Comment</i>
ActiveLocation	CV	The current carrier location that has substrates in the executing state of the processing state model.	ALL	SV	U2	Valid in all data collection events as defined in Table 3.
AlignList	CV	A list of alignment sites information being used by the current active process program.	ALL	DVVAL	L,n 1. <AlignName> : n.	The order in which the alignment name appears in the list is important and is equipment-dependent.
AlignName	CV	Alignment name	ALL	DVVAL	A[1..16]	An item in the AlignList variable.
AnomalyArea	CV	The area within the bounds of an anomaly (in units of micron ²).	ALL	DVVAL	F4	
Anomaly-Attributes	CV	Miscellaneous anomaly information that is equipment-dependent and defined by the equipment supplier.	ALL	DVVAL	L,n 1. <attribute ₁ > : n. <attribute _n >	Mainly used as part of other anomaly-related data (list) (i.e., AnomalySize).
Anomaly-Comment	RSV	Operator-generated comment associated with the anomaly.	A	DVVAL	A[1..80]	
AnomalyData2D	CV	Coordinate data for an anomaly.	A	DVVAL	L,5 1. <AnomalyID> 2. <CoordSys> 3. <Coord2D> 4. <Anomaly-Attributes>	
AnomalyID	CV	A unique anomaly identifier.	A	DVVAL	A[1..16]	
AnomalySize	CV	The X,Y extent of the anomaly in microns. The dimensions of the smallest rectangle that contains the anomaly whose sides are parallel to the X and Y axis.	A	DVVAL	L,2 1. <XExtent> 2. <YExtent>	XExtent and YExtent are of Format F4.
AnomalyTable-Name	CV	Name identifier of anomaly table.	ALL	SV	A[1..80]	
AnomalyTable-Type	CV	Type of anomaly table. (See Section 11.)	ALL	SV	A[1..20]	"TABLE-AREA-DEF", "TABLE-ALIGN-DEF", "TABLE-ANOMALY-DEF", "TABLE-M21-ANOMALY-DEF"
BatchID	CV	The batch identification of the current material inspected/reviewed.	ALL	DVVAL	A[1..16]	
CarrierBuild	CV	ID of the CARRIERBLD ISEM job that the inspection/review data is associated with.	ALL	SV	A[1..80]	See Section 14.
CarrierID	CV	Physical identification of the current material inspected/reviewed.	ALL	DVVAL	A[1..16]	
CarrierNumber	CV	Used to identify carriers in	ALL	DVVAL	A[1..16]	

<i>Variable Name</i>	<i>Type</i>	<i>Description</i>	<i>Level</i>	<i>Class</i>	<i>Format</i>	<i>Comment</i>
		multi-lot runs (batch).				
Classification	RSV	Classification code of an anomaly.	A	DVVAL	A[1..80]	
Coord2D	CV	The two-dimensional coordinate of an anomaly.	ALL	DVVAL	L,2 1. <CoordX> 2. <CoordY>	
CoordSys	CV	The identification for a coordinate system definition. SEMI M20, M20P, or SEMI M21.	ALL	DVVAL	A[1..16]	“M20” “M21” “M20P”
CoordX	CV	The coordinate in the X direction of a site (anomaly, alignment site, or the lower left-hand of an area or element).	ALL	DVVAL	F4	
CoordY	CV	The coordinate in the Y direction of a site (anomaly, alignment site, or the lower left-hand of an area or element).	ALL	DVVAL	F4	
DefaultPriority	CV	The default priority given a material location if none is assigned.	ALL	EC	U2	
DeltaX	CV	The X-axis translation of M20P coordinate system relative to the SEMI M20 coordinate system.	ALL	DVVAL	F4	
DeltaY	CV	The Y-axis translation of M20P coordinate system relative to the SEMI M20 coordinate system.	ALL	DVVAL	F4	
ElementID	CV	The SEMI M21 address for a specific rectangular element on a substrate.	ALL	DVVAL	I4[2]	May refer to a field or die.
ElementList	CV	A list of SEMI M21 elements where processing can be attempted.	ALL	DVVAL	L,n 1. <ElementID> : n.	
Fiducial	CV	The physical feature used to associate a fiducial line used for orientation (i.e., flat or notch on a substrate).	ALL	DVVAL	A[1..16]	“FLAT” “NOTCH”
GroupAnomaly-Count	CV	Anomaly count for the current or last group (i.e., field or die inspected).	G	DVVAL	U2	
GroupArea	CV	Square Area (microns ²) of the last group inspected.	G	DVVAL	F4	
GroupComment	RSV	Operator-generated comment associated with the group.	G	DVVAL	A[1..80]	
GroupID	CV	Inspection/review group identification for the current inspection/review.	ALL	DVVAL	U2	
InspectionPPID	CV	Process program used on the inspection/review tool	ALL	DVVAL	A[1..80]	

<i>Variable Name</i>	<i>Type</i>	<i>Description</i>	<i>Level</i>	<i>Class</i>	<i>Format</i>	<i>Comment</i>
		for the current inspection/review.				
InspectionRunID	CV	Run identification in the current inspection/review.	ALL	DVVAL	A[1..16]	
LotID	CV	Lot identification of the current material inspected/reviewed.	ALL	DVVAL	A[1..16]	
M20Data	CV	The silicon substrate size, fiducial type, and orientation to use.	ALL	DVVAL	L,3 1. <SubstrateSize> 2. <Fiducial> 3. <Orientation>	
M21Data	CV	The data necessary to establish an ISEM SEMI M21 layout on a substrate.	ALL	DVVAL	L,2 1. L,3 1. <M21XSize> 2. <M21YSize> 3. <Tile> 2. L,n 1. L,3 1. <ElementID> 2. <CoordX> 3. <CoordY> : n.	
M21XSize	CV	The value of the SEMI M21 coordinate system in the X-direction.	ALL	DVVAL	F4	
M21YSize	CV	The value of the SEMI M21 coordinate system in the Y-direction.	ALL	DVVAL	F4	
Offset	CV	The distance of the actual or found location of a site relative to its defined or expected location.	ALL	DVVAL	L[2]	Refers to SiteDeltaX, SiteDeltaY which are of Format F4.
OperatorAction	CV	The action taken by the operator on the equipment's operator I/O.	ALL	DVVAL	A[1..80]	
OperatorComment	CV	Operator-generated comment, not associated with any reporting level.	ALL	DVVAL	A[1..80]	(See also Run-Comment, Substrate-Comment, Area-Comment, RegionComment, Site-Comment, and AnomalyComment.)
OperatorID	CV	Identification of the operator of the inspection/review equipment.	ALL	DVVAL	A[1..16]	This information may be added by the host in the ISEM Tables. (See Section 11.)
Orientation	CV	How the wafer is loaded on the equipment.	ALL	EC	F4	"0" degrees indicates that the wafer has the primary fiducial towards the operator.
ProcessBuild-GroupID	CV	Name of the current or last process program executed.	ALL	SV	A[1..80]	See Section 14.
ProcessEquipmentID	CV	Identification of the process equipment used with the	ALL	DVVAL	A[1..16]	This information may be added by the host in

<i>Variable Name</i>	<i>Type</i>	<i>Description</i>	<i>Level</i>	<i>Class</i>	<i>Format</i>	<i>Comment</i>
		current material immediately prior to the inspection/review.				the ISEM Tables. (See Section 11.)
ProcessEquipmentLocation	CV	Location (code) of the process equipment used with the current material immediately prior to the inspection/review.	ALL	DVVAL	A[1..16]	This information may be added by the host in the ISEM Tables. (See Section 11.)
ProcessEquipmentPID	CV	Identification of the process program used with the process equipment used on the current material immediately prior to the inspection/review.	ALL	DVVAL	A[1..80]	This information may be added by the host in the ISEM Tables. (See Section 11.)
ProcessLevel	CV	Identification of the processing level of the current material.	ALL	DVVAL	A[1..16]	This information may be added by the host in the ISEM Tables. (See Section 11.)
ProcessRunID	CV	Run identification for the process prior to current inspection/review.	ALL	DVVAL	A[1..16]	This information may be added by the host in the ISEM Tables. (See Section 11.)
ProductID	CV	The product identification of the current material inspected/reviewed.	ALL	DVVAL	A[1..16]	This information may be added by the host in the ISEM Tables. (See Section 11.)
RegionComment	CV	Operator-generated comment associated with the region.	X	DVVAL	A[1..80]	
RunAnomaly-Count	ISV	Total number of all anomalies found on all substrates in the last run.	R	DVVAL	U2	
RunComment	CV	Operator-generated comment associated with the run.	ALL	DVVAL	A[1..80]	
RunInspected-AreasCount	ISV	The total number of inspected/reviewed areas on all substrates in the last run.	RS	DVVAL	U2	
RunInspection-PPCount	ISV	The total number of process programs used for the current or last run.	RS	DVVAL	U2	
RunSubstrate-Count	CV	The total number of substrates completed in the current inspection run, which remains valid until the next START command.	ALL	DVVAL	U2	
ScaleFactor	CV	A correction factor applied to the translation of one coordinate system to another.	ALL	DVVAL	F4	In most cases, a scaling of 1 (one) is expected.
SiteID	CV	Inspection/Review group identification for the current site inspection/review.	ALL	DVVAL	U2	
SlotID	CV	Carrier slot number from which the current substrate was taken.	ALL	DVVAL	U2	

<i>Variable Name</i>	<i>Type</i>	<i>Description</i>	<i>Level</i>	<i>Class</i>	<i>Format</i>	<i>Comment</i>
SlotList	CV	The list of carrier slots with substrates to be processed.	ALL	DVVAL	L,n 1. <SlotID> : n.	
SubstrateAnomalyCount	ISV	The total number of anomalies for the current substrate.	S	DVVAL	U2	For the most recent inspection.
Substrate-Comment	CV	Operator-generated comment associated with the substrate.	ALL	DVVAL	A[1..80]	
SubstrateID	CV	Substrate identification for the current inspection/review.	ALL	DVVAL	A[1..16]	
Substrate-InspectedAreas-Count	ISV	The total number of inspected areas on the current substrate.	S	DVVAL	U2	
SubstrateRegion-Count	CV	Total area count for the current or last substrate inspected.	ALL	DVVAL	U2	
SubstrateSize	CV	The nominal diameter (in mm) of the current or last substrate inspected/reviewed.	ALL	DVVAL	U2	
SubstrateTotal-AreaInspected	ISV	Total square area inspected/reviewed (micron ²) of the current substrate.	S	DVVAL	F4	
Theta	CV	The rotational difference in radians between a primary and secondary coordinate system.	ALL	DVVAL	F4	
Tile	CV	The layout of the pattern in the substrate.	ALL	DVVAL	A[1..16]	“NTILE” non-tiled, “CTILE” column-tiled, and “RTILE” row-tiled.
XLateData	CV	Variable for the equipment to report offset of the found or actual pattern-based coordinate system relative to the substrate-based coordinate system on the substrate being tested.	ALL	DVVAL	L,4 1. <DeltaX> 2. <DeltaY> 3. <Theta> 4. <ScaleFactor>	

10 Alarm List

Since each model of equipment differs in configuration, it is not practical to provide an exhaustive list of all possible alarms. Instead, the ISEM is requiring the two tables provided as described in SEMI E30 (document section). Alarm List Table, which is intended to provide for equipment configuration-specific alarms and Alarm ID, Alarm Set/Cleared Event Table.

10.1 Alarm List Table

10.1.1 The alarm list table contains examples of alarms that pertain to various configuration aspects of equipment. These examples are intended to illustrate

that alarms pertain to situations in which there exists a potential for exceeding physical safety limits associated with people, equipment, and material being processed as per the SEMI E30 definition of an alarm. (See SEMI E30 for further reference.)

10.2 Alarm ID, Alarm Set/Cleared Event Table

10.2.1 The Alarm ID, Alarm Set/Cleared Event Table documents the association of each ALID to a set and cleared event as required by SEMI E30. (See SEMI E30 for further reference.)

11 ISEM Tables

A fundamental requirement of ISEM equipment is to transfer anomaly and review data between itself and the host. ISEM equipment may also be required to transfer “Area” and “Alignment Site” data needed for run setup. Anomaly and review data sets (as well as area and alignment site data sets) are commonly handled as lists and tables. ISEM equipment shall use tables when transferring this kind of data between itself and the host. List shall be used to refer to sub-sets of this table data.

11.1 ISEM Table Data

11.1.1 ISEM Tables are used to specify area, alignment site, and anomaly coordinate lists for ISEM equipment. ISEM Tables are transferred between the host and the equipment using SECS-II Stream 13 messages (Unformatted Data Set Transfers). For example, an ISEM Table may be used to transfer anomaly data (e.g., “M21” coordinates) generated by inspection equipment to the host, which in turn may then be transferred to review equipment from the host. ISEM Tables also include attributes items that are associated with the table, not with the table data. ISEM Table attributes are

used to include information associated with table data, like the number of columns (NumCols), number of rows (NumRows), and table size (DataLength). (See SEMI E58 for additional information.) Product or process-related information may also be included on the attribute section of the ISEM Tables (e.g., LotID, ProductID, OperatorID, ProcessEquipmentID) (see Table 4). The ISEM does not specify additional table attribute variable items that may be associated with the table.

11.2 TABLE-DEFs

11.2.1 ISEM Tables are documented using TABLE-DEF structures (Figure 5). Each TABLE-DEF structure has a unique name (TableID) and type (TableType). Each column in the TABLE-DEF has a name (e.g., “AREANAME”, “ALIGNNAME”, OR “ANOMALYID”); row names are specific instances or values that correspond to the column headers in TABLE-DEF. A specific TABLE-DEF row is designated by referring to the TABLE-DEF name and the specific row name. ISEM Tables are transferred using standard (SECS-II message) Stream 13 messages (Figure 6), and each ISEM-defined TABLE-DEF item maps into a Stream 13 message item. Align and area data tables are host-defined, and anomaly tables are equipment-defined.

TableType = <TableDef>

TableID = <TableName>

Row Name	Column ₁	Column ₂	Column ₃	...	Column _n
Row ₁ Name					
...					
Row _m Name					

Figure 5
TABLE-DEF Structure



An example of usage of S13,F13 to transfer data sets is shown below:

```

L,7
1. <DATAID>
2. <OBJSPEC>
3. "TableDef"                                <TBLTYP>
4. "TableName"                              <TBLID>
5. L,n                                       # of table attributes
    1. L,2
        1. "NumRows"                        <ATTRID1>
        2. <m>                             <ATTRDATA1>
    2. L,2
        1. "NumCols"                        <ATTRID2>
        2. <n>                             <ATTRDATA2>
    3. L,2
        1. "DataLength"                    <ATTRID3>
        2. <table length>                  <ATTRDATA3>
    4. L,2
        1. "LotID"                         <ATTRID4>
        2. "ABC123"                        <ATTRDATA4>
    n. L,2
        1. "ProductID"                     <ATTRIDn>
        2. "CPUTYPE"                       <ATTRDATAn>
6. L,n                                       # of columns
    1. "AREANAME"                           <COLHDR1>(1st column description)
    .
    n. "ATTRIBUTE5"                         <COLHDRn>(nth column description)
7. L,m                                       # of rows
    1. L,n                                   # of columns
        1. <Item 1,1>                       table item in row 1, column 1
        .
        n. <Item 1,n>                       table item in row 1, column n
    .
    m. L,n                                   # of columns
        1. <Item m,1>                       table item in row m, column 1
        .
        n. <Item m,n>                       table item in row m, column n
  
```

Figure 6
S13,F13 with ISEM TABLE-DEF Data

11.3 Required ISEM Tables — ISEM equipment shall support all three table types: area, align, and anomaly. SEMI M21 anomaly table type may be supported, but it is optional. The ISEM equipment shall be able to store simultaneously at least 3 (three) defined tables of each type supported to guarantee the validity of any table while that table is being transferred (a table transfer transaction is in process).

ISEM align and area tables are stored by the equipment during the current inspection or review run (i.e., until a new remote command PP-SELECT or PP-ASSIGN is sent, or until they are modified with PP-UPDATE).

Table 5 ISEM Table Types (TABLE-DEFs)

Table Type	Req/Opt	Description
"TABLE-AREA-DEF"	R	<i>Area Definition</i> — A set of areas and their attributes, typically the list of areas to be inspected.
"TABLE-ALIGN-DEF"	R	<i>Alignment Site Definition</i> — A set of alignment sites and their attributes.
"TABLE-ANOMALY-DEF"	R	<i>Anomaly Coordinate Data Definition</i> — A set of anomalies and their attributes, with coordinates given in the SEMI M20 or M20P coordinate system.
"TABLE-M21-ANOMALY-DEF"	O	<i>Anomaly SEMI M21 Coordinate Data Definition</i> — A set of anomalies and their attributes, with coordinates given in the SEMI M21 coordinate system.



As indicated in Table 5, the SEMI M21 anomaly definition table “TABLE-M21-ANOMALY-DEF” is optional. The others are required.

ISEM requires that the following columns be included in the TABLE-DEFs. Table 6 defines the column headers and the allowed formats. Anomaly attributes and attribute headings are defined by the supplier, based on equipment capability.

11.3.1 “TABLE-AREA-DEF”

TableType: “TableAreaDef”

TableID: <AreaTableName>

<i>Area Name</i>	<i>Coordx</i>	<i>Coordy</i>	<i>Coordsys</i>	<i>Xtentx</i>	<i>Xtenty</i>	<i>Attribute (1)</i>	<i>...</i>	<i>Attribute</i>

11.3.2 “TABLE-ALIGN-DEF”

TableType: “TableAlignDef”

TableID: <AlignTableName>

<i>AlignName</i>	<i>Coordx</i>	<i>Coordy</i>	<i>Coordsys</i>	<i>Attribute (1)</i>	<i>...</i>	<i>Attribute(N)</i>

11.3.3 “TABLE-ANOMALY-DEF”

TableType: “TableAnomalyDef”

TableID: <AnomalyTableName>

<i>Anomalyid</i>	<i>Coordx</i>	<i>Coordy</i>	<i>Coordsys</i>	<i>AnomalyAttribute</i>	<i>...</i>	<i>AnomalyAttribute</i>

11.3.4 “TABLE-M21-ANOMALY-DEF”

TableType = “TableM21AnomalyDef”

TableID = <M21AnomalyTableName>

<i>Anomalyid</i>	<i>Coordx</i>	<i>Coordy</i>	<i>Elementid</i>	<i>AnomalyAttribute</i>	<i>...</i>	<i>AnomalyAttribute</i>

Defect data shall be transferred between the host and inspection/review/analysis equipment using SEMI E5, S13,F13. Columns in the table are defined by Table 11.3.5 below.

11.3.5 TABLE-STANDARD-DEFECT-DATA-SET-DEF

NOTE 1: Data for each substrate should be reported in column order shown below. Inspection tools should support relevant columns (see NOTE 6). Review and analysis tools should support all columns. Multiple data entries (list format data items) are allowed for a given attribute on a single defect.

NOTE 2: The inspection equipment must add a table attribute called “Substrate Header” (ATTRID). It must be a list that includes the following items in the given order: LotID (A[1..16]), SubstrateID (A[1..16]), ProcessEquipmentID (A[1..16]), substrate center¹ (L[2], CoordX (F4), CoordY (F4)) and centering method² (A[1..16]). Refer to Table 4, Variable Item Dictionary for descriptions of these items.

NOTE 3: See SEMI M21 for (0,0) die location methodology.

1 Vector from the origin of the substrate coordinate system to nominal substrate center location.

2 CoordSys (e.g., “SEMI M20”, “SEMI M21”, “M21P”, etc).



NOTE 4: Die origin is located at lower left-hand corner (LLHC).

NOTE 5: Data format must comply with the ISEM standard.

NOTE 6: In order to signify tool context, anomaly attributes are labeled as follows:

Column name starts with “insp*_” for inspection data, “rev*_” for defect review data, “and “anal*_” for analysis data, where “*” is a numeric string that ensures each set of columns added is uniquely named (e.g., “rev1_” and “rev2_”).

NOTE 7: Inspection, review and analysis tools must add a table attribute called “insp*_ Header,” “rev*_Header” and “anal*_Header” respectively (ATTRID) each time they add data to the table.³ It must be a list that includes the following items in the given order: EquipmentID (A[1..16]), EquipmentType (A[1..16]), OperatorID (A[1..16]), and CLOCK⁴ (A[16]).

Column #	Column Name	Description
1	Insp_Anomaly ID	ID # for the defect
2	Insp_Table specifier	Specifies table with other relevant information
3	Insp_Coordinate X	Intra die X Coordinate wrt LLHC of die in um
4	Insp_Coordinate Y	Intra die Y Coordinate wrt LLHC of die in um
5	Insp_X index	X axis die index wrt center of wafer (COW)
6	Insp_Y index	Y axis die index wrt center of wafer (COW)
7	Insp_X size	Defect size along X axis in microns
8	Insp_Y size	Defect size along Y axis in microns
9	Insp_Defect area	Defect area in square microns
10	Insp_Defect size	Linear measure of defect size in microns
11	Insp_Scatter intensity	Anomaly scattering intensity
12	Insp_Defect class number	Previously defined class number assigned to the defect
13	Insp_Test number	Inspection test in which defect was found
14	Insp_# Optical image count	Number of optical images stored for a given defect
15	Insp_Optical image data	Optical image data specifier
16	Insp_Cluster	= 1 if defect is part of a systematic defect cluster
17	Insp_Cluster class	Systematic defect class name
18	Sampled for SEM	= 1 if defect chosen for SEM review
19	Rev_SEM image data	SEM image data specifier
20	Rev_SEM class	SEM defect class name
21	Rev_Defect height	Defect height in microns
22	Sampled for analysis	= 1 if defect chosen for EDX, = 2 if defect chosen for FIB or other analysis
23	Anal_EDX data	EDX data specifier
24	Anal_FIB data	FIB data specifier

³ Where “*” is a numeric string that corresponds to the one in the column names the header refers to.

⁴ Date and time of the start of inspection, review, or analysis per SEMI E5 CLOCK data item variable.

11.4 TABLE-DEF Column Header Descriptions and Formats

Table 6 Description and Formats for ISEM Table Data

<i>Column Header</i>	<i>Description</i>	<i>Format</i>	<i>Comments</i>
“ALIGNNAME”	The identifier given to an alignment site.	A[1..16]	
ANOMALYATTRIBUTE (n) NOTE: String defined by equipment supplier.	Tool-specific information associated with an ANOMALY for which no specific ISEM data item has been defined.	U2, F4, F8, A[1..16]	Examples: Include information such as magnification, voltage, current, wavelength, brightness, color, height, or chemical spectra. The equipment supplier shall document all attributes that are supported.
“ANOMALYID”	A unique identifier for an anomaly.	A[1..16]	
“AREANAME”	A unique identifier given to an inspection area.	A[1..16]	
ATTRIBUTE(n) NOTE: String defined by equipment supplier.	Tool-specific information associated with an alignment or measurement site for which no specific ISEM data item has been defined.	U2, F4, F8, A[1..16]	Examples: Include information such as magnification, voltage, current, wavelength, number of scans, integration time, or film stack. The equipment supplier shall document all attributes that are supported.
“COORDSYS”	The identification for applicable coordinate system.	A[1..16]	Options are “M20”, “M20P”, and “M21”.
“COORDX”	The x-coordinate for a site.	F4	Units are in microns.
“COORDY”	The y-coordinate for a site.	F4	Units are in microns.
“ELEMENTID”	The SEMI M21 address for a specific rectangular element on a substrate.	I4[2]	
“XTENTX”	The extent in the X-direction of an area to inspect as measured from the lower left-hand corner of the area given by CoordX.	F4	Units in microns.
“XTENTY”	The extent in the Y-direction of an area to inspect measured from the lower left-hand corner of the area given by CoordY.	F4	Units in microns.

12 Process Program Management

12.1 Definition and Rules for ISEM Process Programs

12.1.1 A process program contains information and/or instructions required for the Inspection/Review equipment to process a given run of material. The process program shall supply all of the information required for a remotely executed run to be processed without operator intervention.

12.2 Requirements

12.2.1 The ISEM requires that the SEMI E30 capability of Process Program Management be fully supported for this class of equipment. ISEM requires that the process program have a structure that enables the user to build process programs with default conditions that can be overridden for a run. ISEM requires the ability to vary the quantity of substrates processed, the alignment information used, and the number and/or location of the areas/anomalies to be

inspected/reviewed through the uses of process program variable parameters. The concepts of process program structure and process program variable parameters are discussed in the following sections.

12.3 Process Program Structure

12.3.1 The purpose of this process program structure and the related concepts is to provide flexibility in using process programs to reduce the number of process programs needed. This structure enables the user or host to vary certain parameters of a given process program as needed for any particular run.

12.3.2 Often a process program may be very similar from one run to another and may differ only in a few parameters such as: which substrate slots to run, which areas to inspect, which parameters to run on each substrate, etc. Previously this small variation from run to run would require a large number of process programs to be created and maintained. The flexibility

of the method described in this section will reduce the number of process programs.

12.4 Process Program Variable Parameters — A process program parameter specifies a value that temporarily modifies the value of a process program variable parameter. A process program variable parameter is formally defined within a process program body and contains (1) a variable parameter name that is unique in the body (CPNAME), and (2) a parameter default value for use when the process program is selected for execution without specification of an override value for this variable parameter (CPVAL/CPEVAL).

12.4.1 Overriding Process Program Variable Parameters Default Values

12.4.1.1 Any process-related information that is normally requested from the operator console in manual operation shall have a process program variable parameter identified in the process program and default values assigned in the body of the process program. An equipment would run the process program using the default values unless those values were overridden.

12.4.1.2 These process program variable parameters allow a host to tailor a process program for a specific run of material by temporarily modifying (replacing) the process program default values using a remote command of PP-UPDATE. The modification does not permanently change the process program; the modifications remain in effect only until the next run or

until the next PP-UPDATE remote command is received.

12.4.2 Requirements and Rules

12.4.2.1 ISEM equipment is required to support variable process program parameters. Additionally, ISEM process programs are required to contain variable process program parameters that specify a name for each of the four previously defined ISEM table types. Specifically, parameters for “TABLE-AREA-DEF”, “TABLE-ALIGN-DEF”, “TABLE-ANOMALY-DEF”, and “TABLE-M21-ANOMALY-DEF” table names are required. Only the names that refer to these TABLE-DEFs are required to be included in the process program body. The actual TABLE-DEF data is external to the process program body. The host may always assume that there are variable process program parameters for these four ISEM tables.

12.4.2.2 Before execution of a CARRIERBLD can begin, the presence of all the ISEM Tables that it references shall be verified by the equipment. If they are not all present, an error shall be reported. The equipment shall support data items that may be linked to the event report that specifies the name of missing ISEM Tables. S7,F27 is used for reporting this error condition.

12.4.2.3 The following table summarizes the variable process program parameters that ISEM equipment shall support and the remote command parameters that the host may use to override their values (as defined in Section 13 and Table 9).

Table 7 Required Variable Process Program Parameters

<i>Variable Process Program Parameter and Host Command Parameter Name (CPNAME)</i>	<i>Description</i>
“ALIGNLIST”	A list of location identifiers to be reviewed.
“ANOMALYLIST”	A list of location identifiers to be reviewed.
“AREALIST”	A list of area identifiers to be inspected or reviewed.
“ELEMENTLIST”	A list of array element identifiers to be inspected.
“SLOTLIST”	A list of carrier slot numbers with material to be inspected or to be reviewed.
“SUBSTRATELIST”	A list of substrate IDs with material to be inspected or to be reviewed.
“TABLE-ALIGN-DEF”	A set of alignment site definitions.
“TABLE-ANOMALY-DEF”	A set of (SEMI M20) anomaly location and attribute definitions.
“TABLE-AREA-DEF”	A set of area definitions.
“TABLE-M21-ANOMALY-DEF”	A set of SEMI M21 anomaly location and attribute definitions.

12.4.3 Modifying Process Program Variable Parameters — The remote commands of PP-SELECT, PP-ASSIGN, or PP-UPDATE are used to modify any of the identified process program variable parameters within the process program. The modification is done by including CPNAME/CPVAL pairs within the “PROCESS-BLD-GROUP”, which is part of the remote commands of PP-SELECT or PP-ASSIGN, or by including a different list name in the PP-UPDATE remote command. A CPNAME in a process program shall be identical to the process program variable parameter name as specified in the “PROCESS-BLD-GROUP”. See next section for details of these parameters.

12.5 Use of Process Programs, Remote Commands, and “PROCESS-BLD-GROUP” — This is a brief description of the steps involved in using the process program structure, the TABLE-DEFs, and the modification of process program variable parameters through the use of “PROCESS-BLD-GROUP” and CPNAME/CEPVAL pairs in the enhanced remote command (S2,F49).

- A process program is created with certain items in the body identified as process program variable parameters, along with their default values.
- The host sends to equipment one or more TABLE-DEFs: such as “TABLE-AREA-DEF” and “TABLE-ALIGN-DEF”, along with the names of those tables using Stream 13 commands (See SEMI E5). These tables are now resident on the equipment.
- The host sends an enhanced remote command (S2,F49) to the equipment (either PP-SELECT or PP-ASSIGN) that contains the information needed for processing (“CARRIERBLD”). The “CARRIERBLD” contains a “PROCESS-BLD-GROUP” for each different set of run parameters that are needed during the inspection or review process.
- “PROCESS-BLD-GROUP” shall contain the ProcessBuildGroupID to identify which program is selected.
- If the default list of slots in the process program needs to be changed for this run, then “PROCESS-BLD-GROUP” contains a “MATERIALLIST” and the list of selected slots or substrates, designated by SlotIDs or SubstrateIDs.
- If the default set of inspection areas in the process program needs to be changed for this run, then “PROCESS-BLD-GROUP” contains the name of a specific “TABLE-AREA-DEF” (in AreaTable Name). If not all of the inspection areas given in that “TABLE-AREA-DEF” are needed for this run, then “AREALIST” is used and includes a list of the names of the specific inspection areas which are needed. Those names refer to inspection areas defined in the “TABLE-AREA-DEF”.
- If the default set of alignment sites in the process program needs to be changed for this run, then “PROCESS-BLD-GROUP” contains the name of a specific “TABLE-ALIGN-DEF” (in AlignTable Name). If not all of the alignment sites given in that “TABLE-ALIGN-DEF” are needed for this run, then “ALIGNLIST” is used and includes a list of the names of the specific alignment sites which are needed. Those names refer to alignment sites defined in the “TABLE-ALIGN-DEF”.
- If the SEMI M21 coordinate system is being used and a pattern has been defined with pattern element names and if only certain of those elements need to be inspected for this run, then “PROCESS-BLD-GROUP” contains an “ELEMENTLIST” and the list of selected elements, designated by ElementIDs.
- If certain equipment-specific process program variable parameters need to have different values for this run, then for each needed parameter, “PROCESS-BLD-GROUP” contains a CPNAME (unique name of a specific process program variable parameter) and the new CEPVAL (new value for that parameter).
- The equipment executes the process program with the new values.
- If indicated in the process program, the equipment generates the list of anomalies found and sends it to the host in the format of a “TABLE-ANOMALY-DEF” using the ISEM table definition and Stream 13 transfer messages (see SEMI E5 for format specification).
- The host would send the “TABLE-ANOMALY-DEF” to a review equipment. The host might need to modify part of the table if required by the review equipment. This anomaly table is now resident on the review equipment. (NOTE: The host might choose to not send all of the table of anomalies, but rather a desired selection of them.)
- The host sends the equipment the enhanced remote command (PP-SELECT or PP-ASSIGN) that contains a “CARRIERBLD”. The “CARRIERBLD” has a “PROCESS-BLD-GROUP” for each different set of run parameters that is needed by the review equipment. It is required for the host to use the Enhanced Remote Command S2,F49 for transferring the information

to the equipment. “PROCESS-BLD-GROUP” includes the name of a specific “TABLE-ANOMALY-DEF” in the parameter AnomalyTableName. If not all of the anomalies given in that “TABLE-ANOMALY-DEF” are needed for this run, then either “ANOMALYLIST” or “M21-ANOMALYLIST” is used and includes a list of the names of the specific anomalies which are desired. Those names refer to anomalies defined in the named “TABLE-ANOMALY-DEF”.

- The equipment runs the process program using the selected values and reviewing the specific anomalies indicated.
- The equipment adds review information to the ANOMALYATTRIBUTE list. Then the review equipment sends this modified “TABLE-ANOMALY-DEF” to the host.

13 Remote Commands

The purpose of this section is to identify remote commands, command parameters, and valid commands versus states pertinent to the SEM.

13.1 Requirements

- The equipment shall support the SEMI E30 required remote commands.
- All the remote commands defined by ISEM are required unless they have been qualified by the statement “if the equipment supports this functionality, it shall use this command.” In this case, they are only required if the equipment supports the functionality necessary to support the command. A good example of this is the MAP-CARRIER command. If the equipment does not have the hardware necessary to scan a carrier for the presence of substrates in slots, then the command is not required by the ISEM.
- The alphanumeric strings defined by ISEM for RCMD and CPNAME are required.

Host Command Parameter (CPNAME/CPVAL) — A parameter name/value associated with a particular host command when using stream function (S2,F41) and a (CPNAME/CEPVAL) parameter name/value when using the enhanced remote command (S2,F49). This document specifies unique names (CPNAMEs) and values (CPVALs and CEPVALs) for many command parameters. Note that if there are no associated parameters, a zero length list is sent.

The purpose of the remote commands is to allow host control over the following capabilities:

- Start processing

- Stop processing
- Temporarily suspend processing
- Resume processing
- Abort processing
- Select process programs, material, and/or sites to measure
- Report location of material found

The following remote commands (RCMDs) shall be supported as described below:

NOTE 3: The terms “current cycle” and “safe point” used below are to be defined by the supplier.

13.2 Remote Commands Description

1. **ABORT** — Terminate the current cycle prior to its completion. ABORT has the intent of immediately stopping the process and is used because of abnormal conditions. ABORT makes no guarantee about the subsequent condition of material except as noted in the “ABORTLEVEL” description.
2. **CLEANUP** — De-selection of the current ISEM job (“CARRIERBLD”) and process program (“PROCESS-BLD-GROUP”), including the removal of all material to output locations and any equipment-specific activities needed to transition into the IDLE state. Completion of this command should generate a collection event report. If the equipment supports this functionality, it will use this command.
3. **MAP-CARRIER** — Requests the equipment to provide a list of carrier slots that contain material. MAP-CARRIER has the intent of providing the host with enough information about the location and/or ID of material so it may select material for processing accordingly. Completion of this command shall generate a collection event report. If the equipment supports this functionality, it must use this command.
4. **NEXT-MATERIAL** — Processing of the current substrate is halted at the first safe point and unloaded to the target carrier location. NEXT-MATERIAL has the intent of allowing the host to skip measurement of the current substrate. This is a trigger for processing state transition from WORKING to UNLOAD. If the equipment supports this functionality, it will use this command.
5. **PAUSE** — Suspend processing temporarily at the next safe point. PAUSE has the intent of resuming the process at the same point where it was paused.

RESUME or PP-UPDATE may be used to resume the process.

6. **PP-ASSIGN** — Instructs the equipment that supports queuing to create a new ISEM job (“CARRIERBLD”) for the specified port (“LOCATIONID”) when more than one port is available for processing. If only one port is available, “LOCATIONID” is not required. Priority may optionally be specified with this command. The “PRIORITY” specifies the priority of the newly created job in the ISEM job queue (a value of 0 (zero) assigns the highest priority to the job). Without specifying a priority, the job is queued with the default priority. Jobs with equal priority are queued in the order the PP-ASSIGN commands are received. This command is valid in all PROCESSING states.
7. **PP-SELECT** — Instructs the equipment to make the requested ISEM job(s) (“CARRIERBLD”) available in the execution area. This is a trigger for the processing state transition from IDLE to SETTING UP. The first process program (“PROCESS-BLD-GROUP”) specified in the “CARRIERBLD” is also validated during SETTING UP.
8. **PP-UNASSIGN** — Removes the ISEM job assignment (“CARRIERBLD”) for a carrier or port. The carrier or port is removed from the process queue.
9. **PP-UPDATE** — Provides the ability to alter the current process program being executed during the PAUSED state. The process program variables specified in the PP-UPDATE command will

replace previous definitions in the “PROCESS-BLD-GROUP”. This command will trigger transition to CHECKING for process program parameter verification. A RESUME command is implied with the validation of “all” replaced values to resume the process. If the PP-UPDATE fails, the process program variables present prior to the PP-UPDATE are retained. If no parameters values are specified, the defaults are used.

10. **RESUME** — Resume processing from the point where the process was paused. This is the trigger for processing state transition from PROCESS PAUSE to the previous PROCESS state.
11. **START** — Instructs the equipment to initiate processing. This is the trigger for the processing state transition from READY to LOAD. An “AUTOSTART” command parameter may be included to allow for continuous processing.
12. **STOP** — Complete the current cycle, stop in a safe condition, and return to the IDLE processing state. Stop has the intent of stopping the process entirely. This command can be used to both: stop the current ISEM job or to stop all queued jobs. The equipment is not required to support the continuation of processing.

13.2.1 Remote Commands and Associated Host Command Parameters — This table describes the allowable command parameters (CPNAME) for each remote command (RCMD). Equipment shall support all parameters. The column marked Req/Opt specifies which parameters are required to be sent by the host and which parameters may be optionally sent by the host.

Table 8 Allowable Command Parameters

Remote Command	Parameters		
	CPName	Req/Opt	Comments
ABORT	“ABORTLEVEL”	R	
CLEANUP	“CARRIERID” “LOCATIONID” “SLOTID”	O O O	PORT and SLOT may be used to define a different carrier/slot destination for the substrates.
MAP-CARRIER	“CARRIERID” “LOCATIONID”	R* R*	* One is required.
NEXT-MATERIAL	“CARRIERID” “LOCATIONID” “SLOTID”	O O O	PORT and SLOT may be used to define a different carrier/slot destination for the substrates.
PAUSE	None	NA	None
PP-ASSIGN	“PRIORITY” “CARRIERBLD”*	O R	* More than one “CARRIERBLD” may be specified.
PP-SELECT	“CARRIERBLD”*	R	* More than one “CARRIERBLD” may be specified.
PP-UNASSIGN	“CARRIERBLD”	R	None

PP-UPDATE	“PPBUILDID” “ALIGNLIST” “ANOMALYLIST” “AREALIST” “ELEMENTLIST” “SLOTLIST” “SUBSTRATELIST” “TABLE-ALIGN-DEF” “TABLE-AREA-DEF” “TABLE-ANOMALY-DEF” “TABLE-M21-ANOMALY-DEF”	R R* R* R* R* R* R* R* R* R*	* At least one is required.
RESUME	None	N/A	None
START	“CARRIERBLD”	0	None
STOP	“CARRIERBLD”	0	None

13.2.2 Host Command Parameter Names and Values

Table 9 Host Command Parameters CPNAMES

CPName	Parameter Value		
	Description	Range	Format
“ABORTLEVEL”	ISEM-defined abort levels: HALT — Process halts, and the ABORTING process state is entered. CLEANUP — Process halts, material cleanup is performed, and the ABORTING process state is entered.	“1= HALT” “2 = CLEANUP”	U2
“ALIGNLIST”	L,n 1. AlignName ₁ : n. For the SEMI M20 or M20P coordinate system.		List of A[1..16] data items
“ALIGNNAME”	Alignment name See the “TABLE-ALIGN-DEF” definition for further explanation.		A[1..16]
“ANOMALYLIST”	L,n 1. AnomalyID ₁ : n. For the SEMI M20 or M20P coordinate system.		List of A[1..16] data items
“ANOMALYID”	Anomaly identifier See the “TABLE-ANOMALY-DEF” or the “TABLE-M21-ANOMALY-DEF” definition for further explanation.		A[1..16]
“AREALIST”	L,n 1. AreaName ₁ : n.		List of A[1..16] data items
“AREANAME”	Unique identifier for an area to be inspected. See the “TABLE-AREA-DEF” definition for further explanation.		A[1..16]
“AUTOSTART”	Specifies whether a START command is required from an external source (operator or host) to exit the READY state. 0 = NoAutoStart (A START command required.) 1 = AutoStart (No external START command required to begin execution.)	0–1	U2
“CARRIERID”	Identifier of the carrier that the inspection/review data is associated with.		A[1..16]

CPName	Parameter Value		
	Description	Range	Format
"ELEMENTLIST"	L,n 1. ElementID ₁ : n. For the SEMI M21 coordinate system.		List of A[1..16] data items
"LOCATIONID"	Unique identifier of the location to be used for the "CARRIERBLD" assignment.		U2
"PPBUIDID"	ProcessProgramBuildID		A[1..80]
"PPNAME"	ProcessProgramName		A[1..80]
"PRIORITY"	Assignment priority	0–9 Highest priority corresponds to 0.	U2
"SLOTLIST"	Specifies carrier slots containing substrate for the ISEM job. L,n 1. SlotID ₁ : n. SlotID _n	Zero length list specifies all slots.	List of U2 data items
"STOPLEVEL"	Stop levels defined by the ISEM.	"1 = LOCATIONID" "2 = CARRIERID"	Use defined CPVALs
"SUBSTRATELIST"	Specifies identifiers of substrate for the ISEM job. L,n 1. SubstrateID ₁ : n. SubstrateID _n	Zero length list specifies all substrate (independent of substrate identifier).	List of A[0..16] data items
"CARRIERBLD" E5 Format	L, 3 1. L, 2 ❖ 1. "CARRIERID" A[9] -- CPName 2. CarrierID A[1..16] -- CPValue 2. L, 2 ❖ 1. "LOCATIONID" A[10] -- CPName 2. LocationID U2 -- CPValue 3. L, 2 1. "PROCESS-BLD-LIST" 2. L, n ❖ List of n jobs 1. L, 2 1. "PROCESS-BLD-GROUP" 2. L, m First ISEM job 2. L, 2 1. "PROCESS-BLD-GROUP" 2. L, m Next ISEM job . . . n. L, 2 1. "PROCESS-BLD-GROUP" 2. L, m Last ISEM job	m = 3	List
"PROCESS-BLD-GROUP" (for Inspection)	L,m 1. L,2 1. "PPBUIDID" 2. ProcessBuildGroupID	m ≥ 2	List of m data items

CPName	Parameter Value		
	Description	Range	Format
	2. L,2 1. "PPNAME" 2. ProcessProgramID 3. L,2 1. "SLOTLIST" 2. L,n 1. SlotID₁ : n. Or 1. "SUBSTRATELIST" 2. L,n 1. SubstrateID₁ : n. 4. L,2 1. "AUTOSTART" 2. AutoStart 5. L,2 1. "TABLE-AREA-DEF" 2. AreaTableName 6. L,2 1. "AREANAME" 2. L,n 1. AreaName₁ : n. 7. L,2 1. "TABLE-ALIGN-DEF" 2. AlignTableName 8. L,2 1. "ALIGNNAME" 2. L,n 1. AlignName₁ : n. 9. L,2 1. "ELEMENTLIST" ** 2. L,n 1. ElementID₁ : n. 10. L,2 1. CPNAME* 2. CEPVAL* m. L,2 1. CPNAME* 2. CEPVAL* NOTES: "PPBUILDDID" and "PPNAME" are required. "SLOTLIST", "SUBSTRATELIST", "AREALIST", and "ALIGNLIST" are optional. * Supplier shall define as many of these CPNAME, CEPVAL		

CPName	Parameter Value		
	Description	Range	Format
	pairs as are supported by the equipment. ** "ELEMENTLIST" is required when using the SEMI M21 coordinate system in the definition of an AlignName or AreaName.		
"PROCESS-BLD-GROUP" (for Review Equipment)	L,m 1. L,2 1. "PPBUILDID" 2. ProcessBuildGroupID 2. L,2 1. "PPNAME" 2. ProcessProgramID 3. L,2 1. "SLOTLIST" 2. L,n 1. SlotID₁ : 2. SlotID_n Or 1. "SUBSTRATELIST" 2. L,n 1. SubstrateID₁ : n. SubstrateID_n 4. L,2 1. "AUTOSTART" 2. AutoStart 5. L,2 1. "TABLE-ANOMALY-DEF" 2. AnomalyTableName 6. L,2 1. "ANOMALYNAME" or "M21-ANOMALYNAME" 2. L,n 1. AnomalyID₁ or M21AnomalyID₁ : - n. 7. L,2 1. "ALIGN-TABLE-DEF" 2. AlignTableName 8. L,2 1. "ALIGNNAME" 2. L,n 1. AlignName₁ : - n. 9. L,2 1. "ELEMENTLIST"*** 2. L,n 1. ElementID₁ : - n. 10. L,2	$m \geq 2$	List of m data items

CPName	Parameter Value		
	Description	Range	Format
	1. CPNAME* 2. CEPVAL* m. L,2 1. CPNAME* 2. CEPVAL* NOTES: “PPBUILDID” and “PPNAME” are required. “SLOTLIST”, “SUBSTRATELIST”, “AREALIST”, and “ALIGNLIST” are optional. * Supplier shall define as many of these CPNAME, CEPVAL pairs as are supported by the equipment. ** “ELEMENTLIST” is required when using the SEMI M21 coordinate system in the definition of an AlignName or AreaName.		

NOTE 1: ❖ Required ISEM parameters: “CARRIERID”, “LOCATIONID”, “PROCESS-BLD-GROUP”

13.2.3 *Remote Commands vs. Processing States* — The following table indicates states where the remote commands are allowed. This is indicated with a “X” mark.

Table 10 Remote Commands vs. Processing States

	COMMAND											
	STOP	START	RESUME	PP-UPDATE	PP-SELECT	PAUSE	NEXT-MATERIAL	MAP-CARRIER	PP-ASSIGN	CLEANUP	ABORT	PP-UNASSIGN
PROCESSING STATE												
IDLE					X			X	X			X
ABORTED									X	X		X
PROCESSING ACTIVE												
STOPPING									X		X	
ABORTING												
PAUSE												
ALARM PAUSED	X								X		X	
PROCESS PAUSE												
PAUSING	X								X		X	
PAUSED	X		X	X					X		X	
CHECKING	X								X		X	
PROCESS												
SETTING UP	X					X		X	X		X	X
READY	X	X				X			X		X	X
EXECUTING												
LOAD	X					X			X		X	
UNLOAD	X					X			X		X	
WORKING												
INSPECT	X					X	X		X		X	
ALIGN	X					X	X		X		X	
REVIEW	X					X	X		X		X	



14 Scenarios

14.1 *Run Level Reporting Scenario* — This scenario only has expected events (i.e., no alarms or errors).

COMMENT	HOST	EQUIPMENT	COMMENT
			The equipment is in the IDLE processing state and in The ONLINE REMOTE control state. The host has defined, linked, and enabled RUN level report for CEIDs 2, 3, and 5.
Host sends a PP-SELECT command specifying a "CARRIERBLD"	S2,F49-->		
		<--S2,F50	Command Acknowledge
			The equipment transitions from IDLE to SETTING UP, and material arrives at input port.
		<--S6,F11	SETTING UP -> READY (CEID 3)
Event Report Acknowledge	S6,F12-->		
START	S2,F41-->		
		<--S2,F42	Host Command Acknowledge
			READY -> LOAD. [WHILE] Note End of Run LOAD -> WORKING WORKING -> UNLOAD UNLOAD -> LOAD [END WHILE]
		<--S6,F11	LOAD -> STOPPING (CEID 5)
Event Report Acknowledge	S6,F12-->		
		<--S6,F11	Run Processed Data Valid event.
Event Report Acknowledge	S6,F12-->		
			The equipment transitions from STOPPING to IDLE.



14.2 PP-UPDATE Remote Command Scenario — Host issues the PP-UPDATE remote command.

COMMENT	HOST	EQUIPMENT	COMMENT
START	S2,F41-->	<--S2,F42 <--S6,F11	Positive Acknowledge. READY -> LOAD
Positive Acknowledge.	S6,F12-->		[WHILE] Not End of Run 1) LOAD -> WORKING 2) WORKING -> UNLOAD 3) UNLOAD -> LOAD [END WHILE]
Sometime during the [WHILE]: PAUSE	S2,F41-->	<--S2,F42 <--S6,F11	Positive Acknowledge. Transition to PAUSING
Positive Acknowledge.	S6,F12-->	<--S6,F11	PAUSING -> PAUSED
Positive Acknowledge. PP-UPDATE	S6,F12--> S2,F49-->	<--S2,F50 <--S6,F11	Positive Acknowledge. PAUSED -> CHECKING
Positive Acknowledge.	S6,F12-->	<--S6,F11	CEID is posted. [IF] the updates are valid: Return to the previous process state through history. [ELSE] Return to the PAUSED state. The Process program remains unchanged. [ENDIF]
Positive Acknowledge.	S6,F12-->		



14.3 PP-SELECT Remote Command Scenario

COMMENT	HOST	EQUIPMENT	COMMENT
			The equipment is in the IDLE processing state and in the ONLINE REMOTE control state.
Host sends a TABLE-DEF to the equipment.(See Section 11.)	S13,F13-->		
		<--S13,F14	Table Data Acknowledge
Host sends more tables if needed.	S13,F13-->		
		<--S13,F14	Table Data Acknowledge
Host prepares the remote command to initiate an inspection run, including the "TABLE-AREA-DEF", "TABLE-ALIGN-DEF", and "ELEMENTLIST", if required.			
Host sends a PP-SELECT command specifying a "CARRIERBLD".	S2,F49-->		
		<--S2,F50	Command Acknowledge
			The equipment transitions from IDLE to SETTING UP to READY.
START	S2,F41-->		
		<--S2,F42	Host Command Acknowledge
			READY -> LOAD. [WHILE] Not End of Run LOAD -> WORKING WORKING -> UNLOAD UNLOAD -> LOAD [END WHILE]
			LOAD -> STOPPING
		<--S13,F13	Equipment sends anomaly table with additional data, including the table name using Table Data Send command.
Table Data Acknowledge	S13,F14-->		
			The equipment transitions from STOPPING to IDLE.



14.4 Event Report and ISEM Table Transfer Command Scenario

COMMENT	HOST	EQUIPMENT	COMMENT
			The equipment is in the IDLE processing state and in the ONLINE REMOTE control state.
Host defines report with AnomalyTableName and AnomalyTableType	S2,F33-->		
		<--S2,F34	Define Report Acknowledge
Link report to RunDataComplete event	S2,F35-->		
		<--S2,F36	Link Event Report Acknowledge
Enable event	S2,F37-->		
		<--S2,F38	Enable Event Acknowledge
Host sends an ISEM table to the equipment.	S13,F13-->		
		<--S13,F14	Table Data Acknowledge
Host sends more tables if needed.	S13,F13-->		
		<--S13,F14	Table Data Acknowledge
Host sends a PP-SELECT command specifying a "CARRIERBLD"	S2,F49-->		
		<--S2,F50	Remote Command Acknowledge The equipment transitions from IDLE -> SETTING UP -> READY.
START	S2,F41-->		
		<--S2,F42	Host Command Acknowledge READY -> LOAD. [WHILE] Not End of Run LOAD -> WORKING WORKING -> UNLOAD UNLOAD -> LOAD [END WHILE] LOAD -> STOPPING STOPPING -> IDLE
		<--S6,F11	RunDataComplete event with AnomalyTableName and AnomalyTableType.
Event Report Acknowledge	S6,F12-->		
Host sends Table DataRequest	S13,F15-->		
		<--S13,F16	Equipment sends requested table TBLACK = 0 if no errors.

15 GEM Capabilities

The purpose of this section is to specify any SEMI E30 additional capabilities that are required to be supported by this class of equipment.

15.1 Requirements

15.1.1 This standard requires that the SEMI E30 fundamental requirements and additional capabilities have been implemented on the ISEM equipment with the exception of limits monitoring and trace reporting. If these capabilities are implemented, they shall be implemented as required by the SEMI E30 document. The following SEMI E30 additional capabilities required by ISEM are:

- Dynamic Event Report Configuration
- Variable Data Collection
- Status Data Collection
- Alarm Management
- Remote Control
- Equipment Constants
- Process Program Management

- Spooling
- Trace Data Collection (optional)
- Control (Host-Initiated)

NOTICE: SEMI makes no warranties or representations as to the suitability of the standards set forth herein for any particular application. The determination of the suitability of the standard is solely the responsibility of the user. Users are cautioned to refer to manufacturer's instructions, product labels, product data sheets, and other relevant literature respecting any materials mentioned herein. These standards are subject to change without notice.

The user's attention is called to the possibility that compliance with this standard may require use of copyrighted material or of an invention covered by patent rights. By publication of this standard, SEMI takes no position respecting the validity of any patent rights or copyrights asserted in connection with any item mentioned in this standard. Users of this standard are expressly advised that determination of any such patent rights or copyrights, and the risk of infringement of such rights, are entirely their own responsibility.

RELATED INFORMATION 1

NOTE: This related information is not an official part of SEMI E30.1 and was approved for publication by full letter ballot procedures on September 3, 1999.

R1-1 Defect Classification Code Management

The purpose of this section is to provide a method and specific formats to define, identify, and communicate coordinate systems and site locations on substrates for alignment sites, anomaly locations, and other sites used by the ISEM equipment.

R1-1.1 Classification Codes and Defect Classification — One function of review equipment is to view previously identified anomalies and to associate a defect classification code with each anomaly. A classification code is an identifier for a classification description.

Typically, the review equipment has a set of defect classification codes and their descriptions available to the operator. Then, for each anomaly, the operator selects a particular code to be associated with that anomaly. This action is defect classification.

The set of valid classification codes and their descriptions may change from one run to another. For example, the same main process program could be used with different substrate levels, and each level may use a different set of classification codes. The purpose of this section is to provide the requirements so that a user can both define several sets of classification codes and their descriptions and can also manage these sets on ISEM equipment.

R1-1.2 Requirements

- Each set of classification codes and their descriptions shall have an identifier, known as a classification code set ID.
- Review equipment shall provide a means for the user (the host or the operator) to define a classification code set, consisting of (a) the classification code set ID and (b) the list of classification codes and their descriptions.
- Equipment shall provide a means to manage the various classification code sets.
- A main process program shall include a process program variable that specifies the particular classification code set ID to be used.
- Equipment vendor shall provide documentation to the user regarding how to define and manage classification code sets.

Comment: In one implementation, the equipment considers a classification code set to be a sub-process program or an ISEM table. This would allow the user to identify a classification code set by name (using a PPID) or a table name and thereby managing this sub-process program with the SEMI E30 Process Program Capability or with SEMI E58 ARAMS tables.

R1-2 Reporting Coordinates and Coordinate Systems

The purpose of this section is to provide a method and specific formats to define, identify, and communicate coordinate systems and site locations on substrates for alignment sites, anomaly locations, and other sites used by ISEM equipment.

The ISEM-required formats are intended to minimize the number and type of site location format transformations needing to be supported by both equipment suppliers and users.

All ISEM-required site location formats involve the use of an ISEM-defined right-handed Cartesian coordinate system, established on substrates in an ISEM-defined manner. The scope of the detailed methods in this section are specific to unpatterned and patterned wafers in this release, but the section is intended to be general enough in methodology so that it can be extended to other substrate types in future revisions of ISEM, if required.

The purpose of inspection and review equipment is to locate, evaluate, classify, and report anomalies on substrates. ISEM equipment may deal with either unpatterned or patterned substrates or both. In most cases, the anomaly location is part of the information reported and/or used by ISEM equipment. An anomaly location is reported at a particular site with x,y coordinates in a particular coordinate system. Site coordinates are also used by ISEM equipment for the alignment sites for defining a coordinate system on a substrate. A standard method is needed to define a coordinate system and to report site coordinates for both alignment sites, anomaly locations, and any other reference sites needed by ISEM equipment. A standard method is essential in order to transfer the anomaly site information from one equipment to another.

R1-2.1 Site Location Accuracy — Each equipment has an accuracy with which it can define or locate a site as being within a certain area. This area associated with a site is determined by the equipment accuracy, based on the accuracy of its motion and imaging systems to locate a site, as well as on the accuracy with which it can define the coordinate system on the substrate.

When equipment shall locate a particular site on a substrate based on the expected or design-based location, then the location of a site or feature on an actual substrate is further affected by the accuracy of the equipment which placed the pattern on the substrate.

R1-2.2 Expected or Designed Locations vs. Actual Locations — The placement of patterns, sites, and coordinate systems is designed to be at certain mathematically described locations relative to one another and to an ideal substrate. These are the expected or designed locations. When a pattern is written by equipment onto a specific substrate, the actual placement of the pattern, the pattern-elements, and their features may differ from the expected locations, due to variations in equipment performance and variations in substrate shape and dimensions.

R1-2.3 Substrate Coordinate Systems (Unpatterned) — A substrate coordinate system is a coordinate system which has both origin and axes defined by the shape and dimensions of the substrate and which does not depend on whether there is a pattern on the substrate or whether it is unpatterned. This coordinate system is used to locate or define sites relative to the substrate.

R1-2.4 Substrate Pattern Coordinate System — A substrate pattern coordinate system is a coordinate system which has its origin and axes defined by the pattern as a whole on the substrate. This coordinate system is used to locate or to define sites relative to the pattern on the substrate. The expected or designed location of the pattern on the substrate can be defined in terms of the placement of the origin and axes of the substrate pattern coordinate system relative to those of the substrate coordinate system. The actual location of a pattern on a substrate may differ from the expected location. The actual location is determined by locating two or more alignment sites on the patterned substrate. The alignment sites are specific points of certain features in the pattern. The coordinates of the alignment sites are given in the substrate pattern coordinate system. In many cases, equipment does not align to the specific pattern elements but instead uses the defined locations of the pattern elements within the substrate pattern coordinate system.

R1-2.5 Pattern Element Coordinate System — A pattern-element coordinate system is a coordinate system which has its origin and axes defined by the pattern of one specific rectangular element in a pattern (a defined arrangement) of equal-sized rectangular elements. This coordinate system is used to locate or to define sites relative to that specific pattern-element. The expected or designed location of the pattern-element within a pattern can be defined in terms of the placement of the origin and axes of the pattern-element

coordinate system relative to those of the pattern coordinate system. The actual location of a pattern-element within a pattern on a substrate may differ from the expected location. The actual location is determined by locating two or more alignment sites within the pattern-element. The coordinates of the alignment sites are given in the pattern-element coordinate system.

R1-2.6 Parallel Coordinate Systems — A second coordinate system is considered to be parallel to a first coordinate system if the origin of the second can be defined as a translation from the origin of the first and if the axes of the second are parallel and in the same direction as those of the first.

R1-2.7 Requirements — The following is a list of requirements for ISEM equipment regarding coordinate systems and reporting site locations:

- ISEM equipment shall document whether it deals with coordinate systems based on (a) a substrate, (b) a substrate pattern, or (c) a pattern-element or whether it deals with several of these coordinate systems.
- ISEM equipment shall establish a substrate coordinate system using a standard, documented method. This coordinate system is not based on any pattern on the substrate. This coordinate system shall be a right-hand Cartesian coordinate system and shall be identified by a name.

NOTE: For wafers, this method is defined in SEMI M20 (Specification for Establishing a Wafer Coordinate System), and the coordinate system is named “M20.”

- For equipment dealing with substrate pattern coordinates, the substrate pattern coordinate system shall be established in a standard, documented method relative to the substrate coordinate system (the “unpatterned” coordinate system). This substrate pattern coordinate system shall be a right-hand Cartesian coordinate system and shall be designed to be parallel to the substrate coordinate system. The substrate pattern coordinate system shall be identified by a name. The location of its origin and axes relative to the substrate coordinate system shall be communicated in terms of the substrate coordinate system.

NOTE: For wafers, this method is the one described below, and the substrate pattern coordinate system is named “M20P”, and its origin and axis relative to the SEMI M20 coordinate system are given in terms of “M20” coordinates and are communicated using XlateData.

- For equipment dealing with pattern-element coordinates, the pattern-element coordinate system shall be established in a standard, documented

method relative either to the substrate pattern coordinate system or to another pattern-element coordinate system. The pattern-element coordinate system shall be a right-hand Cartesian coordinate system which is designed to be parallel to the substrate pattern coordinate system. The pattern-element coordinate system shall be identified by a name. The location of its origin and axis relative to the substrate pattern coordinate system shall be communicated in terms of the substrate pattern coordinate system.

NOTE: For wafers, this method is based on SEMI M21, and the coordinate system is named “M21” and its origin and axis relative to the “M20P” coordinate system are given in terms of the M20P coordinates.

- ISEM requires that equipment have the capability to use site location information that is based on the user’s product designs, which the user shall provide in the appropriate ISEM-required format.
- ISEM-compliant equipment shall have the capability to define, locate, and report site information using only the ISEM-defined right-handed Cartesian coordinate system formats. This requirement does not preclude equipment from having additional capability for defining or reporting site location information using other formats.
- Coordinate system name and placement relative to the “higher” coordinate system shall be defined and communicated using the following ISEM data items, in terms of either expected or actual placement: CoordSys, XlateData, and their included data items.
- Alignment site information shall be defined and communicated using the following ISEM items: the variable item AlignList, the “ALIGNLIST”, the Process program class of “TABLE-ALIGN-DEF”, and their included information.
- Areas to be inspected shall be reported using the specific coordinate system defined by the user. The following ISEM items are used to define and communicate area locations: the variable item “AREALIST”, the “AREALIST”, and the Process program class of “TABLE-AREA-DEF”, and their included information.
- The displacement of an actual coordinate system relative to its expected location shall be communicated using the ISEM data item: XlateData and its included data items.
- The displacement of an actual site location relative to its expected site location shall be communicated

using the ISEM data item: Offset and its included data items.

- The equipment vendor shall document the requirements for the ISEM data items used in alignment of a coordinate system.
- The equipment vendor shall provide and document a means for the user to define and communicate a pattern map using SEMI M21 data. A pattern map defines the layout of equal-sized rectangular pattern-elements which make up a pattern. Each pattern-element shall have a name, using the SEMI M21 naming convention.

NOTE: For patterned wafers, the naming method shall be that described in SEMI M21, and the pattern-element information shall be communicated using the ISEM data item of SEMI M21Data.

- For ISEM compliance, inspection equipment shall report various anomaly data; AnomalyID, coordinates, and attributes. Review equipment shall receive this data for anomalies and be able to locate them and perhaps modify the coordinates. Anomaly coordinates shall be reported using ISEM table named “TABLE-ANOMALY-DEF” and its included data.

R1-3 Coordinate System for a Substrate

R1-3.1 SEMI M20 Coordinate System — The SEMI M20 standard (Specification for Establishing a Wafer Coordinate System) describes how to map a right-handed Cartesian coordinate system to a substrate so that its origin is at the center of the substrate, and its negative y-axis bisects the substrate’s primary fiducial. This coordinate system is defined by ISEM to be the “M20” coordinate system. The only information required by equipment in order to establish an “M20” coordinate system is the substrate size and the type of fiducial, which are communicated using the ISEM data items named **SubstrateSize** and **Fiducial**. Another ISEM data item named **Orientation** identifies how the substrate is loaded on the equipment. Note that the SEMI M20 standard requires that the “M20” coordinate system is fixed on the substrate and is not affected by how the substrate is loaded on equipment. Also, as stated in the SEMI M20 standard, an orientation of “0” degrees designates a substrate loaded on equipment, with the primary fiducial towards the operator or “down.”

R1-3.2 M20P Coordinate System — ISEM defines the M20P coordinate system to be one which is aligned to the pattern on the substrate. The M20P coordinate system is useful because in many cases, it is more significant to the user to know the location of an anomaly relative to the pattern on the substrate rather

than relative to the substrate shape and dimensions. ISEM also defines the M20P coordinate system to be one which is designed to be “parallel” to the SEMI M20 coordinate system. In practice, because of experimental errors, both the origins and the axes may differ slightly from their intended values of a simple translation and no rotation. Equipment should be designed to be able to locate the alignment sites, given the various possible experimental errors.

R1-3.3 Establishing an M20P Coordinate System — A minimum of two alignment sites is necessary to establish an M20P coordinate system on a substrate. Additional sites are often used to determine a scaling ratio of the dimensions of the actual coordinate system relative to the dimensions of the expected coordinate system and are reported using the ISEM data item of **ScaleFactor**.

XlateData is used to report actual coordinate system location. Most equipment cannot distinguish whether patterned substrate site location errors are due to the substrate, the layout on the substrate, or the equipment’s ability to locate the sites. However, information that is available through the use of patterned-substrate alignment sites can provide a means for identifying potential equipment problems. For instance, assume that the only pattern-layout location error on a substrate is that due to the establishment of the location of the substrate center and fiducial. For many users and equipment systems, this is a good assumption. If this is the case, then the ISEM data item named **XlateData** can be used to track this error. Although the error may result from multiple sources, being able to track it on various equipment will enable users to apply statistical process control techniques to identify the specific sources.

Offset sites may be found by equipment at actual locations which deviate from their expected locations through either pattern layout errors or equipment “stage” or imaging errors. Again, in a controlled manufacturing process, these combined errors should be normally distributed, and non-normal deviations may indicate possible equipment problems. The actual position of a site relative to its expected position shall be reported through the use of the ISEM data item named **Offset**.

R1-4 Layout of Rectangular Pattern Elements on a Substrate Using SEMI M20 Coordinate System

Equipment shall be capable of routine, automated operation without needing substrate layout information (e.g., field or die maps). However, having the capability to provide substrate layout information to equipment

from the host can be desirable. ISEM defines a means to do this in this section for substrates, based on SEMI M21 (Specification for Assigning Addresses to Rectangular Elements in a Cartesian Array.) The SEMI M21 standard is limited to defining how to assign “addresses” to elements and how to find the “array center” element. It does not specify how the rectangular pattern-elements are located on the substrate. In this section, ISEM defines how these pattern-elements are located on a substrate, using the data item named M21Data, and how to establish within-element coordinate systems. Any additional layout information, such as within-element structure details or element attribute information, is beyond the scope of ISEM.

R1-4.1 ISEM “M21” Layouts

- An “M21” layout consists of an array of equal-sized rectangular pattern-elements with no space between the pattern elements.
- ISEM defines the “M21” layout on a substrate to include all pattern-elements which are either wholly or partially within the circumference of the substrate.
- The ISEM approach is to define the pattern map by specifying the M20P coordinate for the lower left corner of the minimum number of pattern-elements needed to define the layout, along with the pattern-element addresses (names). For a non-tiled layout, the location and name of a single pattern-element is sufficient to establish the “M21” layout. For tiled layouts, the location and name of one pattern-element in each row or column are required. Note that the location of the lower left corner of some pattern-elements may be outside the circumference of the substrate.
- The “M21” pattern-element coordinate system shall have its x and y axes parallel to the respective M20P coordinate system axes and shall have their origins at the lower left corner of each element. The pattern-element coordinate system shall have a name and a specific pattern-element address identifier per SEMI M21.
- Layout definition is supported only for host-to-equipment communications. The user is responsible for ensuring that the pattern-element addresses provided to the equipment agree with the SEMI M21 specification. The equipment need not check this, other than to ensure that there are not conflicts within the provided layout, and shall report results with pattern-element addresses as provided by the user.



- “M21” layouts are established within the M20P coordinate system and need not require any additional alignment site data than is needed to establish the M20P coordinate system. However, as with M20P, additional alignment may be necessary because of errors in either the pattern layout or the equipment’s ability to locate features. Offset shall be used to report the location corrections that result from any within-element alignments.

R1-5 How an M20P Coordinate System Is Established on a Substrate

The following example is fairly basic. For this example, the M20P coordinate system has a zero translation from the SEMI M20 coordinate system. Also, the equipment documentation states that 4 alignment sites are required. The equipment does M20P alignment on two alignment sites and does a low resolution and then a high resolution alignment at each site. Note that the specific alignment point is different at the two resolutions, so the coordinates are slightly different. The alignment sites are defined to the equipment via the process program class named “TABLE-ALIGN-DEF”, as detailed below. The order of the sites in “TABLE-ALIGN-DEF” is not important. The sites are then selected via the CPNAME named “ALIGNLIST”, which is included in the PP-SELECT command. The order of the sites listed in “ALIGNLIST” is important and is as-specified in the equipment’s documentation. The first item is the alignment site for the first low resolution site, the second item is for the first high resolution site, the third item is the second low resolution site, and the fourth is the second high resolution site.

“TABLE-ALIGN-DEF”

<i>AlignName</i>	<i>Coordx</i>	<i>Coordy</i>	<i>Coordsys</i>	<i>Attribute (1)</i>
Coarse1	-60000	-200	“M20P”	
Fine1	-60020	-205	“M20P”	
Coarse2	+60000	+200	“M20P”	
Fine2	+59980	+195	“M20P”	

“ALIGNNAME”

L,4

1. <Coarse1>
2. <Fine1>
3. <Coarse2>
4. <Fine2>

Using this information, the equipment will go to the nominal “M20” location for Coarse1, then “find” where it actually is. The offset between the nominal “M20” location and the actual “M20” location is then used to “find” Fine1. The actual M20 location of Fine1 is saved. The process is then repeated for Coarse2 and Fine2. The equipment can now determine the “M20” to M20P offset from the nominal and actual coordinates. First, a summary of the data:

xN1 = -60020	yN1 = -205	Nominal x and y data for the first fine site
xA1 = -59800	yA1 = -150	Actual x and y data for the first fine site
xN2 = +59980	yN2 = +195	Nominal x and y data for the second fine site
xA2 = +60060	yA2 = +175	Actual x and y data for the second fine site

The equipment first calculates Theta, using, for example, the formula:

$$\Theta = \tan^{-1} \left[\frac{MA - MN}{1 + MA \cdot MN} \right]$$

where MA and MN are, respectively, the slopes of the lines connecting the two actual fine sites and the line connecting the two nominal sites, in “M20” coordinates, calculated as follows:

$$MA = \left[\frac{yA_2 - yA_1}{xA_2 - xA_1} \right] \quad MN = \left[\frac{yN_2 - yN_1}{xN_2 - xN_1} \right]$$

The equipment then calculates ΔX and ΔY , using, for example, the formulas:

$$\Delta X = \left[\frac{C \sin(\Theta) + D \cos(\Theta)}{(\sin(\Theta))^2 + (\cos(\Theta))^2} \right]$$

$$\Delta Y = \left[\frac{C \sin(\Theta) - D \cos(\Theta)}{(\sin(\Theta))^2 + (\cos(\Theta))^2} \right]$$

where C and D , the adjusted site 1 coordinates in a rotation-adjusted coordinate system, are calculated, for example, using the formulas:

$$C = yA_1 - ((xN_1 \sin \Theta) + (yN_1 \cos \Theta))$$

$$D = xA_1 - ((xN_1 \cos \Theta) - (yN_1 \sin \Theta))$$

The equipment can also calculate a $ScaleFactor$ term to indicate the relative ratio between the length of the vector connecting the nominal alignment sites and the length of the vector connecting the actual alignment sites. This can be used, for example, to judge whether there is a problem with the alignment process, since the difference between these two vectors should be small.

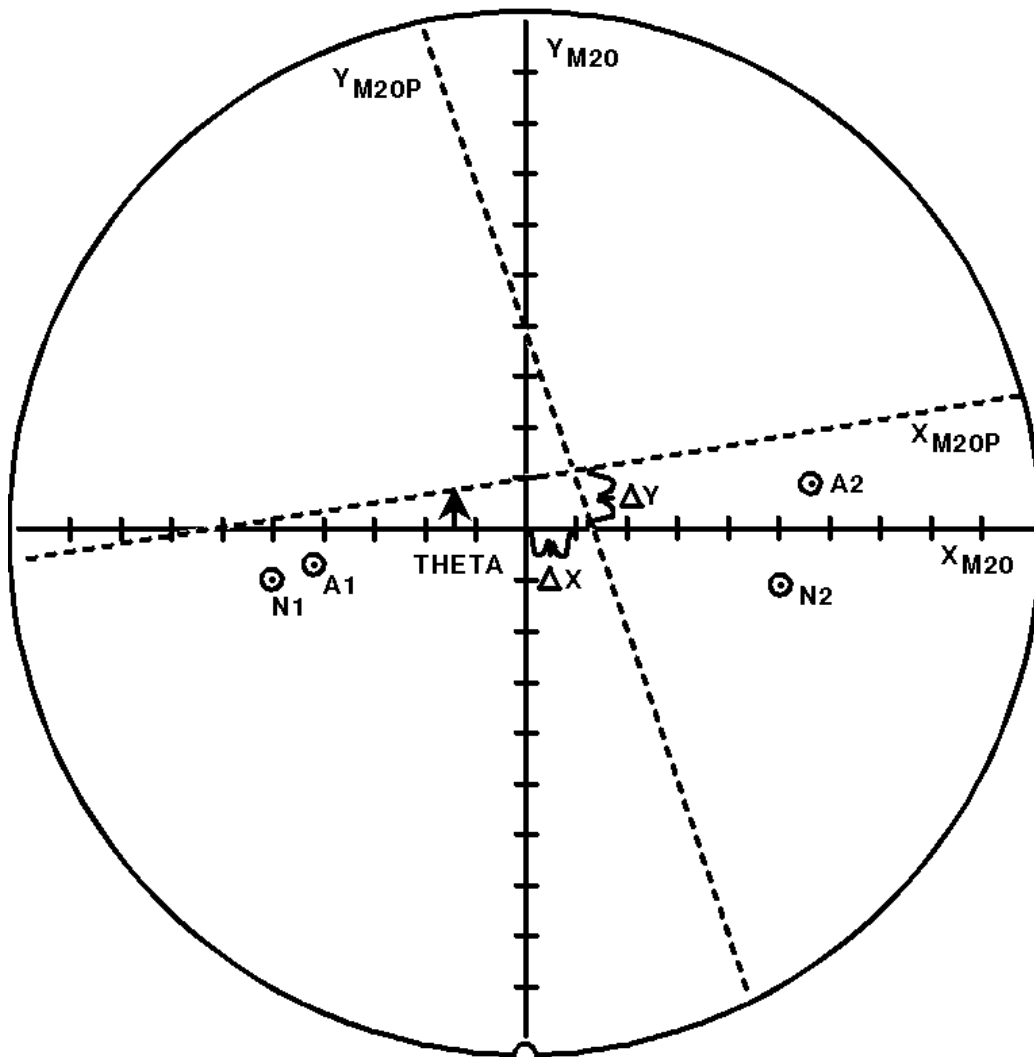
$$ScaleFactor = \frac{VA}{VN}$$

where VA and VN are the length of the vectors connecting the actual and nominal alignment sites, calculated using the formulas:

$$VN = \sqrt{(yN_2 - yN_1)^2 + (xN_2 - xN_1)^2}$$

$$VA = \sqrt{(yA_2 - yA_1)^2 + (xA_2 - xA_1)^2}$$

M20/M20P COORDINATE SYSTEMS EXAMPLE (EXAGGERATED)



N1 = -500000, - 100000
 N2 = 500000, -100000
 A1 = -420000, -70000
 A2 = 560000, 85000

THETA = 8.988°
 ΔX = 70312
 ΔY = 96472

Figure 7
Review Data Management

RELATED INFORMATION 2

APPLICATION NOTES

NOTE: The material contained in these Application Notes is not an official part of SEMI E30.1 and is not intended to modify or supersede the official standard. Rather, these notes describe possible methods for implementing certain ISEM requirements described by the standard and are included as reference material.

R2-1 Using ISEM Table Attributes to Specify Process Related Data Item Variable Values

R2-1.1 Section 11.1 (ISEM Table Data) allows the host to use ISEM Table attributes to specify product and process related information related to the table data. The ISEM Variable Item Dictionary (Table 4) includes seven data items intended to be used for this purpose. They are identified with the comment “This information may be added by the host in the ISEM Tables.” Identifying the value of variable data items in Table attributes is not covered in use of ISEM Tables. One method to accomplish this is to use attribute identifiers (ATTRID n) that are the same identifiers that are used for the equipment variable data items (Table R2-1). (This is very similar to the method specified in Section 12.5 to identify values to override the default values of variable process program parameters using the “PP-SELECT” remote commands).

Table R2-1 ISEM Variable Items and Their Equivalent ISEM Table Attribute Identifiers

Variable Item (Table 4)		ISEM Table Attribute Identifier	Description
Name	Type		
OperatorID	DV	OperatorID	Identification of the operator of the inspection/review equipment.
ProcessEquipmentID	DV	ProcessEquipmentID	Identification of the process equipment used with the current material immediately prior to the inspection/review.
ProcessEquipmentLocation	DV	ProcessEquipmentLocation	Location (code) of the process equipment used with the current material immediately prior to the inspection/review.
ProcessProgramID	DV	ProcessProgramID	Identification of the process program used with the process equipment used on the current material immediately prior to the inspection/review.
ProcessLevel	DV	ProcessLevel	Identification of the processing level of the current material.
ProductID	DV	ProductID	The product identification of the current material inspected/reviewed.
ProcessRunID	DV	ProcessRunID	Run identification for the process prior to current inspection/review.

NOTE 1: The variable item may be identified using any appropriate SECS II data item format.

R2-2 Example ISEM Table with Item Attributes That Specify ISEM Variable Values Using S13,F13 Table Data Send

R2-2.1 Typical values for Data Items are indicated.

L, 8

1. <DATAID>
2. <OBSPEC=null>
3. <TBLTYP=DefectData>
4. <TBLID=null>
5. <TBLCMD=1>

6. L, 2
 1. L, 2
 1. <ATTRID= "ProcessProgramID">
 2. <ATTRDATA= "My Recipe">
 2. L, 2
 1. <ATTRID= "ProcessLevel">
 2. <ATTRDATA= "My Level">
7. L, c
 1. <COLHDR1= "Insp_Anomaly ID">
 2. <COLHDR2= "Insp_Table specifier">
 3. <COLHDR3= "Insp_Coordinate X">
 4. <COLHDR4= "Insp_Coordinate Y">
 - :
 - // etc..as defined by Table 11.3.5
 - c. <COLHDRc>
8. L, r
 1. L, c
 1. <TBLELT11> // A[1..16]
 2. <TBLELT12> // I4
 3. <TBLELT13> // I4
 4. ETC...
 - :
 - c. <TBLELT1c> // A
 - :
 - r. L, c
 1. <TBLELT1r1>
 - :
 - c. <TBLELT1rc>

NOTICE: SEMI makes no warranties or representations as to the suitability of the standard set forth herein for any particular application. The determination of the suitability of the standard is solely the responsibility of the user. Users are cautioned to refer to manufacturer's instructions, product labels, product data sheets, and other relevant literature respecting any materials mentioned herein. These standards are subject to change without notice.

The user's attention is called to the possibility that compliance with this standard may require use of copyrighted material or of an invention covered by patent rights. By publication of this standard, SEMI takes no position respecting the validity of any patent rights or copyrights asserted in connection with any item mentioned in this standard. Users of this standard are expressly advised that determination of any such patent rights or copyrights, and the risk of infringement of such rights, are entirely their own responsibility.

SEMI E30.5-0302

SPECIFICATION FOR METROLOGY SPECIFIC EQUIPMENT MODEL

This specification was technically approved by the Global Information and Control Committee and is the direct responsibility of the North American Global Information and Control Committee. Current edition approved by the North American Information and Control Committee on October 14 and November 27, 2001. Initially available at www.semi.org December 2001; to be published March 2002. Originally published July 2001.

The complete specification for this product includes all general requirements of SEMI E30.

1 Purpose

1.1 This document establishes a Specific Equipment Model (SEM) for Metrology equipment (MSEM). The MSEM consists of equipment characteristics and behaviors that are applicable to this class of equipment and are required to be implemented in addition to the SEMI E30 fundamental requirements and additional capabilities.

2 Scope

2.1 The scope of this document is limited to defining the behavior of Metrology equipment as perceived by a SEMI Equipment Communications Standard II (SECS II/SEMI E5) host that complies with the SEMI E30 model. It defines the view of the equipment through the SECS II link. It does not define the internal operation of the equipment. It includes a specific processing state model as the basis for all equipment behavior of this class.

2.2 This document assumes that the SEMI E30 fundamental requirements and all additional capabilities except those noted in SEMI E30 Capabilities Section in this document have been implemented on the MSEM equipment. This document expands the SEMI E30 Standard requirements and capabilities in the areas of the processing state model, collection events, Process Program management, remote commands, data item variables, and coordinate systems.

2.3 This standard does not purport to address safety issues, if any, associated with its use. It is the responsibility of the users of this standard to establish appropriate safety and health practices and determine the applicability of regulatory limitations prior to use.

3 Limitations

3.1 The intent of this document is to facilitate the integration of Metrology equipment into an automated semiconductor factory. This document accomplishes this by defining an operational model for Metrology equipment as viewed by a factory automation controller. This definition provides a standard host

interface and equipment operational behavior. This document applies specifically to Metrology equipment as used in a semiconductor factory environment. It is possible that this methodology and techniques may apply to other industries.

3.2 MSEM job parameters that specify material (e.g., carrier ID and substrate ID) and material locations (e.g., carrier location ID and carrier slot ID) are intended for metrology equipment for 200 mm and smaller substrate.

4 Referenced Standards

NOTE 1: As listed or revised, all documents cited shall be the latest publications of adopted standards.

4.1 SEMI Standards

SEMI E5 — SEMI Equipment Communications Standard II (SECS II)

SEMI M20 — Specification for Establishing a Wafer Coordinate System

SEMI M21 — Assigning Addresses to Rectangular Elements in a Cartesian Array

SEMI E37.1 — High Speed Messaging Service (HSMS-SS) Single Session

SEMI E58 — Automated Reliability and Availability Standard (ARAMS)

5 Terminology

5.1 Definitions

5.1.1 *alignment, n.* — a procedure in which a coordinate system is established on a substrate or a portion of a substrate.

5.1.2 *alignment mark, n.* — a feature on a substrate selectively used for alignment.

5.1.3 *alignment site, n.* — a point within a feature on a substrate selectively used for alignment.

5.1.4 *cleanup, n.* — deselection of the current Process Program and removal of all material to output locations and any equipment specific activities required to transition the equipment into the IDLE state.

5.1.5 *factory automation controller, n.* — a computer system that provides integration of factory shop control and business systems with semiconductor equipment.

5.1.6 *feature, n.* — a distinctive item in a pattern, or a physical characteristic of the substrate. (e.g., line, point, a wafer flat).

5.1.7 *field, n.* — an exposure repeated in a regular manner on a substrate.

5.1.8 *global alignment, n.* — procedure which establishes a coordinate system for the entire substrate (see alignment). For silicon wafers, this coordinate system is defined in MSEM as the SEMI M20 coordinate system.

5.1.9 *global pattern alignment, n.* — a procedure which establishes a coordinate reference system relative to repeating features on an entire substrate. For silicon wafers, this coordinate system is defined in MSEM as the M20P coordinate system.

5.1.10 *logical port, n.* — one or more physical input or input/output ports that are controlled by the same execution of a Process Program.

5.1.11 *M20P, adj.* — a designation used for the global coordinate system defined within MSEM, that is established relative to a pattern on a silicon wafer.

5.1.12 *material, n.* — a piece or pieces of substrate, one or more substrate, a lot, a batch, or a run.

5.1.13 *metrology equipment, n.* — any equipment that collects and reports information on specific predetermined sites or features on a substrate with consistent data structure, or reports general information about the entire substrate.

5.1.14 *notch, n.* — a cut on the edge of a wafer that is commonly located with respect to a specific crystal plane that adheres to the SEMI M1 standard.

5.1.15 *pattern, n.* — the physical features on a substrate.

5.1.16 *pre-align, n.* — any alignment done prior to placing a substrate on a measurement process location.

5.1.17 *registration, n.* — positioning error between two features on different layers of a substrate.

5.1.18 *safe state, n.* — a state in which the equipment presents no danger to the product or user. This implies that safety interlocks are in place such that the equipment can be serviced without harm to the operator and that the material being processed has been removed from the processing station into an accessible location.

5.1.19 *secondary alignment, n.* — a procedure which improves the accuracy of the coordinate system

mapping on a substrate in a limited area of the substrate.

5.1.20 *site, n.* — a single point on a substrate used for alignment, or the center of an area of the substrate within which measurements are made.

5.2 Abbreviations and Acronyms

5.2.1 *GEM, n.* — Generic Equipment Model

5.2.2 *SEM, n.* — Specific Equipment Model

5.2.3 *SEM, n.* — Scanning Electron Microscope

5.2.4 *TCP/IP, n.* — Transmission Communication Protocol/Internet Protocol.

6 Communication Requirements

6.1 It is required that any MSEM compliant equipment follow the Communications State Model in SEMI E30. In addition MSEM compliant equipment shall support the High Speed Messaging Service (HSMS-SS) communication Standard, and the SECS-I standard for sending SECS II messages over TCP/IP or RS232. The user may determine which of these two lower level transmission protocols is used in each installation. The reason for HSMS-SS requirement is the large volume of data that can be generated by this class of equipment.

7 State Models

7.1 In this section are defined the equipment-specific processing state model and other state models necessary to portray the expected operational states of the equipment to enable host tracking and control in place of a local operator. See SEMI E30 and Other References section for additional information on state charts general rules and utilization.

7.2 *Processing State Model Requirements* — The Processing state model is required to be implemented as defined in the next sections of this document. A state model consists of the following: state model diagram, state definitions and a state transition table. A state model represents the host's view of the equipment, not necessarily the actual equipment operation. All MSEM state model transitions shall be mapped sequentially into the appropriate actual equipment events that satisfy the requirements of those transitions. In certain implementations, the equipment may enter a state and have already satisfied all of the conditions required by the MSEM state model for transition to another state. The equipment makes the required transition without any additional actions in this situation.

7.3 Some equipment may need to include additional states. Additional states may be added, but shall not change the MSEM defined state transitions. All expected transitions between MSEM states must occur.

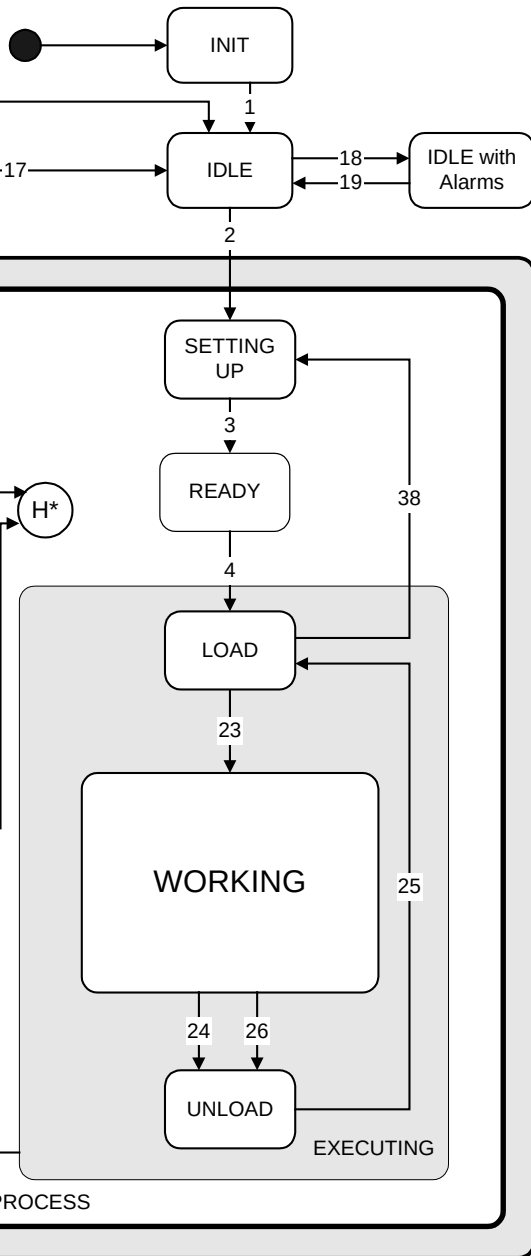


Figure 1
Generic MSEM Processing State Model Diagram

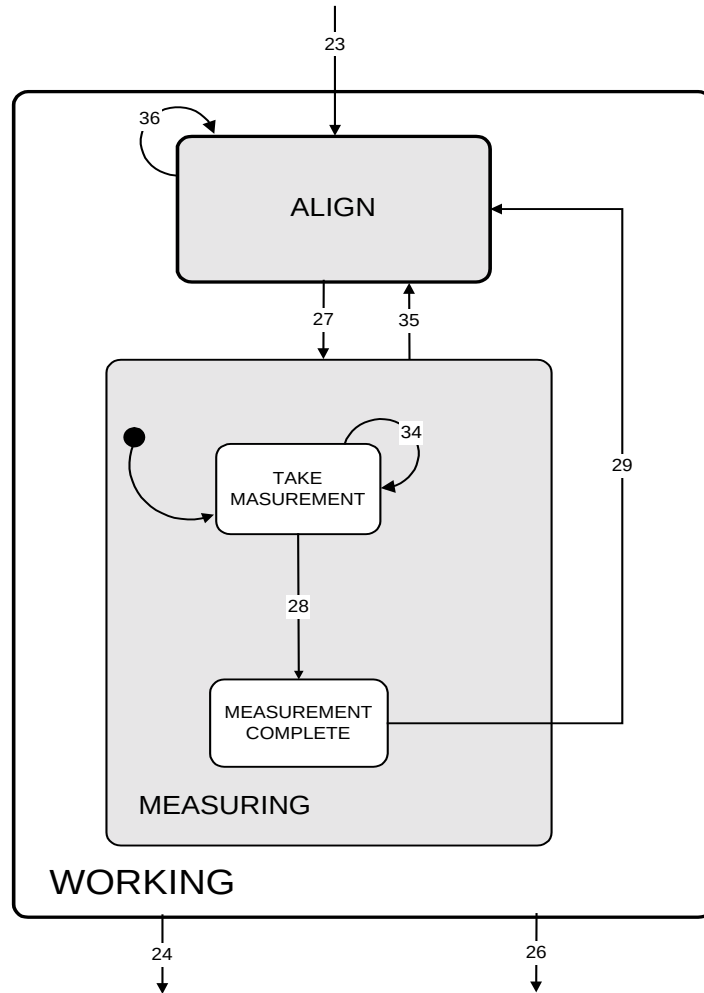


Figure 2
Working State of Processing State Model

7.4 Processing State Model Diagram

7.4.1 Working State of Processing State Model

7.4.2 Working Sub-states of Processing State Model — These states need not to be implemented in all Metrology equipment but if the equipment has the ability to multiple measurements at a site or provide raw scan data to the user this is how it is to be implemented.

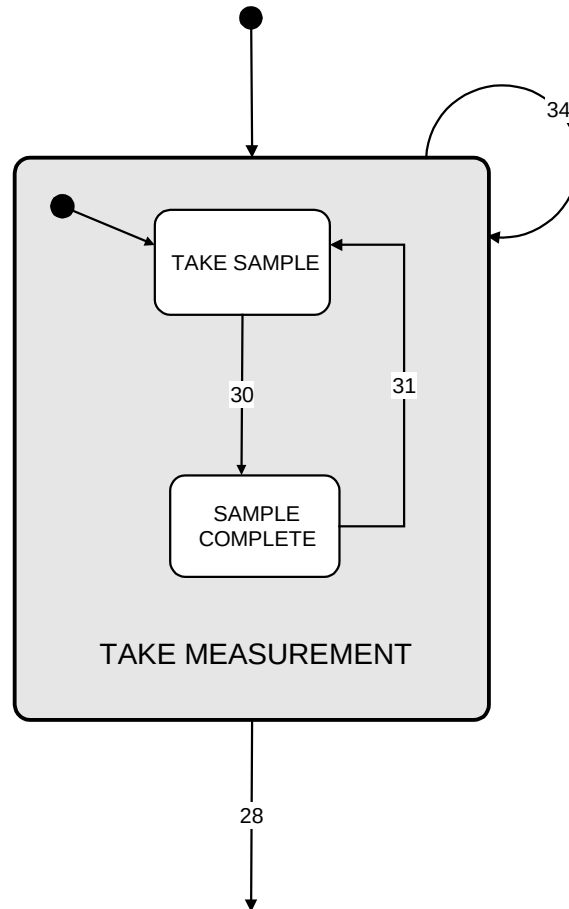


Figure 3
Working Sub-State of Processing State Model

7.5 Processing State Definitions

7.5.1 ABORTED — all activity is suspended as a result of an ABORT command. Any alarm and abort conditions must be cleared before exit from this state. The CLEANUP command is available to the operator or host to transition the equipment from the ABORTED state to IDLE state.

7.5.2 ABORTING (PROCESSING ACTIVE Sub-state) — the equipment has received an ABORT command. All normal activity is suspended. The equipment is taking appropriate action to put the equipment and material in a “safe state” where possible. Data may be invalid or not available.

7.5.3 ALARM PAUSED (PAUSE Sub-state) — an alarm has occurred in the Process or Process Pause states and the equipment is waiting for the alarm to be cleared.

7.5.4 ALIGN (WORKING Sub-state) — the equipment or operator is performing an alignment of the material to the equipment. Within this state the

equipment shall refine or establish its SEMI M20 coordinate system and establish any secondary coordinate systems required.

7.5.5 CHECKING (PROCESS PAUSE Sub-state) — the equipment verifies that the updates made to the Process Program are valid. This is a similar procedure to that which is done in SETTING UP before the equipment is ready to transition to the READY state.

7.5.6 EXECUTING (PROCESS Sub-state) — the equipment is processing material automatically and can continue to do so without external intervention, but normally may include interaction with the host or operator.

7.5.7 IDLE — checks for queued process or awaits a command. IDLE is free of ALARM and error conditions. Any transitions into this state will clear the process area.

7.5.8 IDLE with ALARMS — an alarm has occurred in the IDLE state and the equipment is waiting for all alarms to be cleared.

7.5.9 *INIT* — equipment initialization is occurring.

7.5.10 *LOAD (EXECUTING Sub-state)* — the equipment is determining if processing is complete. If not, then the substrate is being transferred to the equipment processing location, such as the stage. A pre-alignment procedure may be performed prior to the *LOAD* state.

7.5.11 *PAUSE (PROCESS ACTIVE Sub-state)* — *PROCESS* is suspended at the next opportunity. Actions to put the equipment in a “safe state” shall be performed. The equipment is awaiting a command (*RESUME*, *PP-UPDATE*, *STOP* or *ABORT*), or for alarm(s) to be cleared.

7.5.12 *PAUSED (PROCESS PAUSE Sub-state)* — *PROCESS* shall be suspended and the equipment is waiting for a command (*RESUME*, *PP-UPDATE*, *STOP* or *ABORT*).

7.5.13 *PAUSING (PROCESS PAUSE Sub-state)* — *PROCESS* has been suspended at the next opportunity and the equipment is put in a “safe state”.

7.5.14 *PROCESS (PROCESSING ACTIVE Sub-state)* — this state is the parent of those sub-states which refer to the active preparation and execution of a Process Program.

7.5.15 *PROCESSING ACTIVE* — this state is the parent of all sub-states where the context of a Process Program execution exists.

7.5.16 *PROCESS PAUSE (PAUSE Sub-state)* — the equipment is free of alarm conditions in the *PAUSE* state.

7.5.17 *MEASUREMENT COMPLETE (MEASURING Sub-State)* — the equipment has completed collecting data relative to an alignment location.

7.5.18 *MEASURING (WORKING Sub-State)* — the equipment is performing an action between alignments.

7.5.19 *READY (PROCESS Sub-state)* — the equipment is ready to begin processing and is awaiting a *START* command from the operator or host. If an

AUTOSTART is included, then the equipment starts processing immediately.

7.5.20 *SAMPLE COMPLETE (TAKE MEASUREMENT Sub-state)* — the equipment determines if additional samples need to be taken at this site.

7.5.21 *SETTING UP (PROCESS Sub-state)* — the equipment is being set up so that external conditions are satisfied to start processing the material. This includes the receipt of any Process Programs and material to be processed and their validation. Additional information may come from the host during the processing.

7.5.22 *STOPPING (PROCESSING ACTIVE Sub-state)* — the equipment has completed a Process Program or has been instructed to stop processing and shall do so gracefully at the next opportunity. All cleanup necessary is being completed within this state with regard to material, data, control system, etc. Data is normally preserved. Any alarm or error condition is cleared before exit from this state.

7.5.23 *TAKE MEASUREMENT (MEASURING Sub-state)* — the equipment is in the process of collecting data relative to an alignment location. (e.g. Site, Region, Substrate).

7.5.24 *TAKE SAMPLE (TAKE MEASUREMENT Sub-state)* — the equipment collects data from a single structure or of a single sample.

7.5.25 *UNLOAD (EXECUTING Sub-state)* — the substrate is being removed from the processing location.

7.5.26 *WORKING (EXECUTING Sub-state)* — the equipment is processing a specific material.

7.5.27 Processing State Transition Table

Table 1 Processing State Transition Table

<i>Transition #</i>	<i>Current state</i>	<i>Trigger</i>	<i>New State</i>	<i>Actions</i>	<i>Comments</i>
1	INIT	All equipment initialization is complete with no alarms or error conditions.	IDLE	None	If an alarm or error occurs during INIT, the equipment shall remain in this state.
2	IDLE	A process is queued or a command is received.	SETTING UP	The set up procedure is equipment dependent.	Commit has been made to set up.
3	SETTING UP	All setup activity has completed and the equipment is ready to receive a START command.	READY	The equipment is waiting for a START command. Start may be initiated by an operator.	The selected Process Program is available for execution.
4	READY	The equipment receives a START command.	LOAD	The equipment determines if processing is completed. If not, it transfers the next substrate to the processing location.	LOAD is an EXECUTING Substate.
5	EXECUTING	The processing is complete.	STOPPING	None	Equipment specific. Supplier must chose between LOAD or UNLOAD states for completion.
6	PROCESS	The equipment has received a STOP command.	STOPPING	The equipment unloads the material and brings the equipment to a clean and safe state.	Data is typically preserved and is valid.
7	PROCESS	The equipment has received an ABORT command	ABORTING	The equipment is put in a "safe state" if necessary.	Data may be invalid or not available.
8	PROCESS	An alarm occurs.	ALARM PAUSED	PROCESS activity is suspended and the equipment is waiting for all alarms to be cleared.	ALARM PAUSED is a PAUSE Substate.
9	PROCESS	The equipment has received a PAUSE command.	PAUSING	PROCESS shall be suspended at the next opportunity. Actions to put the equipment in a "safe state" shall be performed.	PAUSING is a PAUSE Substate.
10	PROCESS PAUSED	The equipment has received a RESUME command.	Previous PROCESS State	Proceed with the suspended Substate.	PAUSED is a PROCESS PAUSE Substate.
11	CHECKING	The equipment has completed validating any updates made to the current Process Program being executed. Including a PP_UPDATE	Previous PROCESS State	Action is appropriate to the state and the changes made to the Process Program updated.	None

<i>Transition #</i>	<i>Current state</i>	<i>Trigger</i>	<i>New State</i>	<i>Actions</i>	<i>Comments</i>
12	STOPPING	The equipment clean up is complete and the equipment is free of alarms.	IDLE	None	None
13	PAUSE	The equipment has received a STOP command.	STOPPING	The equipment proceeds with clean up.	Normally, data is preserved and is valid.
14	PAUSE	The equipment has received an ABORT command.	ABORTING	Any unsafe condition is resolved if possible.	Data may be invalid or not available.
15	STOPPING	The equipment has received an ABORT command.	ABORTING	Any unsafe condition is resolved if possible.	Data may be invalid or not available.
16	ABORTING	Unsafe conditions have been resolved where possible.	ABORTED	The equipment is waiting for alarm and ABORT conditions to be cleared.	The only state change allowed is to IDLE.
17	ABORTED	All alarms and abort conditions have been cleared.	IDLE	None	If needed the CLEANUP command clears the abort conditions. IDLE is a “clean” state.
18	IDLE	An alarm is set.	IDLE w/ ALARMS	The equipment waits for all alarms to be cleared.	None
19	IDLE w/ ALARMS	All alarms have been cleared.	IDLE	None	IDLE is free of alarms.
20	PAUSING	The equipment has achieved a “safe state”.	PAUSED	The equipment is waiting for a command (RESUME, STOP or ABORT).	None
21	PROCESS PAUSE	An alarm is set.	ALARM PAUSED	The equipment waits for all alarms to be cleared, or a STOP or ABORT command.	None
22	ALARM PAUSED	All alarms are cleared.	PAUSED	The equipment is waiting for a command (RESUME, PP_UPDATE, STOP or ABORT).	None
23	LOAD	Material transfer to processing location is complete and global alignment has been completed.	WORKING	The substrate is being processed.	None
24	WORKING	The processing of the specific material being processed ended normally.	UNLOAD	This material is transferred from the processing location.	“Normal” completion of the substrate.
25	UNLOAD	The material unload is complete.	LOAD	The equipment checks if processing is complete and, if not, transfers the next substrate to the processing location.	None

<i>Transition #</i>	<i>Current state</i>	<i>Trigger</i>	<i>New State</i>	<i>Actions</i>	<i>Comments</i>
26	WORKING	The processing of the specific material being processed ended abnormally.	UNLOAD	This material is transferred from the processing location.	Abnormal exit from WORKING or Next Material command received.
27	ALIGN	The material alignment is complete.	MEASURING	The equipment determines if additional sites are required.	None
28	TAKE MEASURE-MENT	All data collection has been completed for the current site or alignment location.	MEASURE-MENT COMPLETE	The equipment determine if additional sites are required.	Determine if additional alignments are required or unload is required.
29	MEASURE-MENT COMPLETE	Additional sites are required.	ALIGN	The equipment moves and aligns to new data collection site.	None
30	TAKE SAMPLE	Data collection from scan or structure.	SAMPLE COMPLETE	Determine if additional samples are required.	None
31	SAMPLE COMPLETE	Additional samples are required.	TAKE SAMPLE	Start collecting next scan or structure data.	None
32	CHECKING	Validation of Process Program change fails or is cancelled. Includes PP_UPDATE.	PAUSED	The equipment is waiting for a new PP-UPDATE or RESUME command.	Process Program reverts to the conditions that existed prior to the PP-UPDATE.
33	PAUSED	The equipment receives a PP-UPDATE command.	CHECKING	The equipment begins validating the changes made to the Process Programs to be executed.	None
34	TAKE MEASURE-MENT	Another measurement is required at an alignment location.	TAKE MEASURE-MENT	The equipment performs a measurement.	None
35	MEASURING	An error or failure occurred during the measurement and a new alignment is required.	ALIGN	The equipment moves and aligns to new data collection site.	None
36	ALIGN	An error or failure occurred during alignment and a new alignment is required.	ALIGN	The equipment moves and aligns to new data collection site.	None
37	ASSIST	A failure occurred while executing and external assistance was required to continue.	PROCESS PAUSE	The execution is stopped and equipment waits for a resume command.	Resume command must be issued.
38	LOAD	Previous PROCESS-BLD-GROUP program has completed and there are additional process programs assigned to the "CARRIERBLD". See Section 12.	SETTING UP	The equipment performs set up according to specifications of the next process program.	PROCESS-BLD-GROUP may include a AUTOSTART command within its body. Otherwise, the equipment waits for a START command.

Table 2 Processing State Events and Associated Reports

<i>CENAME</i>	<i>From State</i>	<i>To State</i>	<i>Required DVVAL's or Reports</i>
END of RUN	STOPPING	IDLE	RUN complete report available. All jobs specified by PROCESS-BLD-GROUP command parameter(s) in a CARRIERBLD command parameter complete.
END of JOB	LOAD	SETTING UP	JOB complete report available. A job specified by a PROCESS-BLD-GROUP command parameter in a CARRIERBLD command parameter that specifies multiple jobs for the same material completes.
END of SUBSTRATE	UNLOAD	LOAD	SUBSTRATE complete report available.
END of SITE	TAKE MEASURE	MEASURE COMPLETE	SITE complete report available. Only required if supporting sampling.
END of SAMPLE	TAKE SAMPLE	SAMPLE COMPLETE	SAMPLE complete report available. Only required if equipment performs measurements on individual sites.

8 Collection Event List

8.1 The purpose of this section is to identify data collection events for Metrology equipment and define reporting levels.

8.2 *Requirements* — Only those collection events that are not associated with a change of state, and those requiring specific DVVAL's or Reports defined in the SEM, are required to be included in this section. All SEMI E30 required events are required by this SEM.

8.2.1 *Common collection events* — collection events common to, and required on, all equipment of the class being addressed.

8.2.2 *Configuration-specific collection events* — collection events associated with a specific configuration of the equipment class being addressed.

8.3 *Collection Event Tables* — The first table contains processing state event transitions and associated reports. The second table contains additional events or actions and associated reports.

8.3.1 All remote commands (S2F41) that are not preformed by the equipment before responding to the host (S2F42) must generate an event indicating when the task was completed and whether it was successful.

Table 3 Additional Required Collection Events

<i>ACTION or COMPLETED EVENT</i>	<i>Required DVVAL's or Reports</i>
Remote command MAP-CASSETTE completed successfully.	SlotList
Operator hits button on equipment control panel.	OperatorAction

9 Data Item Variables

9.1 The purpose of this section is to define the list of data item variables pertinent to the specific equipment. Values of these variables shall be available to the host via collection event reports and host status queries.

9.2 *Requirement* — all generic variable data items defined in SEMI E30 are required by all Measurement equipment. Data item variables are categorized as follows:

9.2.1 *Common Variables (CV)* — variables common to all equipment of the class being addressed. These variables are covered in Table 4 and must be used to regardless of the measurement tool type.

9.2.2 *Configuration-specific variables (CSV)* — variables associated with a specific configuration of the above equipment class.

9.2.2.1 The following rules should be adhered to in the reporting of these variables.

- The information should be sent using the corresponding formats defined on Table 4 and 5.
- For ASCII variables A[n]. If the actual data does not fill the entire field, blanks shall be used to complete the field.
- For ASCII variables A[1..n]. All fields should be left justified.

9.3 Since MSEM is generic to all equipment it tries not to specify the names of the measurement variables to be sent to the host. MSEM does require that the data be reported at site, wafer and lot levels. It should include the option of reporting data at the individual scan level if supported by the equipment. It is a requirement that the equipment report at least: max, min, mean, standard deviation. Additional statistic calculations may be added by the equipment supplier. For each type of data collected by the equipment at the site level; the data must be summarized and available for reporting at the wafer and lot levels.

9.4 The following rules should be applied when reporting measurement values taken by the tool. The rules specify the units to be used whenever these values are reported.

- All values specified should be reported in floating point format.
- All critical dimensions (lines and spaces) should be in microns.
- All overlay measurements should be in microns.
- All FTIR transmittance values and corresponding wavelengths should be reported in percent units and microns, respectively.
- All film stress values should be reported in teradynes/cm-cm.
- All resistivity measurements should be reported in Ohm-cm.
- All film thickness measurements should be reported in Angstroms.

9.5 Data item variables should be documented in the MSEM data item variable dictionary using the following format:

<i>Variable Name</i>	<i>Category</i>	<i>Description</i>	<i>Class</i>	<i>Format</i>	<i>Comments</i>
----------------------	-----------------	--------------------	--------------	---------------	-----------------

Where:

<i>Variable Name:</i>	A unique name for the data item variable (this name is for reference only).
<i>Category:</i>	Defined as Common or Configuration-Specific variables.
<i>Description:</i>	If class is DVVAL, then the description must contain a statement of when data is valid in terms of MSEM events.
<i>Class:</i>	Data type of the item.
<i>Format:</i>	Acceptable formats are SEMI E5 lists, ASCII, floating point, unsigned integer or signed integer. A description of “ANY”, indicates that any of the above formats are acceptable and is left to the tool vendor to decide. When required use SECS Message Language format.
<i>Comments:</i>	Any additional information pertinent to the variable name.

9.6 Data Item Variable Types

9.6.1 *Equipment Constants (ECV)* — can be changed by the host using S2F15. The operator may have the ability to change some of the values, but the equipment does not change the values on its own. The value of the equipment constant may be queried by the host at any time using the S2F13/14 transaction.

9.6.2 *Status Variables (SV)* — are valid at all times. A SV may not be changed by the host but may be changed by the equipment or the operator. The value of status variables may be queried by the host at anytime using the S1F3/4 or S6F19/20 transactions.

9.6.3 *Data Variables (DVVAL)* — are variables which are valid only upon occurrence of specific collection events. An attempt to read a data variable at the wrong time shall not generate an error, but the data reported may not have relevant meaning.

9.6.4 *Variable Data (V)* — this is a class of variable data which includes all the previously defined types of variables.

Table 4 Data Item Variable List

<i>Variable name</i>	<i>Category</i>	<i>Description</i>	<i>Class</i>	<i>Format</i>	<i>Comments</i>
AlignList	CSV	A list of alignment site information being used by the currently active Process Program.	DVVAL	L,n 1.<AlignName> . . n.	See Table 5 for additional information.
CarrierID	CV	ID of the Carrier that the measurement data is associated with.	DVVAL	A[1..16]	Valid in processing substate.
Coord	CV	The X,Y coordinate for a site in microns.	DVVAL	L,2 1.<CoordX> 2.<CoordY>	See Table 5 for additional information.
Coordsys	CV	The identification for applicable coordinate system.	DVVAL	A[1..16]	Options for silicon wafers are: SEMI M20, M20P, SEMI M21.
Default-Priority	CV	The default priority given a location or carrier ID if none is assigned.	EC	U4	
DeltaX	CSV	The x axis translation between the origins of two coordinate systems, in μm .	DVVAL	F4	Units are in microns.
DeltaY	CSV	The y axis translation between the origins of two coordinate systems, in μm .	DVVAL	F4	Units are in microns.
ElementID	CSV	The M21 address for a specific rectangular element on a patterned silicon wafer.	DVVAL	I4[2]	M21 row number, M21 column number.
ElementList	CSV	A list of M21 elements where measurements were attempted.	DVVAL	L,n 1.<ElementID> . . n.	See Table 5 for additional information.
EquipID	CV	Unique ID of measurement equipment.	SV	A[1..16]	Valid in all sub states.
EquipName	CV	Name of equipment	SV	A[1..16]	Valid in all sub states.
LotID	CV	Lot ID that is associated with the measurement data.	DVVAL	A[1..80]	
M20Data	CV	The silicon wafer size, fiducial type, and orientation to use.	DVVAL	L,3 1.<WaferSize> 2.<Fiducial> 3. <Orientation>	See Table 5 for additional information.

<i>Variable name</i>	<i>Category</i>	<i>Description</i>	<i>Class</i>	<i>Format</i>	<i>Comments</i>
M21Data	CSV	The data necessary to establish an MSEM SEMI M21 layout on a silicon wafer.	DVVAL	L,2 1. L,3 1.<M21XSize> 2.<M21YSize> 3. <Tile> 2. L,n 1. L,3 1.<ElementID> 2.<CoordX> 3.<CoordY> : n	Coord x and Coord y are the X and Y coordinates in the M20P coordinate system of the lower left-hand corner of the element. See Table 5 for additional information.
QueuedJobList	CSV	An ordered list of CarrierID data items (A[1..16]) of material that is queued to run. The first item in the list is the ID of next carrier to run. A soon as a job specified by a PROCESS-BLD-GROUP starts, the CarrierID it specifies is removed from the head of the list.	SV	L	
Offset	CSV	The difference between the defined location of a site and the location at which it is found.	DVVAL	F4[2]	SiteDeltaX, SiteDeltaY
Operator Action	CV	Action taken by operator on equipment's front panel.	DVVAL	A[1..80]	
OperName	CV	Tool Operator name	ECV	A[1..16]	
Orientation	CV	The direction, in degrees, from the equipment's "0" location for a wafer's primary fiducial when initially positioned for measurements.	ECV	F4	This parameter has no effect on the "M20" based wafer coordinate system as discussed in SEMI standard SEMI M20.
PP-Name	CV	The PPID that is being used for measuring.	SV	A[1..80]	
Process SlotList	CV	The list of cassette slots whose contents are to be processed.	DVVAL	L,n 1. <SlotID> . . .	L,0 indicates all slots are to be processed. See Table 5 for additional information.
SiteDeltaX	CSV	The x axis translation between the defined and found locations of a site, in μm .	DVVAL	F4	Units are in microns.
SiteDeltaY	CSV	The y axis translation between the defined and found locations of a site, in μm .	DVVAL	F4	Units are in microns
SiteList	CV	The list of sites where measurements are made.	DVVAL	L,n 1.<SiteName> . . n	See Table 5 for additional information.
SiteName	CV	A unique identifier for a site.	DVVAL	A[1..16]	

<i>Variable name</i>	<i>Category</i>	<i>Description</i>	<i>Class</i>	<i>Format</i>	<i>Comments</i>
SlotID	CV	The cassette slot number.	DVVAL	U4	
WaferID	CSV	Physical wafer identifier.	SV	A[1..24]	
WaferIDRead	CSV	Tool read wafer identifier.	SV	A[1..24]	
WaferSize	CV	The nominal diameter of a silicon wafer.	EC	A[1..16]	mm
XlateData	CSV	Variable for the equipment to report the pattern-based coordinate system offset from the wafer-based coordinate system found on the wafer being tested.	SV	L,4 1 <DeltaX> 2 <DeltaY> 3 <Theta> 4 <ScaleFactor>	See Appendix 1 for an example. See Table 5 for additional information.

9.6.5 Data Item Sub-variable List

Table 5 Data Item Sub-Variable List

<i>Data Item Sub-variables</i>	<i>Description</i>	<i>Format</i>	<i>Comment</i>
AlignName	The identifier given to a alignment site.	A[1..16]	
CoordX	The x coordinate for a site.	F4	Units are in microns.
CoordY	The y coordinate for a site.	F4	Units are in microns.
DeltaX	The x axis translation between the origins of two coordinate systems, in μm .	F4	Units are in microns.
DeltaY	The y axis translation between the origins of two coordinate systems, in μm .	F4	Units are in microns.
ElementID	The M21 address for a specific rectangular element on a patterned silicon wafer.	I4[2]	M21 row number, M21 column number
Fiducial	The type of primary fiducial on a silicon wafer.	A[1..16]	Options are “FLAT” or “NOTCH”
M21XSize	The M21 element size in the x direction, in μm .	F4	Units are in microns.
M21YSize	The M21 element size in the y direction, in μm .	F4	Units are in microns.
ScaleFactor	The scaling factor required by the equipment to adjust from its SEMI M20 coordinate system to the coordinate system established through the use of pattern alignment “alignment site” information. In most cases, a scaling factor of 1 is expected.	F4	
SiteDeltaX	The x axis translation between the defined and found locations of a site, in μm .	F4	Units are in microns.
SiteDeltaY	The y axis translation between the defined and found locations of a site, in μm .	F4	Units are in microns.
SiteName	A unique identifier for a site.	A[1..16]	
THETA (Θ)	The clockwise rotation, in radians, between the SEMI M20 and M20P coordinate system axes.	F4	Radians
Tile	A flag to indicate whether the SEMI M21 layout is tiled, and in which direction.	A[1..16]	Options are: “NTILE” is untitled “CTILE” is column tiled “RTILE” is row tiled

10 Alarm List

10.1 Since each model of equipment differs in configuration, it is not practical to provide an exhaustive list of all possible alarms. Instead, the MSEM is requiring the two tables provided as described in SEMI E30 (Document Section). Alarm List Table which is intended to provide for equipment configuration specific alarms and Alarm ID, Alarm Set/Cleared Event Table.

10.2 Alarm List Table

10.2.1 The alarm list table contains examples of alarms that pertain to various configurational aspects of equipment. These examples are intended to illustrate that alarms pertain to situations in which there exists a potential for exceeding physical safety limits associated with people, equipment, and material being processed as per the SEMI E30 definition of an alarm. See SEMI E30 for further reference.

10.3 Alarm ID, Alarm Set/Cleared Event Table

10.3.1 The Alarm ID, Alarm Set/Cleared Event table documents the association of each ALID to a set and cleared event as required by SEMI E30. See SEMI E30 for further reference.

11 Process Program Management

11.1 *Requirements* — The MSEM requires that the SEMI E30 capability of Process Program Management be fully supported for this class of equipment. The MSEM is also requiring that the Process Program have a structure that enables the user to build Process Programs with default conditions that can be overridden for a run. MSEM is requiring the ability to vary; the quantity of substrates measured, the alignment information used and the number and/or location of the sites to be measured through the uses of Process Program variable parameters. The concepts of Process Program Structure and Process Program variable parameters are discussed in the following sections.

11.2 Process Program Structure

11.2.1 *Definition and Rules for Process Programs* — A Process Program contains information and/or instructions required for the metrology equipment to process a given run of material. Equipment constants can be used to supplement the information contained in a Process Program.

11.2.1.1 The Process Program shall supply all of the information required for a remotely executed run to be processed without operator intervention. Any information that is normally requested from the

operator console in manual operation shall have default values assigned in the body of the Process Program.

11.2.1.2 Process-program parameters are used to tailor a specific run of material and do not permanently modify the Process Program. They will remain in effect only until the next run or until the next PP-UPDATE or PP-SELECT, PP-ASSIGN remote command.

11.2.1.3 MSEM is requiring the ability to define specific Process Program variable parameters to; define what substrates are to be measured (Process SlotList), what sites are to be measured (SiteList or M21SiteList) and what aligns are used by the Process Program (AlignList or M21AlignList). If the MSEM equipment is using the SEMI M21 coordinate system then an additional Process Program variable parameter is required what elements are to be measured (ElementList).

11.2.2 *Definition of Process Program Variable Parameters* — A process-program parameter specifies a value that temporarily modifies the value of a variable parameter in a Process Program. A variable parameter is formally defined within a process-program body and contains:

- a variable parameter name that is unique in the body
- a parameter initial value, known as default value, for use when the Process Program is selected for execution without specification of an override value for this variable parameter.

11.2.3 The Equipment may also support the definition including:

- a parameter restriction that represents one or more conditions that any value specified for the parameter must satisfy.
- a CP-NAME in a remote command must be identical to a variable parameter name in the Process Program specified in the remote command. If the Equipment allows Sub-process Programs, a Sub-process Program reference may also specify parameters.

11.2.4 *Relationship of Process Program Variable Parameter* — The PP-SELECT, PP-ASSIGN or PP-UPDATE remote commands can be used to modify variables within the Process Program. The modification to process-program variables is done by using CP-NAME/CP-VAL pairs within the command.

11.2.5 Sub-Process Programs — Equipment may allow a main Process Program to reference Sub-process Programs. A main Process Program is one that can be referenced by the host. A Sub-process Program is a Process Program that is referenced by a main Process Program or by other Sub-process Programs. No Sub-process Program may reference its main Process Program. If the Equipment supports sub-process-program references, it must be possible for the host to determine which Sub-process Programs are referenced in all Process Programs.

11.2.5.1 Before execution of a main Process Program can begin, the presence of all the Sub-process Programs that it references must be verified by the Equipment. If they are not all present, an error collection event must occur. It must be possible to include in the event report which Sub-process Programs are missing. For a formatted Process Program, the error is reported using an S7F27 message.

11.2.6 Component Descriptions and Allowed Formats

Table 6 Process Program Component Description and Format Table

<i>Component Name</i>	<i>Description</i>	<i>Format</i>	<i>Comments</i>
ALIGN-ATTRIBUTE (n)	Tool specific information associated with alignment site for which no specific MSEM data item has been defined.		Examples include information such as magnification, voltage, current, wavelength, number of scans, integration time, or film stack. The equipment supplier shall document all attributes that are supported.
ALIGN-NAME	The identifier given to a alignment site.	A[1..16]	
COORDSYS	The identification for applicable coordinate system.	A[1..16]	Options for silicon wafers are: SEMI M20, M20P, SEMI M21
COORDX	The x coordinate for a site or the lower left hand corner of an element.	F4	Units are in microns.
COORDY	The y coordinate for a site or the lower left hand corner of an element.	F4	Units are in microns.
ELEMENTID	The element identifier in the SEMI M21 coordinate system	U4[2]	
M21XSIZE	The SEMI M21 element size in the x direction in μm .	F4	
M21YSIZE	The SEMI M21 element size in the y direction in μm .	F4	
SITE-ATTRIBUTE (n)	Tool specific information associated with a measurement site for which no specific MSEM data item has been defined.		Examples include information such as magnification, voltage, current, wavelength, number of scans, integration time, or film stack. The equipment supplier shall document all attributes that are supported.
SITENAME	A unique identifier for a site.	A[1..16]	

11.2.7 DEF-LIST-TABLE Description — DEF-LIST types of tables are used by the equipment in conjunction with the Process Programs. The DEF-LIST table structures provide a means to transfer to equipment a list of a particular type of information. The DEF-LIST table structures are used by the host to send information to equipment for later use in association with a Process Program. Each DEF-LIST that is sent has a unique Table name and each list element in the DEF-LIST has several components including a name (Site-Name, or Align-Name) as well as other components. A specific list row can be accessed for use with a specific Process Program by giving the DEF-LIST table name and the specific row name.

11.2.7.1 There are five types of DEF-LIST tables:

- SITE-DEF-LIST: A list of Sites and their attributes, typically the list of sites to be measured.
- ALIGN-DEF-LIST: A list of alignment sites and their attributes.
- M21-ELEMENT-DEF-LIST: A list of element ID and their low left hand location in M20P or SEMI M20 coordinates.
- M21-ELEMENT-SITE-DEF-LIST: A list of ElementID's and what sites are to be measured within that element.
- M21-ELEMENT-ALIGN-DEF-LIST: a list of ElementID's and the alignment sites to be used at each.

NOTE 2: The attributes are equipment-defined.

11.2.7.2 To support the SiteList, and AlignList, for SEMI M20, M20P and SEMI M21, the following types of tables must be implemented: SITE-DEF-LIST, ALIGN-DEF-LIST, or M21-ELEMENT-SITE- DEF-LIST, M21-ELEMENT-ALIGN-DEF-LIST and M21-ELEMENT-DEF-LIST. A host would send a DEF-LIST to equipment using standard transfer methods for tables as defined by ARAMS (SEMI E58). MSEM requires that the following columns be included in the DEF-LIST tables. The DEF-LIST-TABLE components table defines the DEF-LIST column names and their allowable format is given below.

Table 7 SITE-DEF-LIST

<i>Sitename</i>	<i>Coordx</i>	<i>Coordy</i>	<i>Coordsys</i>	<i>Site- Attribute(n)</i>

11.2.7.3 For SEMI M21 Coordinate system an additional table is required.

Table 8 M21-ELEMENT-SITE-DEF-LIST

<i>Element id</i>	<i>Sitename</i>

Table 9 M21-ELEMENT-DEF-LIST

<i>Element id</i>	<i>Coordx</i>	<i>Coordy</i>	<i>Coordsys</i>	<i>m21xsize</i>	<i>m21ysize</i>

Table 10 ALIGN-DEF-LIST

<i>Align-Name</i>	<i>Coordx</i>	<i>Coordy</i>	<i>Coordsys</i>	<i>Align- Attribute(n)</i>

Table 11 M21-ELEMENT-ALIGN-DEF-LIST

<i>Element id</i>	<i>Align-Name</i>

12 Remote Commands

12.1 The purpose of this section is to identify the MSEM required remote commands, command parameters, and valid commands versus MSEM states.

12.2 *Requirements* — The following capabilities are required:

- StartProcessing
- Stop Processing
- Temporarily Suspend Processing
- Resume Processing
- Abort Processing
- Select Process Programs, Material and/or Sites to Measure
- Report Location of Material Found within a cassette

12.3 All the remote commands defined by MSEM are required unless they have been qualified by the statement “if the equipment supports this functionality it shall use this command”. Then, they are only required if the equipment supports the functionality necessary to support the command. A good example of this is the MAP-CASSETTE command if the equipment does not have the hardware necessary to scan a cassette for the presence of substrates in slots then the command is not required by the MSEM. The alphanumeric strings defined by MSEM for RCMD and CPNAME are required.

12.4 Definitions

12.4.1 *Host Command Parameter (CPNAME/CPVAL)* — a parameter name/value associated with a particular host command (S2,F49). This document will specify unique names (CPNAMEs) and values (CPVALs) for many command parameters. Note that if there are no associated parameters a zero-length list is sent.

12.5 *Remote Commands Description* — this section describes required functionality, suppliers may implement additional commands. The following remote commands (RCMDs) must be supported as described below.

NOTE 2: The terms “current cycle” and “safe point” used below are to be defined by the supplier.

12.5.1 *ABORT* — terminate the current cycle prior to its completion. Abort has the intent of immediately stopping the process and is used because of abnormal conditions. Abort makes no guarantee about the subsequent condition of material except as noted in the CPNAME ABORT-LEVEL description.

12.5.2 *CLEANUP* — Process Program deselection, removal of all material to output locations and any equipment specific activities needed to transition into the IDLE state. Completion of this command should generate a collection event report.

12.5.3 *NEXT-MATERIAL* — processing of the current substrate is halted at the first safe point and unloaded to the target cassette location. Next-material has the intent of allowing the host to skip measurement of the current substrate. This is a trigger for processing state transition from WORKING to UNLOAD.

12.5.4 *PAUSE* — suspend processing temporarily at the next safe point. Pause has the intent of resuming the process at the same point where it was paused. RESUME or PP-UPDATE may be used to resume the process.

12.5.5 *PP-ASSIGN* — instructs the equipment to assign a process to a carrier ID(s)/location(s) and place the exchange station in the process queue with the given priority. Only one assignment is allowed for a exchange station. Without specifying a priority, the material is queued with the default priority. Material with equal priority are queued in the order the PP-ASSIGN commands are received. If the equipment supports the functionality of queuing material, it will use this command.

12.5.6 *PP-SELECT* — instructs the equipment to make the requested Process Program(s) available in the execution area. Additionally, to reduce the number of Process Programs on the equipment, PP-SELECT may define the material to be measured, measurement site locations, and/or the information needed for site alignment; default values shall be used if this information is not specified. This is a trigger for the processing state transition from IDLE to SETTING UP. All Process Programs specified in the command are to be validated

12.5.7 *PP-UPDATE* — provides the ability to alter Process Program variables during the PAUSED processing state. Any CPNAMEs specified in PP-UPDATE will replace the previous definitions. This command will RESUME the process. If the PP-UPDATE fails, the Process Program variables present prior to the PP-UPDATE are retained. If no parameters values are specified, the defaults are used.

12.5.8 *RESUME* — resume processing from the point where the process was paused. This is the trigger for processing state transition 10, from PROCESS PAUSE.

12.5.9 *START* — instructs the equipment to initiate processing. This is the trigger for the processing state transition from READY to LOAD.

12.5.10 *STOP* — complete the current cycle, stop in a safe condition and return to the IDLE processing state. Stop has the intent of stopping the process entirely. The equipment is not required to support the continuation of processing.

12.6 Host Command Parameters Names (CPNAME)

Table 12 Host Command Parameter Table

CPNAME	CPVAL		
	Description	Range	Format
CARRIERBLD	L,2 1. L,2 1. CP-CARRIERID ** 2. CARRIERID or 1. L,2 1. CP- LOCATIONID ** 2. (STORAGE)ID 2. L,N 1. L,2 1. PROCESS-BLD-GROUP 2. L,M : N. L,2 1. PROCESS-BLD-GROUP ** Both CP-CARRIERID and CP-LOCATION may be included if no carrier ID reader is available.		
CP-ABORT-LEVEL	MSEM defined abort levels HALT - halt, goto ABORTED CLEANUP - halt, preform cleanup procedure, goto ABORTED	“HALT” “CLEANUP”	A[7]
CP-ALIGNLIST	L,n 1. CP-ALIGNNAME : n. For the SEMI M20 or M20P Coordinate system		
CP-ALIGN-DEF-LIST	As defined in Section 9		
CP-ALIGNNAME	Alignment name		A[1..80]
CP-AUTOSTART	Defines if a START command is required from an external source (operator or host) to exit the READY state. 0 = START command required, 1 = AUTOSTART no external START command required to begin execution.	0–1	U1
CP-CARRIERID	ID of carrier that the measurement data is associated with.		A[1..16]
CP-ELEMENTID	The M21 address for a specific rectangular pattern on the wafer.		I4[2]

CPNAME	CPVAL		
	Description	Range	Format
CP-ELEMENTLIST	L,n 1. <CP-ELEMENTID> : n.		
CP-LOCATIONID	Unique identifier of the location to be used for the “CARRIERBLD” assignment.		U4
CP-LOTID	lot id		A[1..80]
CP-MATERIALLIST	L,n 1. <CP-SLOTID or CP-WAFERID> : n.		
CP-M21-ALIGNLIST	L,n 1. CP-ALIGNNAME : n. For the SEMI M21 Coordinate system		
CP-M21-SITELIST	L,n 1. CP-SITENAME : n. For the SEMI M21 coordinate system		
CP-PPNAME	Process Program name		A[1..80]
CP-PRIORITY	assignment priority	0–9 0–highest priority	U4
CP-SITELIST	L,n 1. CP-SITENAME : n. For the SEMI M20 or M20P coordinate system		
CP-SITE-DEF-LIST	As defined in Section 9.		
CP-SITENAME	unique identifier for a site		A[1..16]
CP-SLOTID	slot number	1–n	U4
CP-SLOTLIST	Specifies carrier slots that contains substrate to measure. L,n 1. CP-SLOTID : n.	0 indicates all slots	
CP-WAFERID	Physical wafer identifier		A[1..24]

CPNAME	CPVAL		
	Description	Range	Format
PROCESS-BLD-GROUP	L, n 1. L, 2 1. CP-PPBUILDID 2. PPBUILDGROUPID 2. L, 2 1. CP-PPNAME 2. PPNAME 3. L, 2 1. CP-LOTID 2. LOTID 4. L, 2 1. CP-AUTOSTART 2. START 5. L, 2 1. CP-MATERIALLIST 2. L, n 1. CP-SLOTID : n. or 1. CP-WAFERID 2. L, n : n.	$n \geq 2$	List of n data items

CPNAME	CPVAL		
	Description	Range	Format
PROCESS-BUILD-GROUP(continues)	6. L, 2 1. CP-SITE-DEF-LIST 2. SITE-TABLE-NAME 7. L, 2 1. CP-SITELIST 2. L, n 1. CP-SITENAME : n. 8. L, 2 1. CP-ALIGN-DEF-LIST 2. ALIGN-TABLE-NAME 9. L, 2 1. CP-ALIGNLIST 2.L, n 1. CP-ALIGN-NAME : n. 10. L, 2 1. CP-ELEMENTLIST ** 2. L, n 1. CP-ELEMENTID : n. **CP-ELEMENTLIST is required when using the SEMI M21 coordinate system in the definition of an ALIGNNAME or SITENAME . 10. L, 2 1. CPNAME supplier defines Process Program Variable Parameters 2. CPVAL CP-PPNAME is required, CP-SLOTLIST, CP-SITELIST, and CP-ALIGNLIST are optional.		

12.7 Remote Commands and Associated Host Command Parameters — This table describes the allowable command parameters (CPNAME) for each remote command (RCMD). Equipment must support all parameters. The column marked Req/Opt, specifies which parameters are required to be sent by the host and which parameters may be optionally sent by the host.

Table 13 Remote Command and Associated Host Command Parameters Table

Remote Command	Parameter(s)		
	CPNAME	Req/opt	Description
ABORT	CP-ABORT-LEVEL	O	
CLEANUP	CP-LOCATIONID	O	Location and SLOT may be used to define a different cassette / slot destination for the substrates.
	CP-SLOTID	O	
NEXT-MATERIAL	CP-LOCATIONID	O	If no command parameters are specified, then the substrate is returned to its original source carrier and slot. If a CP-SLOT command parameter is specified, then the substrate is placed in that slot of its original source carrier. If a CP-LOCATIONID command parameter is specified, then the substrate is placed in an available slot of the carrier at the specified location. If both CP-SLOT and CP-LOCATION command parameters are specified, then the substrate is placed in specified slot of the carrier at the specified location.
	CP-SLOTID	O	
PAUSE	None	N/A	None
PP-ASSIGN	CP-PRIORITY	O	* More than one CARRIERBLD may be specified. There may be more than one PROCESS-BLD-GROUP in each CARRIERBLD.
	CARRIERBLD*	R	
PP-SELECT	CARRIERBLD*	R	* More than one CARRIERBLD may be specified. There may be more than one PROCESS-BLD-GROUP in each CARRIERBLD.
PP-UNASSIGN	CP-LOCATIONID CP- CARRIERID	R* R*	* At least one is required.
PP-UPDATE	CP-SLOTLIST	R*	* At least one is required. By default use the current active input port.
	CP-SITELIST	R*	
	CP-ALIGNLIST	R*	
RESTAGE	CP-LOCATIONID	R*	* Some equipment may not support this feature.
RESUME	None	N/A	None
START	None	N/A	None
STOP	None	N/A	None

12.8 Remote Command and Processing State Relationship

Table 14 Remote Commands versus Processing States Table

Command									
ABORT									
CLEANUP									
PP-ASSIGN									
NEXT-MATERIAL									
PAUSE									
PP-SELECT									
PP-UPDATE									
RESUME									
START									
STOP									
PROCESSING STATE									
IDLE					X			X	
PROCESSING ACTIVE									
PROCESS									
SETTING UP	X					X		X	X
READY	X	X				X		X	X
EXECUTING								X	
LOAD	X					X		X	X
WORKING									
ALIGN	X					X	X	X	X
MEASURING	X					X	X	X	X
TAKE MEASURE	X					X	X	X	X
TAKE SAMPLE	X					X	X	X	X
SAMPLE COMPLETE	X					X	X	X	X
MEAS COMPLETE	X					X	X	X	X
UNLOAD	X					X		X	X
STOPPING								X	X
PAUSE									
PROCESS PAUSE									
PAUSING	X							X	X
PAUSED	X		X	X				X	X
CHECKING	X							X	X
ALARM PAUSED	X							X	X
ABORTED								X	X

13 Scenarios

13.1 The purpose of this section is to document any MSEM specific scenarios that must be performed by this class of equipment.

13.2 Normal Reporting

COMMENTS	HOST	EQUIPMENT	COMMENTS
			The equipment is in the IDLE processing state and in the ONLINE REMOTE control state. Material arrives on input port 1.
PP-ASSIGN	S2,F63-->	<--S2,F64 <--S6,F11	Positive Acknowledge. IDLE -> SETTING UP
Positive Acknowledge.	S6,F12-->	<--S6,F11	SETTING UP -> READY
Positive Acknowledge. START	S6,F12--> S2,F41-->	<--S2,F42 <--S6,F11	Positive Acknowledge. READY -> LOAD.
Positive Acknowledge.	S6,F12-->		[WHILE] Not End of Run 1) LOAD->WORKING 2) Sample and/or Site level data reporting. 3) WORKING->UNLOAD 4) Substrate level data reporting. 5) UNLOAD->LOAD. [ENDWHILE]. LOAD->STOPPING.
Positive Acknowledge.	S6,F12-->	<--S6,F11	Run level data reporting. STOPPING->IDLE.
Positive Acknowledge.	S6,F12-->	<--S6,F11	

13.3 ABORT/ CLEANUP Remote Command Scenario —Host issues the ABORT/CLEANUP remote command

COMMENTS	HOST	EQUIPMENT	COMMENTS
ABORT	S2,F41-->	<--S2,F42 <--S6,F11	("CLEANUP" not specified) Positive Acknowledge. Transition to ABORTED.
Positive Acknowledge. CLEANUP	S6,F12--> S2,F41-->	<--S2,F42 <--S6,F11	Positive Acknowledge. Equipment performs cleanup activities. CEID is posted. [IF] cleanup is successful: Transition to IDLE [ELSE] Remain in the ABORTED state. [ENDIF].
Positive Acknowledge.	S6,F12-->		

13.4 PP-UPDATE Remote Command Scenario — Host issues the PP-UPDATE remote command.

COMMENTS	HOST	EQUIPMENT	COMMENTS
START	S2,F41-->	<--S2,F42 <--S6,F11	Positive Acknowledge. READY -> LOAD
Positive Acknowledge.	S6,F12-->		[WHILE] Not End of Run 1) LOAD->WORKING 2) WORKING->UNLOAD 3) UNLOAD->LOAD [ENDWHILE]
Sometime during the [WHILE]:			
PAUSE	S2,F41-->	<--S2,F42 <--S6,F11	Positive Acknowledge. Transition to PAUSING
Positive Acknowledge.	S6,F12-->	<--S6,F11	PAUSING -> PAUSED
Positive Acknowledge.	S6,F12-->		
PP-UPDATE	S2,F41-->	<--S2,F42 <--S6,F11	Positive Acknowledge. PAUSED->CHECKING
Positive Acknowledge.	S6,F12-->	<--S6,F11	CEID is posted. [IF] the updates are valid: Return to the previous process state thru history [ELSE] Return to the PAUSED state. The Process Program remains unchanged. [ENDIF]
Positive Acknowledge.	S6,F12-->		

13.5 PP-ASSIGN / PP-UNASSIGN Scenario — Host issues the PP-ASSIGN/PP-UNASSIGN remote commands.

COMMENTS	HOST	EQUIPMENT	COMMENTS
PP-ASSIGN	S2,F41-->	<--S2,F42	Positive Acknowledge.
PP-UNASSIGN	S2,F41-->	<--S2,F42	Positive Acknowledge. This may occur during the STOPPING state.

13.6 The host may setup the material for additional processing without the cassette being removed and replaced on the equipment.

14 SEMI E30 Capabilities

14.1 The purpose of this section is to specify any SEMI E30 Capabilities required by MSEM class equipment.

14.2 *Requirement* — The following SEMI E30 additional capabilities required by MSEM are:

- Dynamic Event Report Configuration
- Variable Data Collection
- Status Data Collection
- Alarm Management
- Remote Control
- Equipment Constants
- Process Program Management
- Spooling
- Trace Data Collection (optional)
- Control (Host Initiated)

15 Related Documents

Harel, D., “Statecharts: A Visual Formalism for Complex Systems,” *Science of Computer Programming* 8 (1987) 231-274.

NOTICE:. SEMI makes no warranties or representations as to the suitability of the standards set forth herein for any particular application. The determination of the suitability of the standard is solely the responsibility of the user. Users are cautioned to refer to manufacturer’s instructions, product labels, product data sheets, and other relevant literature respecting any materials mentioned herein. These standards are subject to change without notice.

The user’s attention is called to the possibility that compliance with this standard may require use of copyrighted material or of an invention covered by patent rights. By publication of this standard, SEMI takes no position respecting the validity of any patent rights or copyrights asserted in connection with any item mentioned in this standard. Users of this standard are expressly advised that determination of any such patent rights or copyrights, and the risk of infringement of such rights, are entirely their own responsibility.

RELATED INFORMATION 1

SEM UNIQUE CAPABILITIES

NOTE: This related information is not an official part of SEMI E30.5 and was derived from work developed in the Metrology Specific Equipment Model Task Force in North America. This related information was approved for publication by full letter ballot on April 30, 2001.

R1-1. *Measurement Site Location* — Metrology equipment most often is used to make measurements and report results at specific sites on a substrate. Unfortunately, equipment suppliers and users have adopted different formats for describing measurement site location information. This has led to three problems which increase the cost of metrology. First, metrology equipment suppliers must provide formats which meet conflicting requirements of their customers, adding to equipment development costs. Second, users must transpose information from different systems to their own format in order to use metrology data. Third, site location information is often an integral part of recipe set up, often requiring that an actual product sample be available for “training” the site location information on the equipment. For users who manufacture many different products on common substrates, multiple metrology “recipes” must be developed where the only differences are site location information. This information is known to the user from product design data, and should not need to be “learned” uniquely on various metrology equipment.

R1-1.1 In order to avoid these problems, MSEM defines specific formats for identifying and reporting site location information on substrates. The MSEM-required formats are intended to minimize the number and type of site location format transformations needing to be supported by both metrology equipment suppliers and users. All MSEM-required site location formats involve the use of an MSEM-defined right-handed Cartesian coordinate system, established on substrates in an MSEM-defined manner. This release of MSEM defines these only for silicon wafer substrates, based on SEMI standard M20. Additional substrate types may be included in future revisions of MSEM, if required.

R1-1.2 MSEM requires that equipment have the capability to use site location information that is based on the user's product designs, which the user must provide in the appropriate MSEM-required format. In other words, equipment shall not require that a sample substrate be used to “train” site locations when users can provide this information from product design data.

R1-1.3 MSEM-compliant equipment shall have the capability to define, locate, measure, and report site information using only the MSEM-defined

right-handed Cartesian coordinate system formats. This requirement does not preclude equipment from having additional capability for defining or reporting site location information using other formats. One such additional format is defined in MSEM for patterned silicon wafer substrates, based on SEMI standard SEMI M21. MSEM-compliant equipment is not required to have this “M21” format capability, but must use the MSEM “M21” format if it is provided.

R1-1.4 Specific MSEM information that defines site location information includes ; the data items AlignList and SiteList; the CPNAMEs CP-ALIGNLIST and CP-SITELIST and the TABLE type named ALIGN-DEF-LIST and SITE-DEF-LIST. Notice that multiple sites can first be Defined using the appropriate Process Program class named ALIGN-DEF-LIST or SITE-DEF-LIST, then selected by using the ALIGN-NAME or SITENAME in an MSEM defined remote command that use CP-ALIGNLIST AND CP-SITELIST. This information is similarly defined for the “M21” coordinate system plus an additional CPNAME of ELEMENTLIST which lists the ElementIDs to be measured on the silicon wafers.

R1-2 *Coordinate Systems For A Silicon Wafer*

R1-2.1 *Requirements* — MSEM defines two right-handed Cartesian coordinate systems for use on silicon wafers. These are identified as the “M20” and “M20P” coordinate systems. Both are based on SEMI standard M20. The SEMI M20 standard describes how to map a right-handed Cartesian coordinate system to a wafer so that its origin is at the center of the wafer, and its negative y-axis bisects the wafer's primary fiducial. This coordinate system is defined by MSEM to be the “M20” coordinate system. MSEM defines the “M20P” coordinate systems to be one which is aligned to the pattern on the wafer. Ideally, there is no difference between the “M20P” and “M20” coordinate system. In the real world, there is a difference, due to experimental errors. This is explained further in the following sections.

R1-2.2 *Implementation* — The only information required by equipment in order to establish an “M20” coordinate system is the wafer size and type of fiducial. This is provided by way of the MSEM data items named WaferSize and Fiducial. Another data item named Orientation provides for control over how the wafer is loaded on equipment. Note that the SEMI

M20 standard requires that the “M20” coordinate system is fixed on the wafer, and is not affected by how the wafer is loaded on equipment. Also, as stated in the SEMI M20 standard, an orientation of “0” is for a wafer loaded on equipment with the primary fiducial towards the operator or “down”.

R1-2.2.1 Often, no other information is required for measurements to be made on a wafer. However, the “M20” coordinate system may not provide sufficient accuracy to locate measurement sites on patterned wafers. This is a result of any of a number of possible reasons, all of which can be described as “experimental errors”. For instance, wafers are not perfectly round, fiducial dimensions may vary, and equipment (both steppers that place patterns on wafers and metrology equipment) capabilities to determine the location of the wafer center and fiducial vary.

R1-2.2.2 In order to deal with these errors, metrology equipment suppliers have developed various strategies, as discussed in the introduction section. Perhaps because the SEMI M20 standard's reference to “origins” of “other coordinate systems” is in the same sentence with “reference points”, many suppliers adopted the location of one (or more) reference point(s) as the origin for their pattern-based coordinate system(s). Without a common reference site, multiple coordinate systems resulted. More nefarious strategies were developed to compensate for “stage errors” on equipment. These usually entail a series of reference points required to “zero in” on the measurement site, and this made comparison of locations determined on different equipment next to impossible.

R1-2.2.3 MSEM defines the “M20P” coordinate system to be one which is identical to the “M20” coordinate system, if there were no experimental errors. In other words, the location of the origin and axes of the “M20P” coordinate system are offset slightly from the origin and axes of the “M20” coordinate system because of experimental errors. If the location of reference points, called “alignment sites” in MSEM, is defined to be the “M20” coordinates where they are “expected” to be, and equipment is designed to be able to “find” the alignment sites, given the various possible experimental errors, the “found” and “expected” SEMI M20 coordinates can be used to determine the “M20-to-M20P” coordinate system transformation, as explained in appendix 1. Alignment site information is specified through the use of the MSEM data items named AlignList, the CPNAME CP-ALIGNLIST and, the table type named ALIGN-DEF-LIST.

R1-2.2.4 Most metrology equipment cannot distinguish whether patterned wafer site location errors are due to the wafer, the layout on the wafer, or the equipment's ability to locate the sites. However, information that is

available through the use of patterned-wafer alignment sites can provide a means for identifying potential metrology equipment problems. For instance, assume that the only pattern-layout location error on a wafer is that due to the establishment of the location of the wafer center and fiducial. For many users and metrology systems, this is a good assumption. If this is the case, then the MSEM data item named XlateData can be used to track this error. Although the error may result from multiple sources, being able to track it on various metrology equipment will enable users to apply statistical process control techniques to identify the specific sources.

R1-2.2.5 Sites specified for measurements or additional alignments may be found by metrology equipment at locations which deviate from their expected locations through either pattern layout errors or metrology equipment “stage” errors. Again, in a controlled manufacturing process, these combined errors should be normally distributed, and non-normal deviations may indicate possible metrology equipment problems. These types of errors shall be reported through the use of the MSEM data item named Offset.

R1-2.2.6 A minimum of two alignment sites are necessary to establish an “M20P” coordinate system on a wafer. Additional sites are often used, as in the example given in appendix 1. The system supplier shall document the requirements for ALIGNLIST information, and detail how any “site-by-site” alignment sites (that is, those needed to obtain better location accuracy in the neighborhood of measurement site) are associated with the specific SITELIST measurement sites.

R1-2.2.7 Alignment site location information may be either provided to or by equipment. Some equipment have the capability to find and report the locations of “reference sites” without prior knowledge of their locations on the wafer. In this case, the equipment shall report the location of both alignment and measurement sites in “M20” coordinates. When alignment site information is provided to equipment to establish an “M20P” coordinate, measurement site locations shall be reported in the “M20P” coordinate system.

R1-2.2.8 MSEM encourages users to define “M20P” locations in reference to the user's device layout design within an ideal SEMI M20 wafer coordinate system. An alternative is to define “M20P” locations by “training” on a “benchmark” system.

R1-3 *Layout of Rectangular Elements on a Silicon Wafer* — Equipment shall be capable of routine, automated operation without needing wafer layout information (e.g., field or die maps). However, having the capability to provide wafer layout information to

metrology equipment from the host can be desirable. In some cases, the amount of information required to define measurement site locations is less if a pattern-element-based format is used. MSEM defines a means to do this in this section, if desired, based on SEMI standard SEMI M21.

R1-3.1 SEMI standard SEMI M21, “Specification for assigning addresses to rectangular elements in a Cartesian array”, is limited (for MSEM purposes) by the fact that nothing is specified about how the rectangular elements are located on the wafer. The SEMI M21 standard details how to assign “addresses” to elements and how to find the “array center” element. In this section, MSEM defines how these elements are located on a wafer, using the data item named “M21Data”, and how to establish within-element coordinate systems.

R1-3.2 There are users who want or require much more layout information than is provided for by SEMI M21, such as within-element structure details or element attribute information. This additional layout information is beyond the scope of MSEM, since it is considered to be information which only is needed to aid operator-interactive use of metrology systems.

R1-3.3 *MSEM “M21” Layout* — The first pattern element layout issue to be addressed is that of determining which SEMI M21 elements are to be included in the MSEM “M21” layout. MSEM defines the “M21” layout to include all elements which either wholly or partially are within the circumference of the wafer. Thus the MSEM “M21” element layout does not correspond to the pattern layout exactly, since some “M21” elements may not contain patterns.

R1-3.4 The second layout issue is that of how best to specify the element locations on the wafer. The MSEM approach is to specify the “M20P” coordinate for the lower left corner of the minimum number of elements needed to define the layout, along with the element addresses. For a non-tiled layout, the location of a single element is sufficient to establish the “M21” layout. For tiled layouts, the location of one element in each row or column is required. Note that the location of the lower left corner of an element may be outside the circumference of the wafer.

R1-3.5 Layout definition is supported only for host-to-equipment communications. The user is

responsible for ensuring that the element addresses provided to the equipment agree with the SEMI M21 specification. The equipment need not check this, other than to ensure that there are not conflicts within the provided layout, and shall report results with element addresses as provided by the user.

R1-3.6 M21 layouts are established within the “M20P” coordinate system, and need not require any additional alignment site data than is needed to establish the “M20P” coordinate system. However, as with “M20P”, additional alignment may be necessary because of errors in either the pattern layout or the equipment's ability to locate features. OFFSET shall be used to report the location corrections that result from any within-element alignments.

R1-3.7 MSEM provides a means for element-based coordinate systems, if required. This capability is provided by the option of specifying “M21” as the coordinate system for ALIGN-DEF-DATA and SITEDEFDATA items. For element-based site locations, MSEM requires that SEMI M21 element coordinate systems have x and y axes parallel to the respective M20-based coordinate system axes, with their origins at the lower left corner of each element.

R1-4 *An Example of How an M20P Coordinate System is Established on a Silicon Wafer*

R1-4.1 The following example is fairly basic. For this example, the equipment does M20P alignment via a repeated two-step process. The first step is done at a low resolution, the second at a high resolution, and the process is done at two positions on the wafer.

R1-4.2 The equipment documentation states that 4 alignment sites are required. These are defined to the equipment via the table type named ALIGN-DEF-LIST, as detailed below. The order of the sites in ALIGN-DEF-LIST is not important. The sites are then selected via the CPNAME item named CP-ALIGNLIST, which is included in the PP-SELECT command. The order of the sites listed in CP-ALIGNLIST is important, and is as specified in the equipment's documentation. The first site is the alignment site for the first low resolution site, the second item is for the first high resolution site, the third item is the second low resolution site, and the fourth is the second high resolution site.

Table R1-1 ALIGN-DEF-LIST

<i>Align-Name</i>	<i>Coordx</i>	<i>Coordy</i>	<i>Coordsys</i>	<i>Align- Attribute(n)</i>
Coarse1	-60000	-200	M20P	
Fine1	-60020	-205	M20P	
Coarse 2	60000	200	M20P	
Fine2	59980	195	M20P	

R1-4.3 ALIGN-DEF-LIST

L,4

1. <Coarse1>
2. <Fine1>
3. <Coarse 2>
4. <Fine2>

R1-4.4 Using this information, the equipment will go to the nominal M20 location for Coarse, then “find” where it actually is. The offset between the nominal M20 location and the actual M20 location is then used to “find” Fine1. The actual M20 location of Fine1 is saved. The process is then repeated for Coarse 2 and Fine2. The equipment can now determine the M20 to M20P offset from the nominal and actual coordinates.

First, a summary of the data:

xN1=-60020	yN1=-205	Nominal x and y data for the first fine site
xA1=-59800	yA1=-150	Actual x and y data for the first fine site
xN2= 59980	yN2= 195	Nominal x and y data for the second fine site
xA2= 60060	yA2= 175	Actual x and y data for the second fine site

R1-4.5 The equipment first calculates THETA (Θ), using, for example, the formula:

$$\Theta = \tan^{-1} \left[\frac{MA - MN}{1 + MAMN} \right]$$

where MA and MN are, respectively, the slopes of the lines connecting the nominal and actual fine sites, in M20 coordinates, calculated as follows:

$$MA = \left[\frac{yA_2 - yA_1}{xA_2 - xA_1} \right] \quad MN = \left[\frac{yN_2 - yN_1}{xN_2 - xN_1} \right]$$

R1-4.6 The equipment then calculates DELTAX and DELTAY, using, for example, the formulas:

$$DELTAX = \left[\frac{C \sin (\Theta) + D \cos (\Theta)}{(\sin (\Theta))^2 + (\cos (\Theta))^2} \right]$$

$$DELTAY = \left[\frac{C \sin (\Theta) - D \cos (\Theta)}{(\sin (\Theta))^2 + (\cos (\Theta))^2} \right]$$



Where C and D are the adjusted site 1 coordinates in a rotation-adjusted coordinate system calculated, for example, using the formulas:

$$C = yA1 - ((xN1 \sin \Theta) + ((yN1 \cos \Theta)$$

$$D = xA1 - ((xN1 \cos \Theta) - ((yN1 \sin \Theta)$$

R1-4.7 The equipment can also calculate a SCALEFACTOR term to indicate the relative difference between the length of the vector connecting the nominal alignment sites and the length of the vector connecting the actual alignment sites. This can be used, for example, to judge whether there is a problem with the alignment process, since the difference between these two vectors should be small.

$$SCALEFACTOR = \frac{VA}{VN}$$

where VA and VN are the length of the vectors connecting the actual and nominal alignment sites, calculated using the formulas:

$$VN = \sqrt{(yN_2 - yN_1)^2 + (xN_2 - xN_1)^2}$$

$$VA = \sqrt{(yA_2 - yA_1)^2 + (xA_2 - xA_1)^2}$$

M20/M20P COORDINATE SYSTEMS EXAMPLE (EXAGGERATED)

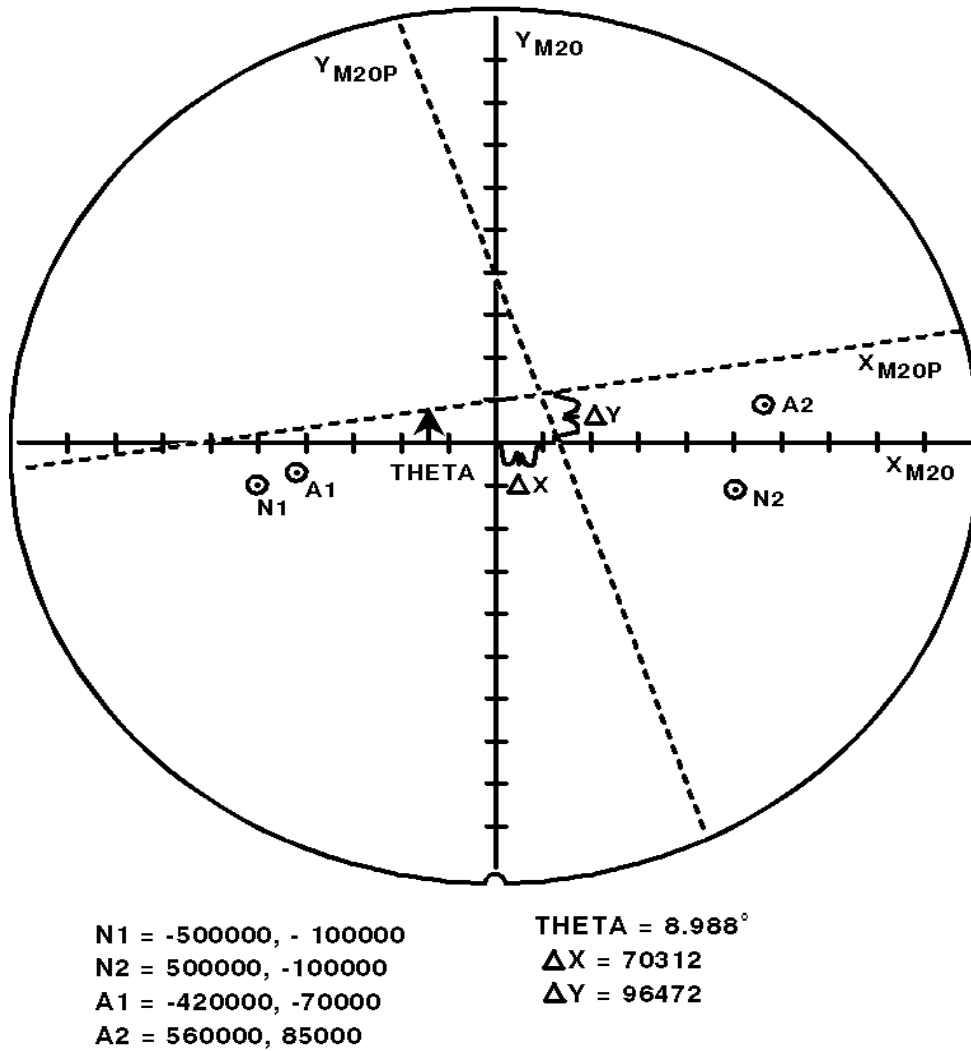


Figure R1-1
M20/M20P Coordinate Systems Example

R1-5 *Multi Measurement Station Metrology Equipment* — MSEM implementation on metrology equipment with multiple measurement stations should follow guidelines established for multi-processing station SEMS such as the Apply and Develop Track Specific Equipment Model (SEMATECH ID #: 95113021AENG Title: SEMATECH Apply/Develop Track Specific Equipment Model (ADTSEM), Version 0.8, January 4, 1996).

NOTICE:. SEMI makes no warranties or representations as to the suitability of the standards set forth herein for any particular application. The determination of the suitability of the standard is solely the responsibility of the user. Users are cautioned to refer to manufacturer's instructions, product labels, product data sheets, and other relevant literature respecting any materials mentioned herein. These standards are subject to change without notice.

The user's attention is called to the possibility that compliance with this standard may require use of copyrighted material or of an invention covered by patent rights. By publication of this standard, SEMI takes no position respecting the validity of any patent rights or copyrights asserted in connection with any item mentioned in this standard. Users of this standard are expressly advised that determination of any such patent rights or copyrights, and the risk of infringement of such rights, are entirely their own responsibility.